# math-foundations-ai999x: capability guide

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Every capability math-foundations-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

## Quick start

```
import math_foundations_ai999x

result = math_foundations_ai999x.solve('quadratic_roots', a=1, b=-5, c=6)
result.show()
# x1 = 2, x2 = 3, discriminant = 1
```

## By topic

| topic | capabilities |
|---|---|
| Algebra / absolute value | Absolute value inequality \|ax + b\| compared with c |
| Algebra / factoring | Factor a quadratic over the integers |
| Algebra / inequalities | Quadratic inequality |
| Algebra / polynomial division | Synthetic division and the remainder theorem, Polynomial long division |
| Algebra / polynomial equations | Rational roots of a polynomial (rational root theorem) |
| Algebra / radical equations | Radical equation sqrt(ax + b) = cx + d |
| Algebra / rational equations | Rational equation a/(x + b) = c/(x + d) |
| Algebra / rational expressions | Partial fractions (two distinct linear factors) |
| Algebra / sequences and series | Sum of an infinite geometric series |
| Algebra / systems of linear equations | System of three linear equations |
| Arithmetic / fractions | Arithmetic with fractions |
| Arithmetic / order of operations | Evaluate an arithmetic expression (order of operations) |
| Arithmetic / percentages | Percent of a number, Whole from a part and its percentage, What percent one number is of another, Percent change (increase or decrease) |

| topic | capabilities |
|---|---|
| Arithmetic / ratios and proportions | Divide a quantity in a given ratio, Solve a proportion, Unit rates and the better buy |
| Arithmetic / rounding | Round to decimal places, Round to significant figures |
| Arithmetic / scientific notation | Scientific notation |
| Complex numbers | Product and modulus of complex numbers |
| Complex numbers / arithmetic | Arithmetic with complex numbers |
| Complex numbers / De Moivre's theorem | Power of a complex number (De Moivre) |
| Complex numbers / polar form | Modulus, argument and polar form of a complex number |
| Complex numbers / roots | The n-th roots of a complex number |
| Conic sections | Centre and radius of a circle from its general equation |
| Coordinate plane | Distance between two points |
| Counting and probability | Combinations and permutations |
| Exponential functions | Compound interest |
| Exponential models | Doubling time of exponential growth |
| Exponentials and logarithms / exponential equations | Exponential equation a*b^(cx + d) = k |
| Exponentials and logarithms / growth and decay | Continuous exponential growth or decay, Amount remaining after radioactive decay (half-life), Doubling time at a percentage growth rate |
| Exponentials and logarithms / laws of logarithms | Laws of logarithms (product, quotient, power) |
| Exponentials and logarithms / logarithmic equations | Logarithmic equation log b(cx + d) = k |
| Exponentials and logarithms / logarithms | Evaluate a logarithm |
| Exponents and logarithms | Solution of an exponential equation |
| Factoring | Factoring a monic quadratic |
| Functions | Average rate of change of a function between two inputs, Composition of two linear functions |
| Functions / composition | Value of a composite function |
| Functions / domain | Domain of a square-root function |
| Functions / evaluation | Evaluate a function at a number |

| topic | capabilities |
| --- | --- |
| Functions / inverse functions | Inverse of a linear function |
| Functions / lines | Intercepts and slope of a line, Distance from a point to a line |
| Functions / piecewise functions | Evaluate a piecewise function |
| Functions / transformations | Image of a point under a graph transformation |
| Inequalities | Solution of a linear inequality |
| Linear and absolute value inequalities | Solution set of \|x - c\| > d, Solution interval of \|x - c\| < d |
| Linear equations | Root of a linear equation |
| Linear functions | Average rate of change between two points, Y-intercept of a line from its slope and a point, Line through two points, Slope of a perpendicular line |
| Logarithms | Solution of a basic logarithmic equation |
| Matrices | Determinant and inverse of a 2 by 2 matrix |
| Matrices and linear systems | Determinant of 2x2 matrix, Matrix scalar multiple definition |
| Other equations | Solutions of an absolute value equation |
| Polynomial equations / cubic and quartic equations | Roots of a cubic equation, Roots of a quartic equation |
| Polynomial equations / quadratic equations | Roots of a quadratic equation, Discriminant of a quadratic |
| Polynomials | Product of two binomials |
| Precalculus / binomial theorem | Coefficient of a term in a binomial expansion |
| Precalculus / conic sections | Focus and directrix of a parabola, Foci, axes and eccentricity of an ellipse, Foci, eccentricity and asymptotes of a hyperbola |
| Precalculus / polar coordinates | Polar to rectangular coordinates, Rectangular to polar coordinates |
| Precalculus / vectors | Dot product and angle between two vectors, Magnitude and direction of a vector |
| Proofs and derivations | Area of rectangle, Associative property of addition, Cramer rule x solution, Determinant cramer two variables, Difference of cubes factorization, Difference of squares, Distributive property verification, Elimination method identity, Elimination solution two variables, Linear equation solution formula, Partial fraction decomposition linear factors, Perfect square trinomial, Product of sums, Quadratic vertex form conversion, Radical equation squaring, Square of difference, Substitution method identity, Substitution solution two variables, Sum of cubes factorization |
| Quadratic functions | Vertex form of a quadratic function |
| Rational expressions | Add rational expressions with same denominator, Multiply rational expressions |
| Rational functions | Asymptotes of a linear rational function |
| Real numbers and algebraic expressions | Perimeter of a rectangle |
| Sequences | Term and partial sum of an arithmetic sequence, Term and partial sum of a geometric sequence |
| Systems of equations | Solution of two linear equations in two unknowns |

| topic | capabilities |
|---|---|
| The coordinate plane and graphs | Midpoint of a line segment |
| Trigonometry / angle measure | Degrees to radians, Radians to degrees |
| Trigonometry / identities | Trigonometric values from one value and the quadrant |
| Trigonometry / oblique triangles | Law of sines (two angles and a side), Law of sines, the ambiguous case (two sides and a non-included angle), Law of cosines (two sides and the included angle), Law of cosines (three sides) |
| Trigonometry / right triangles | Right triangle from its two legs, Right triangle from a leg and the hypotenuse, Right triangle from an acute angle and a side |
| Trigonometry / triangle area | Area of a triangle from its three sides (Heron) |
| Trigonometry / trigonometric equations | Solve a trigonometric equation over one full turn |
| Trigonometry / unit circle | Exact trigonometric values on the unit circle |

## Outcomes

| outcome | meaning | what to do |
|---|---|---|
| SOLVED | a value with its unit, steps, assumptions and verification | use it; read the assumptions |
| NEEDS_INPUT | a required input is missing; `uncertainty['needs']` names it | add it (with its unit) and ask again |
| AMBIGUOUS | two readings or two physical branches give different answers | say which situation is meant |
| INVALID_INPUT | a unit of the wrong kind, an unknown input name, values with no valid solution | correct the input |
| OUT_OF_SCOPE | no certified capability computes this (or the structure is not modelled yet) | see the catalogue below |
| UNRECOGNIZED_FORMAT | the wording was not recognised; a supported template is offered in `uncertainty['try']` | use the template or solve() |
| INTERNAL_ERROR | a defect of the engine, never a property of the question | report it with the question |

## Result fields

| field | meaning |
|---|---|
| value, unit | the result in SI units |
| values | every output of a procedure (x1, x2, discriminant ...) |
| outcome | one of the seven outcomes above |
| steps | each equation used, its inputs (with where each came from) and its result |
| assumptions | conditions and declared defaults the answer relies on |
| verification | residual of every equation, dimensional consistency, independent checks |

| field | meaning |
|---|---|
| latex() | the derivation as LaTeX |
| show() | renders in a notebook, prints in a terminal |

## Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

## Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

| id | operation | inputs | outputs |
|---|---|---|---|
| quadratic_roots | Roots of a quadratic equation | a, b, c | x1, x2, discriminant |
| quadratic_discriminant | Discriminant of a quadratic | a, b, c | discriminant, nature |
| counting_choices | Combinations and permutations | n, k | combinations, permutations |
| cubic_roots | Roots of a cubic equation | a, b, c, d | x1, x2, x3 |
| quartic_roots | Roots of a quartic equation | a, b, c, d, e | x1, x2, x3, x4 |
| fraction_arithmetic | Arithmetic with fractions | a, b, c, d, op | numerator, denominator, value |
| percent_of_number | Percent of a number | p, x | result |
| percent_of_what | Whole from a part and its percentage | part, p | whole |
| what_percent | What percent one number is of another | part, whole | percent |
| percent_change | Percent change (increase or decrease) | old, new | percent_change, difference, direction |
| ratio_split | Divide a quantity in a given ratio | total, a, b, c | part1, part2, part3 |
| proportion_solve | Solve a proportion | p, q, r, position | x |
| unit_rate_compare | Unit rates and the better buy | n1, c1, n2, c2 | unit_rate_1, unit_rate_2, better |
| round_decimal_places | Round to decimal places | x, n | rounded |
| round_significant_figures | Round to significant figures | x, n | rounded |
| scientific_notation | Scientific notation | x | mantissa, exponent |

| id | operation | inputs | outputs |
|---|---|---|---|
| order_of_operations | Evaluate an arithmetic expression (order of operations) | expression | value, numerator, denominator |
| linear_system_3x3 | System of three linear equations | augmented | x, y, z |
| abs_linear_inequality | Absolute value inequality \|ax + b\| compared with c | a, b, c, rel | lower, upper, region, solution |
| rational_equation | Rational equation a/(x + b) = c/(x + d) | a, b, c, d | x |
| radical_equation | Radical equation sqrt(ax + b) = cx + d | a, b, c, d | count, solutions, extraneous |
| synthetic_division | Synthetic division and the remainder theorem | coefficients, r | quotient, remainder |
| polynomial_long_division | Polynomial long division | dividend, divisor | quotient, remainder |
| factor_quadratic_integer | Factor a quadratic over the integers | a, b, c | g, p, q, r, s, factored |
| quadratic_inequality | Quadratic inequality | a, b, c, rel | r1, r2, region, solution |
| infinite_geometric_series | Sum of an infinite geometric series | a, r | sum |
| partial_fractions_linear | Partial fractions (two distinct linear factors) | p, q, r1, r2 | A, B |
| rational_roots | Rational roots of a polynomial (rational root theorem) | coefficients | count, roots, roots_exact |
| evaluate_function | Evaluate a function at a number | expression, x | value, numerator, denominator |
| composite_function_value | Value of a composite function | f, g, x, order | inner, value |
| domain_sqrt_over_linear | Domain of a square-root function | a, b, c, d, root_in_denominator | boundary, side, excluded, domain |
| inverse_linear_function | Inverse of a linear function | a, b, y | inv_slope, inv_intercept, value |
| piecewise_evaluate | Evaluate a piecewise function | e1, left_rel, c, e2, x | value, piece |
| graph_transformation_point | Image of a point under a graph transformation | x0, y0, A, B, h, k | x_new, y_new |
| line_intercepts | Intercepts and slope of a line | a, b, c | x_intercept, y_intercept, slope |
| distance_point_to_line | Distance from a point to a line | x0, y0, a, b, c | distance |
| log_evaluate | Evaluate a logarithm | b, x | value |
| log_laws_combine | Laws of logarithms (product, quotient, power) | u, v, p, q | value |
| continuous_growth | Continuous exponential growth or decay | n0, k, t | amount |

| id | operation | inputs | outputs |
|---|---|---|---|
| half_life_remaining | Amount remaining after radioactive decay (half-life) | n0, half_life, t | remaining, fraction |
| doubling_time_percent | Doubling time at a percentage growth rate | r | doubling_time, rule_of_70_estimate |
| exponential_equation_general | Exponential equation a*b^(cx + d) = k | a, b, c, d, k | x |
| log_equation_linear_argument | Logarithmic equation log_b(cx + d) = k | b, c, d, k | x |
| deg_to_rad | Degrees to radians | deg | radians, pi_multiple, exact |
| rad_to_deg | Radians to degrees | rad | degrees |
| right_triangle_legs | Right triangle from its two legs | a, b | c, A, B |
| right_triangle_leg_hypotenuse | Right triangle from a leg and the hypotenuse | a, c | b, A, B |
| right_triangle_angle_side | Right triangle from an acute angle and a side | A, side, which | opposite, adjacent, hypotenuse, B |
| law_of_sines_aas | Law of sines (two angles and a side) | A, B, a | C, b, c |
| law_of_sines_ssa | Law of sines, the ambiguous case (two sides and a non-included angle) | a, b, A | count, B1, C1, c1, B2, C2, c2 |
| law_of_cosines_sas | Law of cosines (two sides and the included angle) | b, c, A | a, B, C |
| law_of_cosines_sss | Law of cosines (three sides) | a, b, c | A, B, C |
| heron_area | Area of a triangle from its three sides (Heron) | a, b, c | area, s |
| unit_circle_exact | Exact trigonometric values on the unit circle | deg | sin, cos, tan, sin_exact, cos_exact, tan_exact, reference_angle |
| trig_equation_interval | Solve a trigonometric equation over one full turn | func, k, n, unit | count, solutions |
| trig_values_from_one | Trigonometric values from one value and the quadrant | func, value, quadrant | sin, cos, tan, sin2, cos2 |
| vector_dot_angle | Dot product and angle between two vectors | u, v | dot, mag_u, mag_v, angle_deg |
| vector_magnitude_direction | Magnitude and direction of a vector | x, y | magnitude, direction_deg |
| polar_to_rectangular | Polar to rectangular coordinates | r, theta, unit | x, y |
| rectangular_to_polar | Rectangular to polar coordinates | x, y | r, theta_deg, theta_rad |
| parabola_focus_directrix | Focus and directrix of a parabola | orientation, m, h, k | p, focus_x, focus_y, directrix, vertex_x, vertex_y |
| ellipse_parameters | Foci, axes and eccentricity of an ellipse | h, k, A, B | a, b, c, eccentricity, f1x, f1y, f2x, f2y, orientation |

| id | operation | inputs | outputs |
|---|---|---|---|
| hyperbola_parameters | Foci, eccentricity and asymptotes of a hyperbola | h, k, A, B, orientation | a, b, c, eccentricity, asymptote_slope, f1x, f1y, f2x, f2y |
| binomial_coefficient_term | Coefficient of a term in a binomial expansion | a, b, n, k | coefficient, binomial |
| complex_arithmetic | Arithmetic with complex numbers | a, b, c, d, op | re, im |
| complex_modulus_argument | Modulus, argument and polar form of a complex number | a, b | modulus, argument_deg, argument_rad, polar |
| de_moivre_power | Power of a complex number (De Moivre) | a, b, n | re, im, modulus, argument_deg |
| complex_nth_roots | The n-th roots of a complex number | a, b, n | modulus, count, roots |
| abs_value_inequality_lower_bound | Solution set of \|x - c\| > d | c, d | left_limit, right_limit |
| abs_value_inequality_upper_bound | Solution interval of \|x - c\| < d | c, d | lower, upper |
| absolute_value_equation | Solutions of an absolute value equation | a, b, c | x1, x2 |
| add_rational_expressions_same_denom | Add rational expressions with same denominator | num1, num2, den | sum_num, sum_den |
| arithmetic_sequence | Term and partial sum of an arithmetic sequence | a1, d, n | an, sn |
| average_rate_of_change_definition | Average rate of change of a function between two inputs | x_start, f_x_start, x_end, f_x_end | arc |
| average_rate_of_change_linear | Average rate of change between two points | x1, y1, x2, y2 | m |
| binomial_product | Product of two binomials | a, b, c, d | p2, p1, p0 |
| circle_from_general_form | Centre and radius of a circle from its general equation | dd, ee, ff | h, k, rad |
| complex_product_modulus | Product and modulus of complex numbers | a, b, c, d | re, im, modulus |
| compose_linear_functions | Composition of two linear functions | a, b, c, d | m, k |
| compound_interest | Compound interest | principal, r, n, t | amount |
| convert_to_slope_intercept | y-intercept of a line from its slope and a point | m, x1, y1 | b |
| determinant_2x2_formula | Determinant of 2x2 matrix | a, b, c, d | det |
| distance_between_points | Distance between two points | x1, y1, x2, y2 | dist |
| doubling_time | Doubling time of exponential growth | k | td |
| exponential_equation | Solution of an exponential equation | b, y | x |

| id | operation | inputs | outputs |
|---|---|---|---|
| factor_monic_quadratic | Factoring a monic quadratic | b, c | p, q |
| geometric_sequence | Term and partial sum of a geometric sequence | g1, q, n | gn, sn |
| line_through_two_points | Line through two points | x1, y1, x2, y2 | m, k |
| linear_equation_root | Root of a linear equation | a, b, c | x |
| linear_inequality | Solution of a linear inequality | a, b, c | x0, s |
| linear_system_2x2 | Solution of two linear equations in two unknowns | a, b, c, d, e, f | x, y |
| logarithmic_equation | Solution of a basic logarithmic equation | b, c | x |
| matrix_2x2_inverse | Determinant and inverse of a 2 by 2 matrix | a, b, c, d | det, i11, i12, i21, i22 |
| matrix_scalar_multiple_definition | Matrix scalar multiple definition | s, a11, a12, a21, a22 | b11, b12, b21, b22 |
| midpoint_of_segment | Midpoint of a line segment | x1, y1, x2, y2 | mx, my |
| multiply_rational_expressions | Multiply rational expressions | num1, den1, num2, den2 | prod_num, prod_den |
| perimeter_of_rectangle | Perimeter of a rectangle | w, h | p |
| perpendicular_lines_slope_product | Slope of a perpendicular line | m | m_perp |
| rational_function_asymptotes | Asymptotes of a linear rational function | a, b, c, d | va, ha |
| vertex_form | Vertex form of a quadratic function | a, b, c | h, k |

## Roots of a quadratic equation — `quadratic_roots`

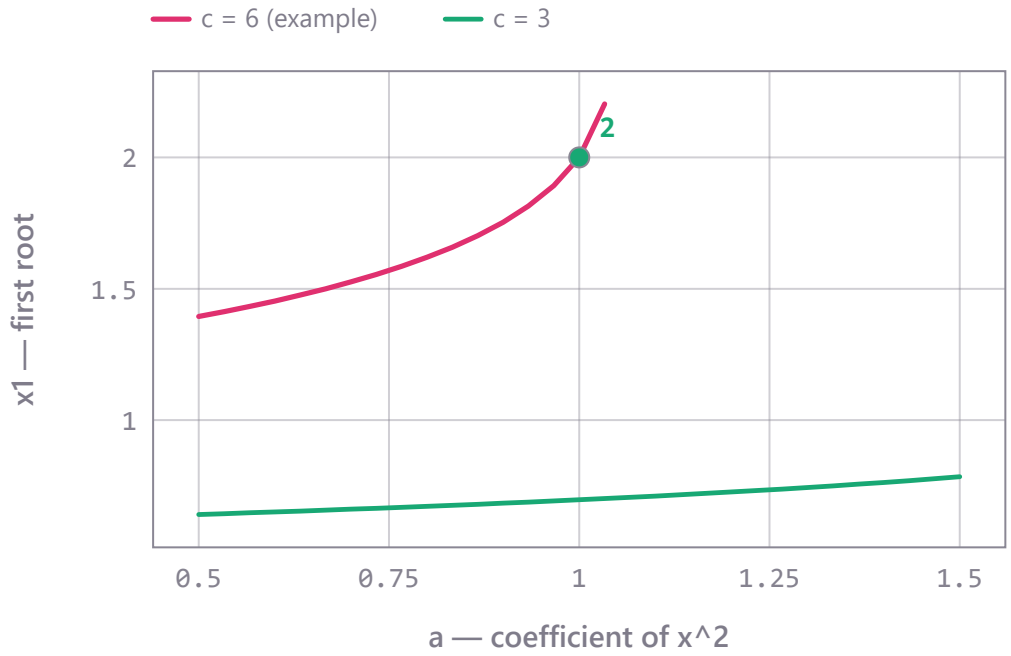
Solve $ax^2 + bx + c = 0$ for x: both roots, real or complex.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of $x^2$ | $a \neq 0$ |
| `b` | coefficient of x | |
| `c` | constant term | |

| output | meaning |
|---|---|
| `x1` | first root (the smaller real root, or the one with positive imaginary part) |
| `x2` | second root |
| `discriminant` | $b^2 - 4ac$ |

- **Method:** quadratic formula in a numerically stable form (q = -(b + sign(b)*sqrt(b^2 - 4ac))/2, $x = q/a$ and c/q)
- **Assumes:** $a \neq 0$ (otherwise the equation is linear); real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x1 (first root (the smaller real root, or the one with positive imaginary part)) against a (coefficient of x^2) from 0.5 to 1.5. One curve for each value of c (constant term): 6, 3. Other inputs as in the worked example: b = -5. 17 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, x1 = 2.

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=-5, c=6)
# -> x1 = 2, x2 = 3, discriminant = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=2, b=3, c=-2)
# -> x1 = -2, x2 = 0.5, discriminant = 25   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=2, c=1)
# -> x1 = -1, x2 = -1, discriminant = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=0, c=4)
# -> x1 = 0 + 2i, x2 = 0 - 2i, discriminant = -16   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_roots', a=1, b=-100000000.0, c=1)
# -> x1 = 1e-08, x2 = 1e+08, discriminant = 1e+16   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve x^2 - 5x + 6 = 0. | SOLVED | x1 = 2, x2 = 3, discriminant = 1 |
| Find the roots of 2x^2 + 3x - 2 = 0. | SOLVED | x1 = -2, x2 = 0.5, discriminant = 25 |

| question | outcome | result |
|---|---|---|
| Solve the quadratic with a = 1, b = -5, c = 6. | SOLVED | x1 = 2, x2 = 3, discriminant = 1 |
| Solve x^2 + 4 = 0. | SOLVED | x1 = 0 + 2i, x2 = 0 - 2i, discriminant = -16 |
| Solve 3x^2 = 12. | SOLVED | x1 = -2, x2 = 2, discriminant = 144 |
| Solve x^5 - 1 = 0. | OUT_OF_SCOPE | Out of scope: a polynomial of degree 5 or higher (equations up to degree 4 are supported). |
| Solve the quadratic with a = 0, b = 2, c = 1. | INVALID_INPUT | Roots of a quadratic equation: a must not be 0: with a = 0 the equation is linear, not quadratic. |
| Solve the quadratic with a = 1, b = 2. | NEEDS_INPUT | quadratic_roots needs c (constant term). Example: solve('quadratic_roots', a=1, b=-5, c=6). |

### Discriminant of a quadratic — `quadratic_discriminant`

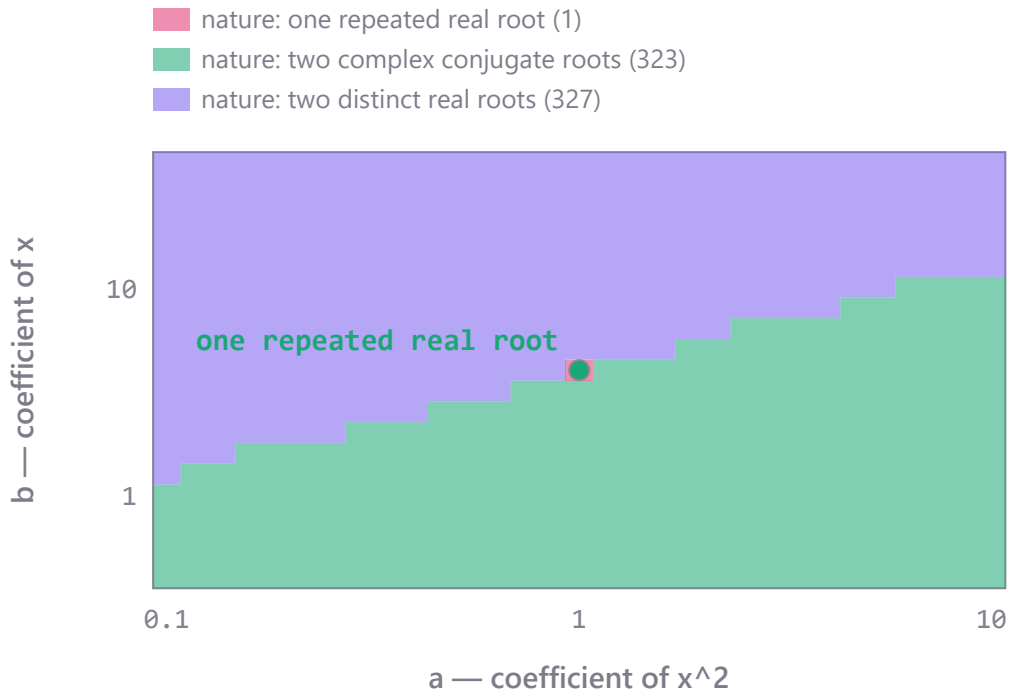
The discriminant $b^2 - 4ac$ of $ax^2 + bx + c$ and the nature of the roots it implies.

$$\Delta = b^2 - 4ac$$

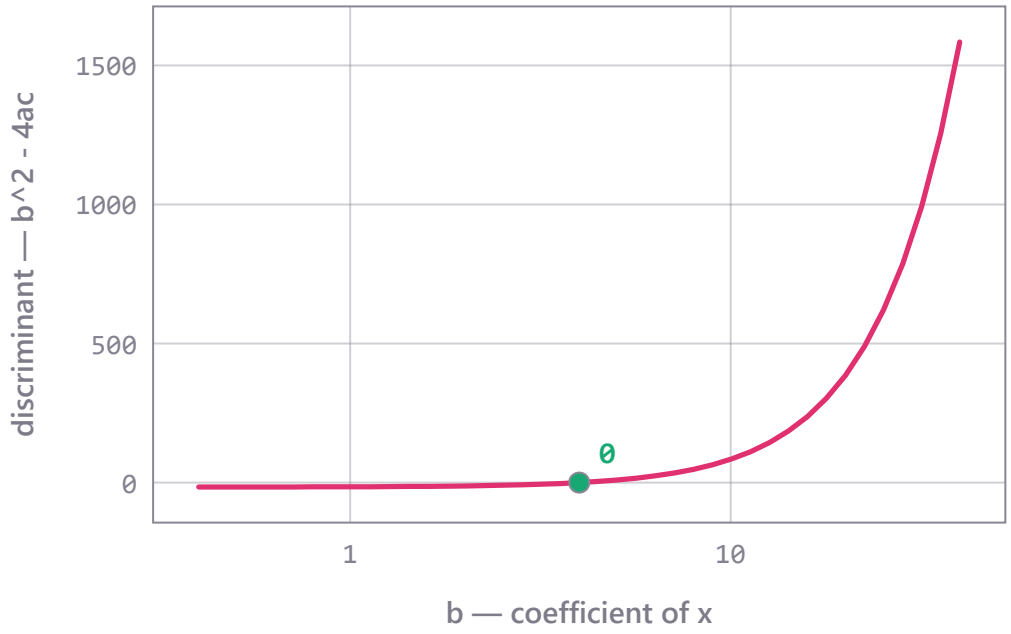
| input | meaning | condition |
|---|---|---|
| `a` | coefficient of $x^2$ | $a \neq 0$ |
| `b` | coefficient of x | |
| `c` | constant term | |

| output | meaning |
|---|---|
| `discriminant` | $b^2 - 4ac$ |
| `nature` | two distinct real / one repeated real / two complex conjugate roots |

- **Method:** direct evaluation of $b^2 - 4ac$ and its sign
- **Assumes:** $a \neq 0$; real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




two distinct real / one repeated real / two complex conjugate roots (nature) over a (coefficient of x^2) from 0.1 to 10 and b (coefficient of x) from 0.4 to 40 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned — one repeated real root: 1 cells, two complex conjugate roots: 323 cells, two distinct real roots: 327 cells. Other inputs as in the worked example: c = 4. Marked: the worked example, a = 1, b = 4 → one repeated real root.




discriminant (b^2 - 4ac) against b (coefficient of x) from 0.4 to 40 on a logarithmic axis. Other inputs as in the worked example: a = 1, c = 4. 41 points computed by math-foundations-ai999x; marked: the worked example, b = 4, discriminant = 0.

```
math_foundations_ai999x.solve('quadratic_discriminant', a=1, b=4, c=4)
# -> discriminant = 0, nature = one repeated real root   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_discriminant', a=2, b=1, c=-3)
# -> discriminant = 25, nature = two distinct real roots   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_discriminant', a=1, b=1, c=1)
# -> discriminant = -3, nature = two complex conjugate roots   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| What is the discriminant of x^2 + 4x + 4? | SOLVED | discriminant = 0, nature = one repeated real root |
| Find the discriminant with a = 2, b = 1, c = -3. | SOLVED | discriminant = 25, nature = two distinct real roots |

## Combinations and permutations — `counting_choices`

The number of ways to choose k items from n without order (combinations) and with order (permutations).

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \quad P(n,k) = \frac{n!}{(n-k)!}$$

| input | meaning | condition |
|---|---|---|
| `n` | number of items to choose from (a whole number) | |
| `k` | number of items chosen (a whole number, k <= n) | |

| output | meaning |
|---|---|
| `combinations` | C(n, k) = n! / (k! (n - k)!) |
| `permutations` | P(n, k) = n! / (n - k)! |

- **Method:** exact integer arithmetic (math.comb, math.perm)
- **Assumes:** n and k whole numbers; $k \le n$; $n \le 170$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




combinations (C(n, k) = n! / (k! (n - k)!)) against n (number of items to choose from (a whole number)) from 1 to 20. Other inputs as in the worked example: k = 3. 18 points computed by math-foundations-ai999x; marked: the worked example, n = 10, combinations = 120.

```
math_foundations_ai999x.solve('counting_choices', n=10, k=3)
# -> combinations = 120, permutations = 720   (SOLVED)
```

```
math_foundations_ai999x.solve('counting_choices', n=5, k=5)
# -> combinations = 1, permutations = 120   (SOLVED)
```

```
math_foundations_ai999x.solve('counting_choices', n=52, k=5)
# -> combinations = 2.59896e+06, permutations = 3.11875e+08   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| How many ways are there to choose 3 books from 10? | SOLVED | combinations = 120, permutations = 720 |
| Compute C(6, 2). | SOLVED | combinations = 15, permutations = 30 |
| How many ways are there to choose 7 books from 4? | INVALID_INPUT | Combinations and permutations: k must not exceed n. |
| Compute C(1000, 3). | INVALID_INPUT | Combinations and permutations: n must be at most 170 (the result would not fit in a floating-point number). |

## Roots of a cubic equation — `cubic_roots`

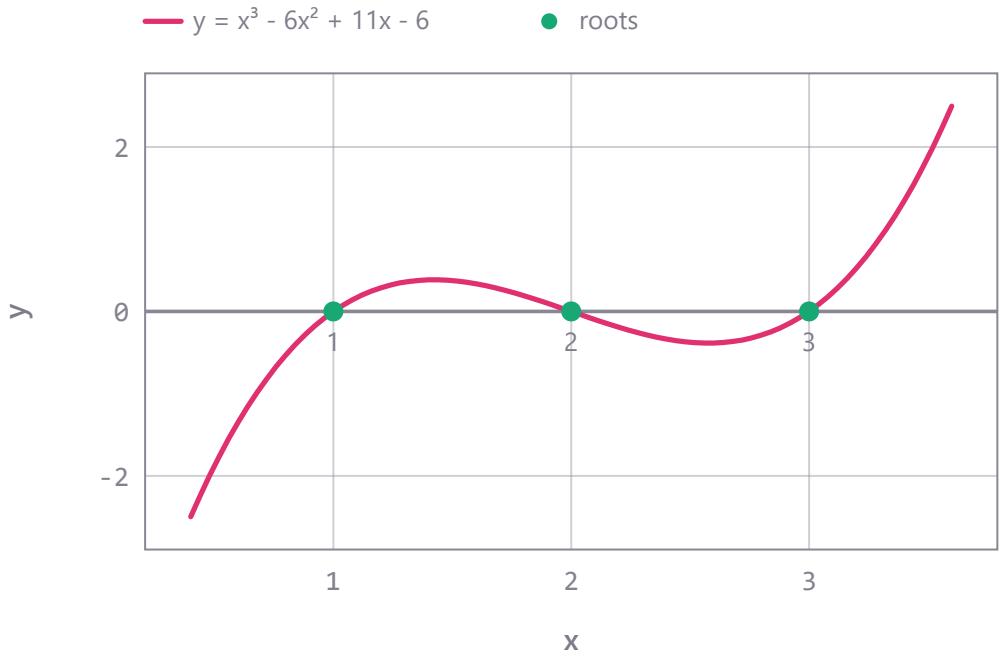
Solve $ax^3 + bx^2 + cx + d = 0$ for x: all three roots, real or complex.

$$ax^3 + bx^2 + cx + d = 0$$

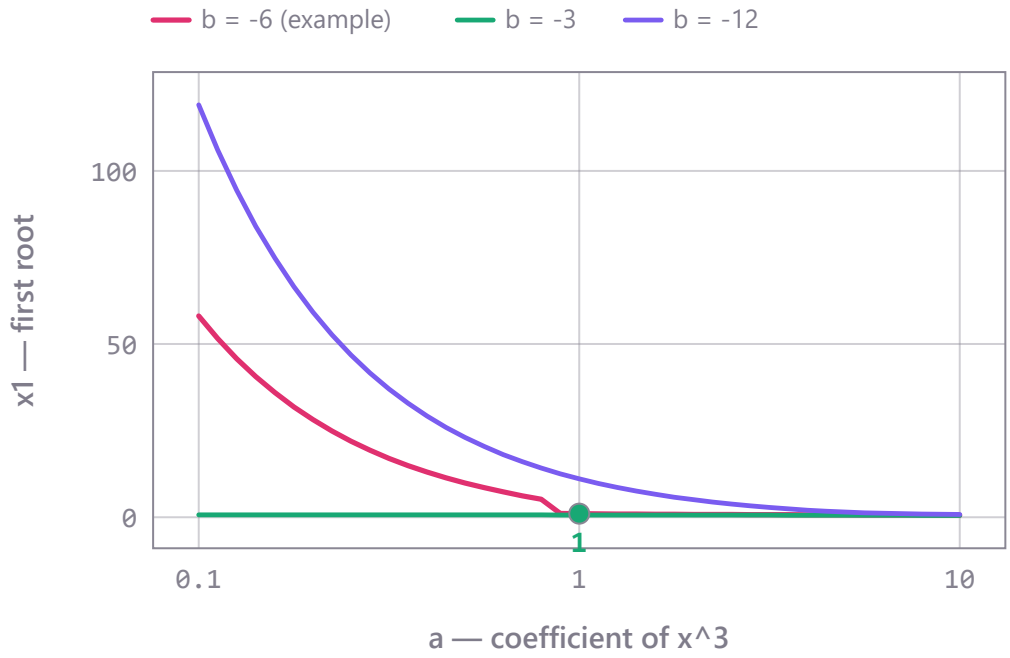
| input | meaning | condition |
|---|---|---|
| `a` | coefficient of $x^3$ | $a \neq 0$ |
| `b` | coefficient of $x^2$ | |
| `c` | coefficient of x | |
| `d` | constant term | |

| output | meaning |
|---|---|
| `x1` | first root (real roots ascending, then complex roots) |
| `x2` | second root |
| `x3` | third root |

- **Method:** high-precision polynomial root finding (Durand-Kerner, 30 digits), each root substituted back
- **Assumes:** $a \neq 0$ (otherwise the equation is not cubic); real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The cubic y = x³ − 6x² + 11x − 6, evaluated by the package at 81 points, and its three real roots as the package returns them: x = 1, 2 and 3.




x1 (first root (real roots ascending, then complex roots)) against a (coefficient of x^3) from 0.1 to 10 on a logarithmic axis. One curve for each value of b (coefficient of x^2): -6, -3, -12. Other inputs as in the worked example: c = 11, d = -6. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, x1 = 1.

```
math_foundations_ai999x.solve('cubic_roots', a=1, b=-6, c=11, d=-6)
# -> x1 = 1, x2 = 2, x3 = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('cubic_roots', a=2, b=-4, c=-22, d=24)
# -> x1 = -3, x2 = 1, x3 = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('cubic_roots', a=1, b=0, c=0, d=-1)
# -> x1 = 1, x2 = -0.5 + 0.866025i, x3 = -0.5 - 0.866025i   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve x^3 - 6x^2 + 11x - 6 = 0. | SOLVED | x1 = 1, x2 = 2, x3 = 3 |
| Solve x^3 - 1 = 0. | SOLVED | x1 = 1, x2 = -0.5 + 0.866025i, x3 = -0.5 - 0.866025i |
| Find the roots of 2x^3 - 4x^2 - 22x + 24 = 0. | SOLVED | x1 = -3, x2 = 1, x3 = 4 |
| Solve the cubic with a = 1, b = 0, c = 0, d = -1. | SOLVED | x1 = 1, x2 = -0.5 + 0.866025i, x3 = -0.5 - 0.866025i |
| Solve the cubic with a = 0, b = 1, c = 2, d = 1. | INVALID_INPUT | Roots of a cubic equation: a must not be 0: with a = 0 the equation has a lower degree. |
| Solve the cubic with a = 1, b = 2. | NEEDS_INPUT | cubic_roots needs c, d (coefficient of x; constant term). Example: solve('cubic_roots', a=1, b=-6, c=11, d=-6). |

## Roots of a quartic equation — `quartic_roots`

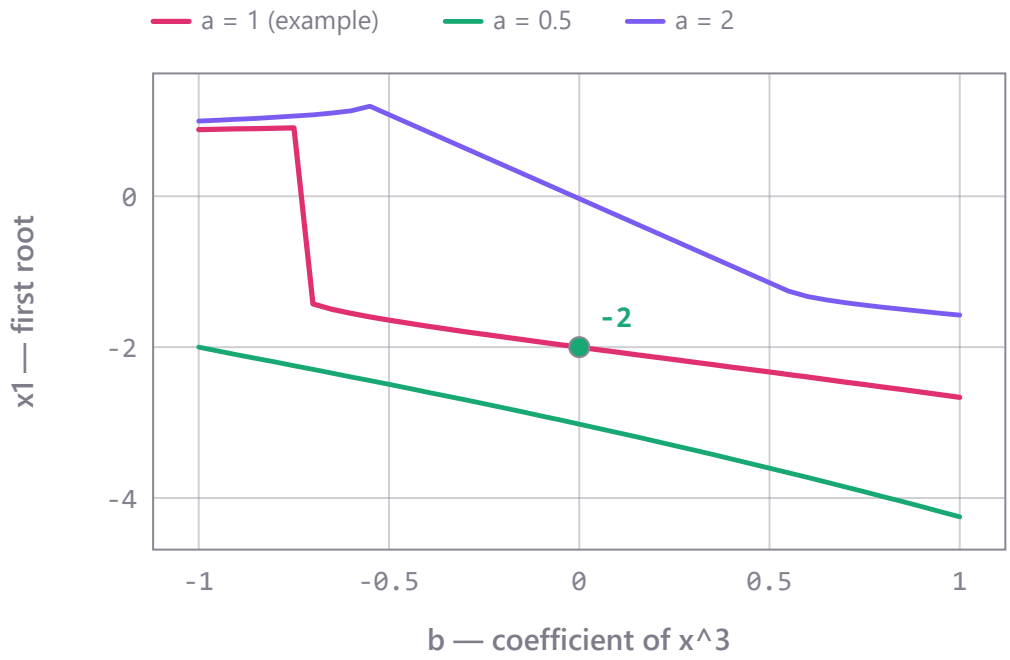
Solve $ax^4 + bx^3 + cx^2 + dx + e = 0$ for x: all four roots, real or complex.

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

| input | meaning | condition |
|---|---|---|
| a | coefficient of $x^4$ | $a \neq 0$ |
| b | coefficient of $x^3$ | |
| c | coefficient of $x^2$ | |
| d | coefficient of x | |
| e | constant term | |

| output | meaning |
|---|---|
| x1 | first root (real roots ascending, then complex roots) |
| x2 | second root |
| x3 | third root |
| x4 | fourth root |

- **Method:** high-precision polynomial root finding (Durand-Kerner, 30 digits), each root substituted back
- **Assumes:** $a \neq 0$ (otherwise the equation is not quartic); real coefficients
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x1 (first root (real roots ascending, then complex roots)) against b (coefficient of x^3) from -1 to 1. One curve for each value of a (coefficient of x^4): 1, 0.5, 2. Other inputs as in the worked example: c = -5, d = 0, e = 4. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 0, x1 = -2.

```
math_foundations_ai999x.solve('quartic_roots', a=1, b=0, c=-5, d=0, e=4)
# -> x1 = -2, x2 = -1, x3 = 1, x4 = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('quartic_roots', a=1, b=0, c=0, d=0, e=-1)
# -> x1 = -1, x2 = 1, x3 = 0 + 1i, x4 = 0 - 1i   (SOLVED)
```

```
math_foundations_ai999x.solve('quartic_roots', a=1, b=-10, c=35, d=-50, e=24)
# -> x1 = 1, x2 = 2, x3 = 3, x4 = 4   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve x^4 - 5x^2 + 4 = 0. | SOLVED | x1 = -2, x2 = -1, x3 = 1, x4 = 2 |
| Find the roots of x^4 - 1 = 0. | SOLVED | x1 = -1, x2 = 1, x3 = 0 + 1i, x4 = 0 - 1i |
| Solve the quartic with a = 1, b = 0, c = -5, d = 0, e = 4. | SOLVED | x1 = -2, x2 = -1, x3 = 1, x4 = 2 |
| Solve the quartic with a = 1, b = 2. | NEEDS_INPUT | quartic_roots needs c, d, e (coefficient of x^2; coefficient of x; constant term). Example: solve('quartic_roots', a=1, |

## Arithmetic with fractions — `fraction_arithmetic`

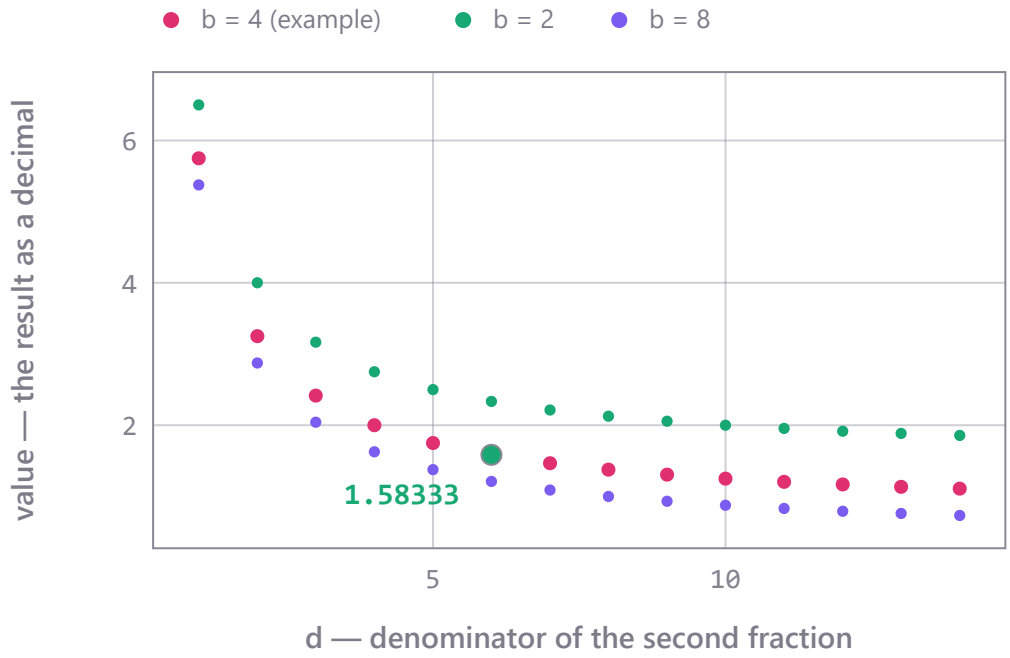
Add, subtract, multiply or divide two fractions a/b and c/d; the result in lowest terms.

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm cb}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

| input | meaning | condition |
|---|---|---|
| `a` | numerator of the first fraction | |
| `b` | denominator of the first fraction (not 0) | |
| `c` | numerator of the second fraction | |
| `d` | denominator of the second fraction (not 0) | |
| `op` | the operation | |

| output | meaning |
|---|---|
| `numerator` | numerator of the result in lowest terms |
| `denominator` | denominator of the result (positive) |
| `value` | the result as a decimal |

- **Method:** common denominator (a/b ± c/d = (ad ± cb)/bd), product of numerators over product of denominators, division as multiplication by the reciprocal; then divide by the greatest common divisor
- **Assumes:** whole-number numerators and denominators; denominators not 0; no division by 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




value (the result as a decimal) against d (denominator of the second fraction (not 0)) from 1 to 14. One curve for each value of b (denominator of the first fraction (not 0)): 4, 2, 8. Other inputs as in the worked example: a = 3, c = 5, op = +. 14 points per curve computed by math-foundations-ai999x; marked: the worked example, d = 6, value = 1.58333.

```
math_foundations_ai999x.solve('fraction_arithmetic', a=3, b=4, c=5, d=6, op='+')
# -> numerator = 19, denominator = 12, value = 1.58333   (SOLVED)
```

```
math_foundations_ai999x.solve('fraction_arithmetic', a=2, b=3, c=9, d=10, op='*')
# -> numerator = 3, denominator = 5, value = 0.6   (SOLVED)
```

```
math_foundations_ai999x.solve('fraction_arithmetic', a=1, b=2, c=3, d=4, op='-')
# -> numerator = -1, denominator = 4, value = -0.25   (SOLVED)
```

```
math_foundations_ai999x.solve('fraction_arithmetic', a=3, b=4, c=9, d=8, op='/')
# -> numerator = 2, denominator = 3, value = 0.666667   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Compute 3/4 + 5/6 as a fraction in lowest terms. | SOLVED | numerator = 19, denominator = 12, value = 1.58333 |
| What is 2/3 times 9/10? | SOLVED | numerator = 3, denominator = 5, value = 0.6 |
| Calculate 7/8 - 1/6. | SOLVED | numerator = 17, denominator = 24, value = 0.708333 |
| Find 5/6 divided by 0/3. | INVALID_INPUT | Arithmetic with fractions: division by the fraction 0. |
| Compute 1/0 + 1/2. | INVALID_INPUT | Arithmetic with fractions: a denominator must not be 0. |

## Percent of a number — `percent_of_number`

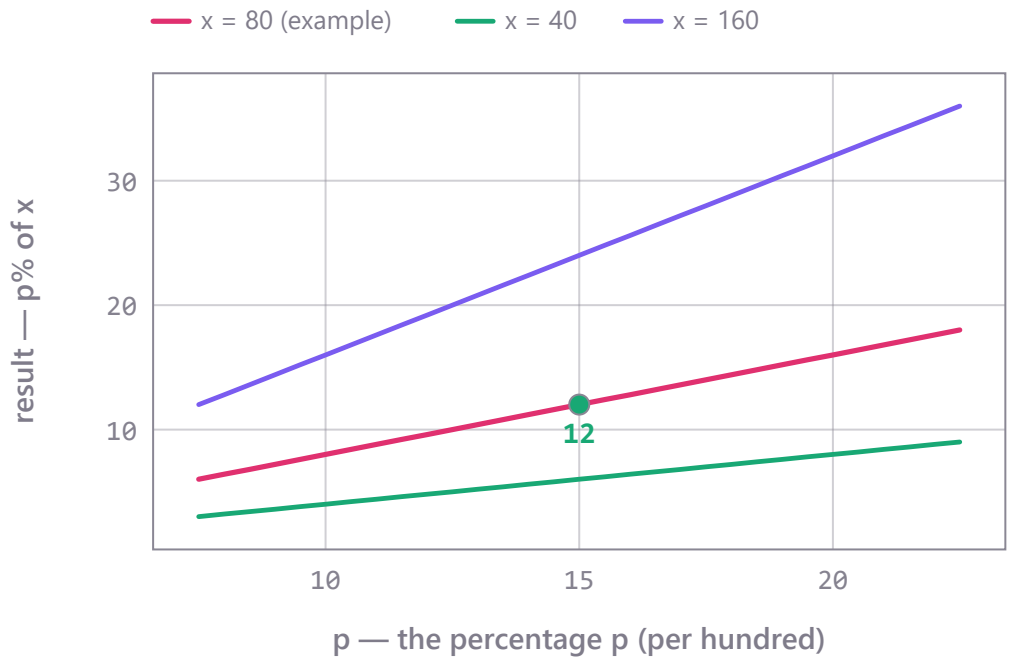
p% of x: the part that p per cent of x amounts to.

$$\frac{p}{100}\,x$$

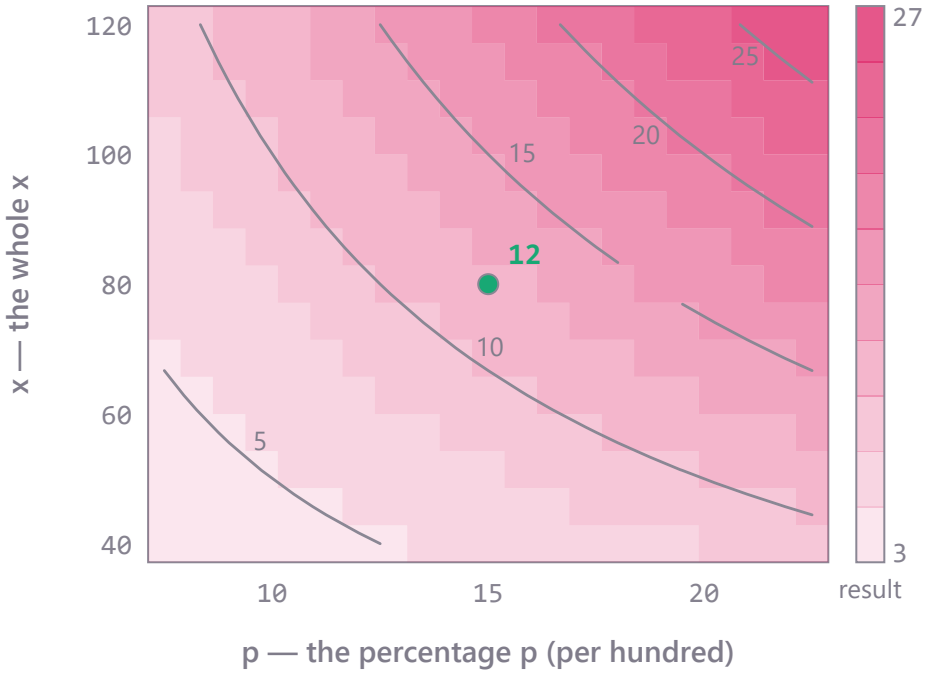
| input | meaning | condition |
|---|---|---|
| `p` | the percentage p (per hundred) | |
| `x` | the whole x | |

| output | meaning |
|---|---|
| `result` | p% of x |

- **Method:** per cent means per hundred: p% of $x$ = $(p/100)$·x
- **Assumes:** finite numbers
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




result (p% of x) against p (the percentage p (per hundred)) from 7.5 to 22.5. One curve for each value of x (the whole x): 80, 40, 160. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, p = 15, result = 12.




result (p% of x) over p (the percentage p (per hundred)) from 7.5 to 22.5 and x (the whole x) from 40 to 120: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 3 to dark 27); iso-lines at result = 5, 10, 15, 20, 25. Marked: the worked example, p = 15, x = 80, result = 12.

```
math_foundations_ai999x.solve('percent_of_number', p=15, x=80)
# -> result = 12   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_of_number', p=12.5, x=240)
# -> result = 30   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_of_number', p=150, x=40)
# -> result = 60   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| What is 15% of 80? | SOLVED | result = 12 |
| Find 12.5 percent of 240. | SOLVED | result = 30 |
| Calculate 150% of 40. | SOLVED | result = 60 |

## Whole from a part and its percentage — `percent_of_what`

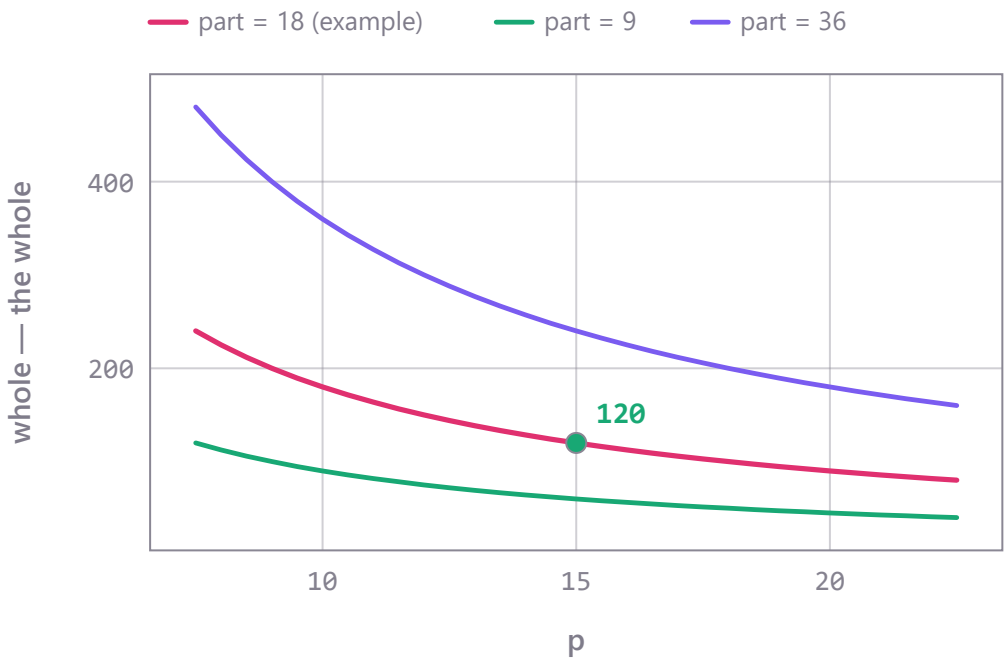
The whole w of which `part` is p per cent (part is p% of what number?).

$$w = \frac{100\,\mathrm{part}}{p}$$

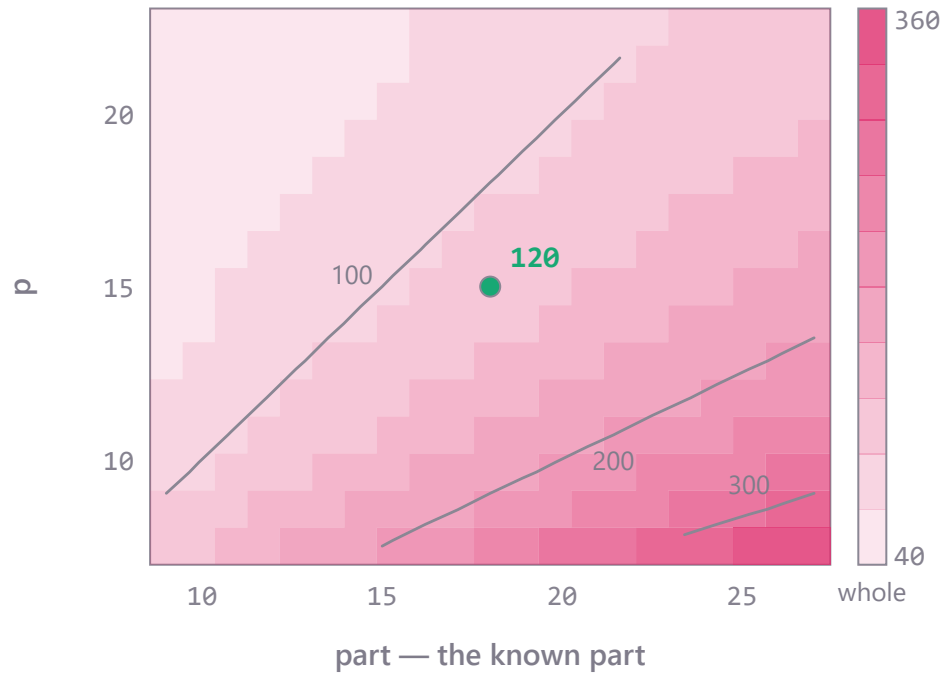
| input | meaning | condition |
|---|---|---|
| `part` | the known part | |
| `p` | the percentage that the part is of the whole | $p \neq 0$ |

| output | meaning |
|---|---|
| `whole` | the whole |

- **Method:** part = (p/100)·whole, so whole = 100·part/p
- **Assumes:** $p \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




whole (the whole) against p (the percentage that the part is of the whole) from 7.5 to 22.5. One curve for each value of part (the known part): 18, 9, 36. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, p = 15, whole = 120.




whole (the whole) over part (the known part) from 9 to 27 and p (the percentage that the part is of the whole) from 7.5 to 22.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 40 to dark 360); iso-lines at whole = 100, 200, 300. Marked: the worked example, part = 18, p = 15, whole = 120.

```
math_foundations_ai999x.solve('percent_of_what', part=18, p=15)
# -> whole = 120   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_of_what', part=26, p=40)
# -> whole = 65   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| 18 is 15% of what number? | SOLVED | whole = 120 |
| 40% of what number is 26? | SOLVED | whole = 65 |
| 5 is 0% of what number? | INVALID_INPUT | Whole from a part and its percentage: p must not be 0 (0% of every number is 0). |

### What percent one number is of another — `what_percent`

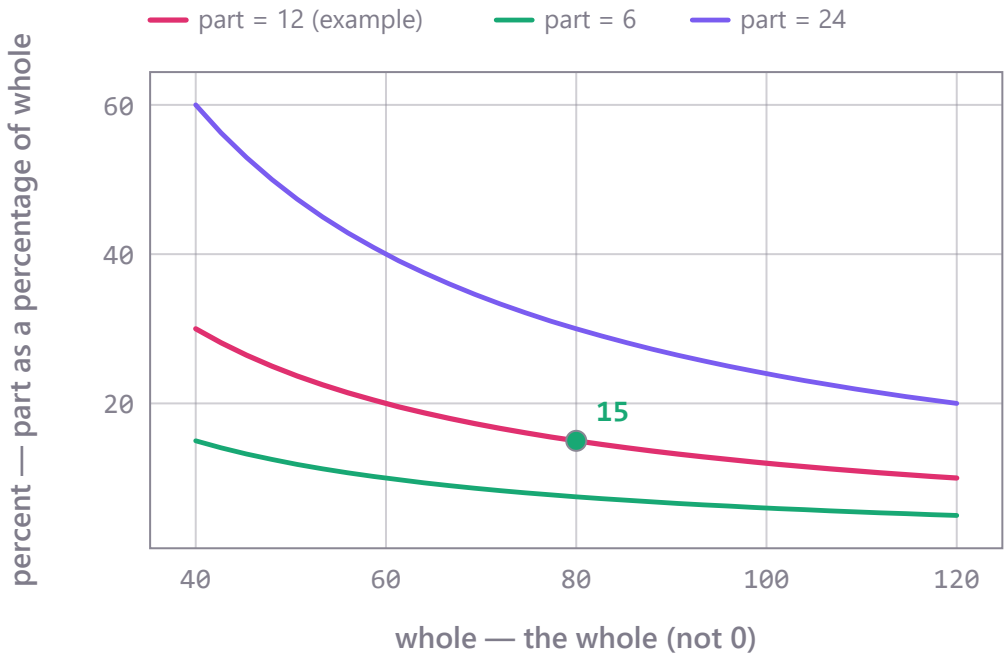
The percentage p such that `part` is p% of `whole`.

$$p = 100\,\frac{\text{part}}{\text{whole}}$$

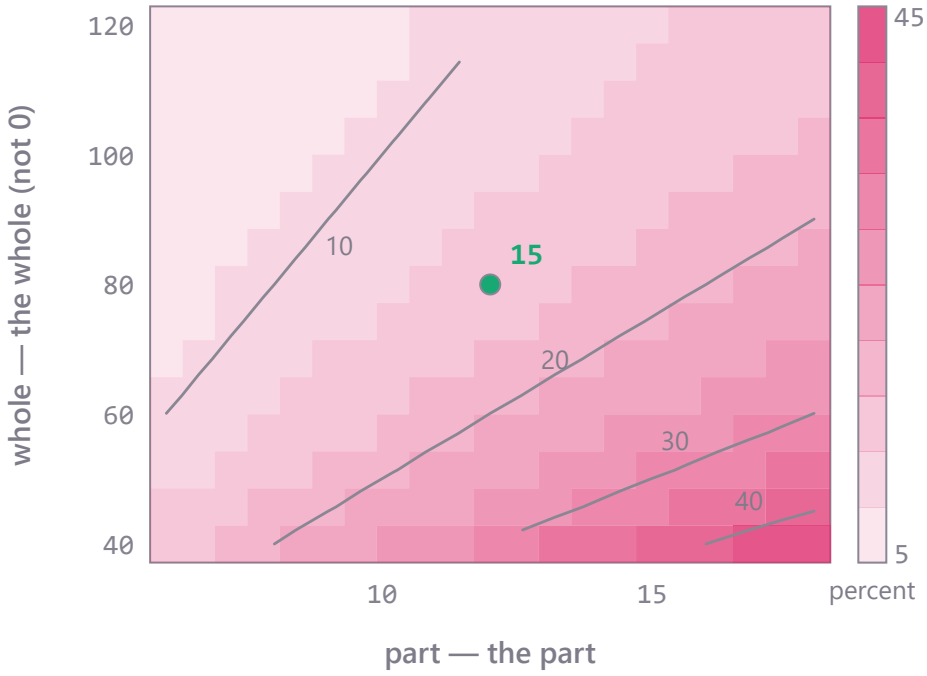
| input | meaning | condition |
|---|---|---|
| `part` | the part | |
| `whole` | the whole (not 0) | whole != 0 |

| output | meaning |
|---|---|
| `percent` | part as a percentage of whole |

- **Method:** $p = 100 \cdot \text{part}/\text{whole}$
- **Assumes:** whole != 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




percent (part as a percentage of whole) against whole (the whole (not 0)) from 40 to 120. One curve for each value of part (the part): 12, 6, 24. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, whole = 80, percent = 15.




percent (part as a percentage of whole) over part (the part) from 6 to 18 and whole (the whole (not 0)) from 40 to 120: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 5 to dark 45); iso-lines at percent = 10, 20, 30, 40. Marked: the worked example, part = 12, whole = 80, percent = 15.

```
math_foundations_ai999x.solve('what_percent', part=12, whole=80)
# -> percent = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('what_percent', part=45, whole=60)
# -> percent = 75   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| What percent of 80 is 12? | SOLVED | percent = 15 |
| 45 is what percent of 60? | SOLVED | percent = 75 |
| What percent of 0 is 5? | INVALID_INPUT | What percent one number is of another: the whole must not be 0. |

### Percent change (increase or decrease) — `percent_change`

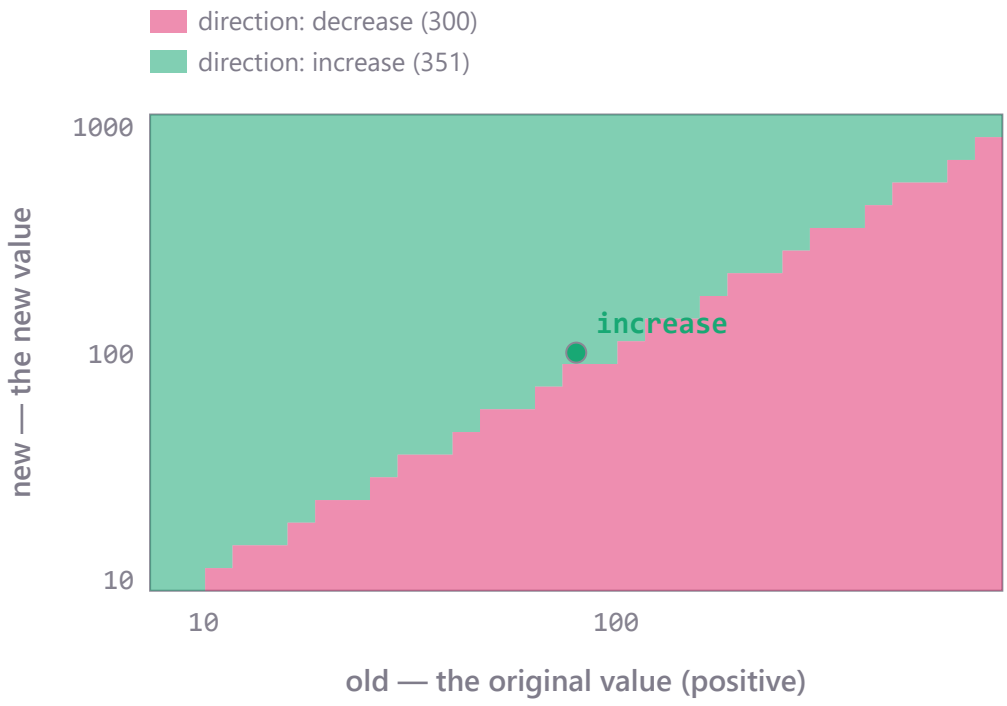
The percent change from an original value to a new value (negative for a decrease).

$$\Delta\% = \frac{\text{new} - \text{old}}{\text{old}} \cdot 100$$

| input | meaning | condition |
|---|---|---|
| `old` | the original value (positive) | |
| `new` | the new value | |

| output | meaning |
|---|---|
| `percent_change` | percent change (negative = decrease) |
| `difference` | new - old |
| `direction` | increase, decrease or no change |

- **Method:** percent change = (new - old)/old · 100
- **Assumes:** old > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




increase, decrease or no change (direction) over old (the original value (positive)) from 8 to 800 and new (the new value) from 10 to 1000 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned — decrease: 300 cells, increase: 351 cells. Marked: the worked example, old = 80, new = 100 → increase.




percent_change (percent change (negative = decrease)) against old (the original value (positive)) from 8 to 800 on a logarithmic axis. One curve for each value of new (the new value): 100, 50, 200. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, old = 80, percent_change = 25.

```
math_foundations_ai999x.solve('percent_change', old=80, new=100)
# -> percent_change = 25, difference = 20, direction = increase   (SOLVED)
```

```
math_foundations_ai999x.solve('percent_change', old=50, new=40)
# -> percent_change = -20, difference = -10, direction = decrease   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the percent change from 80 to 100. | SOLVED | percent_change = 25, difference = 20, direction = increase |
| What is the percent decrease from 50 to 40? | SOLVED | percent_change = -20, difference = -10, direction = decrease |
| A price rises from 40 to 50. What is the percent increase? | SOLVED | percent_change = 25, difference = 10, direction = increase |
| Find the percent change from 0 to 10. | INVALID_INPUT | Percent change (increase or decrease): the original value must be positive (a percent change is measured from it). |

## Divide a quantity in a given ratio — `ratio_split`

Share a total in the ratio a:b (or a:b:c).

$$\text{part}_i = \text{total} \cdot \frac{r_i}{\sum_j r_j}$$

| input | meaning | condition |
|---|---|---|
| `total` | the quantity to share | |

| input | meaning | condition |
|---|---|---|
| `a` | first term of the ratio (positive) | |
| `b` | second term of the ratio (positive) | |
| `c` | third term of the ratio (optional, positive) | |

| output | meaning |
|---|---|
| `part1` | first share |
| `part2` | second share |
| `part3` | third share (a three-term ratio only) |

- **Method:** each share = total · term / (sum of the terms)
- **Assumes:** every term of the ratio positive
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




part1 (first share) against a (first term of the ratio (positive)) from 0.3 to 30 on a logarithmic axis. One curve for each value of total (the quantity to share): 120, 60, 240. Other inputs as in the worked example: b = 5. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 3, part1 = 45.

```
math_foundations_ai999x.solve('ratio_split', total=120, a=3, b=5)
# -> part1 = 45, part2 = 75   (SOLVED)
```

```
math_foundations_ai999x.solve('ratio_split', total=900, a=2, b=3, c=4)
# -> part1 = 200, part2 = 300, part3 = 400   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Divide 120 in the ratio 3:5. | SOLVED | part1 = 45, part2 = 75 |
| Share 900 in the ratio 2:3:4. | SOLVED | part1 = 200, part2 = 300, part3 = 400 |
| Split 50 in the ratio 0:5. | INVALID_INPUT | Divide a quantity in a given ratio: every term of the ratio must be positive. |

### Solve a proportion — `proportion_solve`

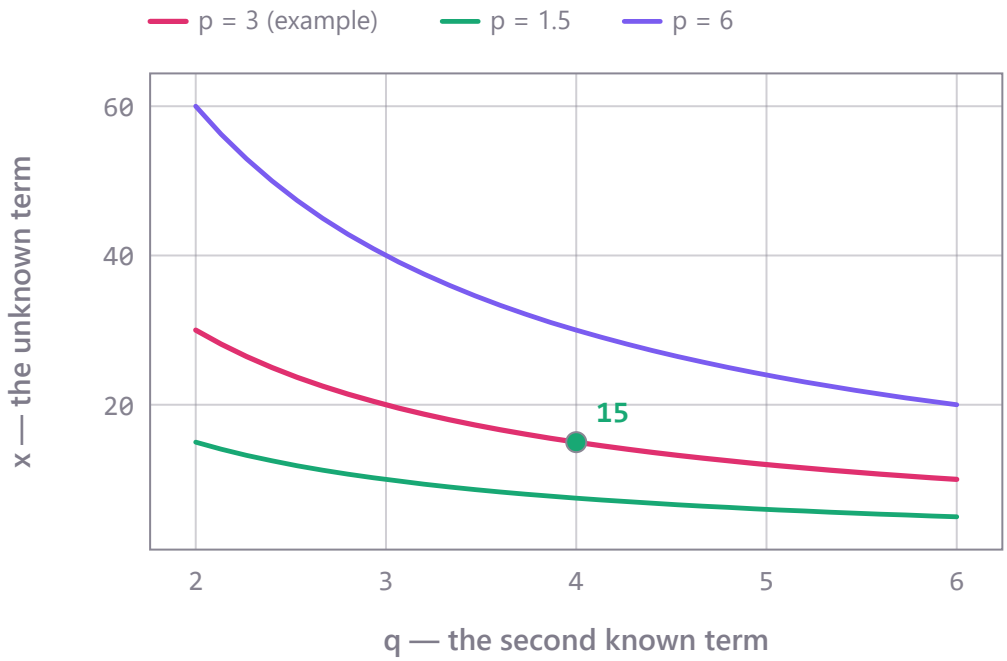
Solve $n_1/d_1 = n_2/d_2$ for the one unknown term x (in any of the four positions).

$$\frac{n_1}{d_1} = \frac{n_2}{d_2} \iff n_1 d_2 = n_2 d_1$$

| input | meaning | condition |
|---|---|---|
| `p` | the first known term (reading n1, d1, n2, d2 in order) | |
| `q` | the second known term | |
| `r` | the third known term | |
| `position` | where the unknown x stands | |

| output | meaning |
|---|---|
| `x` | the unknown term |

- **Method:** cross-multiplication: n1·$d_2 = n_2$·d1, then divide
- **Assumes:** denominators not 0; the proportion has a solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the unknown term) against q (the second known term) from 2 to 6. One curve for each value of p (the first known term (reading n1, d1, n2, d2 in order)): 3, 1.5, 6. Other inputs as in the worked example: r = 20, position = second numerator. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, q = 4, x = 15.

```
math_foundations_ai999x.solve('proportion_solve', p=3, q=4, r=20, position='second numerator')
# -> x = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('proportion_solve', p=6, q=10, r=15, position='first numerator')
# -> x = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('proportion_solve', p=5, q=2, r=7, position='first denominator')
# -> x = 17.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve 3/4 = x/20. | SOLVED | x = 15 |
| Solve the proportion x/6 = 10/15. | SOLVED | x = 4 |
| Solve 5/x = 2/7. | SOLVED | x = 17.5 |
| Solve 3/0 = x/5. | INVALID_INPUT | Solve a proportion: a denominator must not be 0. |

## Unit rates and the better buy — `unit_rate_compare`

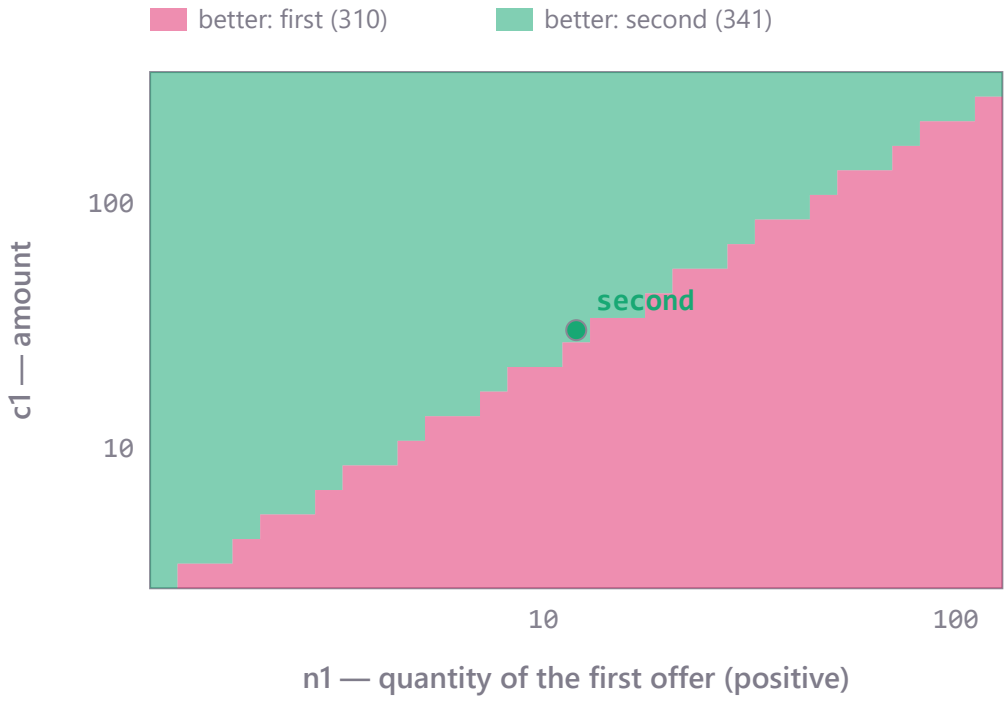
The unit rate (amount per one unit) of an offer, and of a second offer with which is cheaper per unit.

$$\text{unit rate} = \frac{\text{amount}}{\text{quantity}}$$

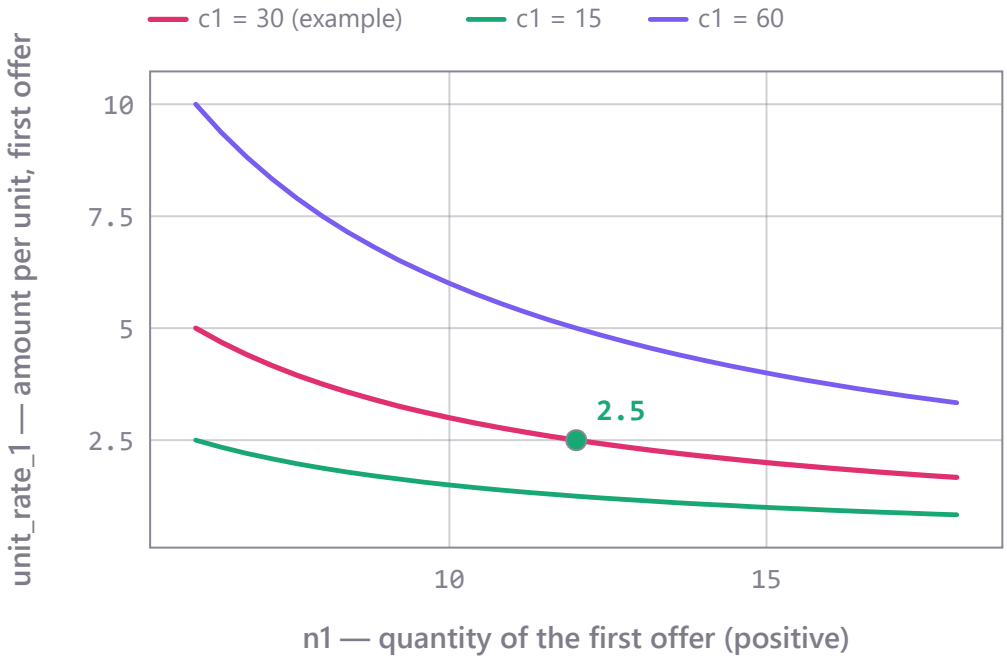
| input | meaning | condition |
|---|---|---|
| `n1` | quantity of the first offer (positive) | |
| `c1` | amount (price, distance, ...) of the first offer | |
| `n2` | quantity of the second offer (optional) | |
| `c2` | amount of the second offer (optional) | |

| output | meaning |
|---|---|
| `unit_rate_1` | amount per unit, first offer |
| `unit_rate_2` | amount per unit, second offer |
| `better` | the offer with the lower amount per unit: first, second or equal |

- **Method:** unit rate = amount / quantity; the lower unit price is the better buy
- **Assumes:** quantities positive; amounts not negative
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




the offer with the lower amount per unit: first, second or equal (better) over n1 (quantity of the first offer (positive)) from 1.2 to 120 and c1 (amount (price, distance, ...) of the first offer) from 3 to 300 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned — first: 310 cells, second: 341 cells. Other inputs as in the worked example: n2 = 20, c2 = 45. Marked: the worked example, n1 = 12, c1 = 30 → second.




unit_rate_1 (amount per unit, first offer) against n1 (quantity of the first offer (positive)) from 6 to 18. One curve for each value of c1 (amount (price, distance, ...) of the first offer): 30, 15, 60. Other inputs as in the worked example: n2 = 20, c2 = 45. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, n1 = 12, unit_rate_1 = 2.5.

```
math_foundations_ai999x.solve('unit_rate_compare', n1=12, c1=30, n2=20, c2=45)
# -> unit_rate_1 = 2.5, unit_rate_2 = 2.25, better = second   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_rate_compare', n1=6, c1=4.5)
# -> unit_rate_1 = 0.75   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Which is the better buy: 12 items for 30 or 20 items for 45? | SOLVED | unit_rate_1 = 2.5, unit_rate_2 = 2.25, better = second |
| Find the unit price of 6 cans for 4.5. | SOLVED | unit_rate_1 = 0.75 |
| Which is the better deal: 3 kg for $7.50 or 5 kg for 12$? | SOLVED | unit_rate_1 = 2.5, unit_rate_2 = 2.4, better = second |
| Find the unit price of 0 items for 5. | INVALID_INPUT | Unit rates and the better buy: quantities must be positive and amounts not negative. |

## Round to decimal places — `round_decimal_places`

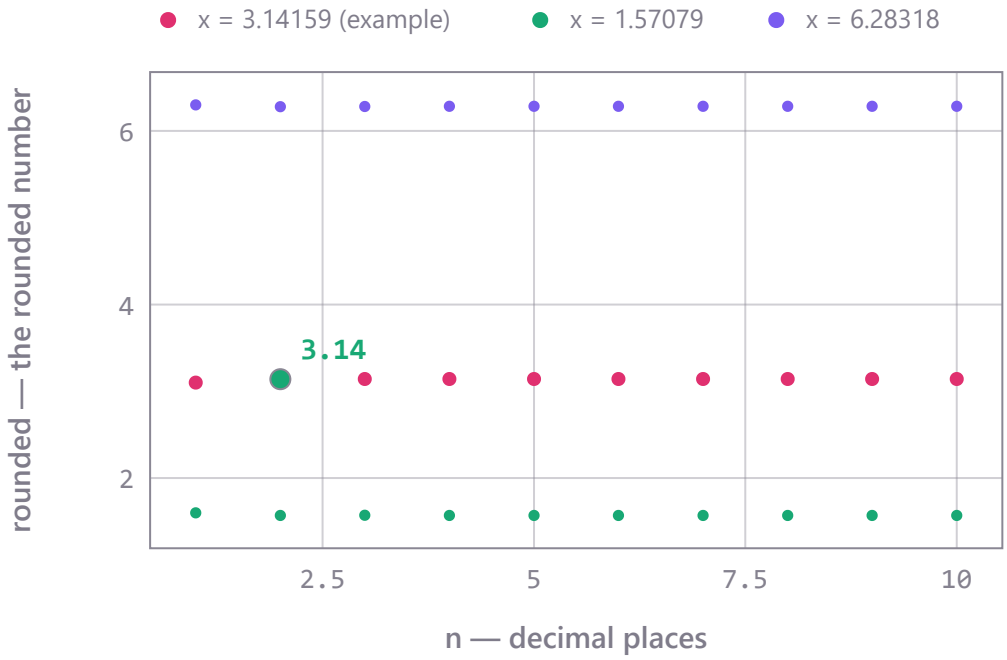
Round a number to n decimal places (or to the nearest tenth, hundredth, ten, hundred ...), halves away from zero.

$$\operatorname{round}(x, n) = \operatorname{sgn}(x)\,\frac{\lfloor |x|\,10^{n} + \tfrac{1}{2} \rfloor}{10^{n}}$$

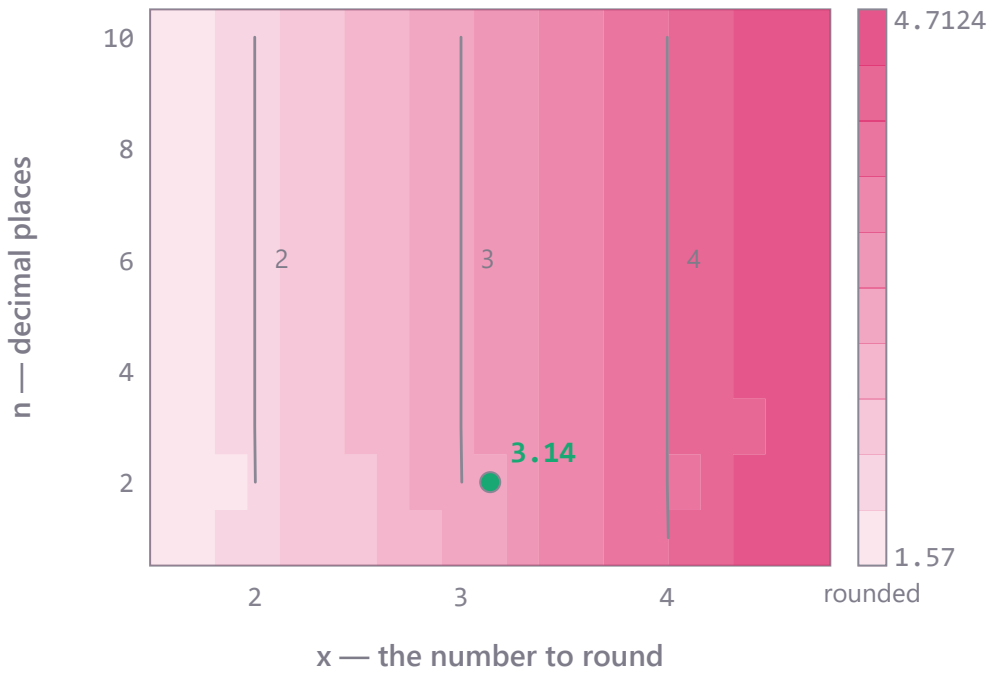
| input | meaning | condition |
|---|---|---|
| `x` | the number to round | |
| `n` | decimal places (negative: tens, hundreds) | |

| output | meaning |
|---|---|
| `rounded` | the rounded number |

- **Method:** exact decimal arithmetic on the number as written; a 5 in the first dropped place rounds away from zero
- **Assumes:** $-9 \le n \le 12$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




rounded (the rounded number) against n (decimal places (negative: tens, hundreds)) from 1 to 10. One curve for each value of x (the number to round): 3.14159, 1.57079, 6.28318. 10 points per curve computed by math-foundations-ai999x; marked: the worked example, n = 2, rounded = 3.14.




rounded (the rounded number) over x (the number to round) from 1.57079 to 4.71239 and n (decimal places (negative: tens, hundreds)) from 1 to 10: each of the 21 × 10 cells is one run of math-foundations-ai999x, shaded by its value (light 1.57 to dark 4.7124); iso-lines at rounded = 2, 3, 4. Marked: the worked example, x = 3.14159, n = 2, rounded = 3.14.

```
math_foundations_ai999x.solve('round_decimal_places', x=3.14159, n=2)
# -> rounded = 3.14   (SOLVED)
```

```
math_foundations_ai999x.solve('round_decimal_places', x=2.675, n=2)
# -> rounded = 2.68   (SOLVED)
```

```
math_foundations_ai999x.solve('round_decimal_places', x=1234.5, n=-2)
# -> rounded = 1200   (SOLVED)
```

```
math_foundations_ai999x.solve('round_decimal_places', x=-0.5, n=0)
# -> rounded = -1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Round 3.14159 to 2 decimal places. | SOLVED | rounded = 3.14 |
| Round 2.675 to the nearest hundredth. | SOLVED | rounded = 2.68 |
| Round 1234.5 to the nearest hundred. | SOLVED | rounded = 1200 |
| Round 7.45 to the nearest whole number. | SOLVED | rounded = 7 |

## Round to significant figures — `round_significant_figures`

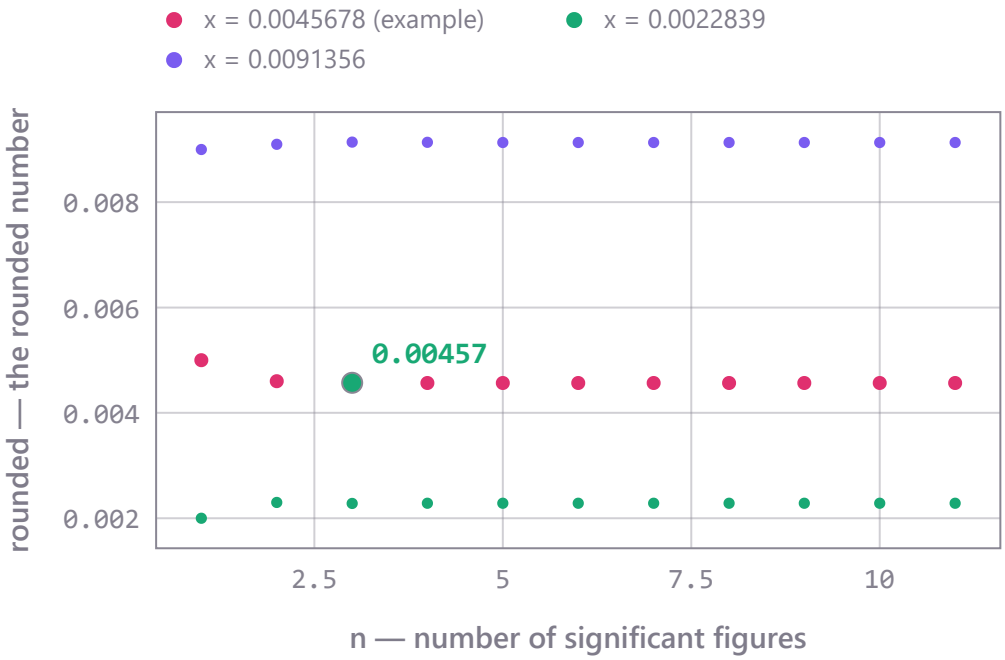
Round a nonzero number to n significant figures, halves away from zero.

$$\mathrm{round}\big(x,\ n - 1 - \lfloor \log_{10} |x| \rfloor\big)$$

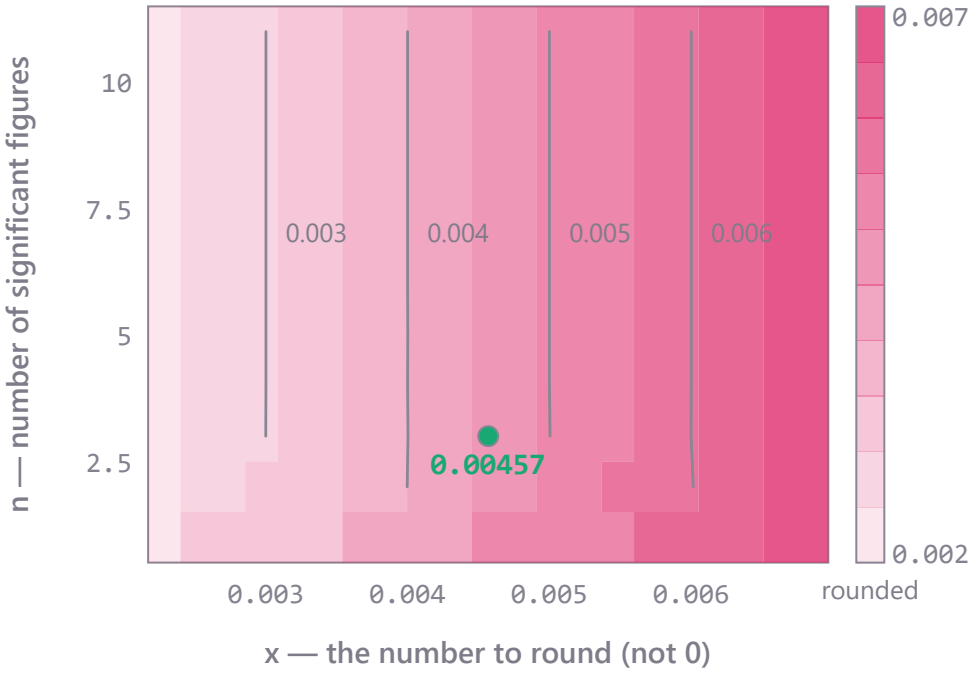
| input | meaning | condition |
|---|---|---|
| `x` | the number to round (not 0) | $x \neq 0$ |
| `n` | number of significant figures | |

| output | meaning |
|---|---|
| `rounded` | the rounded number |

- **Method:** find the exponent e of the leading digit (10^e <= |x| < $10^{e+1}$), round to $10^{e-n+1}$ exactly
- **Assumes:** $x \neq 0$; $1 \leq n \leq 15$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




rounded (the rounded number) against n (number of significant figures) from 1 to 11. One curve for each value of x (the number to round (not 0)): 0.0045678, 0.0022839, 0.0091356. 11 points per curve computed by math-foundations-ai999x; marked: the worked example, n = 3, rounded = 0.00457.




rounded (the rounded number) over x (the number to round (not 0)) from 0.0022839 to 0.0068517 and n (number of significant figures) from 1 to 11: each of the 21 × 11 cells is one run of math-foundations-ai999x, shaded by its value (light 0.002 to dark 0.007); iso-lines at rounded = 0.003, 0.004, 0.005, 0.006. Marked: the worked example, x = 0.0045678, n = 3, rounded = 0.00457.

```
math_foundations_ai999x.solve('round_significant_figures', x=0.0045678, n=3)
# -> rounded = 0.00457   (SOLVED)
```

```
math_foundations_ai999x.solve('round_significant_figures', x=45678, n=2)
# -> rounded = 46000   (SOLVED)
```

```
math_foundations_ai999x.solve('round_significant_figures', x=9.996, n=3)
# -> rounded = 10   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Round 0.0045678 to 3 significant figures. | SOLVED | rounded = 0.00457 |
| Write 45678 to 2 significant figures. | SOLVED | rounded = 46000 |
| Round 0 to 2 significant figures. | INVALID_INPUT | Round to significant figures: 0 has no significant figures to round to. |

## Scientific notation — `scientific_notation`

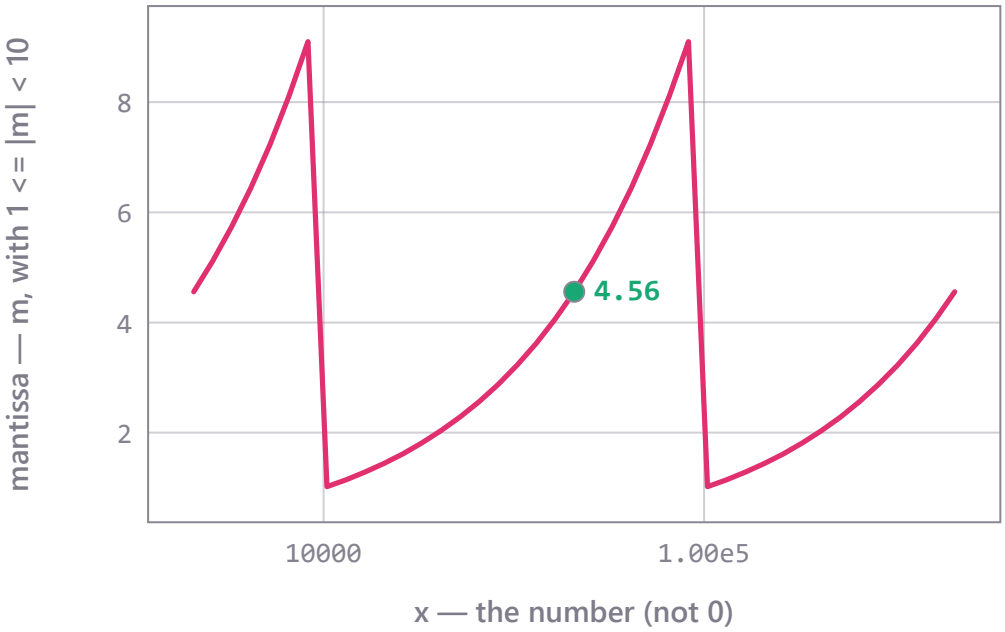
Write a nonzero number as m × $10^e$ with 1 <= |m| < 10.

$$x = m \times 10^e, \quad 1 \le |m| < 10$$

| input | meaning | condition |
|---|---|---|
| `x` | the number (not 0) | $x \neq 0$ |

| output | meaning |
|---|---|
| `mantissa` | m, with 1 <= \|m\| < 10 |
| `exponent` | the integer power of ten e |

- **Method:** exact decimal: e = position of the leading digit, $m = x/10^e$
- **Assumes:** $x \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




mantissa (m, with 1 <= |m| < 10) against x (the number (not 0)) from 4560 to 456000 on a logarithmic axis. 41 points computed by math-foundations-ai999x; marked: the worked example, x = 45600, mantissa = 4.56.

```
math_foundations_ai999x.solve('scientific_notation', x=45600)
# -> mantissa = 4.56, exponent = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('scientific_notation', x=0.00072)
# -> mantissa = 7.2, exponent = -4   (SOLVED)
```

```
math_foundations_ai999x.solve('scientific_notation', x=-3.0)
# -> mantissa = -3, exponent = 0   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Write 45600 in scientific notation. | SOLVED | mantissa = 4.56, exponent = 4 |
| Express 0.00072 in scientific notation. | SOLVED | mantissa = 7.2, exponent = -4 |
| Write 0 in scientific notation. | INVALID_INPUT | Scientific notation: 0 has no scientific notation with a leading digit from 1 to 9. |

## Evaluate an arithmetic expression (order of operations) — `order_of_operations`

The exact value of an arithmetic expression with + - * / ^ and parentheses, following the order of operations.

$$( \ ) \ \rightarrow \ a^b \ \rightarrow \ \times \div \ \rightarrow \ + -$$

| input | meaning | condition |
|---|---|---|
| `expression` | the expression, e.g. $3 + 4(2-5)^2/6$ | |

| output | meaning |
|---|---|
| `value` | the value |
| `numerator` | numerator of the exact value in lowest terms |
| `denominator` | denominator of the exact value |

- **Method:** parentheses first, then powers (right to left), then multiplication and division, then addition and subtraction (left to right); exact rational arithmetic
- **Assumes:** numbers, + - * / ^ and parentheses only; no division by 0; integer exponents up to 64
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The worked example of Evaluate an arithmetic expression (order of operations) as inputs and results: stated expression = 3 + 4 * (2 - 5)^2 / 6; returned by math-foundations-ai999x: value = 9, numerator = 9, denominator = 1.

```
math_foundations_ai999x.solve('order_of_operations', expression='3 + 4 * (2 - 5)^2 / 6')
# -> value = 9, numerator = 9, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('order_of_operations', expression='8 - 2 * 3^2 + 10 / 5')
# -> value = -8, numerator = -8, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('order_of_operations', expression='2^3^2')
# -> value = 512, numerator = 512, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('order_of_operations', expression='1/3 + 1/6')
# -> value = 0.5, numerator = 1, denominator = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate 3 + 4 * (2 - 5)^2 / 6. | SOLVED | value = 9, numerator = 9, denominator = 1 |
| Use the order of operations to evaluate 8 - 2 * 3^2 + 10 / 5. | SOLVED | value = -8, numerator = -8, denominator = 1 |
| Evaluate -2^2 + 3 * (4 - 1). | SOLVED | value = 5, numerator = 5, denominator = 1 |
| Evaluate 6 / (3 - 3) + 1. | INVALID_INPUT | Evaluate an arithmetic expression (order of operations): division by zero. |

### System of three linear equations — `linear_system_3x3`

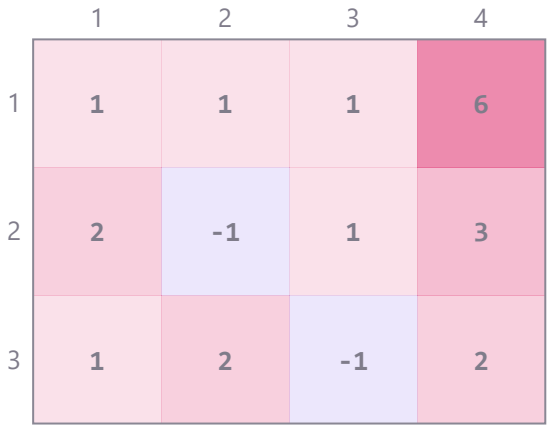
Solve three linear equations in x, y and z (a unique solution, or a refusal when there is none).

$$A\mathbf{x} = \mathbf{d}, \quad \mathbf{x} = A^{-1}\mathbf{d}$$

| input | meaning | condition |
|---|---|---|
| `augmented` | augmented matrix, one row a b c d per equation $ax + by + cz = d$ | |

| output | meaning |
|---|---|
| `x` | x |
| `y` | y |
| `z` | z |

- **Method:** Gaussian elimination in exact rational arithmetic
- **Assumes:** determinant of the coefficients != 0 (a unique solution)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




augmented (input)

The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): augmented (input). Returned with them by math-foundations-ai999x: x = 1, y = 2, z = 3.

```
math_foundations_ai999x.solve('linear_system_3x3', augmented='1 1 1 6; 2 -1 1 3; 1 2 -1 2')
# -> x = 1, y = 2, z = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_system_3x3', augmented=[[2, 1, -1, 8], [-3, -1, 2, -11], [-2,
1, 2, -3]])
# -> x = 2, y = 3, z = -1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve the system x + y + z = 6, 2x - y + z = 3, x + 2y - z = 2. | SOLVED | x = 1, y = 2, z = 3 |
| Solve the system 2x + y - z = 8; -3x - y + 2z = -11; -2x + y + 2z = -3. | SOLVED | x = 2, y = 3, z = -1 |
| Solve the system x + y + z = 1, 2x + 2y + 2z = 2, x - y = 0. | INVALID_INPUT | System of three linear equations: the system has no unique solution (its determinant is 0). |

### Absolute value inequality |ax + b| compared with c — `abs_linear_inequality`

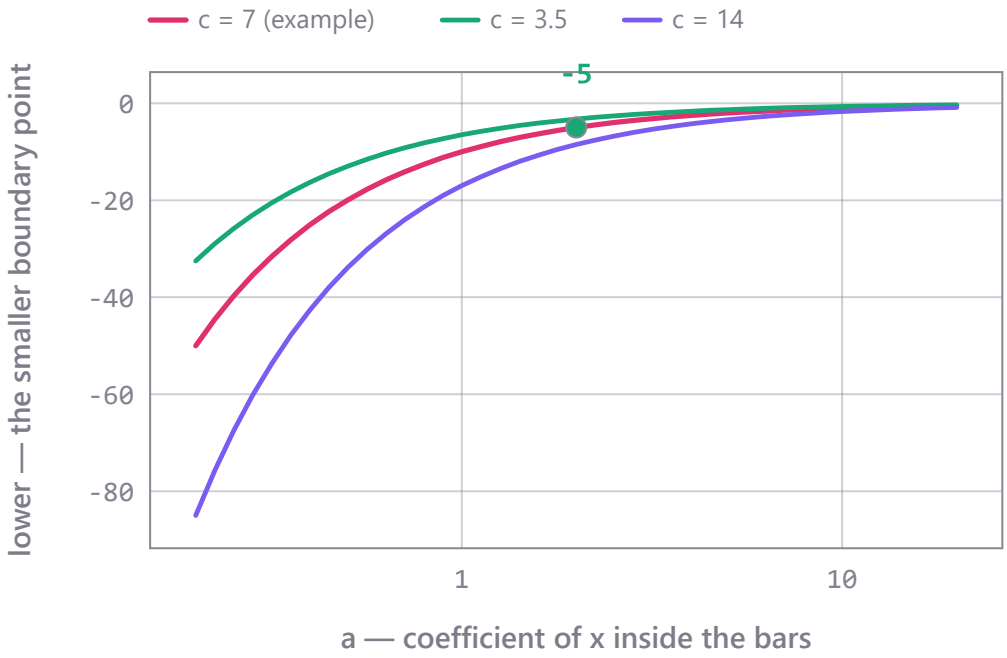
Solve $|ax + b|$ < c, <= c, > c $or \geq c\ (c > 0)$: the two boundary points and whether x lies between or outside them.

$$|ax + b| < c \iff -c < ax + b < c, \qquad |ax + b| > c \iff ax + b < -c \text{ or } ax + b > c$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x inside the bars | $a \neq 0$ |
| `b` | constant inside the bars | |
| `c` | the bound (positive) | |
| `rel` | the comparison | |

| output | meaning |
|---|---|
| `lower` | the smaller boundary point |
| `upper` | the larger boundary point |
| `region` | between (for < and <=) or outside (for > and >=) |
| `solution` | the solution set |

- **Method:** |u| < c means -c < u < c; |u| > c means $u < -c$ or $u > c$; solve each linear piece for x
- **Assumes:** $a \neq 0$; $c > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




lower (the smaller boundary point) against a (coefficient of x inside the bars) from 0.2 to 20 on a logarithmic axis. One curve for each value of c (the bound (positive)): 7, 3.5, 14. Other inputs as in the worked example: b = 3, rel = <. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, lower = -5.

```
math_foundations_ai999x.solve('abs_linear_inequality', a=2, b=3, c=7, rel='<')
# -> lower = -5, upper = 2, region = between, solution = -5 < x < 2   (SOLVED)
```

```
math_foundations_ai999x.solve('abs_linear_inequality', a=3, b=-1, c=5, rel='>=')
# -> lower = -1.33333, upper = 2, region = outside, solution = x <= -1.333333333 or x >= 2
(SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve \|2x + 3\| < 7. | SOLVED | lower = -5, upper = 2, region = between, solution = -5 < x < 2 |
| Solve the inequality \|3x - 1\| >= 5. | SOLVED | lower = -1.33333, upper = 2, region = outside, solution = x <= -1.333333333 or x >= 2 |
| Solve \|x + 4\| <= 0. | INVALID_INPUT | Absolute value inequality \|ax + b\| compared with c: c must be positive (with c <= 0 the solution set is empty, a point o |

## Rational equation a/(x + b) = c/(x + d) — `rational_equation`

Solve $a/(x+b) = c/(x+d)$ for x, rejecting a candidate that makes a denominator 0.

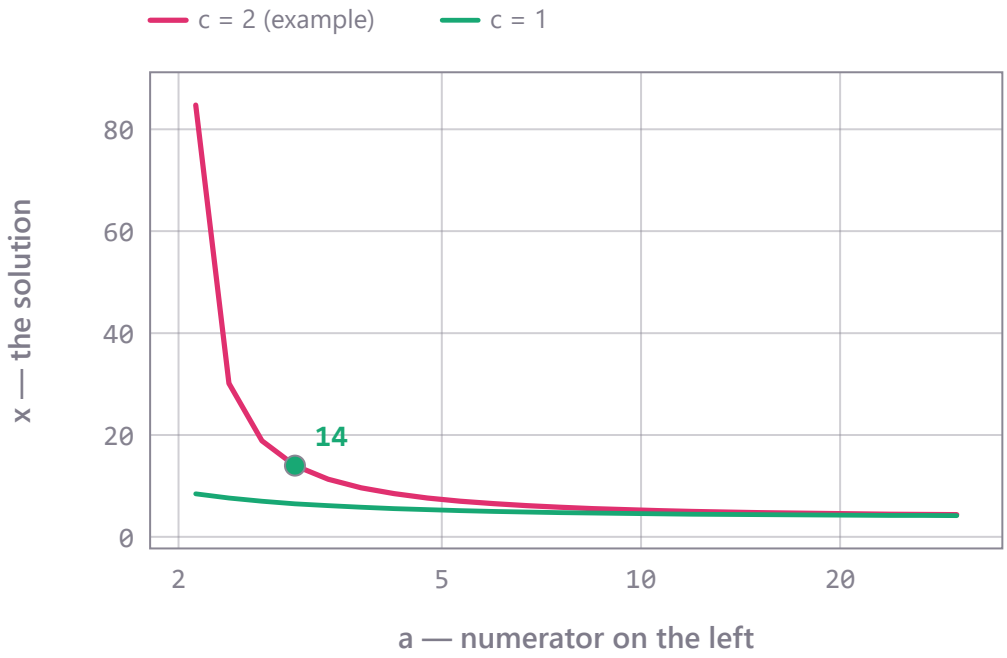
$$\frac{a}{x+b} = \frac{c}{x+d} \;\Rightarrow\; x = \frac{cb - ad}{a - c}$$

| input | meaning | condition |
|---|---|---|
| a | numerator on the left | |
| b | constant in the left denominator x + b | |
| c | numerator on the right | |

| input | meaning | condition |
|---|---|---|
| d | constant in the right denominator x + d | |

| output | meaning |
|---|---|
| x | the solution |

- **Method:** multiply by both denominators: a(x + d) = c(x + b), solve the linear equation, check the excluded values $x = -b$, $x = -d$
- **Assumes:** $a \neq c$; the candidate is not an excluded value
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the solution) against a (numerator on the left) from 2.12384 to 30 on a logarithmic axis. One curve for each value of c (numerator on the right): 2, 1. Other inputs as in the worked example: b = 1, d = -4. 24 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 3, x = 14.

```
math_foundations_ai999x.solve('rational_equation', a=3, b=1, c=2, d=-4)
# -> x = 14   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_equation', a=5, b=-2, c=3, d=6)
# -> x = -18   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Solve 3/(x + 1) = 2/(x - 4). | SOLVED | x = 14 |
| Solve 5/(x - 2) = 3/(x + 6). | SOLVED | x = -18 |
| Solve 2/(x + 1) = 2/(x + 3). | INVALID_INPUT | Rational equation a/(x + b) = c/(x + d): no unique solution (equal numerators: no solution or the same expression on bot |

| question | outcome | result |
|---|---|---|
| Solve 0/(x + 1) = 4/(x - 2). | INVALID_INPUT | Rational equation a/(x + b) = c/(x + d): no solution: the only candidate x = -1 makes a denominator 0 (extraneous). |

### Radical equation sqrt(ax + b) = cx + d — `radical_equation`

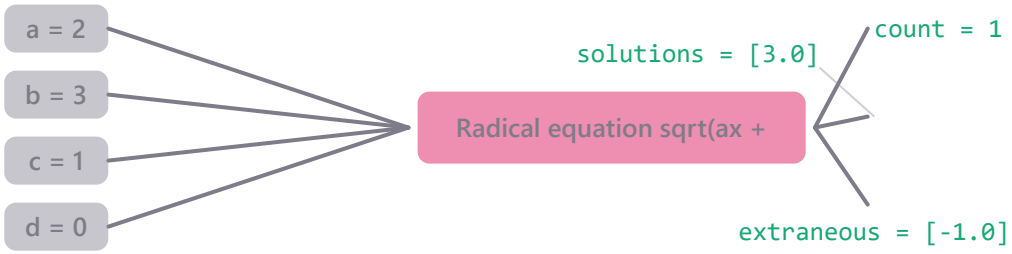
Solve sqrt(a*x + b) = c*x + d: square, solve, and keep only the roots that satisfy the original equation.

$$\sqrt{ax+b} = cx+d \;\Rightarrow\; ax+b = (cx+d)^2,\; cx+d \geq 0$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x under the root | $a \neq 0$ |
| `b` | constant under the root | |
| `c` | coefficient of x on the right | |
| `d` | constant on the right | |

| output | meaning |
|---|---|
| `count` | number of solutions (0, 1 or 2) |
| `solutions` | the solutions, ascending |
| `extraneous` | roots of the squared equation that fail the original (extraneous) |

- **Method:** square both sides (a x + b = (c x + d)^2), solve the quadratic exactly, keep a root only if c $x+d \geq 0$
- **Assumes:** $a \neq 0$; real numbers
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The worked example of Radical equation sqrt(ax + b) = cx + d as inputs and results: stated a = 2, b = 3, c = 1, d = 0; returned by math-foundations-ai999x: count = 1, solutions = [3.0], extraneous = [-1.0].

```
math_foundations_ai999x.solve('radical_equation', a=2, b=3, c=1, d=0)
# -> count = 1, solutions = [3.0], extraneous = [-1.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('radical_equation', a=1, b=7, c=1, d=-5)
# -> count = 1, solutions = [9.0], extraneous = [2.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('radical_equation', a=1, b=-1, c=1, d=-1)
# -> count = 2, solutions = [1.0, 2.0], extraneous = []   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve sqrt(2x + 3) = x. | SOLVED | count = 1, solutions = [3.0], extraneous = [-1.0] |
| Solve √(x + 7) = x - 5. | SOLVED | count = 1, solutions = [9.0], extraneous = [2.0] |
| Solve sqrt(x - 1) = x - 1. | SOLVED | count = 2, solutions = [1.0, 2.0], extraneous = [] |
| Solve sqrt(x + 4) = -2. | SOLVED | count = 0, solutions = [], extraneous = [0.0] |

## Synthetic division and the remainder theorem — `synthetic_division`

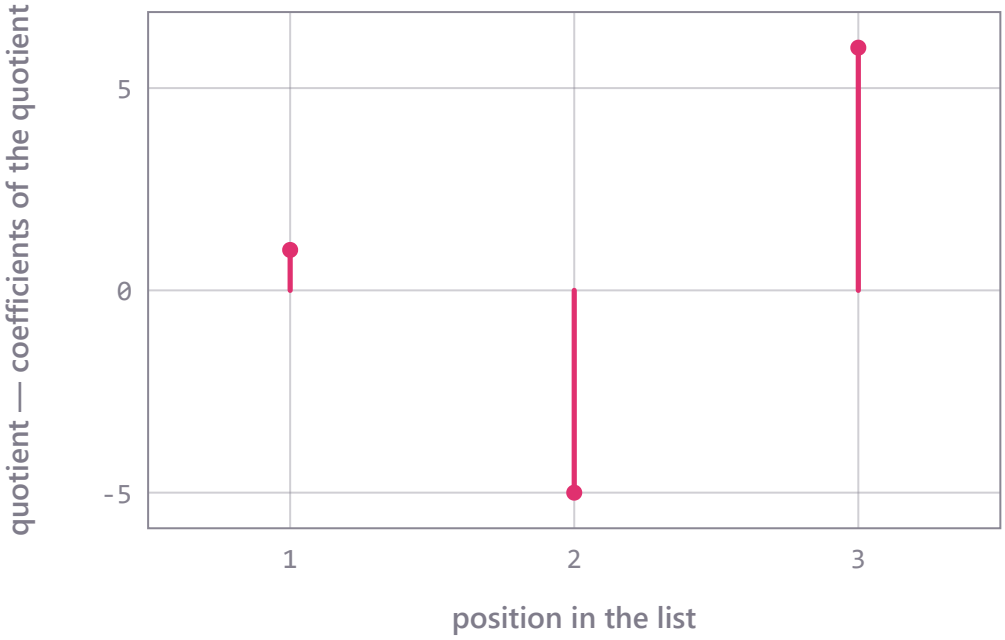
Divide a polynomial by x - r: the quotient's coefficients and the remainder, which equals p(r).

$$p(x) = (x - r)\,q(x) + p(r)$$

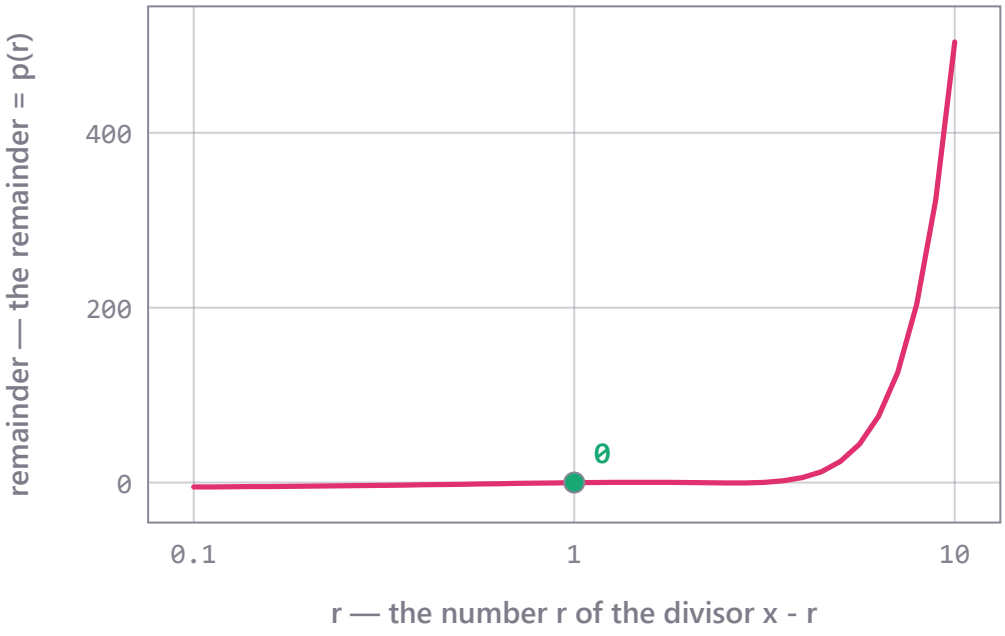
| input | meaning | condition |
|---|---|---|
| `coefficients` | coefficients of the dividend, highest power first (2x^3 - 5x + 1 -> 2, 0, -5, 1) | |
| `r` | the number r of the divisor x - r | |

| output | meaning |
|---|---|
| `quotient` | coefficients of the quotient, highest power first |
| `remainder` | the remainder = p(r) |

- **Method:** synthetic division (Horner's scheme): bring down, multiply by r, add; the last number is the remainder p(r)
- **Assumes:** dividend of degree >= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




coefficients of the quotient, highest power first (quotient) returned by math-foundations-ai999x for the worked example, one value for each position in the list (3 values, first 1, last 6, largest 6). Inputs: coefficients = 1, -6, 11, -6, r = 1.




remainder (the remainder = p(r)) against r (the number r of the divisor x - r) from 0.1 to 10 on a logarithmic axis. Other inputs as in the worked example: coefficients = 1, -6, 11, -6. 41 points computed by math-foundations-ai999x; marked: the worked example, r = 1, remainder = 0.

```
math_foundations_ai999x.solve('synthetic_division', coefficients='1, -6, 11, -6', r=1)
# -> quotient = [1.0, -5.0, 6.0], remainder = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('synthetic_division', coefficients=[1, 0, -3, 0, 2], r=3)
# -> quotient = [1.0, 3.0, 6.0, 18.0], remainder = 56   (SOLVED)
```

```
math_foundations_ai999x.solve('synthetic_division', coefficients=[2, 0, 3, -4], r=-2)
# -> quotient = [2.0, -4.0, 11.0], remainder = -26   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Divide x^3 - 6x^2 + 11x - 6 by x - 1. | SOLVED | quotient = [1.0, -5.0, 6.0], remainder = 0 |
| Find the remainder when x^4 - 3x^2 + 2 is divided by x - 3. | SOLVED | quotient = [1.0, 3.0, 6.0, 18.0], remainder = 56 |
| Use synthetic division to divide 2x^3 + 3x - 4 by x + 2. | SOLVED | quotient = [2.0, -4.0, 11.0], remainder = -26 |

## Polynomial long division — `polynomial_long_division`

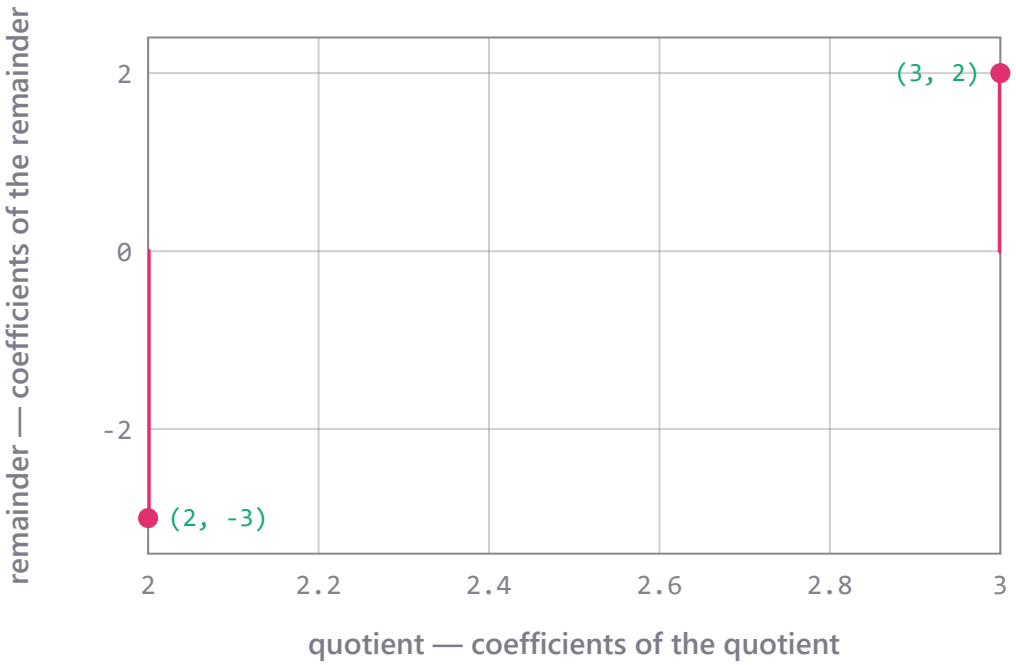
Divide one polynomial by another: quotient and remainder, with $p = qd + r$ and deg r < deg d.

$$p(x) = q(x)\,d(x) + r(x), \quad \deg r < \deg d$$

| input | meaning | condition |
|---|---|---|
| `dividend` | coefficients of the dividend, highest power first | |
| `divisor` | coefficients of the divisor, highest power first (degree >= 1) | |

| output | meaning |
|---|---|
| `quotient` | coefficients of the quotient, highest power first |
| `remainder` | coefficients of the remainder, highest power first |

- **Method:** long division: divide leading terms, multiply back, subtract, repeat; exact rational arithmetic
- **Assumes:** divisor of degree >= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




coefficients of the remainder, highest power first (remainder) at each coefficients of the quotient, highest power first (quotient), both returned by math-foundations-ai999x for the worked example (dividend = 2, 3, -1, 5, divisor = 1, 0, 1): 2 → -3; 3 → 2.

```
math_foundations_ai999x.solve('polynomial_long_division', dividend='2, 3, -1, 5', divisor='1, 0, 1')
# -> quotient = [2.0, 3.0], remainder = [-3.0, 2.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('polynomial_long_division', dividend=[6, -5, 0, 4], divisor=[2, -3])
# -> quotient = [3.0, 2.0, 3.0], remainder = [13.0]   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Divide 2x^3 + 3x^2 - x + 5 by x^2 + 1. | SOLVED | quotient = [2.0, 3.0], remainder = [-3.0, 2.0] |
| Use long division to divide 6x^3 - 5x^2 + 4 by 2x - 3. | SOLVED | quotient = [3.0, 2.0, 3.0], remainder = [13.0] |
| Find the quotient and remainder when x^4 - 1 is divided by x^2 + x + 1. | SOLVED | quotient = [1.0, -1.0, 0.0], remainder = [1.0, -1.0] |

## Factor a quadratic over the integers — `factor_quadratic_integer`

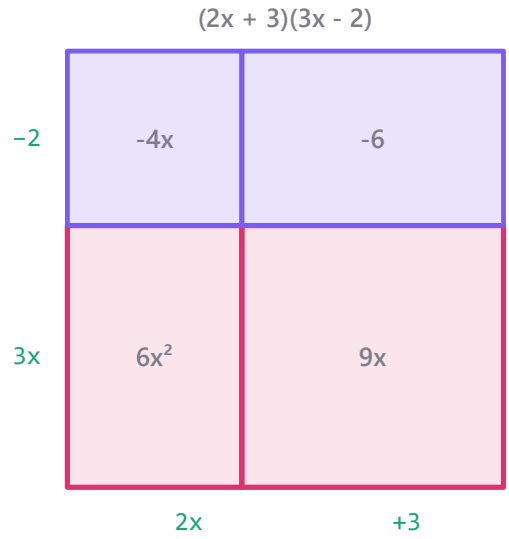
Factor $ax^2 + bx + c$ with whole-number coefficients as g(px + q)(rx + s), or say that it does not factor over the integers.

$$ax^2 + bx + c = g\,(px + q)(rx + s)$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of $x^2$ (not 0) | |
| `b` | coefficient of x | |
| `c` | constant term | |

| output | meaning |
|---|---|
| `g` | common factor (sign of a) |
| `p` | x-coefficient of the first factor |
| `q` | constant of the first factor |
| `r` | x-coefficient of the second factor |
| `s` | constant of the second factor |
| `factored` | the factored form |

- **Method:** take out the greatest common factor; the rational roots -q/p and -s/r of the primitive quadratic give its factors (Gauss's lemma)
- **Assumes:** whole-number coefficients; $a \neq 0$; the discriminant is a perfect square
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




Area model of $6x^2$ + 5x + -6 (as stated) with the factors the package returned, (2x + 3)(3x - 2): the sides of the rectangle are (2x + 3) and (3x − 2); each cell is the product of its side terms (cells of negative sign in the second colour), and together they make the quadratic. Side lengths drawn by magnitude (x = 1). Marked: the factored form.

```
math_foundations_ai999x.solve('factor_quadratic_integer', a=6, b=5, c=-6)
# -> g = 1, p = 2, q = 3, r = 3, s = -2, factored = (2x + 3)(3x - 2)   (SOLVED)
```

```
math_foundations_ai999x.solve('factor_quadratic_integer', a=4, b=0, c=-9)
# -> g = 1, p = 2, q = 3, r = 2, s = -3, factored = (2x + 3)(2x - 3)   (SOLVED)
```

```
math_foundations_ai999x.solve('factor_quadratic_integer', a=2, b=0, c=-8)
# -> g = 2, p = 1, q = 2, r = 1, s = -2, factored = 2(x + 2)(x - 2)   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Factor 6x^2 + 5x - 6 over the integers. | SOLVED | g = 1, p = 2, q = 3, r = 3, s = -2, factored = (2x + 3)(3x - 2) |
| Factor 4x^2 - 9. | SOLVED | g = 1, p = 2, q = 3, r = 2, s = -3, factored = (2x + 3)(2x - 3) |
| Factor completely 2x^2 - 8. | SOLVED | g = 2, p = 1, q = 2, r = 1, s = -2, factored = 2(x + 2)(x - 2) |
| Factor 2x^2 + 3x + 5 over the integers. | INVALID_INPUT | Factor a quadratic over the integers: this quadratic does not factor over the integers (its discriminant is not a perfec |

## Quadratic inequality — `quadratic_inequality`

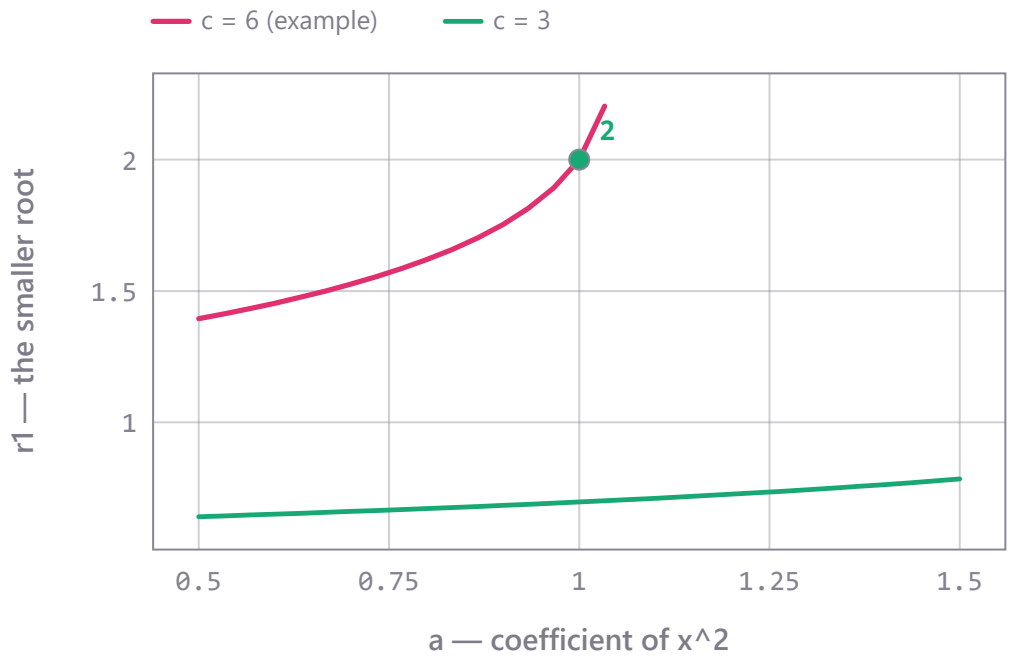
Solve $ax^2 + bx + c < 0$ (or <=, >, >=) when the quadratic has two distinct real roots.

$$a(x - r_1)(x - r_2) < 0 \iff r_1 < x < r_2 \ \ (a > 0)$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of $x^2$ | $a \neq 0$ |
| `b` | coefficient of x | |
| `c` | constant term | |
| `rel` | the comparison with 0 | |

| output | meaning |
|---|---|
| `r1` | the smaller root |
| `r2` | the larger root |
| `region` | between or outside the roots |
| `solution` | the solution set |

- **Method:** the roots split the line into three intervals; the sign of a decides whether the quadratic is negative between or outside them
- **Assumes:** $a \neq 0$; $b^2 - 4ac > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




r1 (the smaller root) against a (coefficient of x^2) from 0.5 to 1.5. One curve for each value of c (constant term): 6, 3. Other inputs as in the worked example: b = -5, rel = <. 17 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, r1 = 2.

```
math_foundations_ai999x.solve('quadratic_inequality', a=1, b=-5, c=6, rel='<')
# -> r1 = 2, r2 = 3, region = between, solution = 2 < x < 3   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_inequality', a=2, b=-1, c=-3, rel='>=')
# -> r1 = -1, r2 = 1.5, region = outside, solution = x <= -1 or x >= 1.5   (SOLVED)
```

```
math_foundations_ai999x.solve('quadratic_inequality', a=-1, b=0, c=4, rel='>')
# -> r1 = -2, r2 = 2, region = between, solution = -2 < x < 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve x^2 - 5x + 6 < 0. | SOLVED | r1 = 2, r2 = 3, region = between, solution = 2 < x < 3 |
| Solve the quadratic inequality 2x^2 - x - 3 >= 0. | SOLVED | r1 = -1, r2 = 1.5, region = outside, solution = x <= -1 or x >= 1.5 |
| Solve -x^2 + 4 > 0. | SOLVED | r1 = -2, r2 = 2, region = between, solution = -2 < x < 2 |
| Solve x^2 + 1 < 0. | OUT_OF_SCOPE | Out of scope: a quadratic inequality whose quadratic has no two distinct real roots. |

## Sum of an infinite geometric series — `infinite_geometric_series`

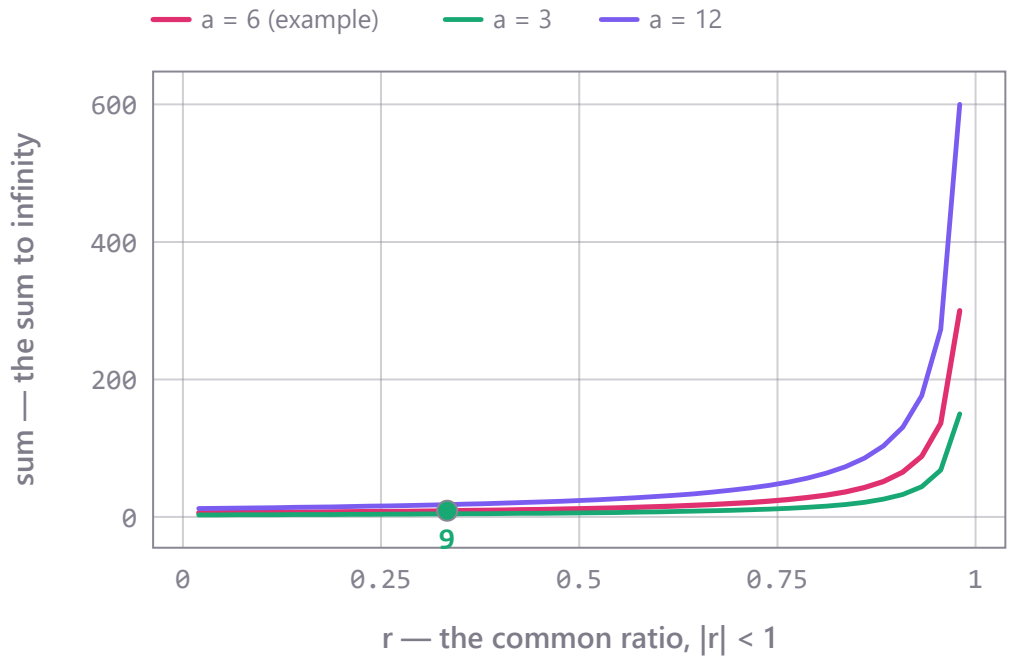
$a + ar + ar^2 + \ldots = a/(1 - r)$ for |r| < 1 (a refusal when the series diverges).

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, \quad |r| < 1$$

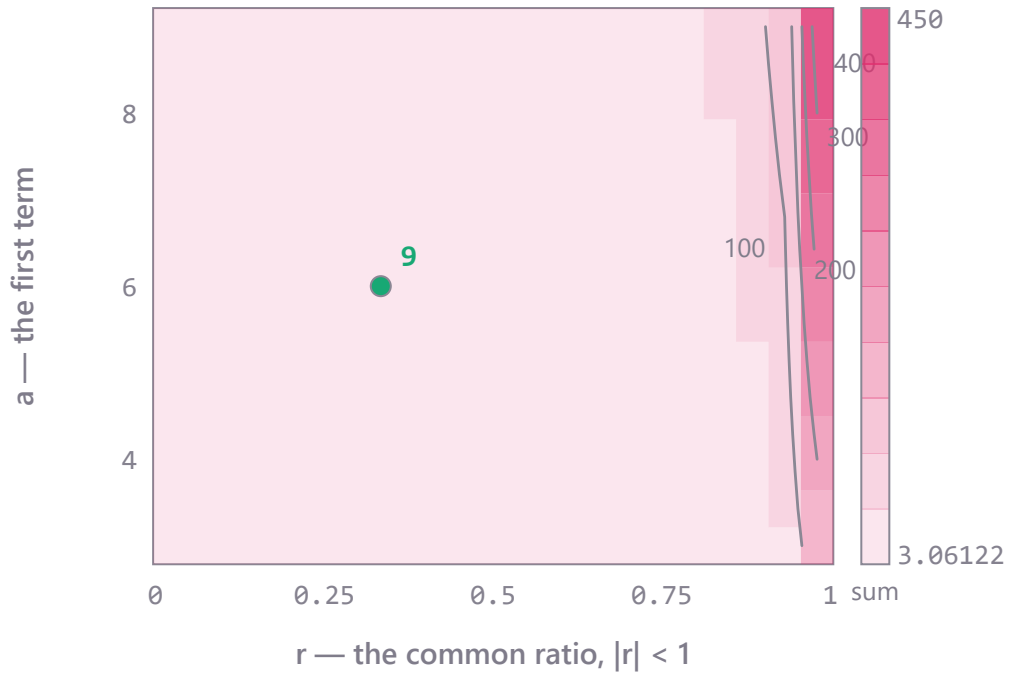
| input | meaning | condition |
|---|---|---|
| `a` | the first term | |
| `r` | the common ratio, \|r\| < 1 | |

| output | meaning |
|---|---|
| `sum` | the sum to infinity |

- **Method:** $S_n = a$(1 - r^n)/(1 - r) and $r^n$ -> 0 when |r| < 1
- **Assumes:** |r| < 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




sum (the sum to infinity) against r (the common ratio, |r| < 1) from 0.02 to 0.98. One curve for each value of a (the first term): 6, 3, 12. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, r = 0.333333, sum = 9.




sum (the sum to infinity) over r (the common ratio, |r| < 1) from 0.02 to 0.98 and a (the first term) from 3 to 9: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 3.06122 to dark 450); iso-lines at sum = 100, 200, 300, 400. Marked: the worked example, r = 0.333333, a = 6, sum = 9.

```
math_foundations_ai999x.solve('infinite_geometric_series', a=6, r=0.3333333333333333)
# -> sum = 9   (SOLVED)
```

```
math_foundations_ai999x.solve('infinite_geometric_series', a=8, r=0.5)
# -> sum = 16   (SOLVED)
```

```
math_foundations_ai999x.solve('infinite_geometric_series', a=9, r=-0.3333333333333333)
# -> sum = 6.75   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the sum of the infinite geometric series with first term 6 and ratio 1/3. | SOLVED | sum = 9 |
| Find the sum to infinity of 8 + 4 + 2 + 1 + ... | SOLVED | sum = 16 |
| Find the sum to infinity of 9 - 3 + 1 - ... | SOLVED | sum = 6.75 |
| Find the sum of the infinite geometric series with first term 2 and ratio 3/2. | INVALID_INPUT | Sum of an infinite geometric series: the series diverges unless \|r\| < 1. |

## Partial fractions (two distinct linear factors) — `partial_fractions_linear`

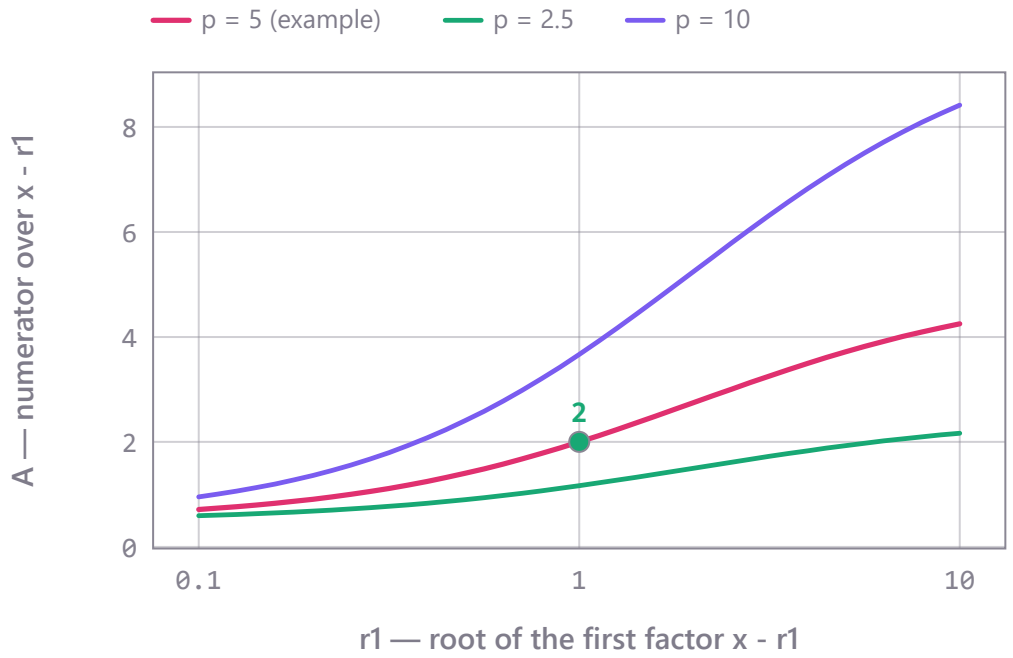
Write $(px+q)/((x-r_1)(x-r_2))$ as $A/(x-r_1)+B/(x-r_2)$.

$$\frac{px+q}{(x-r_1)(x-r_2)} = \frac{A}{x-r_1} + \frac{B}{x-r_2}, \quad A = \frac{pr_1+q}{r_1-r_2}$$

| input | meaning | condition |
|---|---|---|
| `p` | coefficient of x in the numerator | |
| `q` | constant of the numerator | |
| `r1` | root of the first factor $x - r_1$ | |
| `r2` | root of the second factor $x - r_2$ $(r_2 \neq r_1)$ | |

| output | meaning |
|---|---|
| `A` | numerator over $x - r_1$ |
| `B` | numerator over $x - r_2$ |

- **Method:** cover-up rule: A = (p r1 + q)/(r1 - r2), B = (p r2 + q)/(r2 - r1)
- **Assumes:** $r_1 \neq r_2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




A (numerator over x - r1) against r1 (root of the first factor x - r1) from 0.1 to 10 on a logarithmic axis. One curve for each value of p (coefficient of x in the numerator): 5, 2.5, 10. Other inputs as in the worked example: q = 1, r2 = -2. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, r1 = 1, A = 2.

```
math_foundations_ai999x.solve('partial_fractions_linear', p=5, q=1, r1=1, r2=-2)
# -> A = 2, B = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('partial_fractions_linear', p=3, q=5, r1=-1, r2=2)
# -> A = -0.666667, B = 3.66667   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Decompose (5x + 1)/((x - 1)(x + 2)) into partial fractions. | SOLVED | A = 2, B = 3 |

| question | outcome | result |
|---|---|---|
| Find the partial fraction decomposition of (3x + 5)/(x^2 - x - 2). | SOLVED | A = -0.666667, B = 3.66667 |
| Decompose (2x + 3)/((x - 4)(x - 4)) into partial fractions. | INVALID_INPUT | Partial fractions (two distinct linear factors): the two linear factors must be different (repeated factors need another |

### Rational roots of a polynomial (rational root theorem) — `rational_roots`

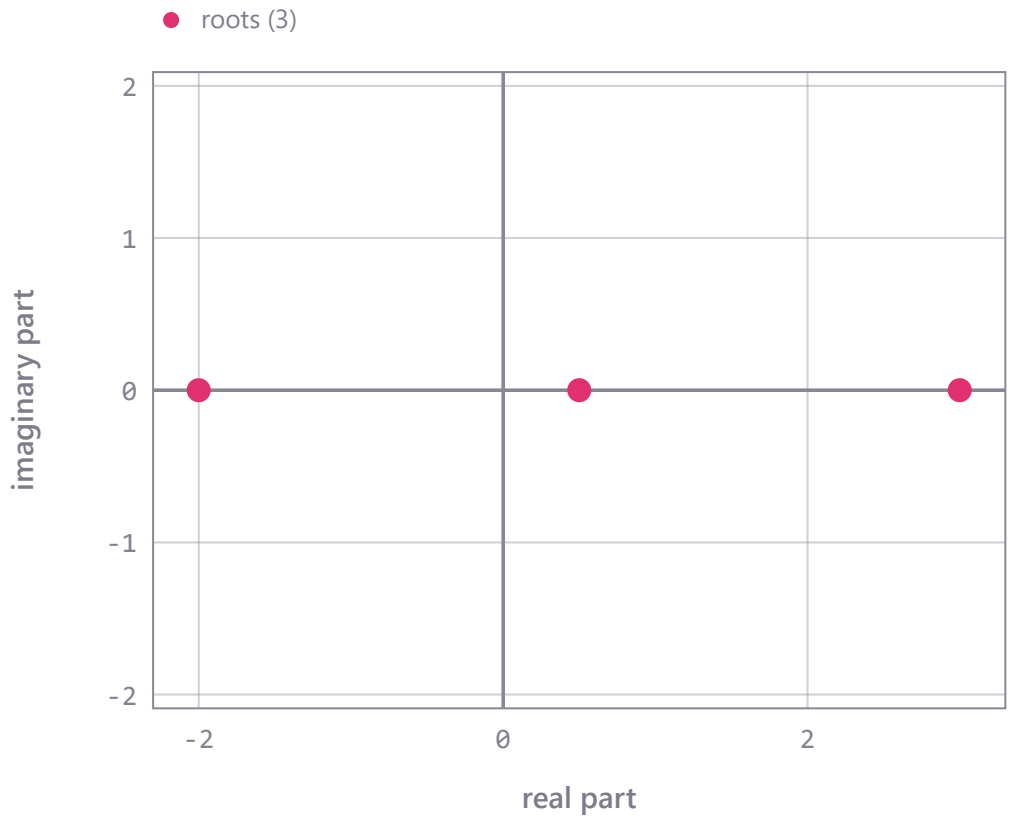
Every rational root of a polynomial with whole-number coefficients (degree up to 8).

$$a_n x^n + \cdots + a_0 = 0,\ x = \tfrac{p}{q} \Rightarrow p \mid a_0,\ q \mid a_n$$

| input | meaning | condition |
|---|---|---|
| `coefficients` | whole-number coefficients, highest power first | |

| output | meaning |
|---|---|
| `count` | number of distinct rational roots |
| `roots` | the rational roots, ascending |
| `roots_exact` | the rational roots as fractions |

- **Method:** rational root theorem: a root p/q in lowest terms has p dividing the constant term and q dividing the leading coefficient; test every candidate exactly
- **Assumes:** whole-number coefficients; degree 1 to 8
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




the rational roots, ascending (roots) returned by math-foundations-ai999x for the worked example (coefficients = 2, -3, -11, 6), drawn in the complex plane: -2 + 0i, 0.5 + 0i, 3 + 0i.

```
math_foundations_ai999x.solve('rational_roots', coefficients='2, -3, -11, 6')
# -> count = 3, roots = [-2.0, 0.5, 3.0], roots_exact = ['-2', '1/2', '3']   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_roots', coefficients=[1, 0, -7, 6])
# -> count = 3, roots = [-3.0, 1.0, 2.0], roots_exact = ['-3', '1', '2']   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_roots', coefficients=[1, 0, -2])
# -> count = 0, roots = [], roots_exact = []   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find all rational roots of 2x^3 - 3x^2 - 11x + 6 = 0. | SOLVED | count = 3, roots = [-2.0, 0.5, 3.0], roots_exact = ['-2', '1/2', '3'] |
| List the rational zeros of x^3 - 7x + 6. | SOLVED | count = 3, roots = [-3.0, 1.0, 2.0], roots_exact = ['-3', '1', '2'] |
| Find the rational roots of x^2 - 2. | SOLVED | count = 0, roots = [], roots_exact = [] |
| Find all rational roots of 0.5x^2 - 2. | UNRECOGNIZED_FORMAT | I could not match this question to a documented form, so I will not guess. Every form: math_foundations_ai999x.templates |

## Evaluate a function at a number — `evaluate_function`

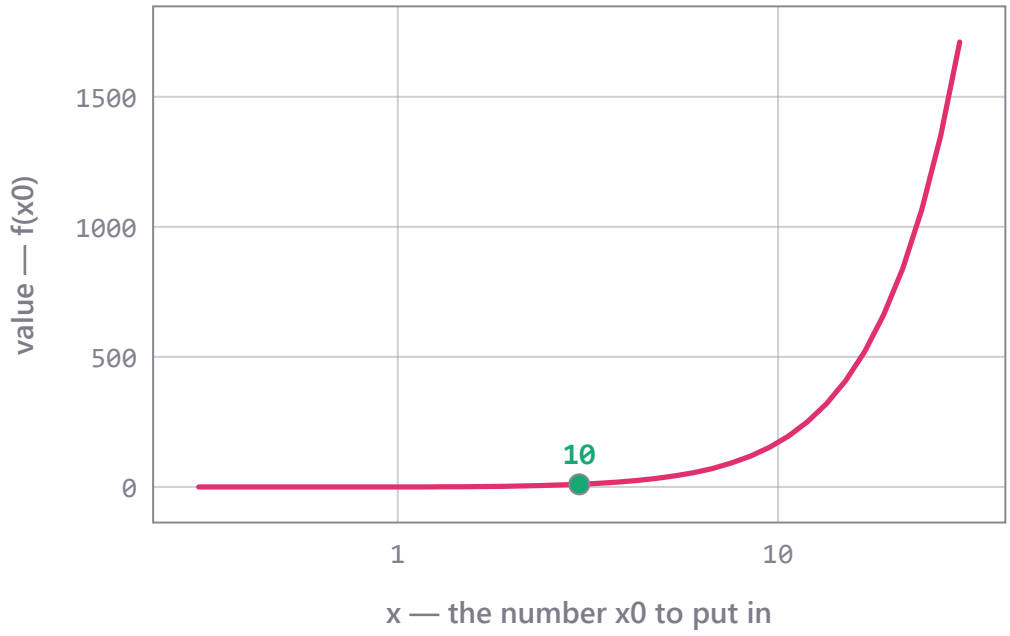
The value f(x0) of a function given by a formula in x (exact when it is rational).

$$f(x_0)$$

| input | meaning | condition |
|---|---|---|
| `expression` | the formula of f in x: numbers, x, + - * / ^, parentheses, sqrt, abs, ln | |
| `x` | the number x0 to put in | |

| output | meaning |
|---|---|
| `value` | f(x0) |
| `numerator` | numerator of the exact value |
| `denominator` | denominator of the exact value |

- **Method:** substitute x0 for x and evaluate with the order of operations, in exact rational arithmetic
- **Assumes:** x0 in the domain of f (no division by 0, no root of a negative number, no logarithm of a number <= 0)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




value (f(x0)) against x (the number x0 to put in) from 0.3 to 30 on a logarithmic axis. Other inputs as in the worked example: expression = 2x^2 - 3x + 1. 41 points computed by math-foundations-ai999x; marked: the worked example, x = 3, value = 10.

```
math_foundations_ai999x.solve('evaluate_function', expression='2x^2 - 3x + 1', x=3)
# -> value = 10, numerator = 10, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('evaluate_function', expression='x^3 - 2x + 5', x=-2)
# -> value = 1, numerator = 1, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('evaluate_function', expression='(x + 1)/(x - 3)', x=2)
# -> value = -3, numerator = -3, denominator = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('evaluate_function', expression='sqrt(x + 7)', x=2)
# -> value = 3, numerator = 3, denominator = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate f(3) for f(x) = 2x^2 - 3x + 1. | SOLVED | value = 10, numerator = 10, denominator = 1 |
| If f(x) = x^3 - 2x + 5, find f(-2). | SOLVED | value = 1, numerator = 1, denominator = 1 |
| Find f(1/2) when f(x) = 4x^2 - 1. | SOLVED | value = 0, numerator = 0, denominator = 1 |
| Find f(3) when f(x) = (x + 1)/(x - 3). | INVALID_INPUT | Evaluate a function at a number: f(3) is not defined: division by zero. |

## Value of a composite function — `composite_function_value`

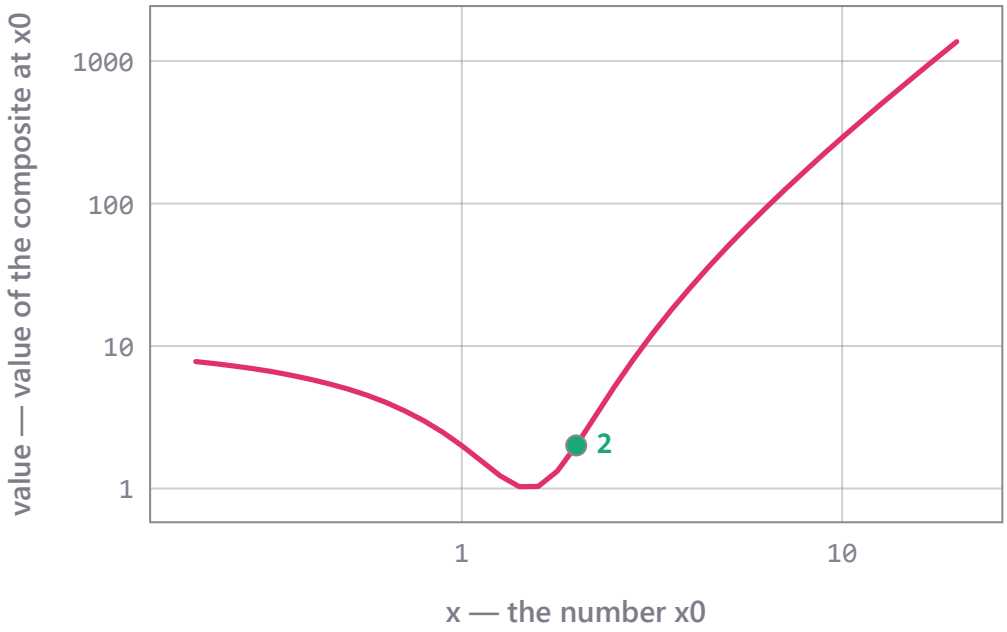
f(g(x0)) or g(f(x0)) for two functions given by formulas.

$$(f \circ g)(x_0) = f\big(g(x_0)\big)$$

| input | meaning | condition |
|---|---|---|
| `f` | the formula of f in x | |
| `g` | the formula of g in x | |
| `x` | the number x0 | |
| `order` | which composition | |

| output | meaning |
|---|---|
| `inner` | value of the inner function at x0 |
| `value` | value of the composite at x0 |

- **Method:** evaluate the inner function at x0, then the outer function at that value
- **Assumes:** each value in the domain of the function applied to it
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




value (value of the composite at x0) against x (the number x0) from 0.2 to 20 on a logarithmic axis, logarithmic output axis. Other inputs as in the worked example: f = x^2 + 1, g = 2x - 3, order = f(g(x)). 41 points computed by math-foundations-ai999x; marked: the worked example, x = 2, value = 2.

```
math_foundations_ai999x.solve('composite_function_value', f='x^2 + 1', g='2x - 3', x=2,
order='f(g(x))')
# -> inner = 1, value = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('composite_function_value', f='3x + 2', g='x^2', x=-1,
order='g(f(x))')
# -> inner = -1, value = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| If f(x) = x^2 + 1 and g(x) = 2x - 3, find f(g(2)). | SOLVED | inner = 1, value = 2 |
| If f(x) = x^2 + 1 and g(x) = 2x - 3, find g(f(2)). | SOLVED | inner = 5, value = 7 |
| Let f(x) = 3x + 2 and g(x) = x^2, evaluate g(f(-1)). | SOLVED | inner = -1, value = 1 |
| If f(x) = 1/x and g(x) = x - 2, find f(g(2)). | INVALID_INPUT | Value of a composite function: the composition is not defined at x = 2: division by zero. |

## Domain of a square-root function — `domain_sqrt_over_linear`

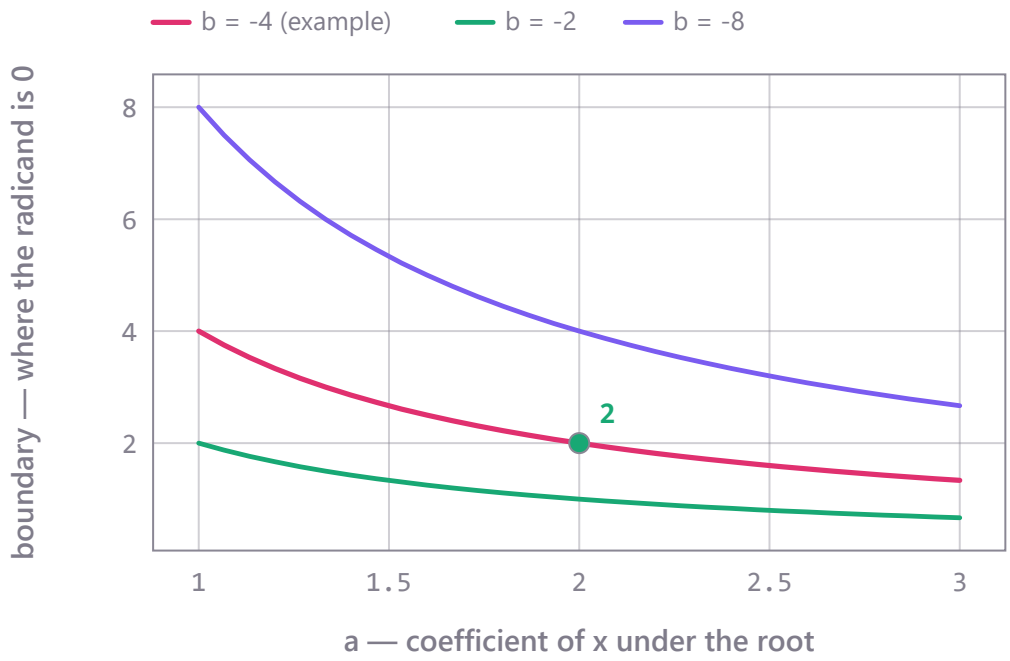
The domain of f(x) = sqrt(a*x + b)/(c*x + d) (or $1/sqrt$(a*x + b)): where the radicand is allowed and the denominator is not 0.

$$ax + b \geq 0, \quad cx + d \neq 0$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x under the root | $a \neq 0$ |
| `b` | constant under the root | |
| `c` | coefficient of x in the denominator (0 for none) | |
| `d` | constant in the denominator (1 for none) | |
| `root_in_denominator` | true when the root itself is in the denominator (the radicand must then be positive) | |

| output | meaning |
|---|---|
| `boundary` | where the radicand is 0 |
| `side` | the inequality x must satisfy with the boundary |
| `excluded` | a point removed because the denominator is 0 there (None if none) |
| `domain` | the domain in interval notation |

- **Method:** radicand $ax + b \geq 0$ (> 0 when the root is in a denominator); remove the zero of $cx + d$
- **Assumes:** $a \neq 0$; the denominator is not identically 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




boundary (where the radicand is 0) against a (coefficient of x under the root) from 1 to 3. One curve for each value of b (constant under the root): -4, -2, -8. Other inputs as in the worked example: c = 0, d = 1. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, boundary = 2.

```
math_foundations_ai999x.solve('domain_sqrt_over_linear', a=2, b=-4, c=0, d=1)
# -> boundary = 2, side = >=, excluded = none, domain = [2, inf)   (SOLVED)
```

```
math_foundations_ai999x.solve('domain_sqrt_over_linear', a=1, b=3, c=1, d=-1)
# -> boundary = -3, side = >=, excluded = 1, domain = [-3, 1) U (1, inf)   (SOLVED)
```

```
math_foundations_ai999x.solve('domain_sqrt_over_linear', a=-1, b=5, c=0, d=1,
root_in_denominator=True)
# -> boundary = 5, side = <, excluded = none, domain = (-inf, 5)   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the domain of f(x) = sqrt(2x - 4). | SOLVED | boundary = 2, side = >=, excluded = none, domain = [2, inf) |
| Find the domain of f(x) = sqrt(x + 3)/(x - 1). | SOLVED | boundary = -3, side = >=, excluded = 1, domain = [-3, 1) U (1, inf) |
| Find the domain of f(x) = 1/sqrt(5 - x). | SOLVED | boundary = 5, side = <, excluded = none, domain = (-inf, 5) |
| Find the domain of f(x) = sqrt(x - 4)/(x - 2). | SOLVED | boundary = 4, side = >=, excluded = none, domain = [4, inf) |

## Inverse of a linear function — `inverse_linear_function`

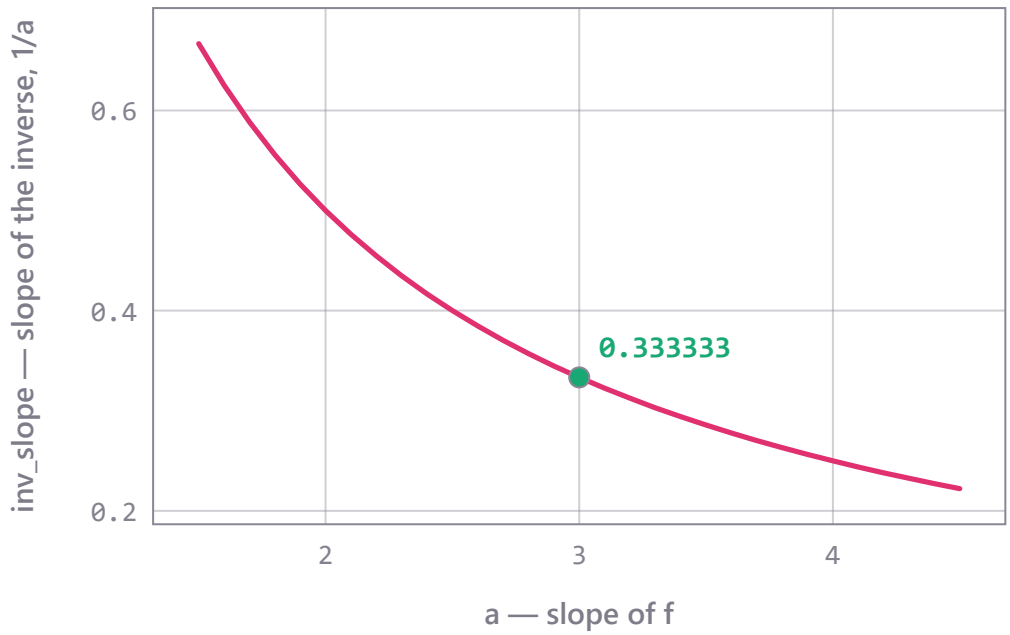
For f(x) = a*x + b $(a \neq 0)$: $f^{-1}$(x) = x/a - b/a, and $f^{-1}$(y) for a given y.

$$f^{-1}(x) = \frac{x - b}{a}$$

| input | meaning | condition |
|---|---|---|
| `a` | slope of f | $a \neq 0$ |
| `b` | intercept of f | |
| `y` | a value to invert (optional) | |

| output | meaning |
|---|---|
| `inv_slope` | slope of the inverse, $1/a$ |
| `inv_intercept` | intercept of the inverse, -b/a |
| `value` | $f^{-1}$(y) |

- **Method:** swap x and y in $y = ax + b$ and solve for y
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




inv_slope (slope of the inverse, 1/a) against a (slope of f) from 1.5 to 4.5. Other inputs as in the worked example: b = -6. 31 points computed by math-foundations-ai999x; marked: the worked example, a = 3, inv_slope = 0.333333.

```
math_foundations_ai999x.solve('inverse_linear_function', a=3, b=-6)
# -> inv_slope = 0.333333, inv_intercept = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('inverse_linear_function', a=3, b=-6, y=9)
# -> inv_slope = 0.333333, inv_intercept = 2, value = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('inverse_linear_function', a=-2, b=5)
# -> inv_slope = -0.5, inv_intercept = 2.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the inverse of f(x) = 3x - 6. | SOLVED | inv_slope = 0.333333, inv_intercept = 2 |
| Find f^-1(9) if f(x) = 3x - 6. | SOLVED | inv_slope = 0.333333, inv_intercept = 2, value = 5 |
| Find the inverse function of f(x) = -2x + 5. | SOLVED | inv_slope = -0.5, inv_intercept = 2.5 |

## Evaluate a piecewise function — `piecewise_evaluate`

f(x0) for f(x) = e1 when $x < c$ (or x <= c) and e2 otherwise.

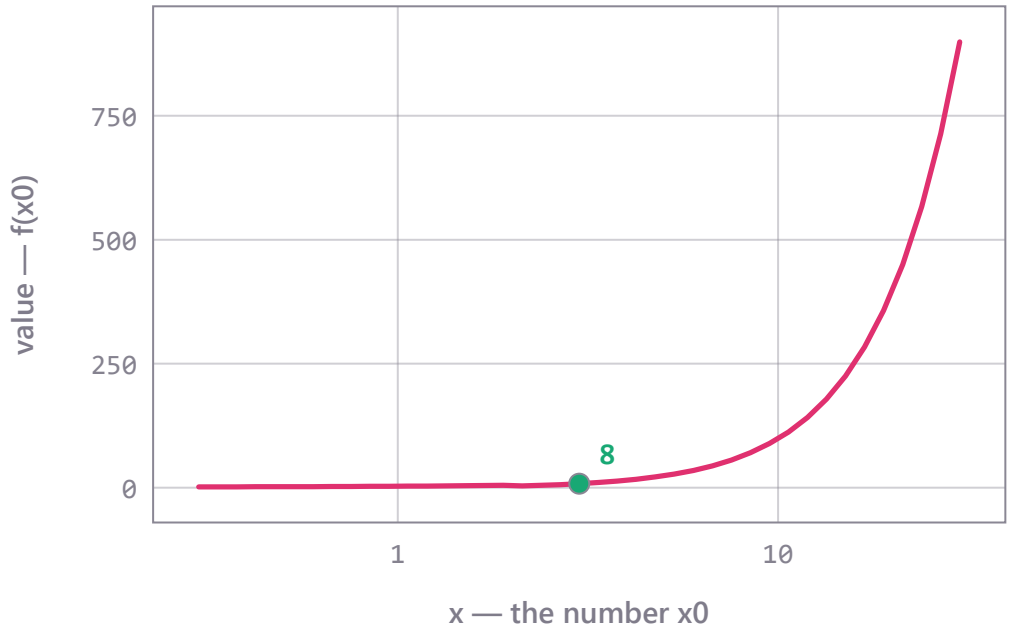
$$f(x) = \begin{cases} e_1(x) & x < c \\ e_2(x) & x \geq c \end{cases}$$

| input | meaning | condition |
|---|---|---|
| `e1` | formula of the left piece | |
| `left_rel` | the left piece holds for $x < c$ or $x \leq c$ | |
| `c` | the break point c | |

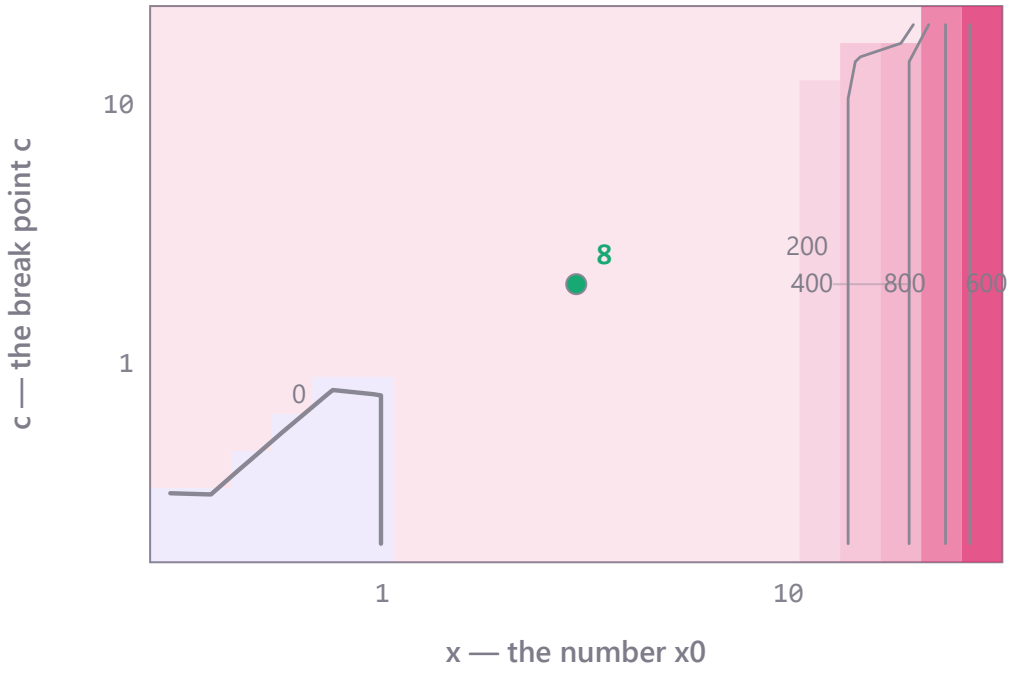
| input | meaning | condition |
|---|---|---|
| `e2` | formula of the right piece | |
| `x` | the number x0 | |

| output | meaning |
|---|---|
| `value` | f(x0) |
| `piece` | which piece applies (1 = left, 2 = right) |

- **Method:** decide which condition x0 satisfies (the break point belongs to the piece whose condition includes it), then evaluate that piece
- **Assumes:** the two conditions split the line at c (one < and one >=, or <= and >); x0 in the domain of its piece
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




value (f(x0)) against x (the number x0) from 0.3 to 30 on a logarithmic axis. Other inputs as in the worked example: e1 = 2x + 1, left_rel = <, c = 2, e2 = x^2 - 1. 41 points computed by math-foundations-ai999x; marked: the worked example, x = 3, value = 8.




value (f(x0)) over x (the number x0) from 0.3 to 30 and c (the break point c) from 0.2 to 20 (logarithmic axes): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its sign and magnitude (second colour negative); iso-lines at value = 0, 200, 400, 600, 800. Other inputs as in the worked example: e1 = 2x + 1, left_rel = <, e2 = x^2 - 1. Marked: the worked example, x = 3, c = 2, value = 8.

```
math_foundations_ai999x.solve('piecewise_evaluate', e1='2x + 1', left_rel='<', c=2, e2='x^2 - 1',
x=3)
# -> value = 8, piece = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('piecewise_evaluate', e1='3 - x', left_rel='<=', c=0, e2='x^2 + 4',
x=0)
# -> value = 3, piece = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate f(3) for f(x) = 2x + 1 if x < 2, x^2 - 1 if x >= 2. | SOLVED | value = 8, piece = 2 |
| If f(x) = 3 - x for x <= 0 and x^2 + 4 for x > 0, find f(0). | SOLVED | value = 3, piece = 1 |
| Evaluate f(2) for f(x) = 2x + 1 if x < 2, x^2 - 1 if x >= 2. | SOLVED | value = 3, piece = 2 |
| Evaluate f(1) for f(x) = x if x < 2, x^2 if x > 3. | OUT_OF_SCOPE | Out of scope: a piecewise function whose two pieces do not split the line at one point. |

## Image of a point under a graph transformation — `graph_transformation_point`

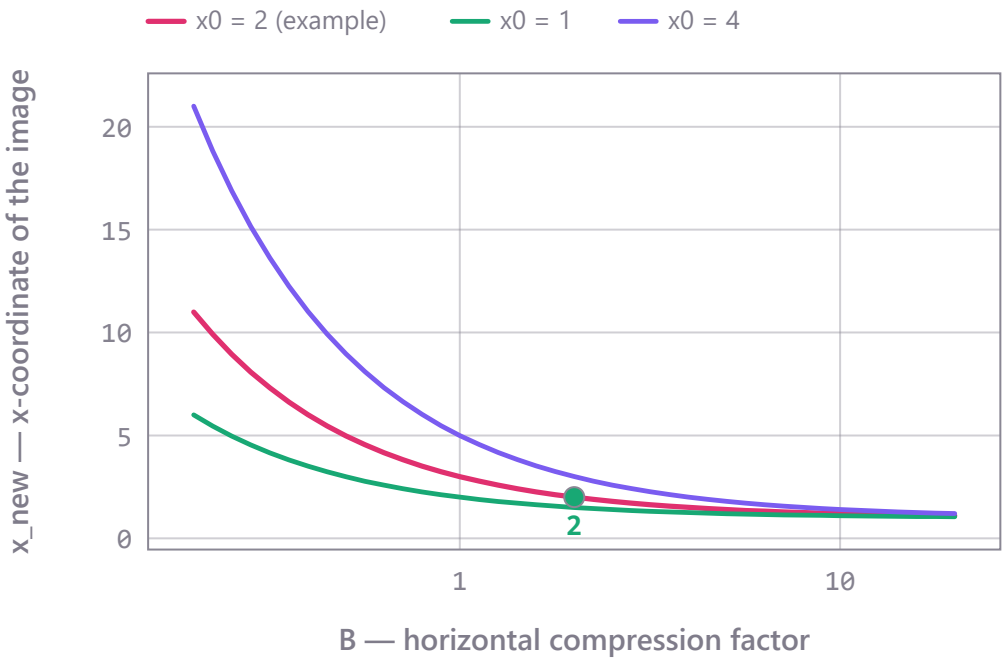
If (x0, y0) is on $y = f(x)$, the corresponding point on $y = Af(B(x - h)) + k$.

$$(x_0, y_0) \mapsto \left(\frac{x_0}{B} + h,\ Ay_0 + k\right)$$

| input | meaning | condition |
|---|---|---|
| `x0` | x-coordinate of the point on $y = f(x)$ | |
| `y0` | y-coordinate of the point | |
| `A` | vertical stretch (negative: reflection in the x-axis) | |
| `B` | horizontal compression factor (negative: reflection in the y-axis) | |
| `h` | horizontal shift | |
| `k` | vertical shift | |

| output | meaning |
|---|---|
| `x_new` | x-coordinate of the image |
| `y_new` | y-coordinate of the image |

- **Method:** B(x - h) must equal x0, so $x = x_0/B + h$; the height becomes $Ay_0 + k$
- **Assumes:** $A \neq 0$; $B \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x_new (x-coordinate of the image) against B (horizontal compression factor (negative: reflection in the y-axis)) from 0.2 to 20 on a logarithmic axis. One curve for each value of x0 (x-coordinate of the point on y = f(x)): 2, 1, 4. Other inputs as in the worked example: y0 = 5, A = 3, h = 1, k = 4. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, B = 2, x_new = 2.

```
math_foundations_ai999x.solve('graph_transformation_point', x0=2, y0=5, A=3, B=2, h=1, k=4)
# -> x_new = 2, y_new = 19   (SOLVED)
```

```
math_foundations_ai999x.solve('graph_transformation_point', x0=-4, y0=6, A=-1, B=1, h=-3, k=0)
# -> x_new = -7, y_new = -6   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| The point (2, 5) is on the graph of y = f(x). Find the corresponding point on the graph of y = 3f(2(x - 1)) + 4. | SOLVED | x_new = 2, y_new = 19 |
| The point (-4, 6) is on the graph of y = f(x). Find the corresponding point on the graph of y = -f(x + 3). | SOLVED | x_new = -7, y_new = -6 |
| The point (3, 1) is on the graph of y = f(x). Find the corresponding point on the graph of y = f(3x) - 2. | SOLVED | x_new = 1, y_new = -1 |

### Intercepts and slope of a line — `line_intercepts`

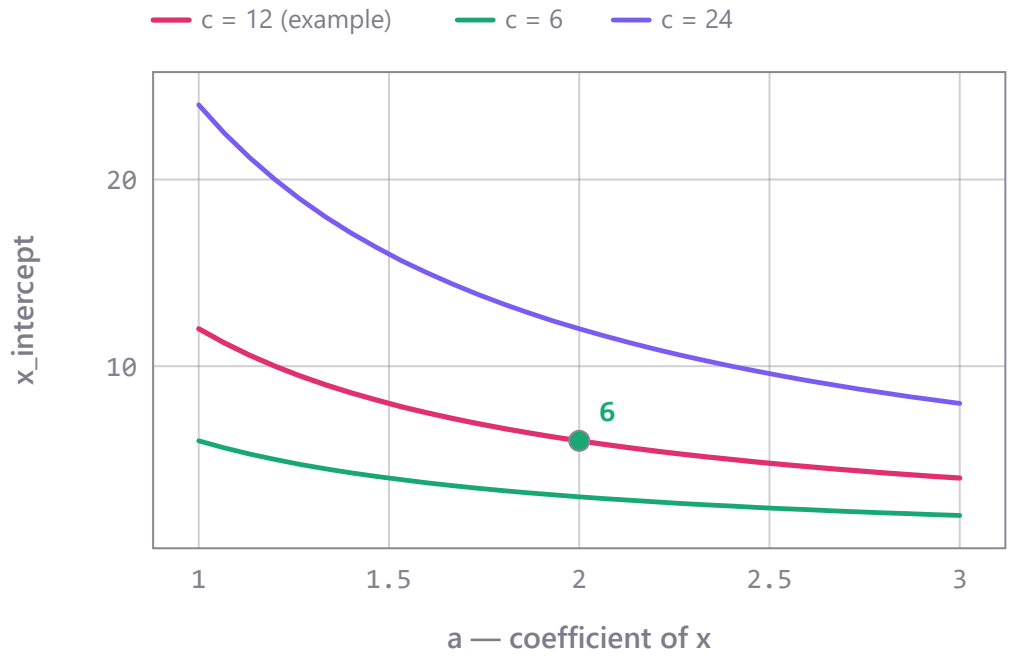
The x- and y-intercepts and the slope of the line $ax + by = c$.

$$ax + by = c: \quad x_0 = \frac{c}{a},\ y_0 = \frac{c}{b},\ m = -\frac{a}{b}$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x | $a \neq 0$ |
| `b` | coefficient of y | $b \neq 0$ |
| `c` | right-hand side | |

| output | meaning |
|---|---|
| `x_intercept` | where the line meets the x-axis $(y = 0)$ |
| `y_intercept` | where it meets the y-axis $(x = 0)$ |
| `slope` | the slope -a/b |

- **Method:** set $y = 0$ for the x-intercept and $x = 0$ for the y-intercept
- **Assumes:** $a \neq 0$; $b \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x_intercept (where the line meets the x-axis (y = 0)) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 12, 6, 24. Other inputs as in the worked example: b = 3. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x_intercept = 6.

```
math_foundations_ai999x.solve('line_intercepts', a=2, b=3, c=12)
# -> x_intercept = 6, y_intercept = 4, slope = -0.666667   (SOLVED)
```

```
math_foundations_ai999x.solve('line_intercepts', a=-2, b=1, c=-6)
# -> x_intercept = 3, y_intercept = -6, slope = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the x- and y-intercepts of 2x + 3y = 12. | SOLVED | x_intercept = 6, y_intercept = 4, slope = -0.666667 |
| Find the intercepts of the line y = 2x - 6. | SOLVED | x_intercept = 3, y_intercept = -6, slope = 2 |
| Find the intercepts of the line y = 5. | INVALID_INPUT | Intercepts and slope of a line: a and b must both be nonzero (otherwise the line is parallel to an axis and misses one o |

## Distance from a point to a line — `distance_point_to_line`

The perpendicular distance from (x0, y0) to the line $ax + by + c = 0$.

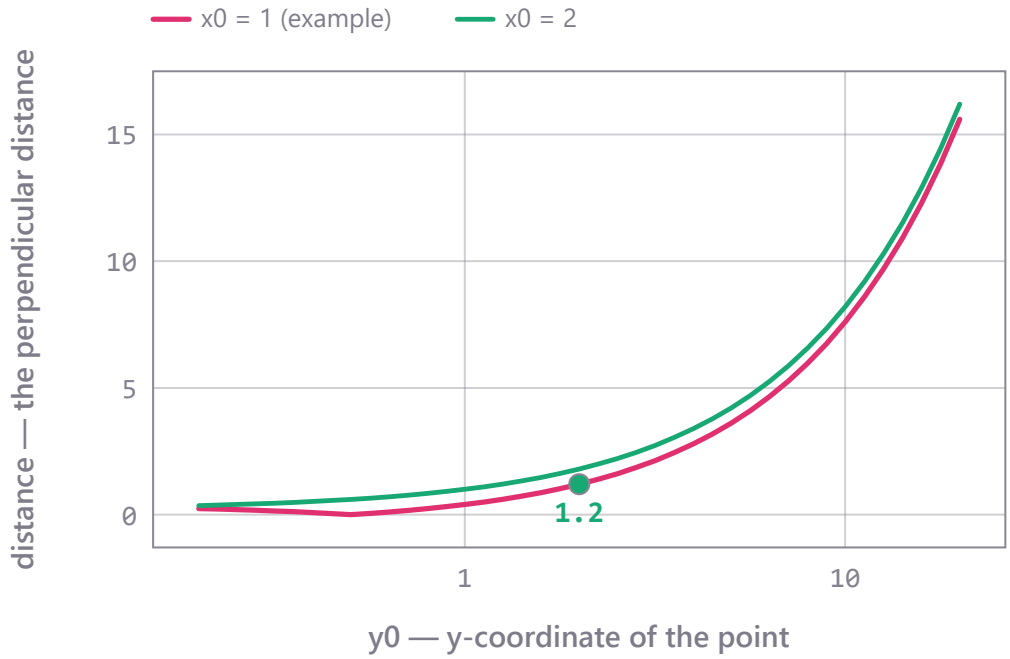
$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

| input | meaning | condition |
|---|---|---|
| `x0` | x-coordinate of the point | |
| `y0` | y-coordinate of the point | |
| `a` | coefficient of x | |

| input | meaning | condition |
|---|---|---|
| `b` | coefficient of y | |
| `c` | constant term | |

| output | meaning |
|---|---|
| `distance` | the perpendicular distance |

- **Method:** |a $x_0$ + b $y_0$ + c| / sqrt(a^2 + b^2)
- **Assumes:** a and b not both 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




distance (the perpendicular distance) against y0 (y-coordinate of the point) from 0.2 to 20 on a logarithmic axis. One curve for each value of x0 (x-coordinate of the point): 1, 2. Other inputs as in the worked example: a = 3, b = 4, c = -5. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, y0 = 2, distance = 1.2.

```
math_foundations_ai999x.solve('distance_point_to_line', x0=1, y0=2, a=3, b=4, c=-5)
# -> distance = 1.2   (SOLVED)
```

```
math_foundations_ai999x.solve('distance_point_to_line', x0=0, y0=0, a=-2, b=1, c=-5)
# -> distance = 2.23607   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the distance from the point (1, 2) to the line 3x + 4y - 5 = 0. | SOLVED | distance = 1.2 |
| Find the distance from the point (0, 0) to the line y = 2x + 5. | SOLVED | distance = 2.23607 |

## Evaluate a logarithm — `log_evaluate`

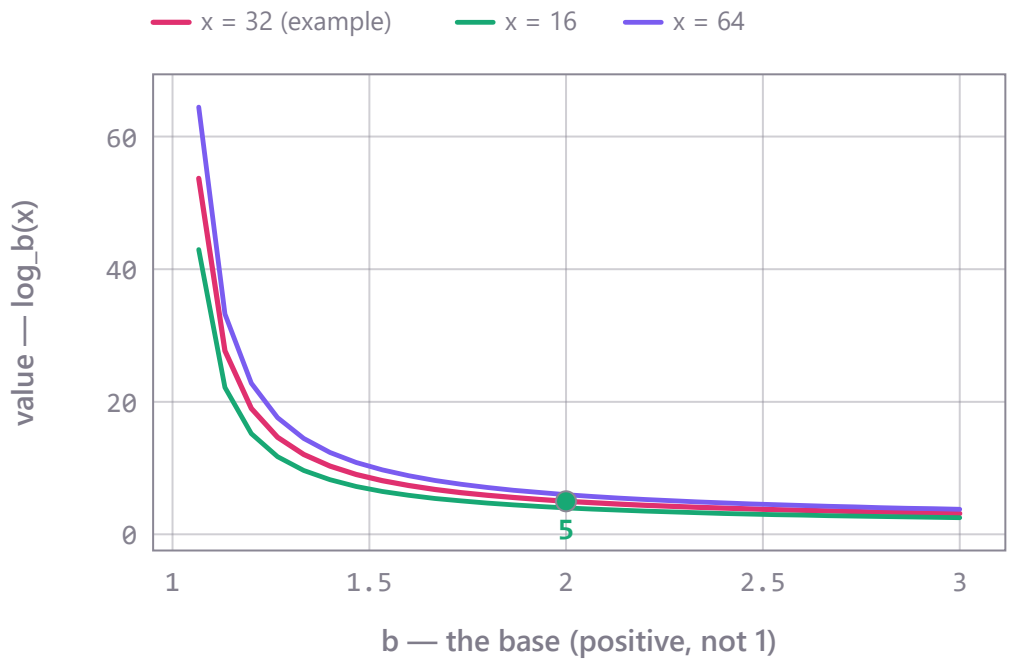
log_b(x) for a positive base $b \neq 1$ and $x > 0$ (log is base 10, ln base e).

$$\log_b x = \frac{\ln x}{\ln b}$$

| input | meaning | condition |
|---|---|---|
| `b` | the base (positive, not 1) | |
| `x` | the number (positive) | |

| output | meaning |
|---|---|
| `value` | log_b(x) |

- **Method:** change of base: log_b(x) = ln(x)/ln(b)
- **Assumes:** $b > 0$; $b \neq 1$; $x > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




value (log_b(x)) against b (the base (positive, not 1)) from 1 to 3. One curve for each value of x (the number (positive)): 32, 16, 64. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, value = 5.

```
math_foundations_ai999x.solve('log_evaluate', b=2, x=32)
# -> value = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=3, x=81)
# -> value = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=2.718281828459045, x=20)
# -> value = 2.99573   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=10, x=1000)
# -> value = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('log_evaluate', b=2, x=0.125)
# -> value = -3   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Evaluate log base 2 of 32. | SOLVED | value = 5 |
| Find log_3(81). | SOLVED | value = 4 |
| Compute ln(20). | SOLVED | value = 2.99573 |
| Find log(1000). | SOLVED | value = 3 |
| Evaluate log base 2 of -8. | INVALID_INPUT | Evaluate a logarithm: the logarithm is defined only for positive numbers. |
| Evaluate log base 1 of 5. | INVALID_INPUT | Evaluate a logarithm: the base must be positive and not 1. |

## Laws of logarithms (product, quotient, power) — `log_laws_combine`

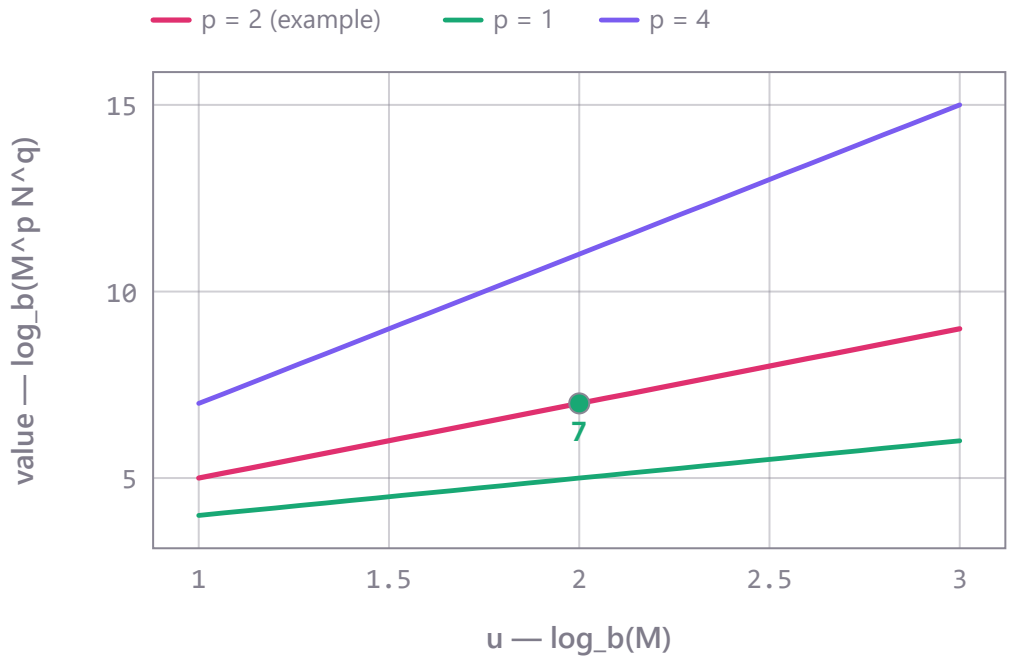
Given log_b(M) = u and log_b(N) = v, the value of log_b(M^p * N^q) (quotients and roots included).

$$\log_b(M^p N^q) = p \log_b M + q \log_b N$$

| input | meaning | condition |
|---|---|---|
| `u` | log_b(M) | |
| `v` | log_b(N) | |
| `p` | power of M | |
| `q` | power of N (negative: N in the denominator) | |

| output | meaning |
|---|---|
| `value` | log_b($M^p$ N^q) |

- **Method:** product law log(MN) = log M + log N, quotient law, power law log(M^p) = p log M
- **Assumes:** M, $N > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




value (log_b(M^p N^q)) against u (log_b(M)) from 1 to 3. One curve for each value of p (power of M): 2, 1, 4. Other inputs as in the worked example: v = 3, q = 1. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, u = 2, value = 7.

```
math_foundations_ai999x.solve('log_laws_combine', u=2, v=3, p=2, q=1)
# -> value = 7   (SOLVED)
```

```
math_foundations_ai999x.solve('log_laws_combine', u=1.5, v=4, p=3, q=-1)
# -> value = 0.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Given log_b(M) = 2 and log_b(N) = 3, find log_b(M^2 N). | SOLVED | value = 7 |
| If log_b(M) = 1.5 and log_b(N) = 4, find log_b(M^3/N). | SOLVED | value = 0.5 |
| Given log_b(M) = 6 and log_b(N) = 2, find log_b(sqrt(M)/N^2). | SOLVED | value = -1 |

## Continuous exponential growth or decay — `continuous_growth`

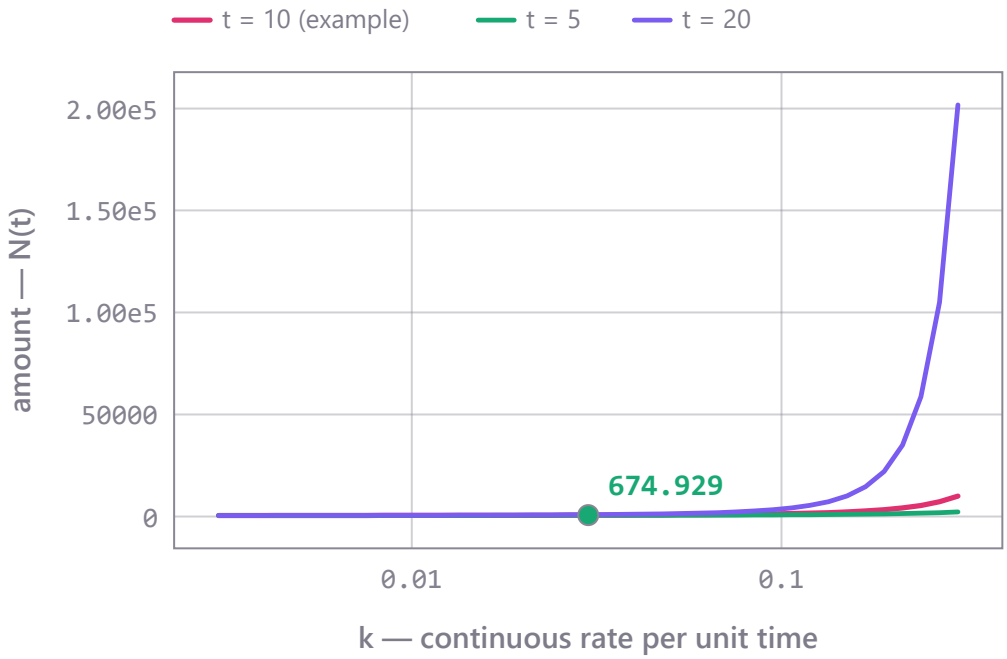
N($t$) $= N_0 e^{(k}$ t) for continuous growth $(k > 0)$ or decay $(k < 0)$, including continuous compounding.

$$N(t) = N_0\, e^{kt}$$

| input | meaning | condition |
|---|---|---|
| `n0` | initial amount N0 | |
| `k` | continuous rate per unit time, as a decimal (0.03 for 3%; negative for decay) | |
| `t` | elapsed time, in the rate's unit | |

| output | meaning |
|---|---|
| `amount` | N(t) |

- **Method:** the solution of $dN/dt = kN$: $N = N_0\ e^{kt}$
- **Assumes:** |k t| <= 700
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




amount (N(t)) against k (continuous rate per unit time, as a decimal (0.03 for 3%; negative for decay)) from 0.003 to 0.3 on a logarithmic axis. One curve for each value of t (elapsed time, in the rate's unit): 10, 5, 20. Other inputs as in the worked example: n0 = 500. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, k = 0.03, amount = 674.929.

```
math_foundations_ai999x.solve('continuous_growth', n0=500, k=0.03, t=10)
# -> amount = 674.929   (SOLVED)
```

```
math_foundations_ai999x.solve('continuous_growth', n0=1000, k=0.05, t=3)
# -> amount = 1161.83   (SOLVED)
```

```
math_foundations_ai999x.solve('continuous_growth', n0=200, k=-0.05, t=8)
# -> amount = 134.064   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| A population of 500 grows continuously at 3% per year. Find the population after 10 years. | SOLVED | amount = 674.929 |
| Find the amount when 1000 is compounded continuously at 5% for 3 years. | SOLVED | amount = 1161.83 |
| A quantity of 200 decays continuously at a rate of 5% per hour. Find the amount after 8 hours. | SOLVED | amount = 134.064 |
| A population of 500 grows continuously at 3% per year. Find | UNRECOGNIZED_FORMAT | I could not match this question to a documented form, so I will not guess. Every form: |

| question | outcome | result |
|---|---|---|
| the population after 10 days. | | math_foundations_ai999x.templates |

### Amount remaining after radioactive decay (half-life) — `half_life_remaining`

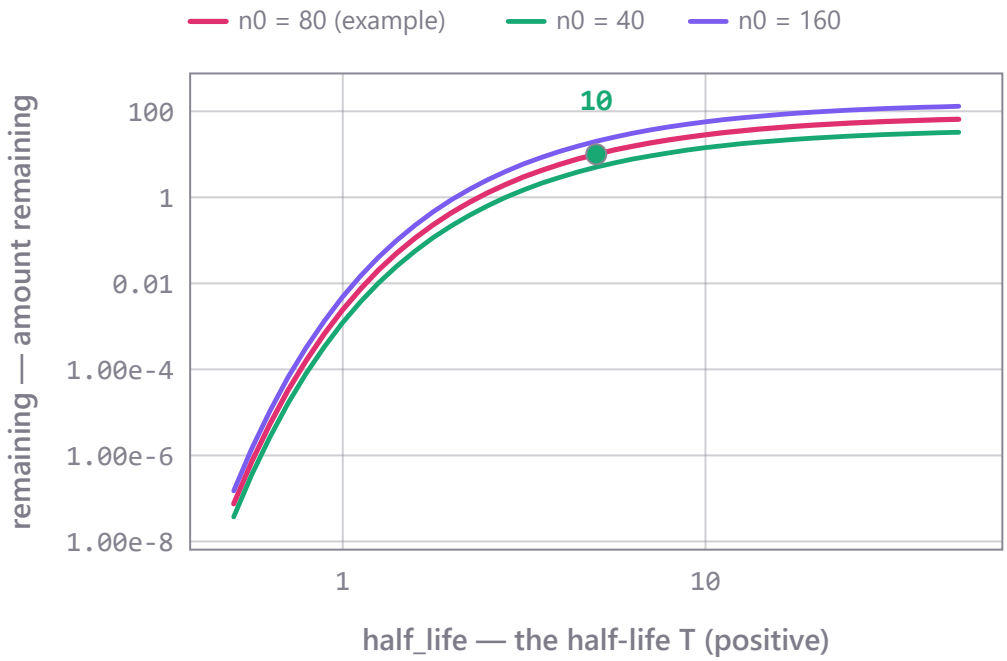
The amount left after time t of a substance with half-life T: $N_0(1/2)^{t/T}$.

$$N(t) = N_0\left(\tfrac{1}{2}\right)^{t/T}$$

| input | meaning | condition |
|---|---|---|
| `n0` | initial amount (positive) | |
| `half_life` | the half-life T (positive) | |
| `t` | elapsed time, in the unit of T | |

| output | meaning |
|---|---|
| `remaining` | amount remaining (in the unit of N0) |
| `fraction` | fraction remaining |

- **Method:** each half-life halves the amount: $N = N_0\ (1/2)^{t/T}$
- **Assumes:** $n_0 > 0$; $T > 0$; $t \geq 0$; t and T in the same unit
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




remaining (amount remaining (in the unit of N0)) against half_life (the half-life T (positive)) from 0.5 to 50 on a logarithmic axis, logarithmic output axis. One curve for each value of n0 (initial amount (positive)): 80, 40, 160. Other inputs as in the worked example: t = 15. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, half_life = 5, remaining = 10.

```
math_foundations_ai999x.solve('half_life_remaining', n0=80, half_life=5, t=15)
# -> remaining = 10, fraction = 0.125   (SOLVED)
```

```
math_foundations_ai999x.solve('half_life_remaining', n0=200, half_life=8, t=20)
# -> remaining = 35.3553, fraction = 0.176777   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| An 80 g sample has a half-life of 5 years. How much remains after 15 years? | SOLVED | remaining = 10, fraction = 0.125 |
| A sample of 200 mg of iodine-131 has a half-life of 8 days. How much is left after 20 days? | SOLVED | remaining = 35.3553, fraction = 0.176777 |
| An 80 g sample has a half-life of 5 years. How much remains after 15 days? | UNRECOGNIZED_FORMAT | I could not match this question to a documented form, so I will not guess. Every form: math_foundations_ai999x.templates |

## Doubling time at a percentage growth rate — `doubling_time_percent`

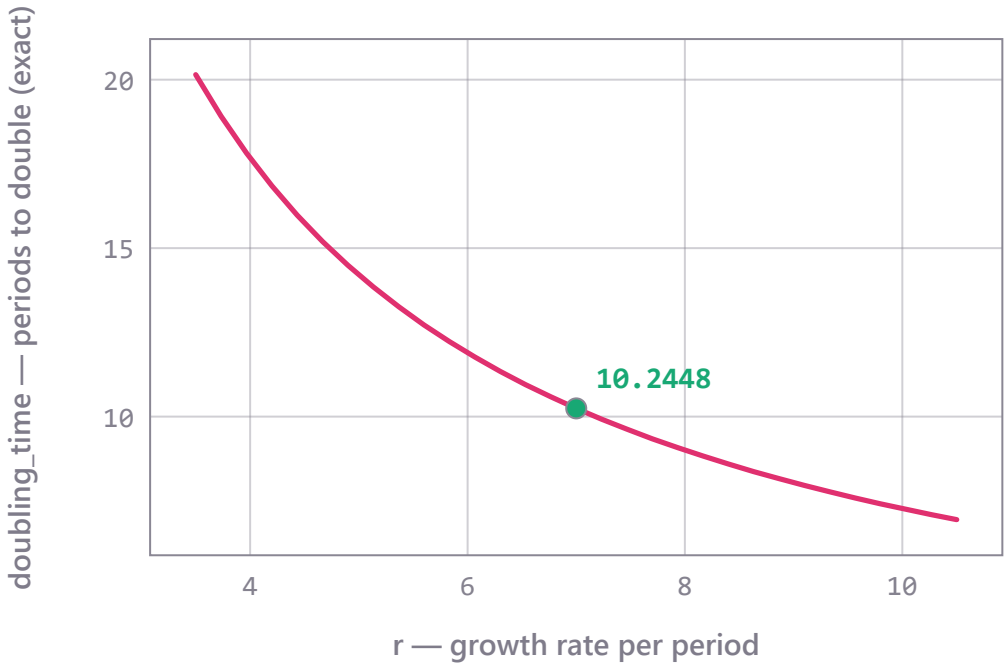
The time for a quantity growing by r% per period (compounded once per period) to double.

$$T = \frac{\ln 2}{\ln\left(1 + \frac{r}{100}\right)}$$

| input | meaning | condition |
|---|---|---|
| `r` | growth rate per period, in percent (positive) | |

| output | meaning |
|---|---|
| `doubling_time` | periods to double (exact) |
| `rule_of_70_estimate` | the rule-of-70 approximation $70/r$ |

- **Method:** solve $(1 + r/100)^T = 2$: $T = ln\,2/ln$(1 + r/100)
- **Assumes:** $r > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




doubling_time (periods to double (exact)) against r (growth rate per period, in percent (positive)) from 3.5 to 10.5. 31 points computed by math-foundations-ai999x; marked: the worked example, r = 7, doubling_time = 10.2448.

```
math_foundations_ai999x.solve('doubling_time_percent', r=7)
# -> doubling_time = 10.2448, rule_of_70_estimate = 10   (SOLVED)
```

```
math_foundations_ai999x.solve('doubling_time_percent', r=100)
# -> doubling_time = 1, rule_of_70_estimate = 0.7   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| How long does it take an investment to double at 7% interest compounded annually? | SOLVED | doubling_time = 10.2448, rule_of_70_estimate = 10 |
| What is the doubling time at a growth rate of 3% per year? | SOLVED | doubling_time = 23.4498, rule_of_70_estimate = 23.3333 |

## Exponential equation a*b^(cx + d) = k — `exponential_equation_general`

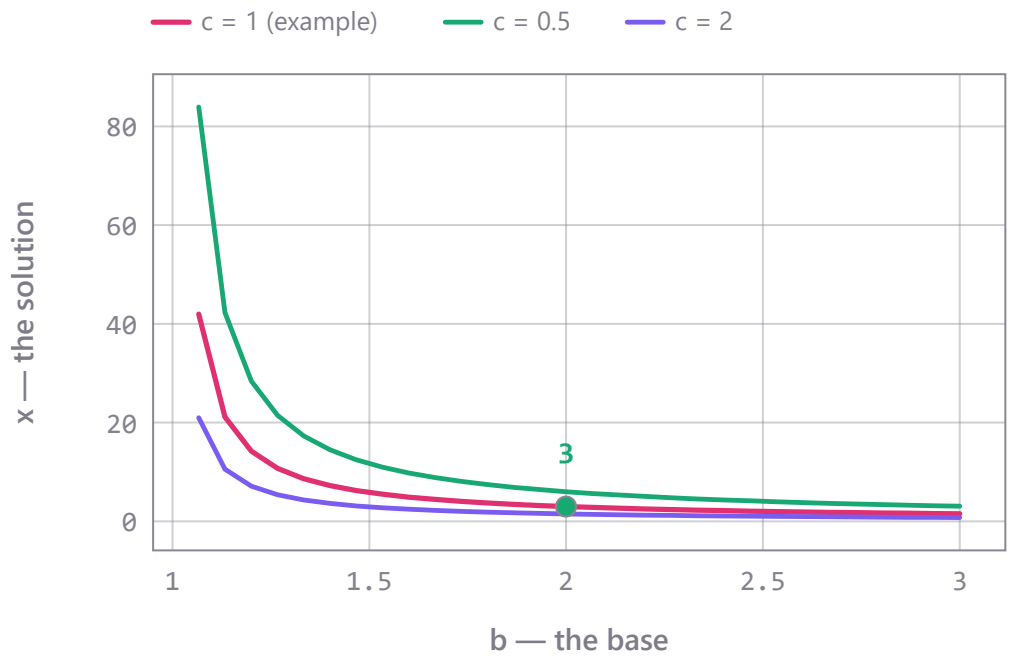
Solve $ab^{cx+d} = k$ for x by taking logarithms.

$$x = \frac{\log_b(k/a) - d}{c}$$

| input | meaning | condition |
|---|---|---|
| a | the coefficient a | $a \neq 0$ |
| b | the base (positive, not 1; e for the natural base) | |
| c | coefficient of x in the exponent | $c \neq 0$ |
| d | constant in the exponent | |
| k | the right-hand side | |

| output | meaning |
|---|---|
| x | the solution |

- **Method:** divide by a, take log base b: c $x + d = log_b$(k/a)
- **Assumes:** $a \neq 0$; $c \neq 0$; $b > 0$, $b \neq 1$; $k/a > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the solution) against b (the base (positive, not 1; e for the natural base)) from 1 to 3. One curve for each value of c (coefficient of x in the exponent): 1, 0.5, 2. Other inputs as in the worked example: a = 3, d = 1, k = 48. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 3.

```
math_foundations_ai999x.solve('exponential_equation_general', a=3, b=2, c=1, d=1, k=48)
# -> x = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('exponential_equation_general', a=5, b=2.718281828459045, c=2, d=0,
k=20)
# -> x = 0.693147   (SOLVED)
```

```
math_foundations_ai999x.solve('exponential_equation_general', a=1, b=2, c=3, d=-1, k=16)
# -> x = 1.66667   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Solve 3 * 2^(x + 1) = 48. | SOLVED | x = 3 |
| Solve 5e^(2x) = 20. | SOLVED | x = 0.693147 |
| Solve 2^(3x - 1) = 16. | SOLVED | x = 1.66667 |
| Solve 4 * 3^x = 324. | SOLVED | x = 4 |

| question | outcome | result |
|---|---|---|
| Solve 2 * 3^(x - 1) = -6. | INVALID_INPUT | Exponential equation a*b^(cx + d) = k: no real solution: b^(cx + d) is positive, so k/a must be positive. |

### Logarithmic equation log_b(cx + d) = k — `log_equation_linear_argument`

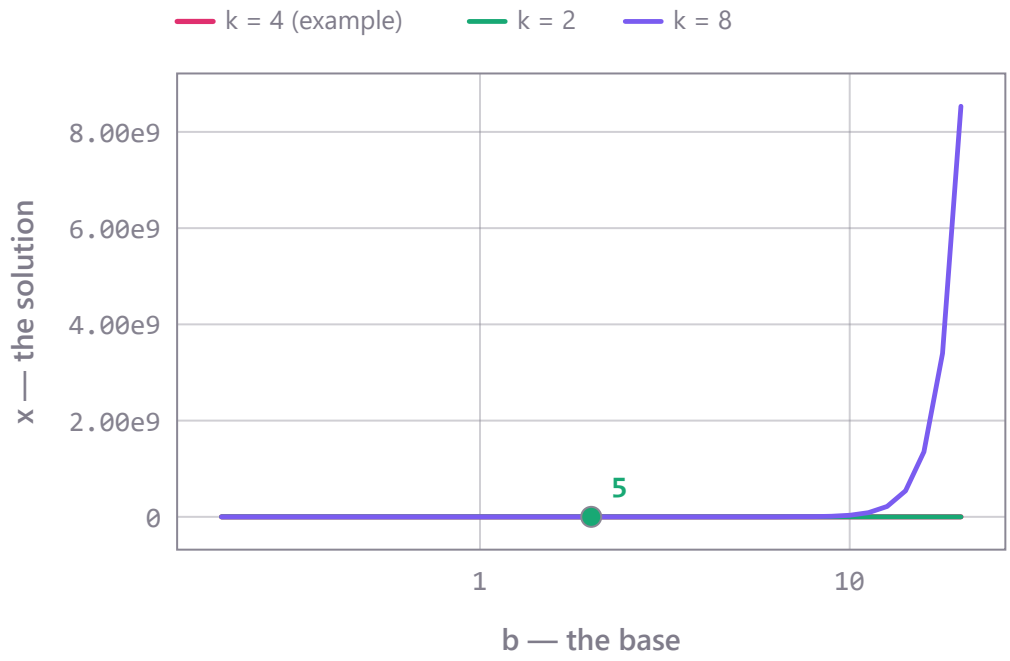
Solve log_b(c*x + d) = k for x by rewriting it in exponential form.

$$\log_b(cx + d) = k \iff cx + d = b^k$$

| input | meaning | condition |
|---|---|---|
| b | the base (positive, not 1; 10 for log, e for ln) | |
| c | coefficient of x in the argument | $c \neq 0$ |
| d | constant in the argument | |
| k | the right-hand side | |

| output | meaning |
|---|---|
| x | the solution |

- **Method:** log_b(u) = k means $u = b^k$; the argument $b^k$ is positive, so the solution is always admissible
- **Assumes:** $b > 0$, $b \neq 1$; $c \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the solution) against b (the base (positive, not 1; 10 for log, e for ln)) from 0.2 to 20 on a logarithmic axis. One curve for each value of k (the right-hand side): 4, 2, 8. Other inputs as in the worked example: c = 3, d = 1. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 5.

```
math_foundations_ai999x.solve('log_equation_linear_argument', b=2, c=3, d=1, k=4)
# -> x = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('log_equation_linear_argument', b=2.718281828459045, c=2, d=-1, k=3)
# -> x = 10.5428   (SOLVED)
```

```
math_foundations_ai999x.solve('log_equation_linear_argument', b=10, c=5, d=0, k=2)
# -> x = 20   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve log base 2 of (3x + 1) = 4. | SOLVED | x = 5 |
| Solve log_3(x - 4) = 2. | SOLVED | x = 13 |
| Solve ln(2x - 1) = 3. | SOLVED | x = 10.5428 |
| Solve log(5x) = 2. | SOLVED | x = 20 |
| Solve for x in the equation log_3(x - 1) = 2. | SOLVED | x = 10 |
| Solve log base 1 of (2x) = 3. | INVALID_INPUT | Logarithmic equation log_b(cx + d) = k: the base must be positive and not 1. |

## Degrees to radians — `deg_to_rad`

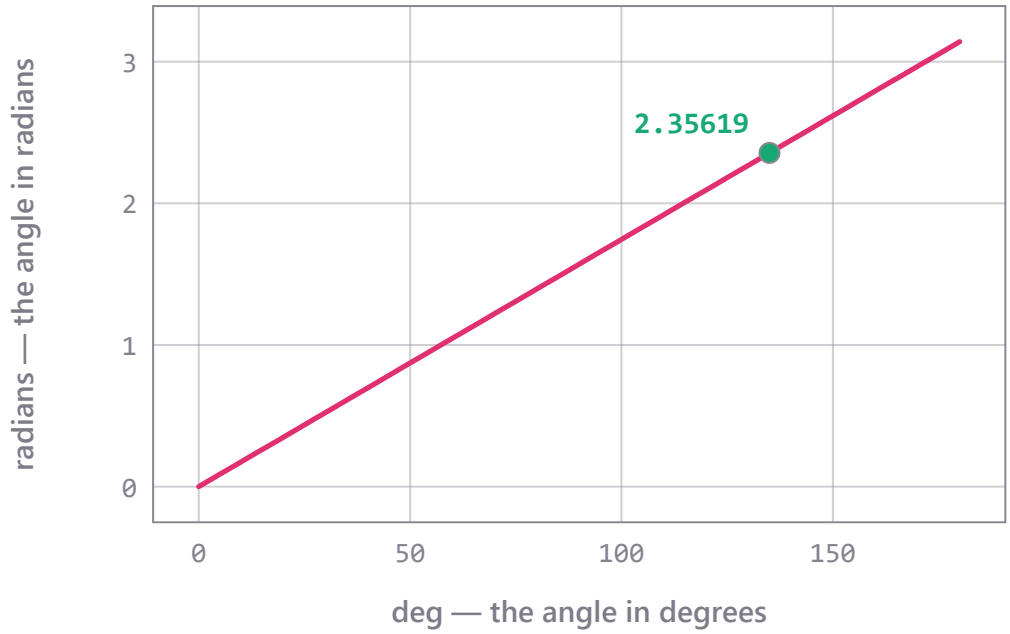
Convert an angle in degrees to radians (also as a multiple of pi).

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$$

| input | meaning | condition |
|---|---|---|
| `deg` | the angle in degrees | |

| output | meaning |
|---|---|
| `radians` | the angle in radians |
| `pi_multiple` | the angle as a multiple of pi |
| `exact` | exact form, e.g. $3pi/4$ |

- **Method:** multiply by $pi/180$
- **Assumes:** a finite angle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




radians (the angle in radians) against deg (the angle in degrees) from 0 to 180. 41 points computed by math-foundations-ai999x; marked: the worked example, deg = 135, radians = 2.35619.

```
math_foundations_ai999x.solve('deg_to_rad', deg=135)
# -> radians = 2.35619, pi_multiple = 0.75, exact = 3pi/4   (SOLVED)
```

```
math_foundations_ai999x.solve('deg_to_rad', deg=210)
# -> radians = 3.66519, pi_multiple = 1.16667, exact = 7pi/6   (SOLVED)
```

```
math_foundations_ai999x.solve('deg_to_rad', deg=-90)
# -> radians = -1.5708, pi_multiple = -0.5, exact = -pi/2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Convert 135 degrees to radians. | SOLVED | radians = 2.35619, pi_multiple = 0.75, exact = 3pi/4 |
| What is 210° in radians? | SOLVED | radians = 3.66519, pi_multiple = 1.16667, exact = 7pi/6 |
| Express 1 degree in radians. | SOLVED | radians = 0.0174533, pi_multiple = 0.00555556, exact = pi/180 |

## Radians to degrees — `rad_to_deg`

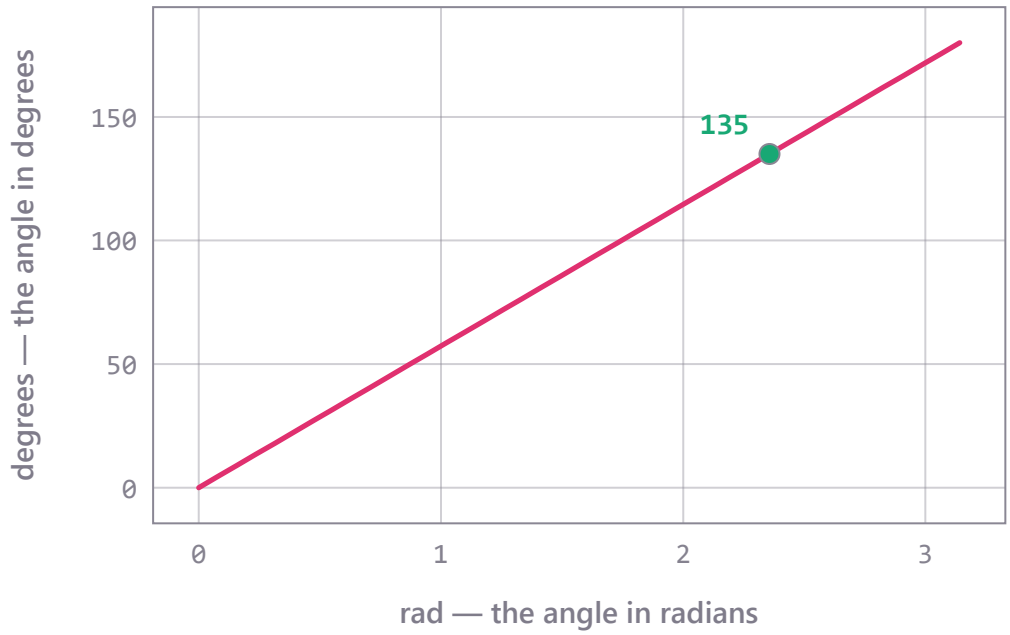
Convert an angle in radians (a number, or a multiple of pi such as 3pi/4) to degrees.

$$\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$$

| input | meaning | condition |
|---|---|---|
| `rad` | the angle in radians | |

| output | meaning |
|---|---|
| `degrees` | the angle in degrees |

- **Method:** multiply by $180/pi$
- **Assumes:** a finite angle
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




degrees (the angle in degrees) against rad (the angle in radians) from 0 to 3.14159. 41 points computed by math-foundations-ai999x; marked: the worked example, rad = 2.35619, degrees = 135.

```
math_foundations_ai999x.solve('rad_to_deg', rad=2.356194490192345)
# -> degrees = 135   (SOLVED)
```

```
math_foundations_ai999x.solve('rad_to_deg', rad=2.5)
# -> degrees = 143.239   (SOLVED)
```

```
math_foundations_ai999x.solve('rad_to_deg', rad=3.141592653589793)
# -> degrees = 180   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Convert 3pi/4 radians to degrees. | SOLVED | degrees = 135 |
| Convert 2.5 radians to degrees. | SOLVED | degrees = 143.239 |
| What is 5π/6 in degrees? | SOLVED | degrees = 150 |

## Right triangle from its two legs — `right_triangle_legs`

Given the legs a and b of a right triangle: the hypotenuse c and the acute angles A (opposite a) and B.

$$c = \sqrt{a^2 + b^2}, \quad \tan A = \frac{a}{b}$$

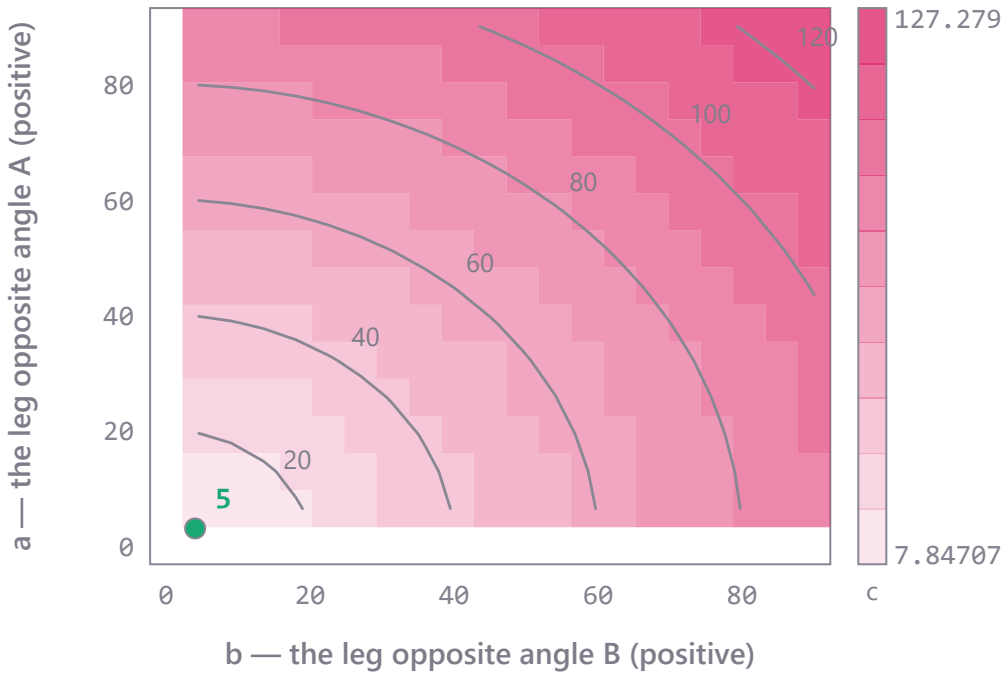
| input | meaning | condition |
|---|---|---|
| a | the leg opposite angle A (positive) | |
| b | the leg opposite angle B (positive) | |

| output | meaning |
|---|---|
| c | the hypotenuse |
| A | angle opposite a, degrees |
| B | angle opposite b, degrees |

- **Method:** Pythagoras $c = \sqrt{a^2 + b^2}$; tan $A = a/b$; $B = 90 - A$
- **Assumes:** $a > 0$; $b > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




c (the hypotenuse) against a (the leg opposite angle A (positive)) from 0 to 90. One curve for each value of b (the leg opposite angle B (positive)): 4, 22.5, 67.5. 40 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 3, c = 5.




c (the hypotenuse) over b (the leg opposite angle B (positive)) from 0 to 90 and a (the leg opposite angle A (positive)) from 0 to 90: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 7.84707 to dark 127.279); iso-lines at c = 20, 40, 60, 80, 100, 120; blank cells: inputs the capability refuses. Marked: the worked example, b = 4, a = 3, c = 5.

```
math_foundations_ai999x.solve('right_triangle_legs', a=3, b=4)
# -> c = 5, A = 36.8699, B = 53.1301   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_legs', a=5, b=12)
# -> c = 13, A = 22.6199, B = 67.3801   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_legs', a=1, b=1)
# -> c = 1.41421, A = 45, B = 45   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| A right triangle has legs 3 and 4. Find the hypotenuse and the acute angles. | SOLVED | c = 5, A = 36.8699, B = 53.1301 |
| Solve the right triangle with legs a = 5 and b = 12. | SOLVED | c = 13, A = 22.6199, B = 67.3801 |
| Solve the right triangle with legs a = 0 and b = 12. | INVALID_INPUT | Right triangle from its two legs: the legs must be positive. |

## Right triangle from a leg and the hypotenuse — `right_triangle_leg_hypotenuse`

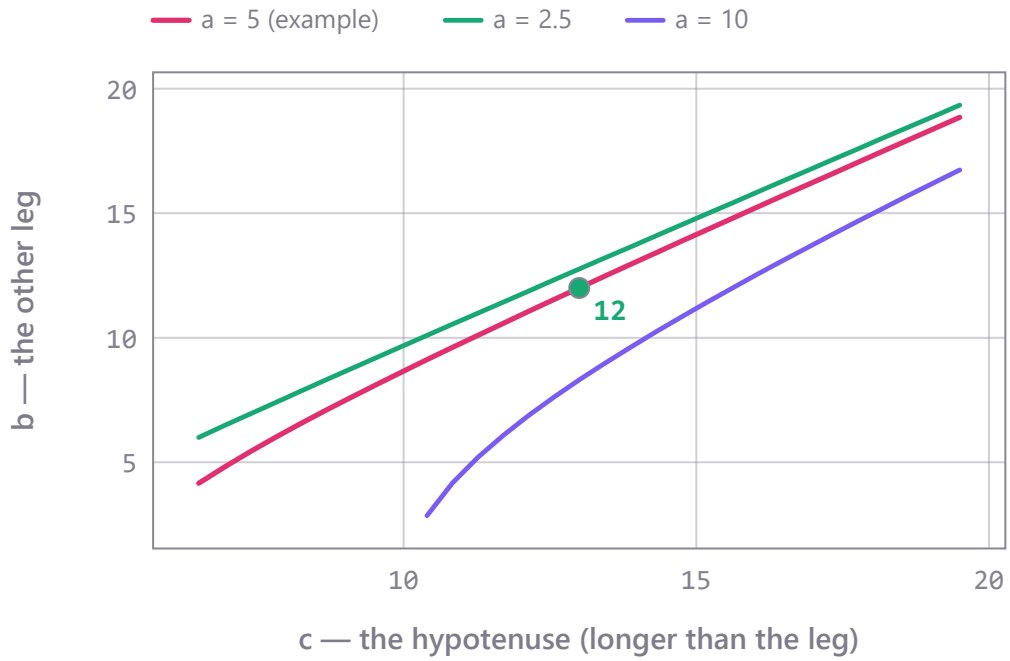
Given a leg a and the hypotenuse c of a right triangle: the other leg b and the acute angles A (opposite a) and B.

$$b = \sqrt{c^2 - a^2}, \quad \sin A = \frac{a}{c}$$

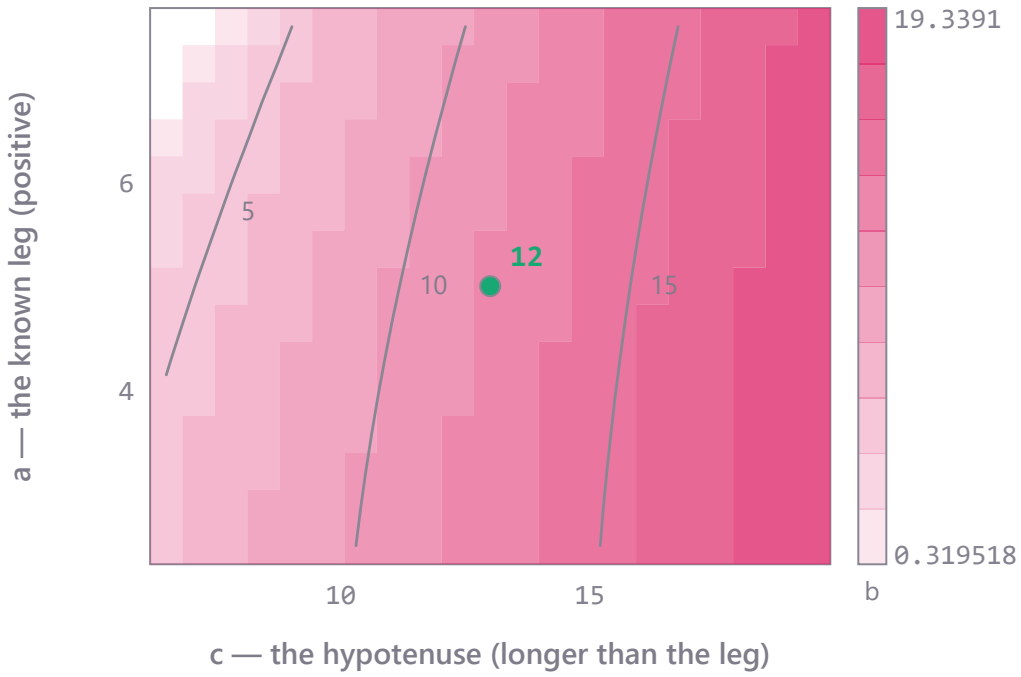
| input | meaning | condition |
|---|---|---|
| `a` | the known leg (positive) | |
| `c` | the hypotenuse (longer than the leg) | |

| output | meaning |
|---|---|
| `b` | the other leg |
| `A` | angle opposite a, degrees |
| `B` | angle opposite b, degrees |

- **Method:** $b = \sqrt{c^2 - a^2}$; sin $A = a/c$, $B = 90 - A$
- **Assumes:** $0 < a < c$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




b (the other leg) against c (the hypotenuse (longer than the leg)) from 6.5 to 19.5. One curve for each value of a (the known leg (positive)): 5, 2.5, 10. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 13, b = 12.




b (the other leg) over c (the hypotenuse (longer than the leg)) from 6.5 to 19.5 and a (the known leg (positive)) from 2.5 to 7.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0.319518 to dark 19.3391); iso-lines at b = 5, 10, 15; blank cells: inputs the capability refuses. Marked: the worked example, c = 13, a = 5, b = 12.

```
math_foundations_ai999x.solve('right_triangle_leg_hypotenuse', a=5, c=13)
# -> b = 12, A = 22.6199, B = 67.3801   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_leg_hypotenuse', a=3, c=6)
# -> b = 5.19615, A = 30, B = 60   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| A right triangle has hypotenuse 13 and one leg 5. Find the other leg and the acute angles. | SOLVED | b = 12, A = 22.6199, B = 67.3801 |
| A right triangle has hypotenuse 6 and one leg 3. Find the other leg and the angles. | SOLVED | b = 5.19615, A = 30, B = 60 |
| A right triangle has hypotenuse 5 and one leg 7. Find the other leg. | INVALID_INPUT | Right triangle from a leg and the hypotenuse: the leg must be positive and shorter than the hypotenuse. |

## Right triangle from an acute angle and a side — `right_triangle_angle_side`

Given an acute angle A and one side (the hypotenuse, the side opposite A or the side adjacent to A): every side and the other angle.

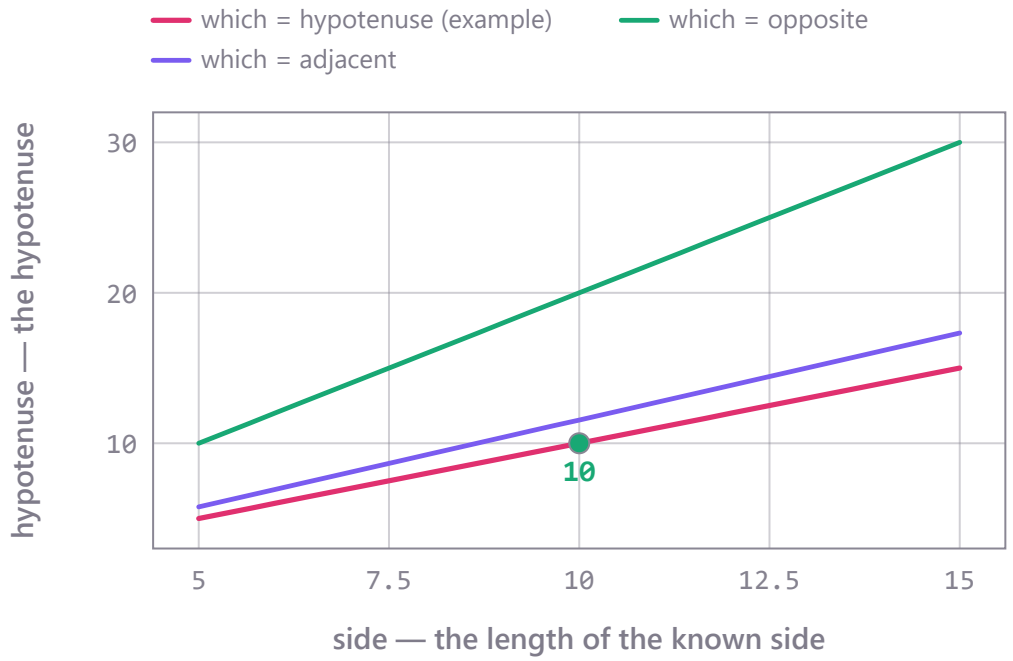
$$\sin A = \frac{\text{opp}}{\text{hyp}},\ \cos A = \frac{\text{adj}}{\text{hyp}},\ \tan A = \frac{\text{opp}}{\text{adj}}$$

| input | meaning | condition |
|---|---|---|
| `A` | the acute angle A in degrees | |

| input | meaning | condition |
|---|---|---|
| `side` | the length of the known side | |
| `which` | which side is known | |

| output | meaning |
|---|---|
| `opposite` | side opposite A |
| `adjacent` | side adjacent to A |
| `hypotenuse` | the hypotenuse |
| `B` | the other acute angle, degrees |

- **Method:** SOH-CAH-TOA: sin A = opp/hyp, cos A = adj/hyp, tan A = opp/adj
- **Assumes:** $0 < A < 90$; side > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




hypotenuse (the hypotenuse) against side (the length of the known side) from 5 to 15. One curve for each value of which (which side is known): hypotenuse, opposite, adjacent. Other inputs as in the worked example: A = 30. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, side = 10, hypotenuse = 10.

```
math_foundations_ai999x.solve('right_triangle_angle_side', A=30, side=10, which='hypotenuse')
# -> opposite = 5, adjacent = 8.66025, hypotenuse = 10, B = 60   (SOLVED)
```

```
math_foundations_ai999x.solve('right_triangle_angle_side', A=45, side=7, which='opposite')
# -> opposite = 7, adjacent = 7, hypotenuse = 9.89949, B = 45   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| In a right triangle, angle A is 30 degrees and the hypotenuse is 10. Find the other sides. | SOLVED | opposite = 5, adjacent = 8.66025, hypotenuse = 10, B = 60 |

| question | outcome | result |
|---|---|---|
| In a right triangle, angle A is 45 degrees and the side opposite A is 7. Find the other sides. | SOLVED | opposite = 7, adjacent = 7, hypotenuse = 9.89949, B = 45 |
| In a right triangle, angle A is 95 degrees and the hypotenuse is 10. Find the other sides. | INVALID_INPUT | Right triangle from an acute angle and a side: the acute angle must be between 0 and 90 degrees. |

### Law of sines (two angles and a side) — `law_of_sines_aas`

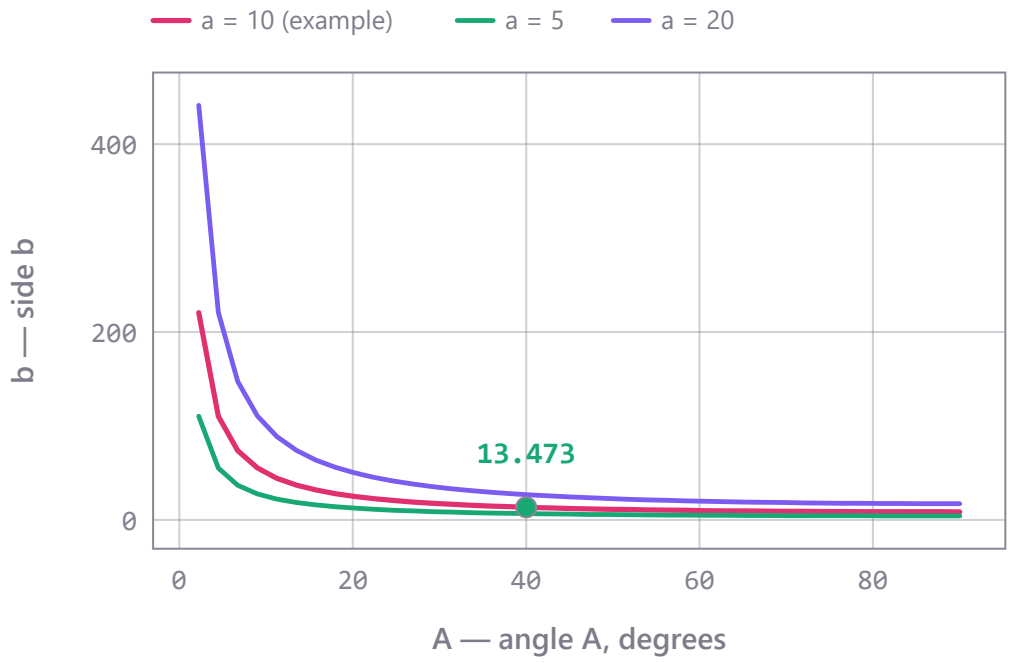
Given angles A and B and the side a opposite A: the third angle C and the sides b and c.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

| input | meaning | condition |
|---|---|---|
| A | angle A, degrees | |
| B | angle B, degrees | |
| a | side a, opposite A | |

| output | meaning |
|---|---|
| C | angle C, degrees |
| b | side b |
| c | side c |

- **Method:** $C = 180 - A - B$; law of sines a/sin $A = b/sin\ B = c/sin$ C
- **Assumes:** A, $B > 0$; $A + B < 180$; $a > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




b (side b) against A (angle A, degrees) from 0 to 90. One curve for each value of a (side a, opposite A): 10, 5, 20. Other inputs as in the worked example: B = 60. 40 points per curve computed by math-foundations-ai999x; marked: the worked example, A = 40, b = 13.473.

```
math_foundations_ai999x.solve('law_of_sines_aas', A=40, B=60, a=10)
# -> C = 80, b = 13.473, c = 15.3209   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_sines_aas', A=30, B=90, a=5)
# -> C = 60, b = 10, c = 8.66025   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| In triangle ABC, A = 40 degrees, B = 60 degrees and a = 10. Solve the triangle. | SOLVED | C = 80, b = 13.473, c = 15.3209 |
| In triangle ABC, A = 30°, B = 90° and a = 5. Find the remaining sides and angles. | SOLVED | C = 60, b = 10, c = 8.66025 |
| In triangle ABC, A = 100 degrees, B = 90 degrees and a = 5. Solve the triangle. | INVALID_INPUT | Law of sines (two angles and a side): the two angles must be positive with a sum below 180 degrees. |

### Law of sines, the ambiguous case (two sides and a non-included angle) — `law_of_sines_ssa`

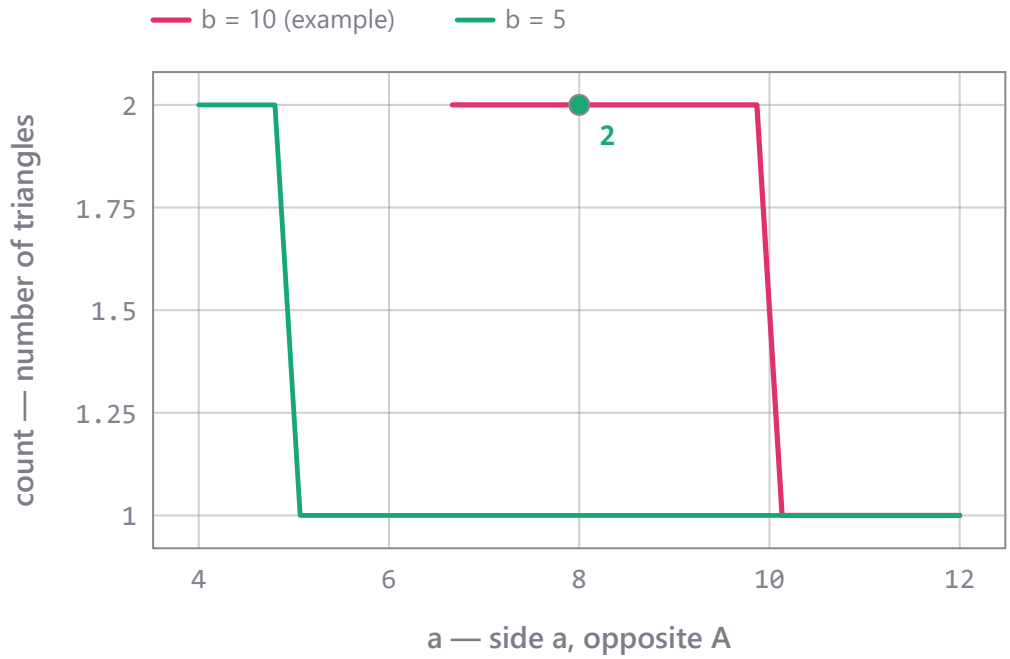
Given sides a and b and the angle A opposite a: how many triangles exist (0, 1 or 2) and every one of them.

$$\sin B = \frac{b \sin A}{a}, \quad B_2 = 180^\circ - B_1$$

| input | meaning | condition |
|---|---|---|
| `a` | side a, opposite A | |
| `b` | side b | |
| `A` | angle A, degrees | |

| output | meaning |
|---|---|
| `count` | number of triangles (1 or 2; none is refused) |
| `B1` | angle B of the first triangle (the acute one) |
| `C1` | angle C of the first triangle |
| `c1` | side c of the first triangle |
| `B2` | angle B of the second triangle (obtuse), when there are two |
| `C2` | angle C of the second triangle |
| `c2` | side c of the second triangle |

- **Method:** sin $B = b$ sin A / a; B and $180 - B$ both give a triangle when $A + B < 180$ (the ambiguous case)
- **Assumes:** a, $b > 0$; $0 < A < 180$; a triangle exists (else refused)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




count (number of triangles (1 or 2; none is refused)) against a (side a, opposite A) from 4 to 12. One curve for each value of b (side b): 10, 5. Other inputs as in the worked example: A = 40. 21 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 8, count = 2.

```
math_foundations_ai999x.solve('law_of_sines_ssa', a=8, b=10, A=40)
# -> count = 2, B1 = 53.4641, C1 = 86.5359, c1 = 12.423, B2 = 126.536, C2 = 13.4641, c2 = 2.89784
(SOLVED)
```

```
math_foundations_ai999x.solve('law_of_sines_ssa', a=12, b=10, A=40)
# -> count = 1, B1 = 32.3884, C1 = 107.612, c1 = 17.7937   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_sines_ssa', a=5, b=10, A=30)
# -> count = 1, B1 = 90, C1 = 60, c1 = 8.66025   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| In triangle ABC, a = 8, b = 10 and A = 40 degrees. Solve the triangle. | SOLVED | count = 2, B1 = 53.4641, C1 = 86.5359, c1 = 12.423, B2 = 126.536, C2 = 13.4641, c2 = 2.89784 |
| In triangle ABC, a = 12, b = 10 and A = 40°. Solve the triangle. | SOLVED | count = 1, B1 = 32.3884, C1 = 107.612, c1 = 17.7937 |
| In triangle ABC, a = 3, b = 10 and A = 40 degrees. Solve the triangle. | INVALID_INPUT | Law of sines, the ambiguous case (two sides and a non-included angle): no triangle exists: side a is shorter than the he |

## Law of cosines (two sides and the included angle) — `law_of_cosines_sas`

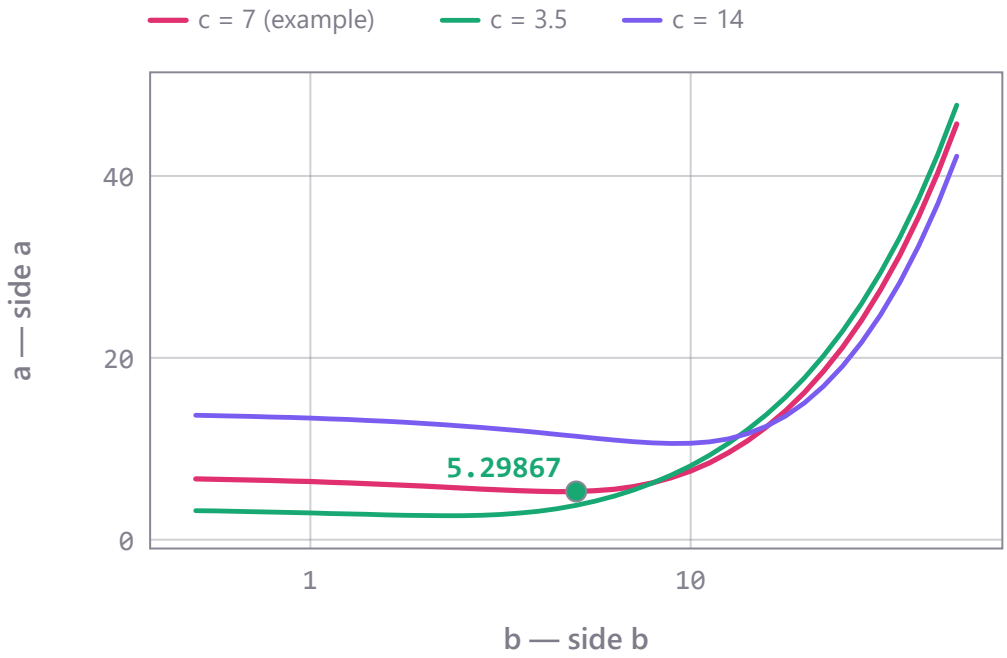
Given sides b and c and the included angle A: the third side a and the angles B and C.

$$a^2 = b^2 + c^2 - 2bc\cos A$$

| input | meaning | condition |
|---|---|---|
| `b` | side b | |
| `c` | side c | |
| `A` | the included angle A, degrees | |

| output | meaning |
|---|---|
| `a` | side a |
| `B` | angle B, degrees |
| `C` | angle C, degrees |

- **Method:** $a^2 = b^2 + c^2 - 2bc$ cos A; B from the law of cosines (no ambiguity)
- **Assumes:** b, $c > 0$; $0 < A < 180$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




a (side a) against b (side b) from 0.5 to 50 on a logarithmic axis. One curve for each value of c (side c): 7, 3.5, 14. Other inputs as in the worked example: A = 49. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 5, a = 5.29867.

```
math_foundations_ai999x.solve('law_of_cosines_sas', b=5, c=7, A=49)
# -> a = 5.29867, B = 45.4117, C = 85.5883   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_cosines_sas', b=3, c=4, A=90)
# -> a = 5, B = 36.8699, C = 53.1301   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| In triangle ABC, b = 5, c = 7 and A = 49 degrees. Solve the triangle. | SOLVED | a = 5.29867, B = 45.4117, C = 85.5883 |

| question | outcome | result |
|---|---|---|
| In triangle ABC, b = 3, c = 4 and A = 90°. Find the remaining sides and angles. | SOLVED | a = 5, B = 36.8699, C = 53.1301 |
| In triangle ABC, b = 3, c = 4 and A = 190 degrees. Solve the triangle. | INVALID_INPUT | Law of cosines (two sides and the included angle): the sides must be positive and A between 0 and 180 degrees. |

### Law of cosines (three sides) — `law_of_cosines_sss`

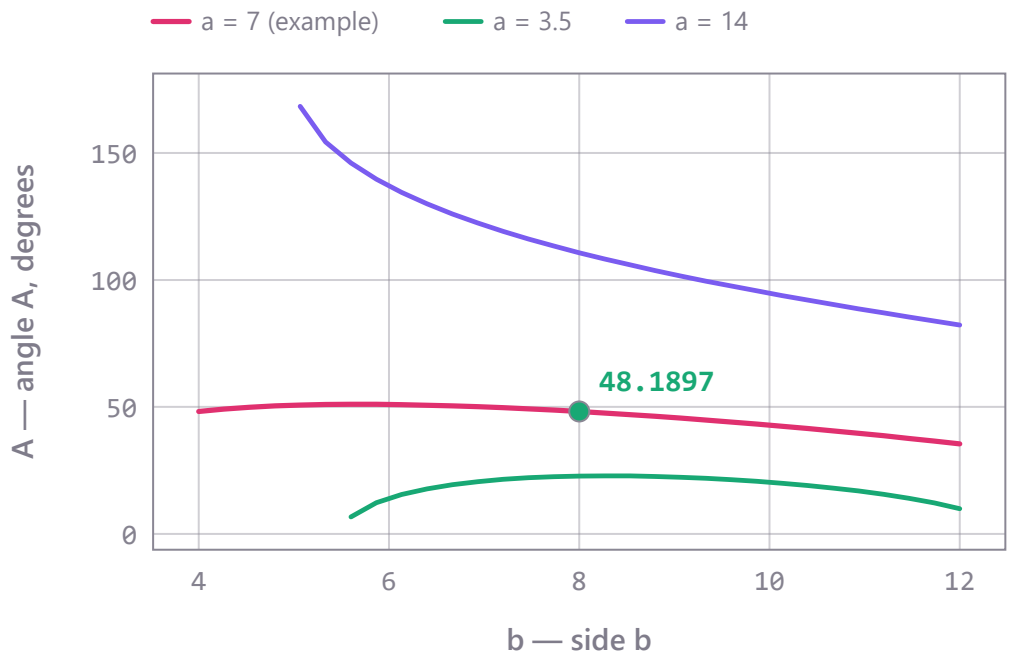
Given the three sides a, b, c: the angles A, B, C.

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

| input | meaning | condition |
|---|---|---|
| a | side a | |
| b | side b | |
| c | side c | |

| output | meaning |
|---|---|
| A | angle A, degrees |
| B | angle B, degrees |
| C | angle C, degrees |

- **Method:** cos $A = (b^2 + c^2 - a^2)/(2bc)$, likewise B; $C = 180 - A - B$
- **Assumes:** positive sides; triangle inequality (each side shorter than the sum of the other two)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




A (angle A, degrees) against b (side b) from 4 to 12. One curve for each value of a (side a): 7, 3.5, 14. Other inputs as in the worked example: c = 9. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 8, A = 48.1897.

```
math_foundations_ai999x.solve('law_of_cosines_sss', a=7, b=8, c=9)
# -> A = 48.1897, B = 58.4119, C = 73.3985   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_cosines_sss', a=3, b=4, c=5)
# -> A = 36.8699, B = 53.1301, C = 90   (SOLVED)
```

```
math_foundations_ai999x.solve('law_of_cosines_sss', a=2, b=2, c=2)
# -> A = 60, B = 60, C = 60   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| In triangle ABC, a = 7, b = 8 and c = 9. Find the angles. | SOLVED | A = 48.1897, B = 58.4119, C = 73.3985 |
| In triangle ABC, a = 3, b = 4 and c = 5. Solve the triangle. | SOLVED | A = 36.8699, B = 53.1301, C = 90 |
| In triangle ABC, a = 1, b = 2 and c = 5. Find the angles. | INVALID_INPUT | Law of cosines (three sides): these lengths do not form a triangle (each side must be shorter than the sum of the other |

## Area of a triangle from its three sides (Heron) — `heron_area`

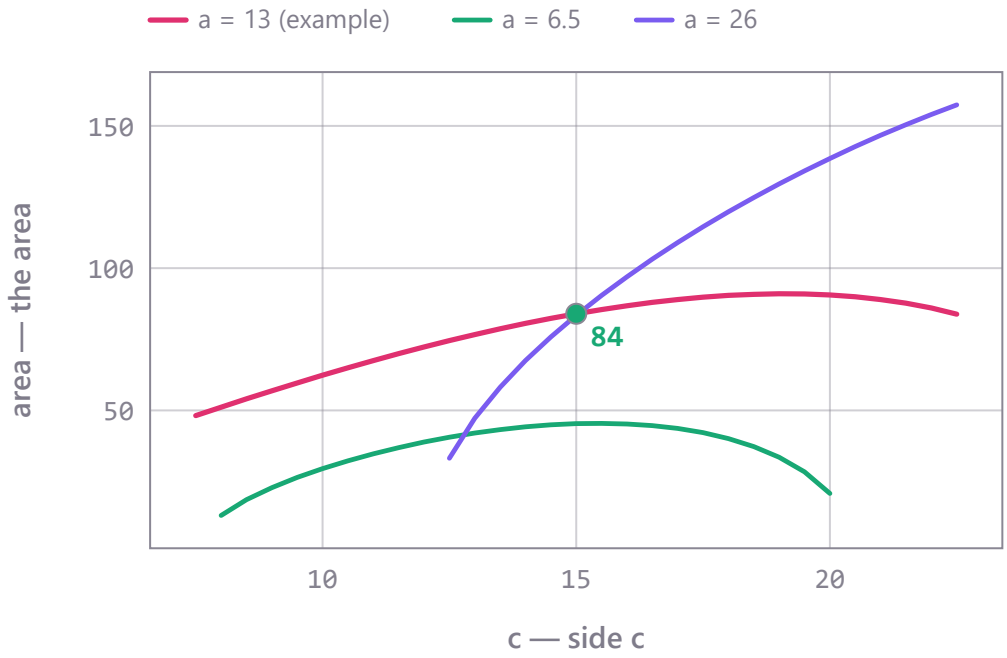
The area of a triangle with sides a, b, c, and its semi-perimeter s.

$$K = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

| input | meaning | condition |
|---|---|---|
| `a` | side a | |
| `b` | side b | |
| `c` | side c | |

| output | meaning |
|---|---|
| `area` | the area |
| `s` | the semi-perimeter $(a+b+c)/2$ |

- **Method:** Heron's formula, in its numerically stable arrangement
- **Assumes:** positive sides; triangle inequality
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




area (the area) against c (side c) from 7.5 to 22.5. One curve for each value of a (side a): 13, 6.5, 26. Other inputs as in the worked example: b = 14. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 15, area = 84.

```
math_foundations_ai999x.solve('heron_area', a=13, b=14, c=15)
# -> area = 84, s = 21   (SOLVED)
```

```
math_foundations_ai999x.solve('heron_area', a=3, b=4, c=5)
# -> area = 6, s = 6   (SOLVED)
```

```
math_foundations_ai999x.solve('heron_area', a=5, b=6, c=7)
# -> area = 14.6969, s = 9   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the area of a triangle with sides 13, 14 and 15. | SOLVED | area = 84, s = 21 |
| Use Heron's formula to find the area of a triangle with sides 5, 6 and 7. | SOLVED | area = 14.6969, s = 9 |
| Find the area of a triangle with sides 1, 2 and 3. | INVALID_INPUT | Area of a triangle from its three sides (Heron): these lengths do not form a triangle (each side must be shorter than th |

## Exact trigonometric values on the unit circle — `unit_circle_exact`

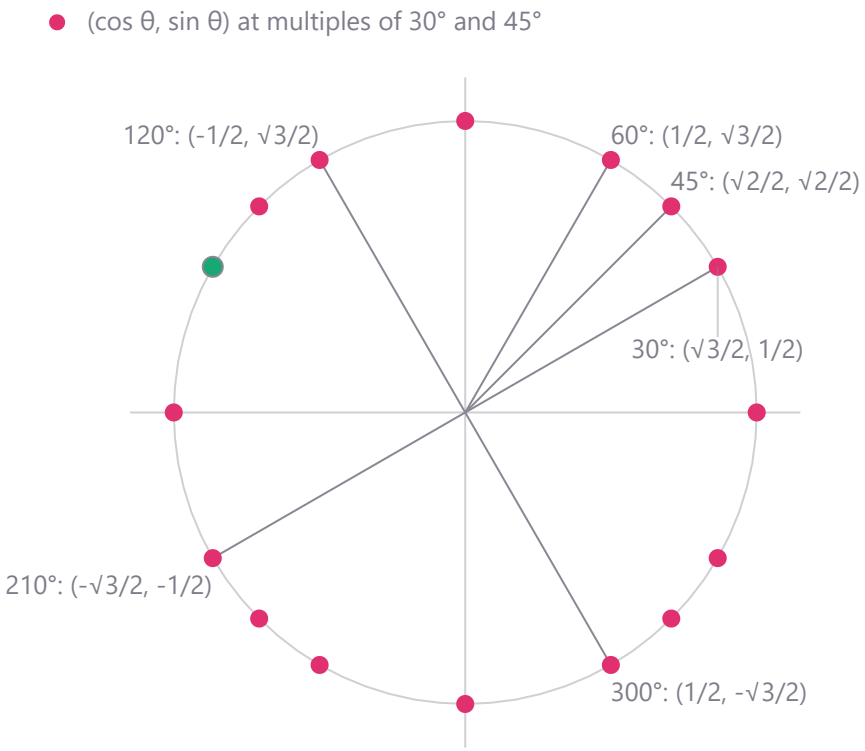
The exact sin, cos and tan of an angle that is a multiple of 30 or 45 degrees (pi/6, pi/4), with the reference angle.

$$\sin 30^\circ = \tfrac{1}{2},\ \sin 45^\circ = \tfrac{\sqrt{2}}{2},\ \sin 60^\circ = \tfrac{\sqrt{3}}{2}$$

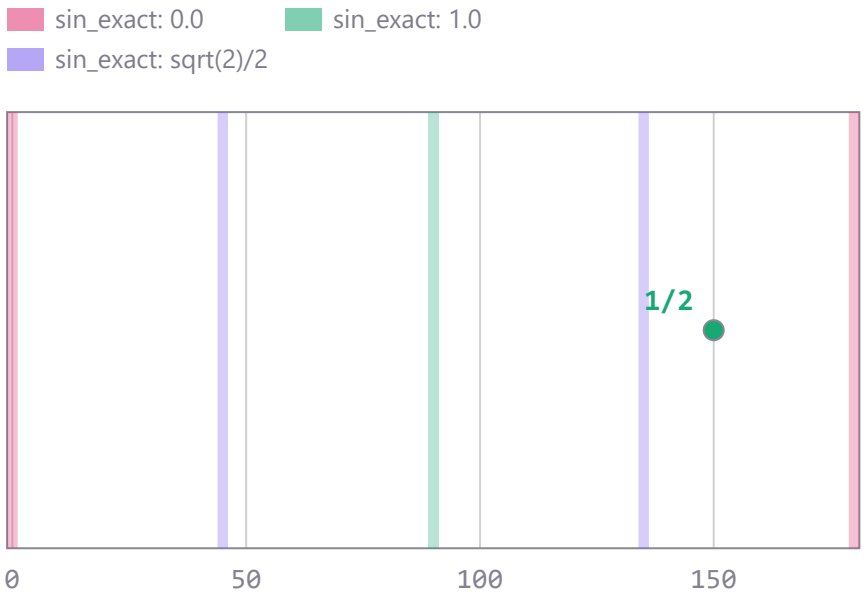
| input | meaning | condition |
|---|---|---|
| `deg` | the angle in degrees (a multiple of 30 or 45) | |

| output | meaning |
|---|---|
| `sin` | sin of the angle |
| `cos` | cos of the angle |
| `tan` | tan of the angle (None where undefined) |
| `sin_exact` | exact sin |
| `cos_exact` | exact cos |
| `tan_exact` | exact tan |
| `reference_angle` | the reference angle, degrees |

- **Method:** the reference angle's exact values from the $30-60-90$ and $45-45-90$ triangles, with the signs of the quadrant (ASTC)
- **Assumes:** a whole multiple of 30 or 45 degrees
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The unit circle with the exact cosine and sine the package returns at the multiples of 30° and 45° (16 angles; six labelled); for example cos 150° = -√3/2 = -0.866025 and sin 150° = 1/2 = 0.5 (marked).




exact sin (sin_exact) along deg (the angle in degrees (a multiple of 30 or 45)) from 0 to 180: 81 runs of math-foundations-ai999x, each band coloured by the verdict returned (0.0, 1.0, sqrt(2)/2). Marked: the worked example, deg = 150 → 1/2.

```
math_foundations_ai999x.solve('unit_circle_exact', deg=150)
# -> sin = 0.5, cos = -0.866025, tan = -0.57735, sin_exact = 1/2, cos_exact = -sqrt(3)/2, tan_exact
= -sqrt(3)/3, reference_angle = 30   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_circle_exact', deg=225)
# -> sin = -0.707107, cos = -0.707107, tan = 1, sin_exact = -sqrt(2)/2, cos_exact = -sqrt(2)/2,
tan_exact = 1, reference_angle = 45   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_circle_exact', deg=90)
# -> sin = 1, cos = 0, tan = none, sin_exact = 1, cos_exact = 0, tan_exact = undefined,
reference_angle = 90   (SOLVED)
```

```
math_foundations_ai999x.solve('unit_circle_exact', deg=-60)
# -> sin = -0.866025, cos = 0.5, tan = -1.73205, sin_exact = -sqrt(3)/2, cos_exact = 1/2, tan_exact
= -sqrt(3), reference_angle = 60   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the exact values of sin, cos and tan of 150 degrees. | SOLVED | sin = 0.5, cos = -0.866025, tan = -0.57735, sin_exact = 1/2, cos_exact = -sqrt(3)/2, tan_exact = -sqrt(3)/3, reference_angle = 30 |
| Find the exact value of sin(5pi/6). | SOLVED | sin = 0.5, cos = -0.866025, tan = -0.57735, sin_exact = 1/2, cos_exact = -sqrt(3)/2, tan_exact = -sqrt(3)/3, reference_angle = 30 |
| What is cos(225°) exactly? | SOLVED | sin = -0.707107, cos = -0.707107, tan = 1, sin_exact = -sqrt(2)/2, cos_exact = -sqrt(2)/2, tan_exact = 1, reference_angle = 45 |
| Find the exact value of tan(20 degrees). | INVALID_INPUT | Exact trigonometric values on the unit circle: exact values are given for multiples of 30 or 45 degrees (pi/6 or pi/4). |

### Solve a trigonometric equation over one full turn — `trig_equation_interval`

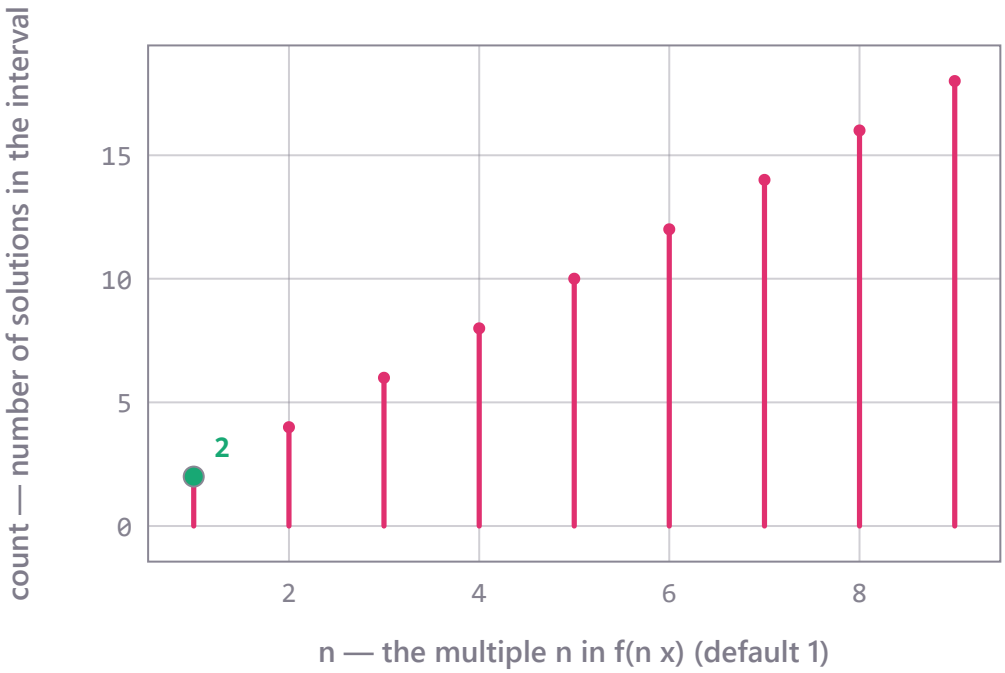
Every solution of $asin$(n x) = k (or cos, tan) with $0 \le x < 360$ degrees or $0 \le x < 2$ pi.

$$\sin u = k \Rightarrow u = \arcsin k,\ 180^\circ - \arcsin k \pmod{360^\circ}, \quad u = nx$$

| input | meaning | condition |
|---|---|---|
| `func` | sin, cos or tan | |
| `k` | the value the function must take (after dividing by its coefficient) | |
| `n` | the multiple n in f(n x) (default 1) | |
| `unit` | degrees ([0, 360)) or radians ([0, 2pi)) | |

| output | meaning |
|---|---|
| `count` | number of solutions in the interval |
| `solutions` | the solutions, ascending |

- **Method:** principal value (asin, acos or atan), the second solution of the period (180 - a, -a, a + 180), then every period: x = (u + 360 m)/n inside the interval
- **Assumes:** n a whole number from 1 to 12
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




count (number of solutions in the interval) against n (the multiple n in f(n x) (default 1)) from 1 to 9. Other inputs as in the worked example: func = sin, k = 0.5, unit = degrees. 9 points computed by math-foundations-ai999x; marked: the worked example, n = 1, count = 2.

```
math_foundations_ai999x.solve('trig_equation_interval', func='sin', k=0.5, n=1, unit='degrees')
# -> count = 2, solutions = [30.000000000000004, 150.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='cos', k=0.5, n=1, unit='radians')
# -> count = 2, solutions = [1.0471975511965979, 5.235987755982989]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='tan', k=1, n=2, unit='degrees')
# -> count = 4, solutions = [22.5, 112.5, 202.5, 292.5]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='sin', k=1, n=1, unit='degrees')
# -> count = 1, solutions = [90.0]   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_equation_interval', func='cos', k=2, n=1, unit='degrees')
# -> count = 0, solutions = []   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve sin(x) = 0.5 for 0 <= x < 360 degrees. | SOLVED | count = 2, solutions = [30.000000000000004, 150.0] |
| Solve 2cos(x) = 1 on [0, 2pi). | SOLVED | count = 2, solutions = [1.0471975511965979, 5.235987755982989] |
| Solve tan(2x) = 1 for 0 <= x < 360 degrees. | SOLVED | count = 4, solutions = [22.5, 112.5, 202.5, 292.5] |
| Solve sin(x) = 0 for 0 ≤ x < 2π. | SOLVED | count = 2, solutions = [0.0, 3.141592653589793] |
| Solve 3sin(x) = 4 for 0 <= x < 360 degrees. | SOLVED | count = 0, solutions = [] |

## Trigonometric values from one value and the quadrant — `trig_values_from_one`

Given sin, cos or tan of an angle and its quadrant: sin, cos, tan, sin 2t and cos 2t (Pythagorean and double-angle identities).

$$\cos t = \pm\sqrt{1-\sin^2 t}, \quad \sin 2t = 2\sin t\cos t, \quad \cos 2t = \cos^2 t - \sin^2 t$$

| input | meaning | condition |
|---|---|---|
| `func` | the function whose value is known | |
| `value` | its value | |
| `quadrant` | the quadrant of the angle (1 to 4) | |

| output | meaning |
|---|---|
| `sin` | sin t |
| `cos` | cos t |
| `tan` | tan t |
| `sin2` | sin 2t |
| `cos2` | cos 2t |

- **Method:** $sin^2 + cos^2 = 1$ with the quadrant's signs (ASTC); sin $2t = 2$ sin t cos t; cos $2t = cos^2 t - sin^2$ t
- **Assumes:** the value's sign agrees with the quadrant; 0 < |sin|, |cos| < 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




cos (cos t) against value (its value) from 0.3 to 0.9. Other inputs as in the worked example: func = sin, quadrant = 2. 31 points computed by math-foundations-ai999x; marked: the worked example, value = 0.6, cos = -0.8.

```
math_foundations_ai999x.solve('trig_values_from_one', func='sin', value=0.6, quadrant=2)
# -> sin = 0.6, cos = -0.8, tan = -0.75, sin2 = -0.96, cos2 = 0.28   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_values_from_one', func='cos', value=-0.38461538461538464,
quadrant=3)
# -> sin = -0.923077, cos = -0.384615, tan = 2.4, sin2 = 0.710059, cos2 = -0.704142   (SOLVED)
```

```
math_foundations_ai999x.solve('trig_values_from_one', func='tan', value=-2, quadrant=4)
# -> sin = -0.894427, cos = 0.447214, tan = -2, sin2 = -0.8, cos2 = -0.6   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| If sin(t) = 3/5 and t is in quadrant II, find cos(t), tan(t), sin(2t) and cos(2t). | SOLVED | sin = 0.6, cos = -0.8, tan = -0.75, sin2 = -0.96, cos2 = 0.28 |
| Given cos(x) = -5/13 and x lies in the third quadrant, find sin(x) and tan(x). | SOLVED | sin = -0.923077, cos = -0.384615, tan = 2.4, sin2 = 0.710059, cos2 = -0.704142 |
| If sin(t) = 3/5 and t is in quadrant III, find cos(t). | INVALID_INPUT | Trigonometric values from one value and the quadrant: sine is negative in quadrant 3. |

## Dot product and angle between two vectors — `vector_dot_angle`

The dot product, the magnitudes and the angle between two vectors in the plane or in space.

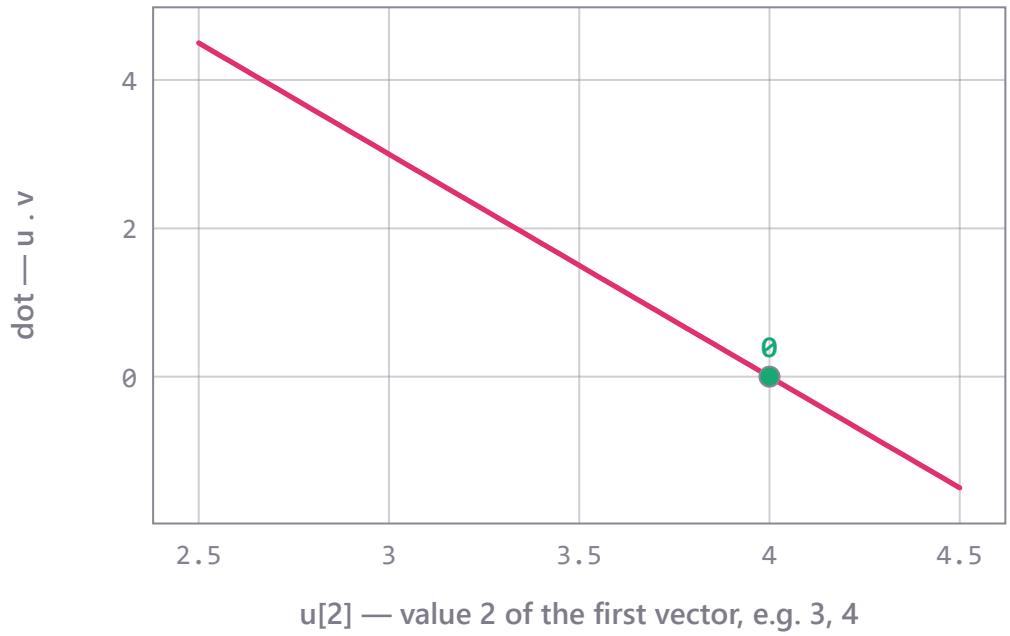
$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$$

| input | meaning | condition |
|---|---|---|
| `u` | the first vector, e.g. 3, 4 | |

| input | meaning | condition |
|---|---|---|
| `v` | the second vector | |

| output | meaning |
|---|---|
| `dot` | u . v |
| `mag_u` | \|u\| |
| `mag_v` | \|v\| |
| `angle_deg` | the angle between them, degrees (0 to 180) |

- **Method:** u . v = sum of products of components; the angle from atan2(|u x v|, u . v) (stable for nearly parallel vectors)
- **Assumes:** same dimension (2 or 3); neither vector is zero
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




dot (u . v) as value 2 of u (the first vector, e.g. 3, 4) moves from 2.5 to 4.5, the other values as given (3, 4); 41 runs of math-foundations-ai999x. Other inputs: v = 4, -3. Marked: the worked example, u[2] = 4, dot = 0.

```
math_foundations_ai999x.solve('vector_dot_angle', u='3, 4', v='4, -3')
# -> dot = 0, mag_u = 5, mag_v = 5, angle_deg = 90   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_dot_angle', u=[1, 2, 3], v=[4, -5, 6])
# -> dot = 12, mag_u = 3.74166, mag_v = 8.77496, angle_deg = 68.5624   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_dot_angle', u=[1, 0], v=[1, 1])
# -> dot = 1, mag_u = 1, mag_v = 1.41421, angle_deg = 45   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the angle between u = <3, 4> and v = <4, -3>. | SOLVED | dot = 0, mag_u = 5, mag_v = 5, angle_deg = 90 |
| Find the dot product of <1, 2, 3> and <4, -5, 6>. | SOLVED | dot = 12, mag_u = 3.74166, mag_v = 8.77496, angle_deg = 68.5624 |
| Find the dot product and the angle between the vectors (1, 0) and (1, 1). | SOLVED | dot = 1, mag_u = 1, mag_v = 1.41421, angle_deg = 45 |
| Find the angle between u = <0, 0> and v = <1, 2>. | INVALID_INPUT | Dot product and angle between two vectors: the zero vector has no direction, so the angle is not defined. |

### Magnitude and direction of a vector — `vector_magnitude_direction`

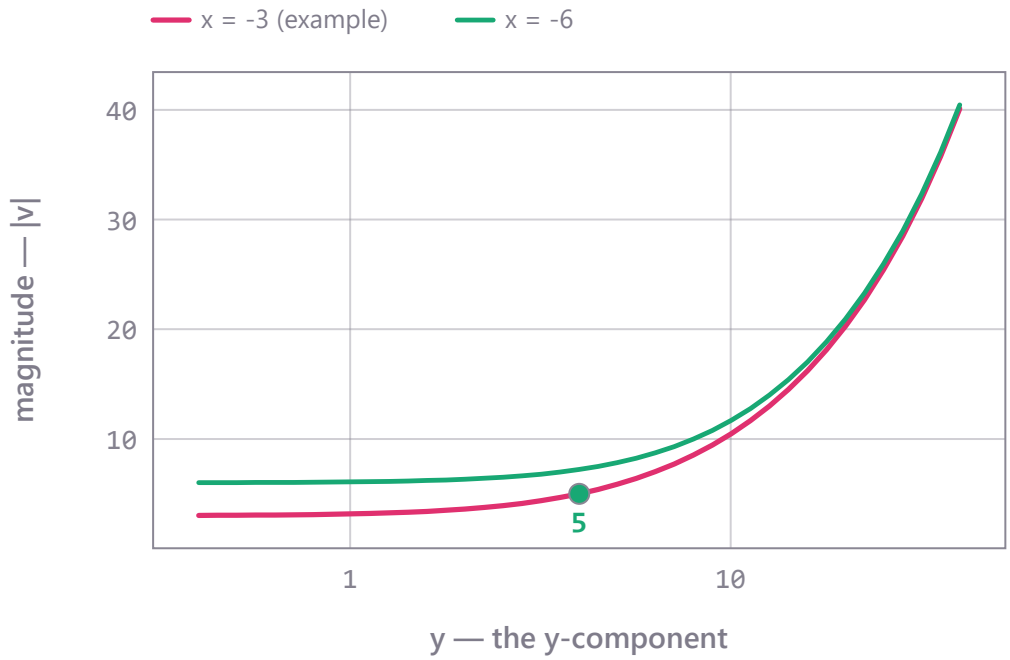
The length of the plane vector <x, y> and its direction angle from the positive x-axis (0 to 360 degrees).

$$|\mathbf{v}| = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan2}(y, x)$$

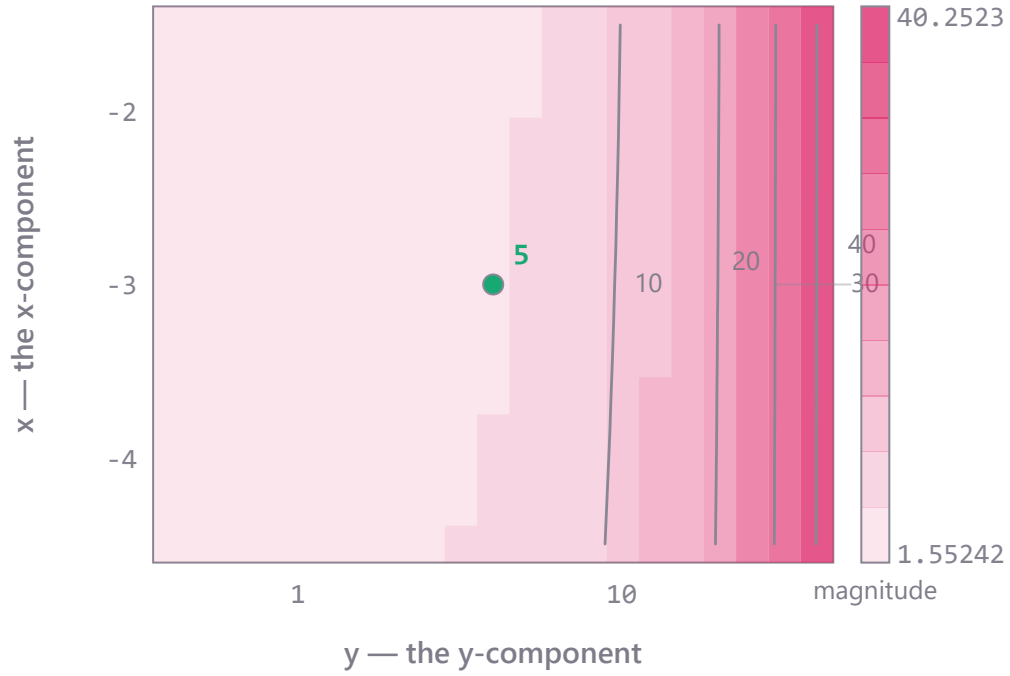
| input | meaning | condition |
|---|---|---|
| `x` | the x-component | |
| `y` | the y-component | |

| output | meaning |
|---|---|
| `magnitude` | \|v\| |
| `direction_deg` | the direction angle, degrees in [0, 360) |

- **Method:** |v| = sqrt(x^2 + y^2); the angle in the quadrant of (x, y)
- **Assumes:** not the zero vector
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




magnitude (|v|) against y (the y-component) from 0.4 to 40 on a logarithmic axis. One curve for each value of x (the x-component): -3, -6. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, y = 4, magnitude = 5.




magnitude (|v|) over y (the y-component) from 0.4 to 40 and x (the x-component) from -4.5 to -1.5 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 1.55242 to dark 40.2523); iso-lines at magnitude = 10, 20, 30, 40. Marked: the worked example, y = 4, x = -3, magnitude = 5.

```
math_foundations_ai999x.solve('vector_magnitude_direction', x=-3, y=4)
# -> magnitude = 5, direction_deg = 126.87   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_magnitude_direction', x=5, y=-12)
# -> magnitude = 13, direction_deg = 292.62   (SOLVED)
```

```
math_foundations_ai999x.solve('vector_magnitude_direction', x=0, y=-2)
# -> magnitude = 2, direction_deg = 270   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the magnitude and direction of the vector <-3, 4>. | SOLVED | magnitude = 5, direction_deg = 126.87 |
| Find the magnitude and direction angle of v = <0, -2>. | SOLVED | magnitude = 2, direction_deg = 270 |
| Find the magnitude and direction of the vector <0, 0>. | INVALID_INPUT | Magnitude and direction of a vector: the zero vector has no direction. |

## Polar to rectangular coordinates — `polar_to_rectangular`

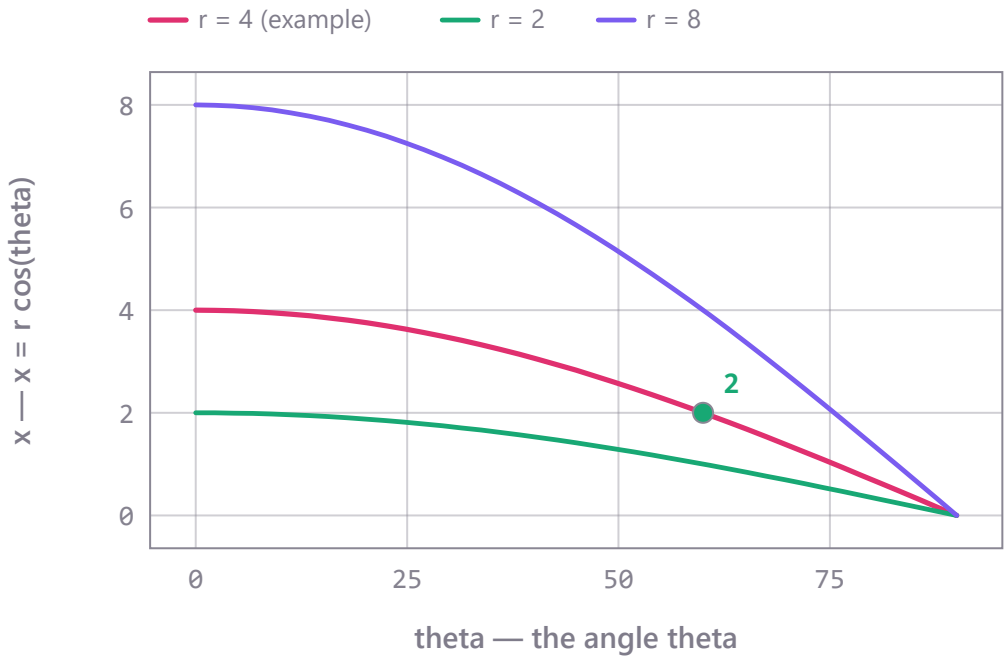
The point (x, y) with polar coordinates (r, theta).

$$x = r\cos\theta, \quad y = r\sin\theta$$

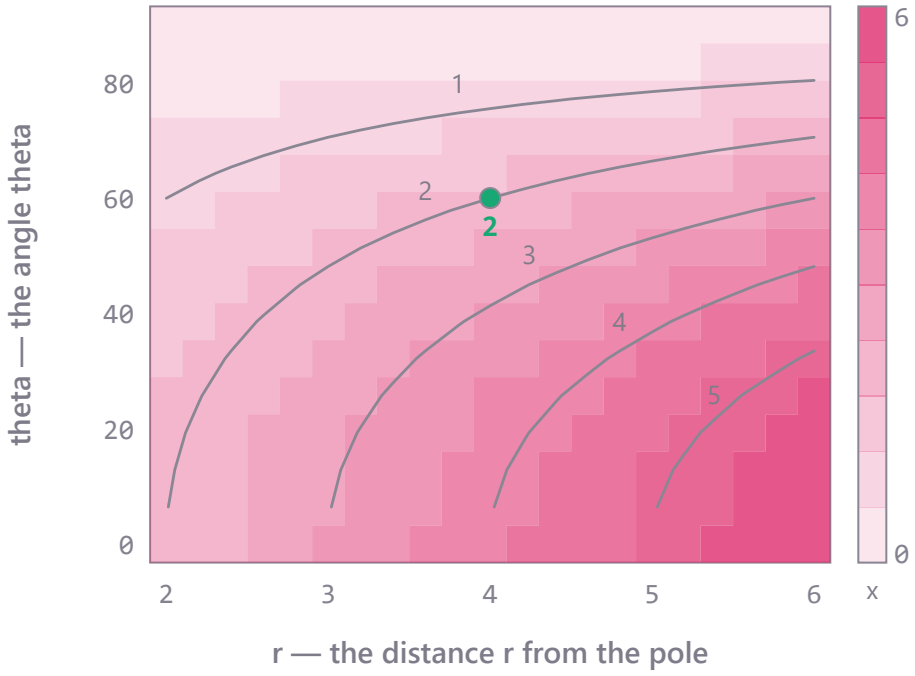
| input | meaning | condition |
|---|---|---|
| `r` | the distance r from the pole | |
| `theta` | the angle theta | |
| `unit` | degrees or radians | |

| output | meaning |
|---|---|
| `x` | $x = r$ cos(theta) |
| `y` | $y = r$ sin(theta) |

- **Method:** $x = r$ cos(theta), $y = r$ sin(theta)
- **Assumes:** the angle's unit stated
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (x = r cos(theta)) against theta (the angle theta) from 0 to 90. One curve for each value of r (the distance r from the pole): 4, 2, 8. Other inputs as in the worked example: unit = degrees. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, theta = 60, x = 2.




x (x = r cos(theta)) over r (the distance r from the pole) from 2 to 6 and theta (the angle theta) from 0 to 90: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0 to dark 6); iso-lines at x = 1, 2, 3, 4, 5. Other inputs as in the worked example: unit = degrees. Marked: the worked example, r = 4, theta = 60, x = 2.

```
math_foundations_ai999x.solve('polar_to_rectangular', r=4, theta=60, unit='degrees')
# -> x = 2, y = 3.4641   (SOLVED)
```

```
math_foundations_ai999x.solve('polar_to_rectangular', r=2, theta=2.356194490192345, unit='radians')
# -> x = -1.41421, y = 1.41421   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Convert the polar point (4, 60 degrees) to rectangular coordinates. | SOLVED | x = 2, y = 3.4641 |
| Convert (2, 3pi/4) from polar to rectangular coordinates. | SOLVED | x = -1.41421, y = 1.41421 |
| Convert the polar point (4, 60) to rectangular coordinates. | UNRECOGNIZED_FORMAT | I could not match this question to a documented form, so I will not guess. Every form: math_foundations_ai999x.templates |

### Rectangular to polar coordinates — `rectangular_to_polar`

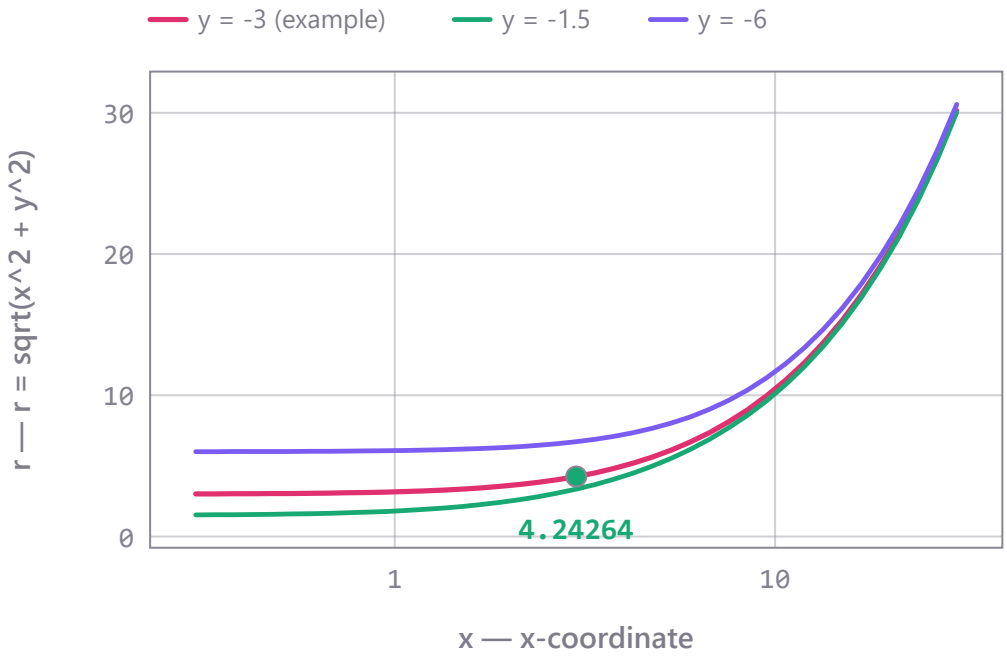
Polar coordinates (r, theta) of the point (x, y), with $r > 0$ and theta in [0, 360) degrees / [0, 2 pi).

$$r = \sqrt{x^2 + y^2}, \quad \tan\theta = \frac{y}{x}$$

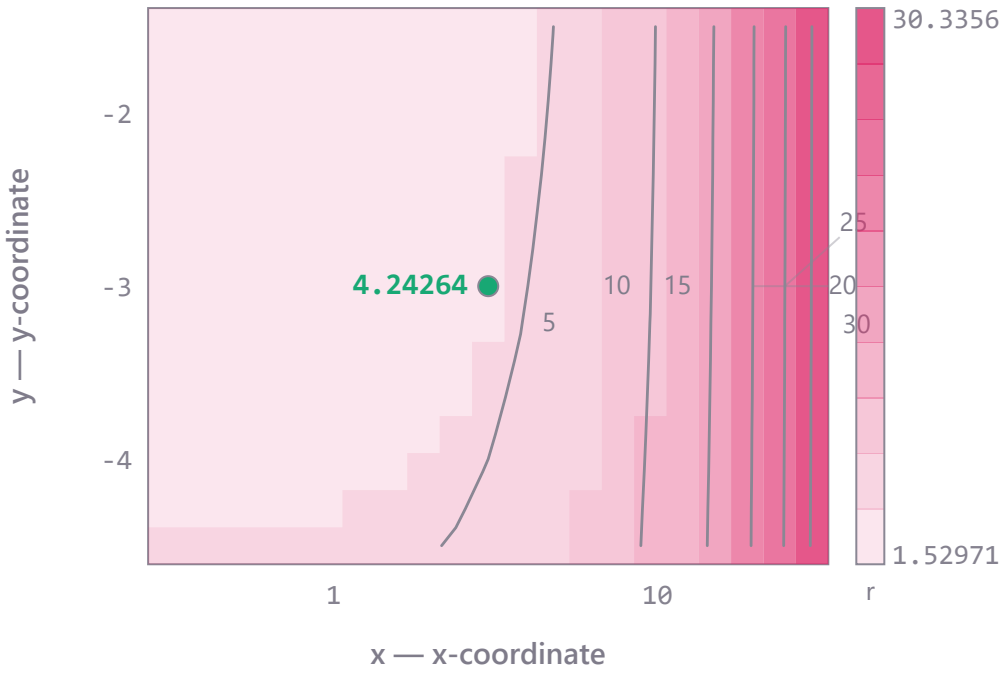
| input | meaning | condition |
|---|---|---|
| `x` | x-coordinate | |
| `y` | y-coordinate | |

| output | meaning |
|---|---|
| `r` | $r = \sqrt{x^2 + y^2}$ |
| `theta_deg` | theta in degrees, [0, 360) |
| `theta_rad` | theta in radians, [0, 2 pi) |

- **Method:** $r = \sqrt{x^2 + y^2}$; theta in the quadrant of (x, y)
- **Assumes:** not the origin
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




r (r = sqrt(x^2 + y^2)) against x (x-coordinate) from 0.3 to 30 on a logarithmic axis. One curve for each value of y (y-coordinate): -3, -1.5, -6. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, x = 3, r = 4.24264.




r (r = sqrt(x^2 + y^2)) over x (x-coordinate) from 0.3 to 30 and y (y-coordinate) from -4.5 to -1.5 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 1.52971 to dark 30.3356); iso-lines at r = 5, 10, 15, 20, 25, 30. Marked: the worked example, x = 3, y = -3, r = 4.24264.

```
math_foundations_ai999x.solve('rectangular_to_polar', x=3, y=-3)
# -> r = 4.24264, theta_deg = 315, theta_rad = 5.49779   (SOLVED)
```

```
math_foundations_ai999x.solve('rectangular_to_polar', x=-1, y=1)
# -> r = 1.41421, theta_deg = 135, theta_rad = 2.35619   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Convert the point (3, -3) to polar coordinates. | SOLVED | r = 4.24264, theta_deg = 315, theta_rad = 5.49779 |
| Find the polar coordinates of (-1, 1). | SOLVED | r = 1.41421, theta_deg = 135, theta_rad = 2.35619 |
| Convert the point (0, 0) to polar coordinates. | INVALID_INPUT | Rectangular to polar coordinates: the origin has r = 0 and no defined angle. |

## Focus and directrix of a parabola — `parabola_focus_directrix`

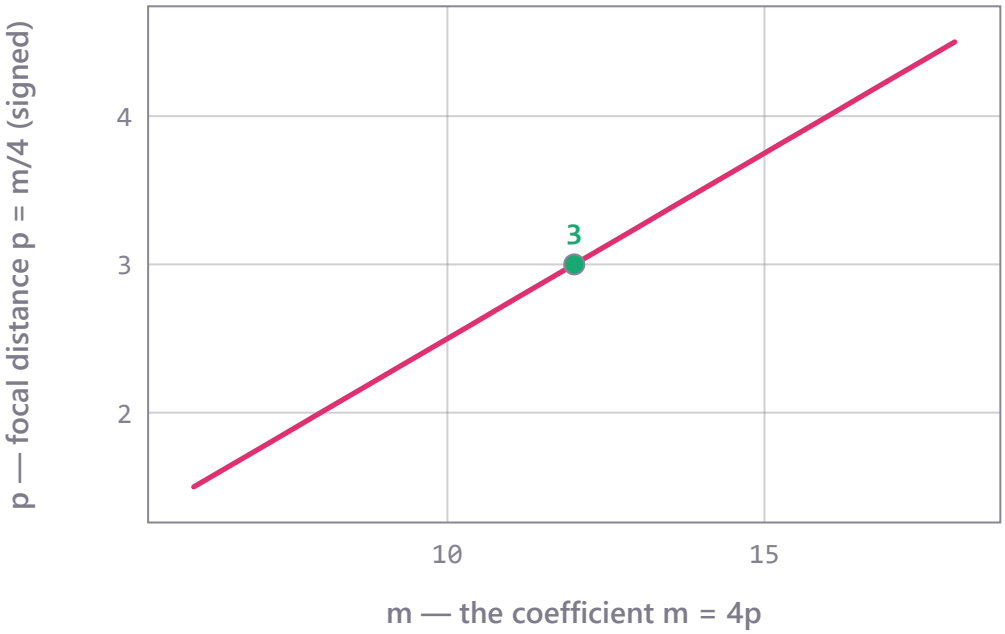
Vertex, focal distance p, focus and directrix of $(x-h)^2 = 4p$(y - k) or $(y-k)^2 = 4p$(x - h).

$$(x-h)^2 = 4p(y-k):\ F = (h, k+p),\ y = k-p$$

| input | meaning | condition |
|---|---|---|
| `orientation` | vertical axis ($(x-h)^2 = m$(y-k)) or horizontal | |
| `m` | the coefficient $m = 4p$ | $m \neq 0$ |
| `h` | x of the vertex | |
| `k` | y of the vertex | |

| output | meaning |
|---|---|
| `p` | focal distance $p = m/4$ (signed) |
| `focus_x` | x of the focus |
| `focus_y` | y of the focus |
| `directrix` | the directrix: y = this (vertical axis) or x = this (horizontal axis) |
| `vertex_x` | x of the vertex |
| `vertex_y` | y of the vertex |

- **Method:** compare with $(x - h)^2 = 4p$(y - k): $p = m/4$, focus at distance p from the vertex along the axis
- **Assumes:** $m \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




p (focal distance p = m/4 (signed)) against m (the coefficient m = 4p) from 6 to 18. Other inputs as in the worked example: orientation = vertical. 31 points computed by math-foundations-ai999x; marked: the worked example, m = 12, p = 3.

```
math_foundations_ai999x.solve('parabola_focus_directrix', orientation='vertical', m=12)
# -> p = 3, focus_x = 0, focus_y = 3, directrix = -3, vertex_x = 0, vertex_y = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('parabola_focus_directrix', orientation='horizontal', m=-8, h=3, k=1)
# -> p = -2, focus_x = 1, focus_y = 1, directrix = 5, vertex_x = 3, vertex_y = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('parabola_focus_directrix', orientation='vertical', m=0.5)
# -> p = 0.125, focus_x = 0, focus_y = 0.125, directrix = -0.125, vertex_x = 0, vertex_y = 0
(SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the focus and directrix of the parabola x^2 = 12y. | SOLVED | p = 3, focus_x = 0, focus_y = 3, directrix = -3, vertex_x = 0, vertex_y = 0 |
| Find the focus and directrix of (y - 1)^2 = -8(x - 3). | SOLVED | p = -2, focus_x = 1, focus_y = 1, directrix = 5, vertex_x = 3, vertex_y = 1 |
| Find the focus and directrix of y = 2x^2. | SOLVED | p = 0.125, focus_x = 0, focus_y = 0.125, directrix = -0.125, vertex_x = 0, vertex_y = 0 |
| Find the focus and directrix of the parabola (x + 2)^2 = 4(y - 5). | SOLVED | p = 1, focus_x = -2, focus_y = 6, directrix = 4, vertex_x = -2, vertex_y = 5 |

### Foci, axes and eccentricity of an ellipse — `ellipse_parameters`

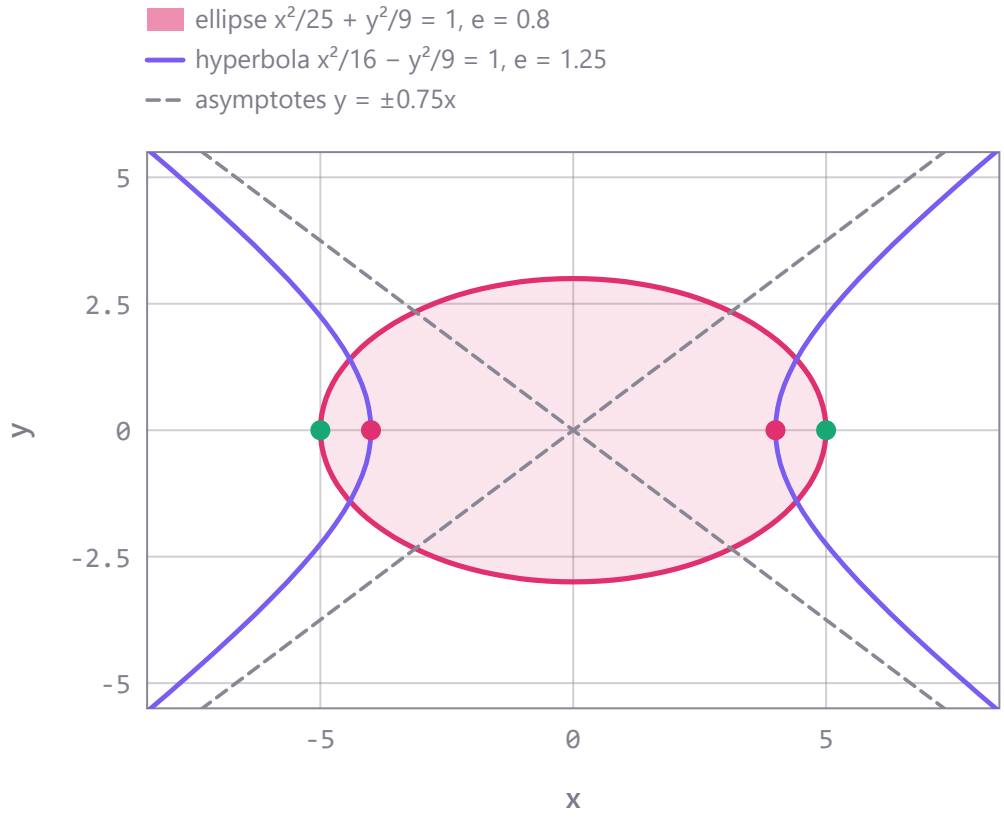
For $(x-h)^2/A + (y-k)^2/B = 1$: semi-axes a and b, focal distance c, eccentricity and the two foci.

$$\frac{(x-h)^2}{A} + \frac{(y-k)^2}{B} = 1, \quad c^2 = a^2 - b^2, \quad e = \frac{c}{a}$$

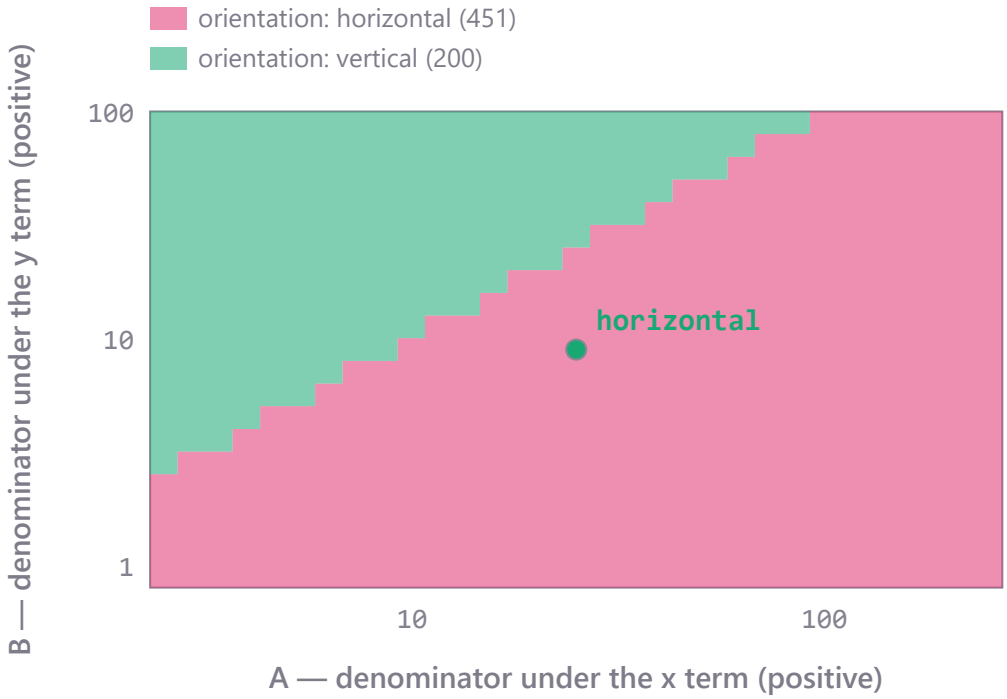
| input | meaning | condition |
|---|---|---|
| `h` | x of the centre | |
| `k` | y of the centre | |
| `A` | denominator under the x term (positive) | |
| `B` | denominator under the y term (positive) | |

| output | meaning |
|---|---|
| `a` | semi-major axis |
| `b` | semi-minor axis |
| `c` | centre-to-focus distance |
| `eccentricity` | $e = c/a$ |
| `f1x` | first focus x |
| `f1y` | first focus y |
| `f2x` | second focus x |
| `f2y` | second focus y |
| `orientation` | the major axis is horizontal or vertical |

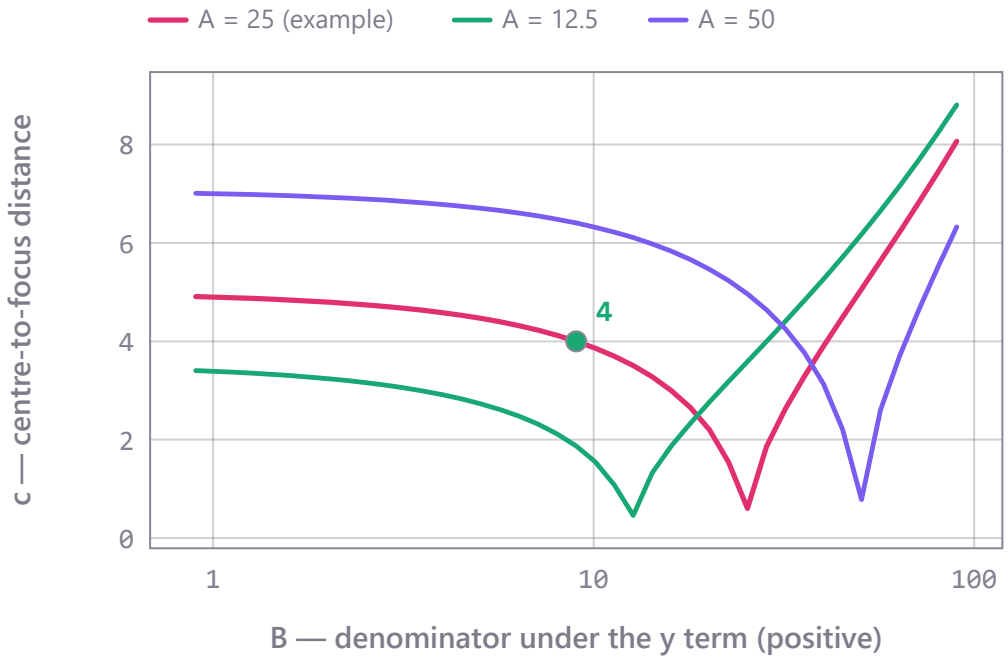
- **Method:** a^2 = the larger denominator, b^2 = the smaller, $c^2 = a^2 - b^2$; foci on the major axis
- **Assumes:** A, $B > 0$; $A \neq B$ (otherwise a circle)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




An ellipse and a hyperbola drawn from the parameters the package returns: ellipse a = 5, b = 3, foci (±4, 0), eccentricity 0.8; hyperbola a = 4, b = 3, foci (±5, 0), eccentricity 1.25, asymptotes of slope ±0.75.




the major axis is horizontal or vertical (orientation) over A (denominator under the x term (positive)) from 2.5 to 250 and B (denominator under the y term (positive)) from 0.9 to 90 (logarithmic axes): each of the 31 × 21 cells is one run of math-foundations-ai999x, coloured by the verdict it returned — horizontal: 451 cells, vertical: 200 cells. Other inputs as in the worked example: h = 0, k = 0. Marked: the worked example, A = 25, B = 9 → horizontal.




c (centre-to-focus distance) against B (denominator under the y term (positive)) from 0.9 to 90 on a logarithmic axis. One curve for each value of A (denominator under the x term (positive)): 25, 12.5, 50. Other inputs as in the worked example: h = 0, k = 0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, B = 9, c = 4.

```
math_foundations_ai999x.solve('ellipse_parameters', h=0, k=0, A=25, B=9)
# -> a = 5, b = 3, c = 4, eccentricity = 0.8, f1x = -4, f1y = 0, f2x = 4, f2y = 0, orientation = horizontal   (SOLVED)
```

```
math_foundations_ai999x.solve('ellipse_parameters', h=1, k=-2, A=16, B=25)
# -> a = 5, b = 4, c = 3, eccentricity = 0.6, f1x = 1, f1y = -5, f2x = 1, f2y = 1, orientation = vertical   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the foci and eccentricity of the ellipse x^2/25 + y^2/9 = 1. | SOLVED | a = 5, b = 3, c = 4, eccentricity = 0.8, f1x = -4, f1y = 0, f2x = 4, f2y = 0, orientation = horizontal |
| Find the foci of (x - 1)^2/16 + (y + 2)^2/25 = 1. | SOLVED | a = 5, b = 4, c = 3, eccentricity = 0.6, f1x = 1, f1y = -5, f2x = 1, f2y = 1, orientation = vertical |
| Find the foci of the ellipse x^2/4 + y^2/4 = 1. | INVALID_INPUT | Foci, axes and eccentricity of an ellipse: equal denominators give a circle (no distinct foci). |

## Foci, eccentricity and asymptotes of a hyperbola — `hyperbola_parameters`

For $(x-h)^2/A - (y-k)^2/B = 1$ (or the vertical form): a, b, c, eccentricity, the asymptotes' slope and the foci.

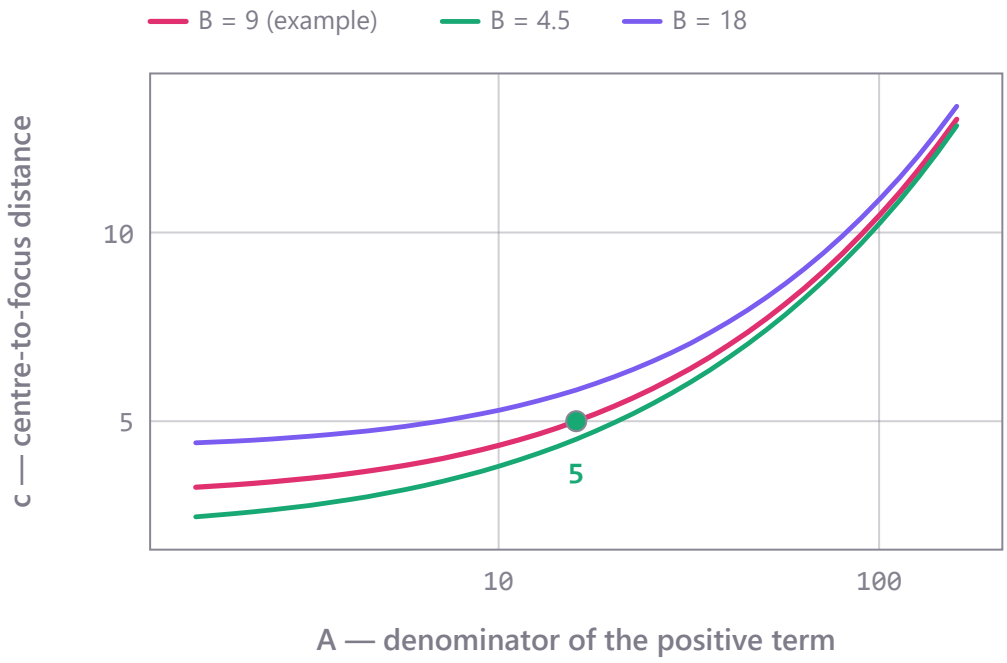
$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \quad c^2 = a^2 + b^2$$

| input | meaning | condition |
|---|---|---|
| `h` | x of the centre | |

| input | meaning | condition |
|---|---|---|
| `k` | y of the centre | |
| `A` | denominator of the positive term | |
| `B` | denominator of the negative term | |
| `orientation` | horizontal (x term positive) or vertical | |

| output | meaning |
|---|---|
| `a` | $a = \sqrt{A}$ |
| `b` | $b = \sqrt{B}$ |
| `c` | centre-to-focus distance |
| `eccentricity` | $e = c/a$ |
| `asymptote_slope` | slope of the asymptotes (plus or minus this) |
| `f1x` | first focus x |
| `f1y` | first focus y |
| `f2x` | second focus x |
| `f2y` | second focus y |

- **Method:** $c^2 = a^2 + b^2$; asymptotes y - k = ±(b/a)(x - h) (horizontal) or ±(a/b) (vertical)
- **Assumes:** A, $B > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




c (centre-to-focus distance) against A (denominator of the positive term) from 1.6 to 160 on a logarithmic axis. One curve for each value of B (denominator of the negative term): 9, 4.5, 18. Other inputs as in the worked example: h = 0, k = 0, orientation = horizontal. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, A = 16, c = 5.

```
math_foundations_ai999x.solve('hyperbola_parameters', h=0, k=0, A=16, B=9,
orientation='horizontal')
# -> a = 4, b = 3, c = 5, eccentricity = 1.25, asymptote_slope = 0.75, f1x = -5, f1y = 0, f2x = 5,
f2y = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('hyperbola_parameters', h=-1, k=2, A=9, B=16, orientation='vertical')
# -> a = 3, b = 4, c = 5, eccentricity = 1.66667, asymptote_slope = 0.75, f1x = -1, f1y = -3, f2x =
-1, f2y = 7   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the foci and asymptotes of the hyperbola x^2/16 - y^2/9 = 1. | SOLVED | a = 4, b = 3, c = 5, eccentricity = 1.25, asymptote_slope = 0.75, f1x = -5, f1y = 0, f2x = 5, f2y = 0 |
| Find the foci of (y - 2)^2/9 - (x + 1)^2/16 = 1. | SOLVED | a = 3, b = 4, c = 5, eccentricity = 1.66667, asymptote_slope = 0.75, f1x = -1, f1y = -3, f2x = -1, f2y = 7 |

## Coefficient of a term in a binomial expansion — `binomial_coefficient_term`

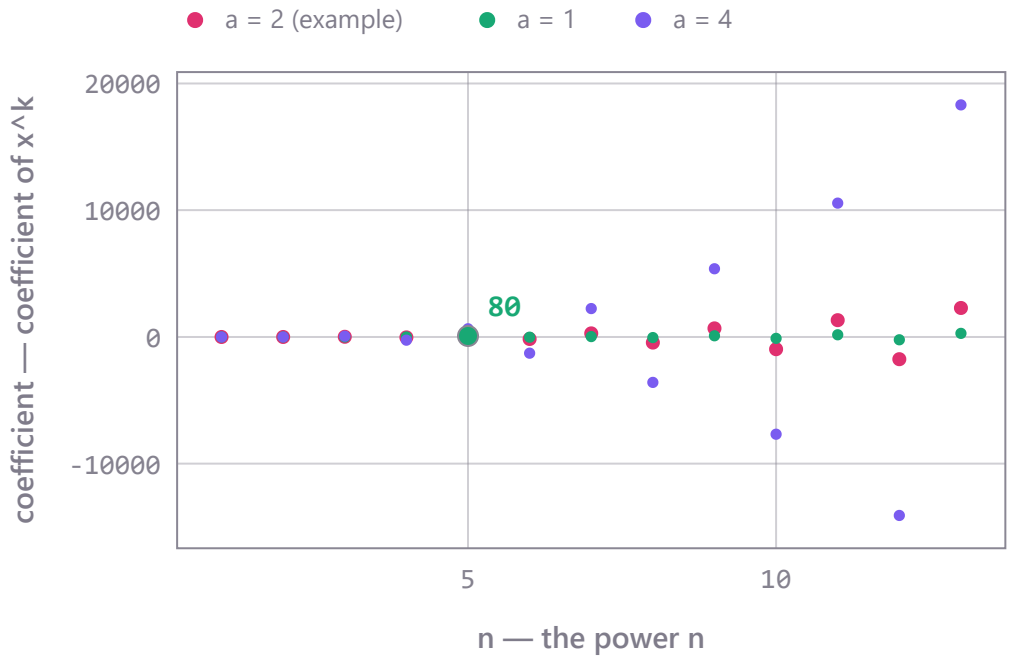
The coefficient of $x^k$ in $(ax + b)^n$, by the binomial theorem.

$$(ax + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} x^k$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x in the binomial | |
| `b` | constant of the binomial | |
| `n` | the power n | |
| `k` | the power of x whose coefficient is wanted | |

| output | meaning |
|---|---|
| `coefficient` | coefficient of $x^k$ |
| `binomial` | the binomial coefficient C(n, k) |

- **Method:** binomial theorem: the $x^k$ term is C(n, k) $(ax)^k\, b^{n-k}$
- **Assumes:** $0 \le n \le 60$; $k \ge 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




coefficient (coefficient of x^k) against n (the power n) from 1 to 13. One curve for each value of a (coefficient of x in the binomial): 2, 1, 4. Other inputs as in the worked example: b = -1, k = 3. 13 points per curve computed by math-foundations-ai999x; marked: the worked example, n = 5, coefficient = 80.

```
math_foundations_ai999x.solve('binomial_coefficient_term', a=2, b=-1, n=5, k=3)
# -> coefficient = 80, binomial = 10   (SOLVED)
```

```
math_foundations_ai999x.solve('binomial_coefficient_term', a=1, b=3, n=6, k=2)
# -> coefficient = 1215, binomial = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('binomial_coefficient_term', a=-2, b=3, n=4, k=0)
# -> coefficient = 81, binomial = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the coefficient of x^3 in (2x - 1)^5. | SOLVED | coefficient = 80, binomial = 10 |
| What is the coefficient of x^2 in the expansion of (x + 3)^6? | SOLVED | coefficient = 1215, binomial = 15 |
| Find the constant term in (3 - 2x)^4. | SOLVED | coefficient = 81, binomial = 1 |
| Find the coefficient of x in (x + 2)^3. | SOLVED | coefficient = 12, binomial = 3 |

## Arithmetic with complex numbers — `complex_arithmetic`

(a + bi) + - * / (c + di) in the form $re + imi$.

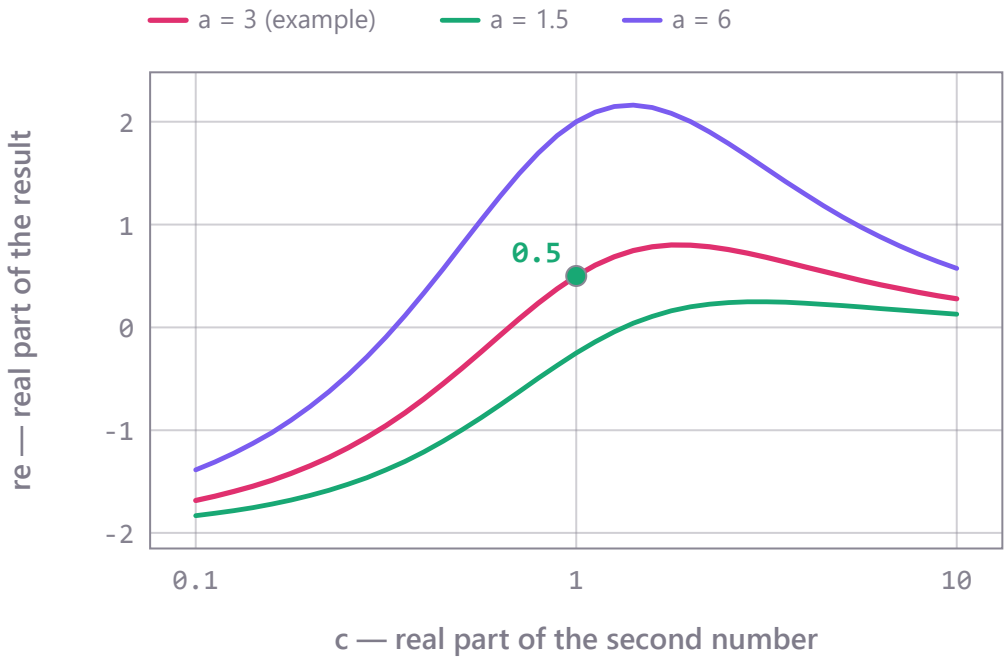
$$\frac{a+bi}{c+di} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$$

| input | meaning | condition |
|---|---|---|
| a | real part of the first number | |
| b | imaginary part of the first number | |

| input | meaning | condition |
|---|---|---|
| `c` | real part of the second number | |
| `d` | imaginary part of the second number | |
| `op` | the operation | |

| output | meaning |
|---|---|
| `re` | real part of the result |
| `im` | imaginary part of the result |

- **Method:** $i^2 = -1$; division multiplies numerator and denominator by the conjugate c - di
- **Assumes:** no division by 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




re (real part of the result) against c (real part of the second number) from 0.1 to 10 on a logarithmic axis. One curve for each value of a (real part of the first number): 3, 1.5, 6. Other inputs as in the worked example: b = 2, d = -1, op = /. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 1, re = 0.5.

```
math_foundations_ai999x.solve('complex_arithmetic', a=3, b=2, c=1, d=-1, op='/')
# -> re = 0.5, im = 2.5   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_arithmetic', a=2, b=-3, c=4, d=1, op='*')
# -> re = 11, im = -10   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_arithmetic', a=2, b=-3, c=4, d=1, op='+')
# -> re = 6, im = -2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Compute (3 + 2i) / (1 - i). | SOLVED | re = 0.5, im = 2.5 |

| question | outcome | result |
|---|---|---|
| Simplify (2 - 3i)(4 + i). | SOLVED | re = 11, im = -10 |
| Find (2 - 3i) + (4 + i). | SOLVED | re = 6, im = -2 |
| Compute (5 + i) - (2 - 4i). | SOLVED | re = 3, im = 5 |
| Compute (1 + i) / (0 + 0i). | INVALID_INPUT | Arithmetic with complex numbers: division by 0. |

### Modulus, argument and polar form of a complex number — `complex_modulus_argument`

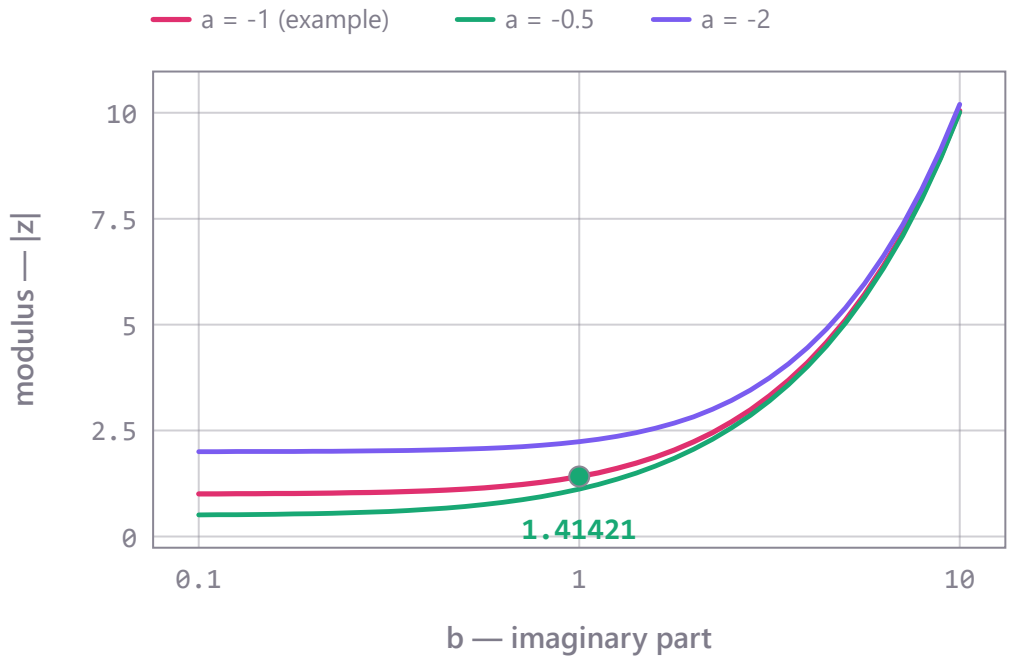
|z| and the principal argument (-180, 180] degrees of $z = a + bi$, and z in polar form.

$$z = r(\cos\theta + i\sin\theta), \quad r = \sqrt{a^2 + b^2}$$

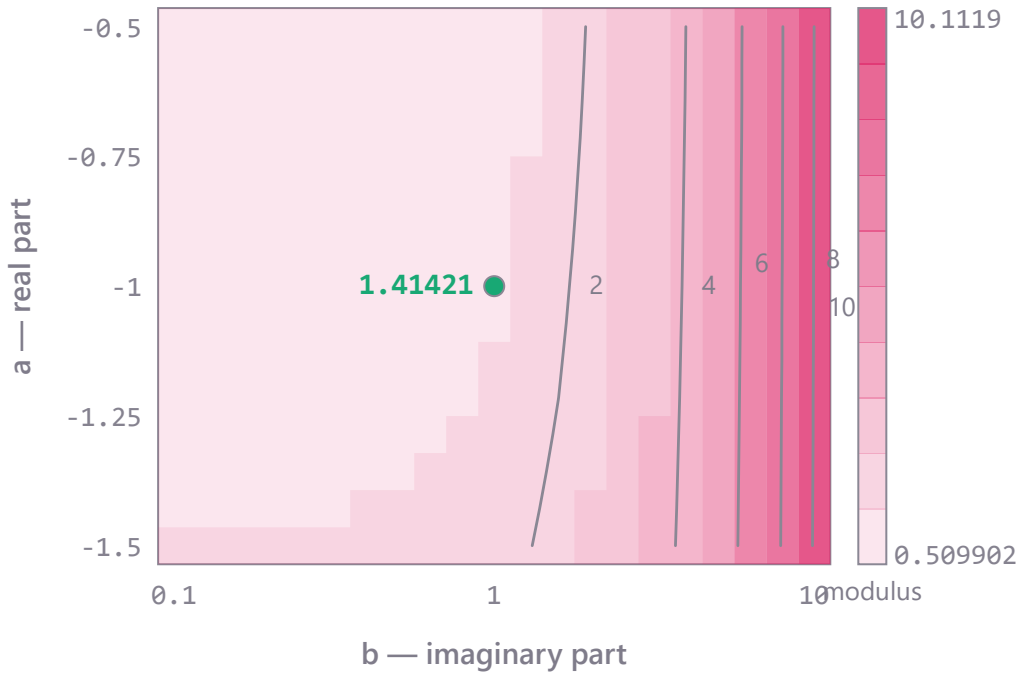
| input | meaning | condition |
|---|---|---|
| `a` | real part | |
| `b` | imaginary part | |

| output | meaning |
|---|---|
| `modulus` | |z| |
| `argument_deg` | principal argument, degrees in (-180, 180] |
| `argument_rad` | principal argument, radians |
| `polar` | $z = r$(cos t + i sin t) |

- **Method:** |z| = sqrt(a^2 + b^2); the argument in the quadrant of (a, b)
- **Assumes:** $z \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




modulus (|z|) against b (imaginary part) from 0.1 to 10 on a logarithmic axis. One curve for each value of a (real part): -1, -0.5, -2. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 1, modulus = 1.41421.




modulus (|z|) over b (imaginary part) from 0.1 to 10 and a (real part) from -1.5 to -0.5 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0.509902 to dark 10.1119); iso-lines at modulus = 2, 4, 6, 8, 10. Marked: the worked example, b = 1, a = -1, modulus = 1.41421.

```
math_foundations_ai999x.solve('complex_modulus_argument', a=-1, b=1)
# -> modulus = 1.41421, argument_deg = 135, argument_rad = 2.35619, polar = 1.414213562(cos(135
deg) + i sin(135 deg))   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_modulus_argument', a=1, b=-1)
# -> modulus = 1.41421, argument_deg = -45, argument_rad = -0.785398, polar = 1.414213562(cos(-45
deg) + i sin(-45 deg))   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_modulus_argument', a=3, b=-4)
# -> modulus = 5, argument_deg = -53.1301, argument_rad = -0.927295, polar = 5(cos(-53.13010235
deg) + i sin(-53.13010235 deg))   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_modulus_argument', a=-2, b=0)
# -> modulus = 2, argument_deg = 180, argument_rad = 3.14159, polar = 2(cos(180 deg) + i sin(180
deg))   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the modulus and argument of -1 + i. | SOLVED | modulus = 1.41421, argument_deg = 135, argument_rad = 2.35619, polar = 1.414213562(cos(135 deg) + i sin(135 deg)) |
| Write 1 - i in polar form. | SOLVED | modulus = 1.41421, argument_deg = -45, argument_rad = -0.785398, polar = 1.414213562(cos(-45 deg) + i sin(-45 deg)) |
| Find the modulus and argument of z = 3 - 4i. | SOLVED | modulus = 5, argument_deg = -53.1301, argument_rad = -0.927295, polar = 5(cos(-53.13010235 deg) + i sin(-53.13010235 deg)) |
| Find the modulus and argument of 0. | INVALID_INPUT | Modulus, argument and polar form of a complex number: 0 has modulus 0 and no argument. |

### Power of a complex number (De Moivre) — `de_moivre_power`

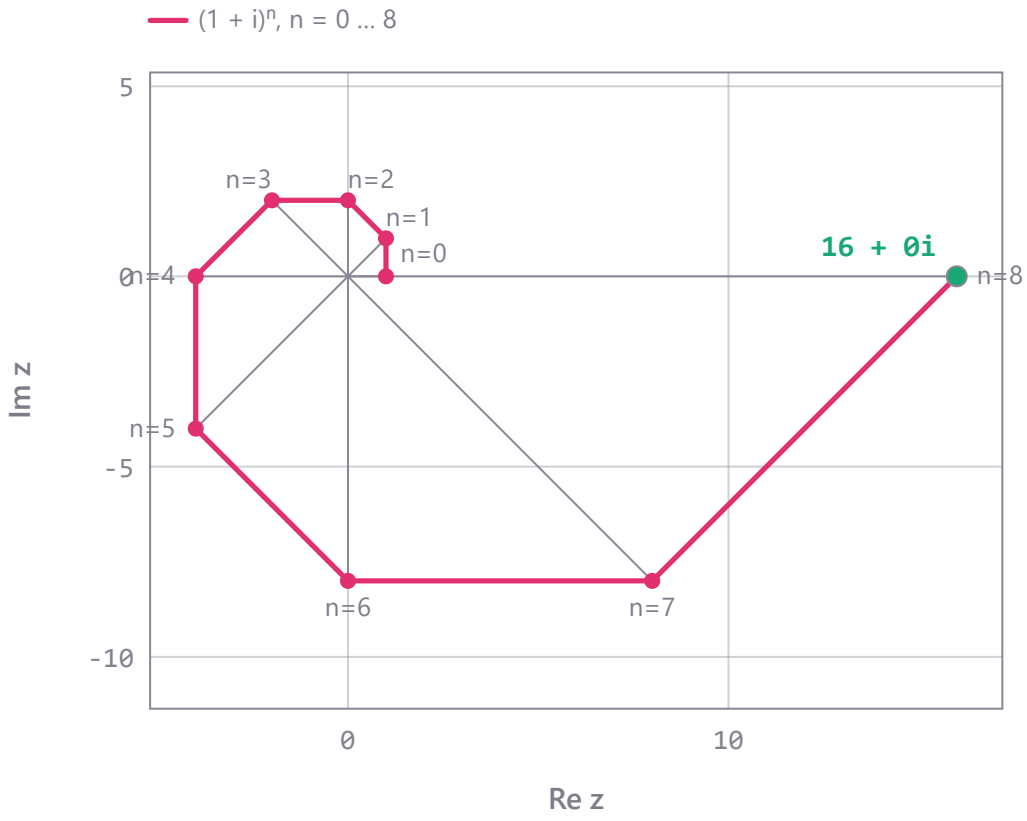
$(a + bi)^n$ for a whole number n (|n| <= 40): real and imaginary parts, modulus and argument.

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$$

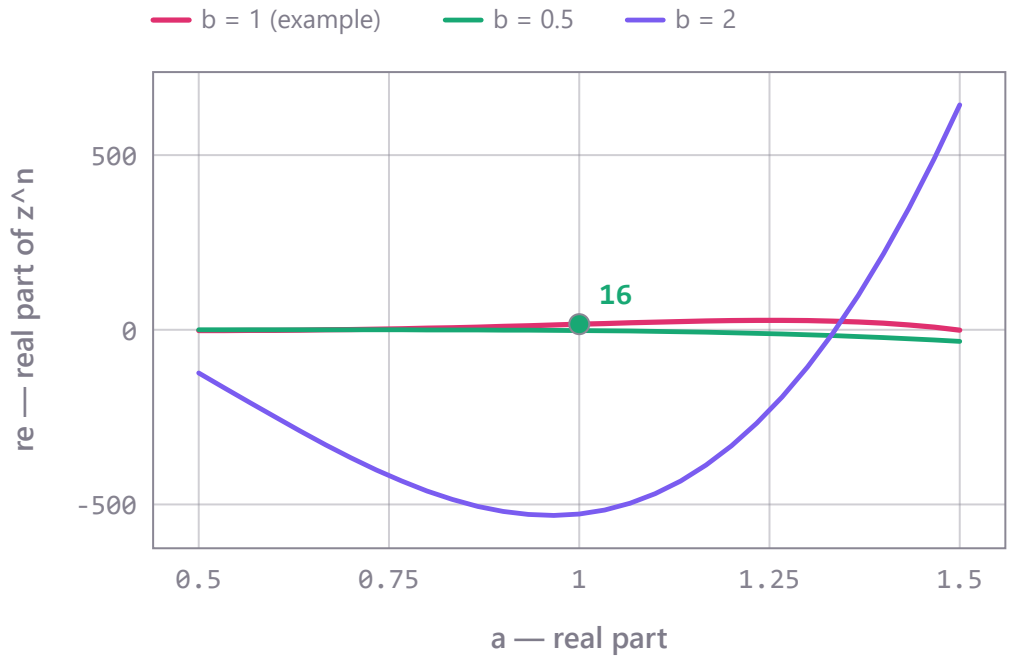
| input | meaning | condition |
|---|---|---|
| `a` | real part | |
| `b` | imaginary part | |
| `n` | the power n | |

| output | meaning |
|---|---|
| `re` | real part of $z^n$ |
| `im` | imaginary part of $z^n$ |
| `modulus` | $\lvert z^n\rvert$ = \|z\|^n |
| `argument_deg` | argument of $z^n$, degrees |

- **Method:** De Moivre: $z^n = r^n$ (cos n t + i sin n t); computed exactly (binary exponentiation in rational arithmetic)
- **Assumes:** |n| <= 40; $z \neq 0$ when $n \leq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The powers (1 + i)$^n$ for n = 0 … 8, each computed by the package: every step turns by 45° and stretches by √2; marked: n = 8, (1 + i)$^8$ = 16 + 0i, modulus 16, argument 0°.




re (real part of z^n) against a (real part) from 0.5 to 1.5. One curve for each value of b (imaginary part): 1, 0.5, 2. Other inputs as in the worked example: n = 8. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, re = 16.

```
math_foundations_ai999x.solve('de_moivre_power', a=1, b=1, n=8)
# -> re = 16, im = 0, modulus = 16, argument_deg = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('de_moivre_power', a=0, b=2, n=3)
# -> re = 0, im = -8, modulus = 8, argument_deg = -90   (SOLVED)
```

```
math_foundations_ai999x.solve('de_moivre_power', a=1, b=-1, n=-2)
# -> re = 0, im = 0.5, modulus = 0.5, argument_deg = 90   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Use De Moivre's theorem to compute (1 + i)^8. | SOLVED | re = 16, im = 0, modulus = 16, argument_deg = 0 |
| Compute (sqrt(3) + i)^6. | SOLVED | re = -64, im = 0, modulus = 64, argument_deg = 180 |
| Find (2i)^3. | SOLVED | re = 0, im = -8, modulus = 8, argument_deg = -90 |
| Compute (0 + 0i)^-2. | INVALID_INPUT | Power of a complex number (De Moivre): 0 to a power that is not positive is undefined. |

## The n-th roots of a complex number — `complex_nth_roots`

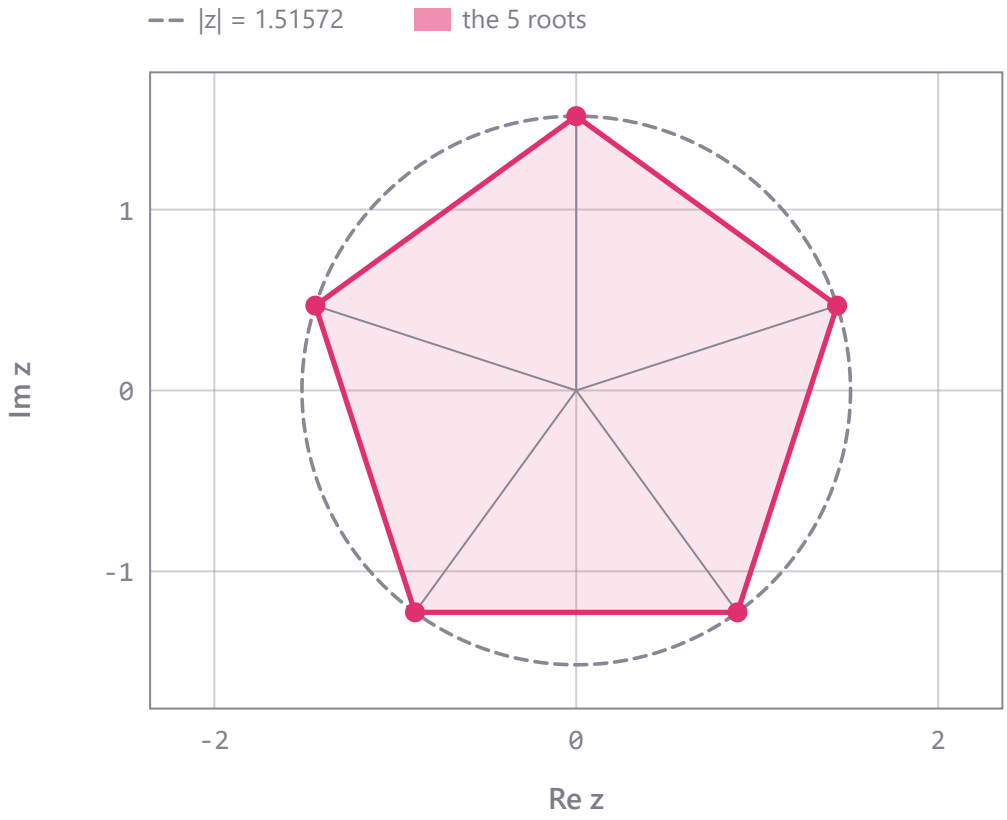
All n complex n-th roots of $z = a + bi$ (n from 2 to 12), ordered by argument from 0 to 360 degrees.

$$w_k = \sqrt[n]{r}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right)$$

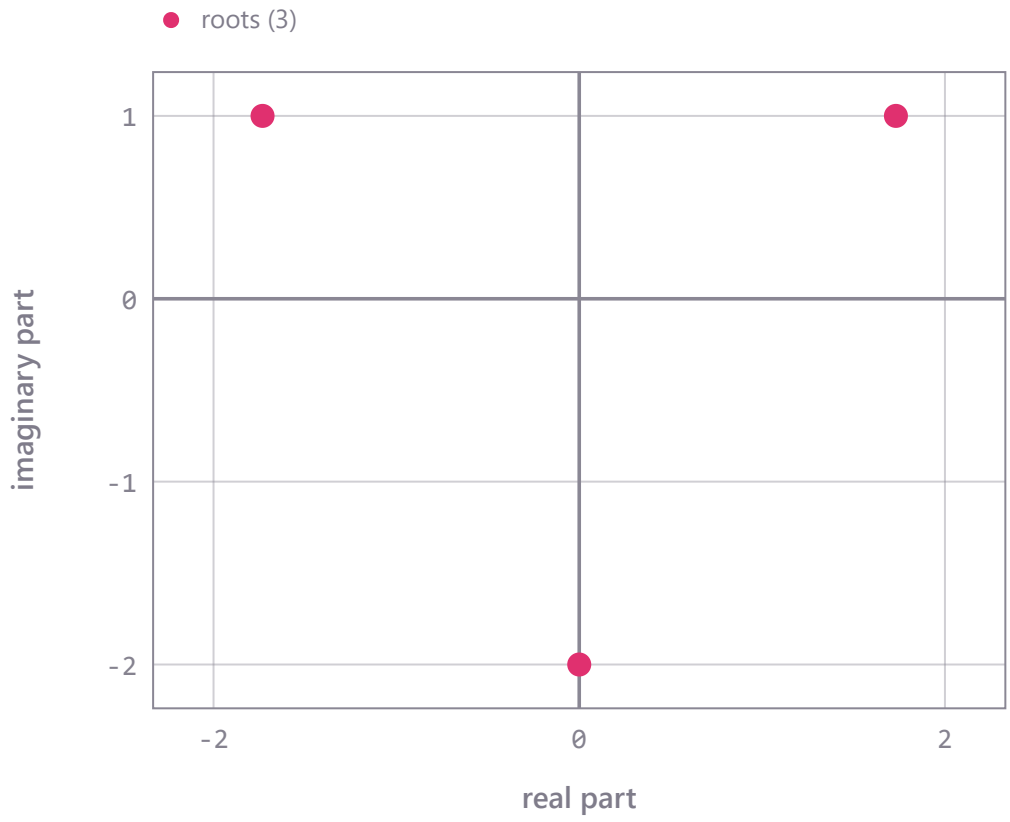
| input | meaning | condition |
|---|---|---|
| `a` | real part of z | |
| `b` | imaginary part of z | |
| `n` | which roots (2 = square roots) | |

| output | meaning |
|---|---|
| `modulus` | the common modulus \|z\|^$(1/n)$ |
| `count` | the number of roots, n |
| `roots` | the roots, by argument in [0, 360) |

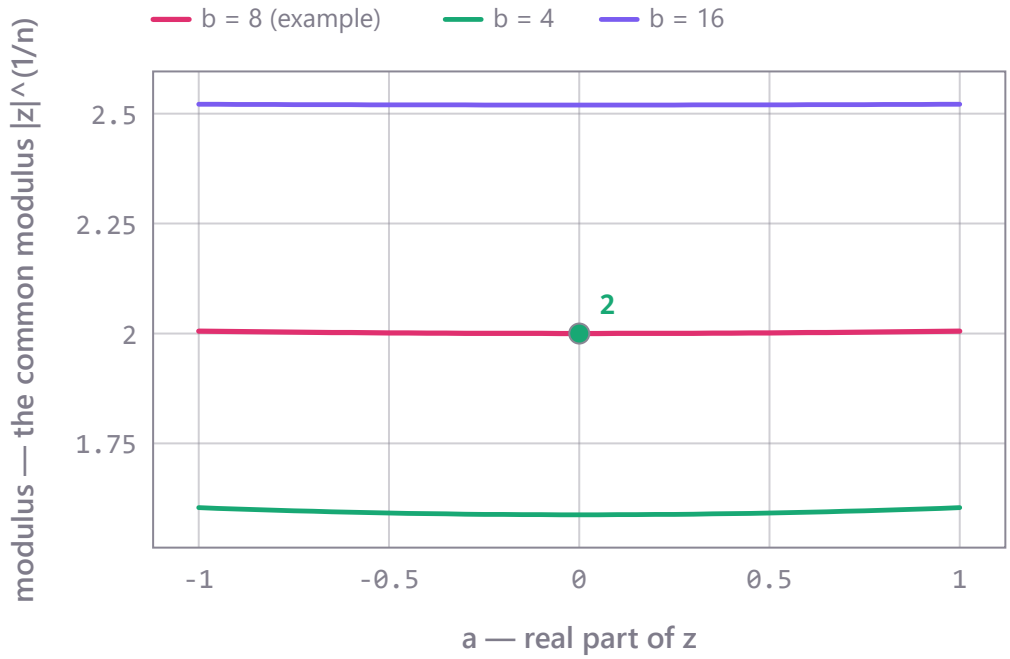
- **Method:** w_k = |z|^$(1/n)$ (cos((t + 2 pi k)/n) + i sin((t + 2 pi k)/n)), $k = 0 \,..\, n-1$
- **Assumes:** $z \neq 0$; $2 \leq n \leq 12$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The 5 complex roots of $z^5 = 0 + 8i$ returned by the package — -0.890916 − 1.22624i, 0.890916 − 1.22624i, 1.44153 + 0.468382i, 0 + 1.51572i, -1.44153 + 0.468382i — the vertices of a regular pentagon on the circle of radius 1.51572 (= 8^(1/5)).




the roots, by argument in [0, 360) (roots) returned by math-foundations-ai999x for the worked example (a = 0, b = 8, n = 3), drawn in the complex plane: 1.73205 + 1i, -1.73205 + 1i, 0 − 2i.




modulus (the common modulus |z|^(1/n)) against a (real part of z) from -1 to 1. One curve for each value of b (imaginary part of z): 8, 4, 16. Other inputs as in the worked example: n = 3. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 0, modulus = 2.

```
math_foundations_ai999x.solve('complex_nth_roots', a=0, b=8, n=3)
# -> modulus = 2, count = 3, roots = [(1.73205080757+1j), (-1.73205080757+1j), -2j]   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_nth_roots', a=-16, b=0, n=4)
# -> modulus = 2, count = 4, roots = [(1.41421356237+1.41421356237j),
(-1.41421356237+1.41421356237j), (-1.41421356237-1.41421356237j), (1.41421356237-1.41421356237j)]
(SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the cube roots of 8i. | SOLVED | modulus = 2, count = 3, roots = [(1.73205080757+1j), (-1.73205080757+1j), -2j] |
| Find all the fourth roots of -16. | SOLVED | modulus = 2, count = 4, roots = [(1.41421356237+1.41421356237j), (-1.41421356237+1.41421356237j), (-1.41421356237-1.41421356237j), (1.41421356237-1.41421356237j)] |
| Find the square roots of 9. | SOLVED | modulus = 3, count = 2, roots = [(3+0j), (-3+0j)] |
| Find the cube roots of 0. | INVALID_INPUT | The n-th roots of a complex number: 0 has only the root 0. |

## Solution set of |x - c| > d — `abs_value_inequality_lower_bound`

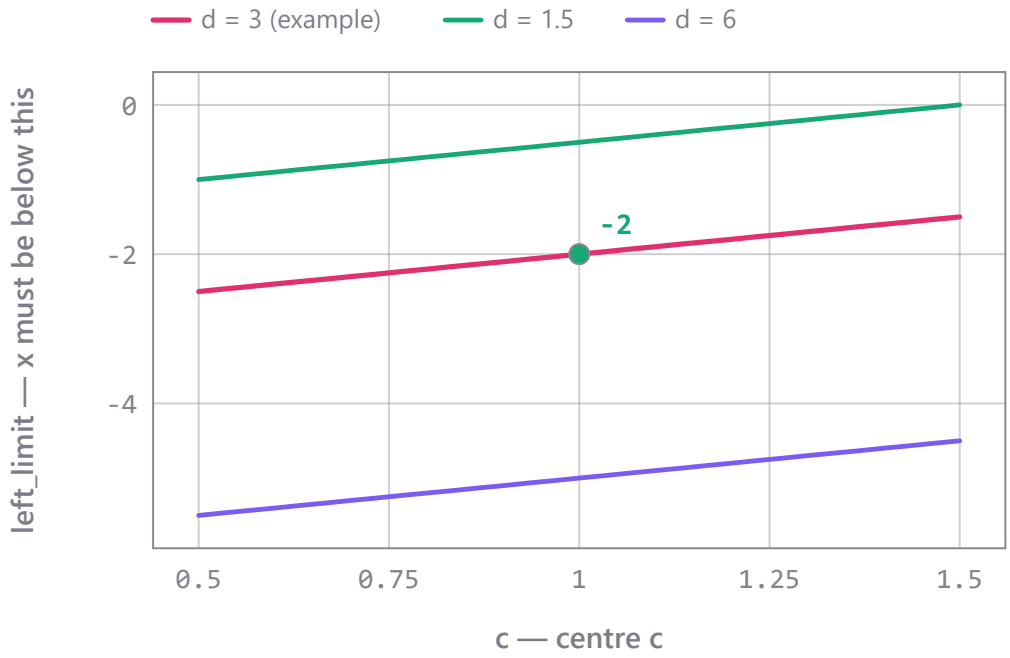
For |x - c| > d with $d \geq 0$, the solutions are x < left_limit or x > right_limit.

$$\text{right limit} - \text{left limit} = 2d, \quad \text{right limit} + \text{left limit} = 2c$$

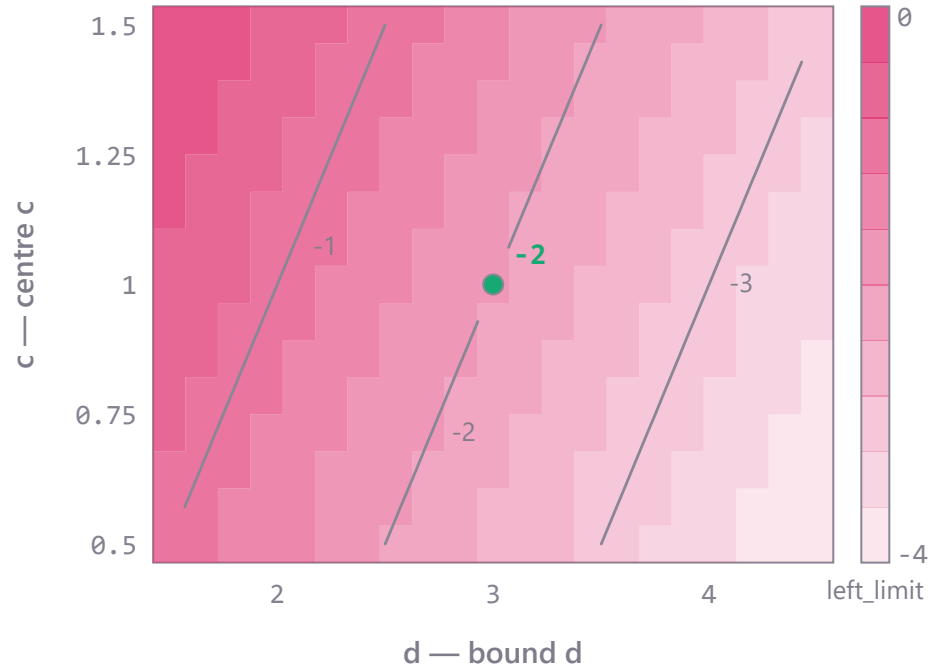
| input | meaning | condition |
|---|---|---|
| `c` | centre c | |
| `d` | bound d | $d \geq 0$ |

| output | meaning |
|---|---|
| `left_limit` | x must be below this |
| `right_limit` | or above this |

- **Method:** |x - c| > d means $x - c < -d$ or $x - c > d$
- **Assumes:** $d \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




left_limit (x must be below this) against c (centre c) from 0.5 to 1.5. One curve for each value of d (bound d): 3, 1.5, 6. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 1, left_limit = -2.




left_limit (x must be below this) over d (bound d) from 1.5 to 4.5 and c (centre c) from 0.5 to 1.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light -4 to dark 0); iso-lines at left_limit = -3, -2, -1. Marked: the worked example, d = 3, c = 1, left_limit = -2.

```
math_foundations_ai999x.solve('abs_value_inequality_lower_bound', c=1.0, d=3.0)
# -> left_limit = -2, right_limit = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('abs_value_inequality_lower_bound', c=5.0, d=0.0)
# -> left_limit = 5, right_limit = 5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve \|x - 1\| > 3. | SOLVED | left_limit = -2, right_limit = 4 |
| For which x is \|x - 6\| greater than 2? | SOLVED | left_limit = 4, right_limit = 8 |

### Solution interval of |x - c| < d — `abs_value_inequality_upper_bound`

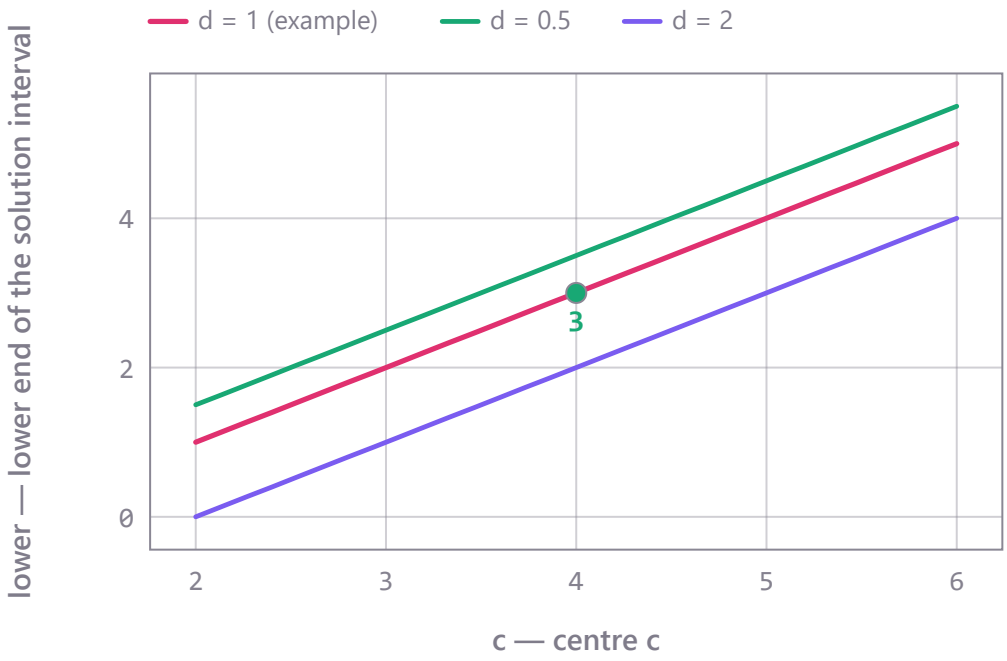
For |x - c| < d with $d > 0$, the solutions are lower < x < upper.

$$\text{upper} - \text{lower} = 2d, \quad \text{upper} + \text{lower} = 2c$$

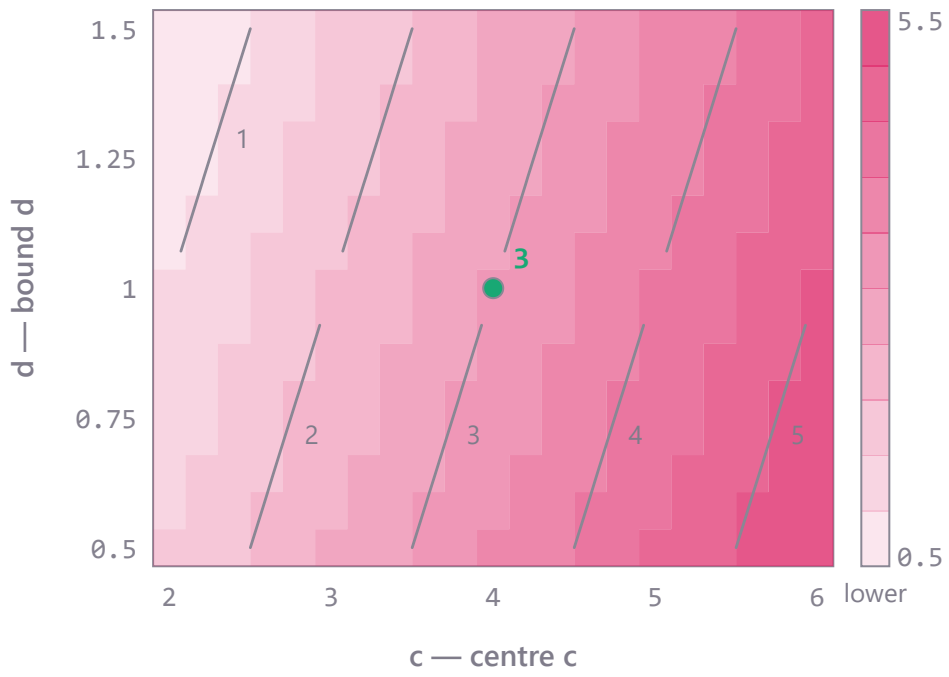
| input | meaning | condition |
|---|---|---|
| `c` | centre c | |
| `d` | bound d | $d > 0$ |

| output | meaning |
|---|---|
| `lower` | lower end of the solution interval (excluded) |
| `upper` | upper end of the solution interval (excluded) |

- **Method:** |x - c| < d means -d < x - c < d
- **Assumes:** $d > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




lower (lower end of the solution interval (excluded)) against c (centre c) from 2 to 6. One curve for each value of d (bound d): 1, 0.5, 2. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 4, lower = 3.




lower (lower end of the solution interval (excluded)) over c (centre c) from 2 to 6 and d (bound d) from 0.5 to 1.5: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 0.5 to dark 5.5); iso-lines at lower = 1, 2, 3, 4, 5. Marked: the worked example, c = 4, d = 1, lower = 3.

```
math_foundations_ai999x.solve('abs_value_inequality_upper_bound', c=4.0, d=1.0)
# -> lower = 3, upper = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('abs_value_inequality_upper_bound', c=0.0, d=2.5)
# -> lower = -2.5, upper = 2.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve \|x - 4\| < 1. | SOLVED | lower = 3, upper = 5 |
| Solve \|x - 2\| < 0. | INVALID_INPUT | Solution interval of \|x - c\| < d: the condition d > 0 does not hold for these inputs. |

## Solutions of an absolute value equation — `absolute_value_equation`

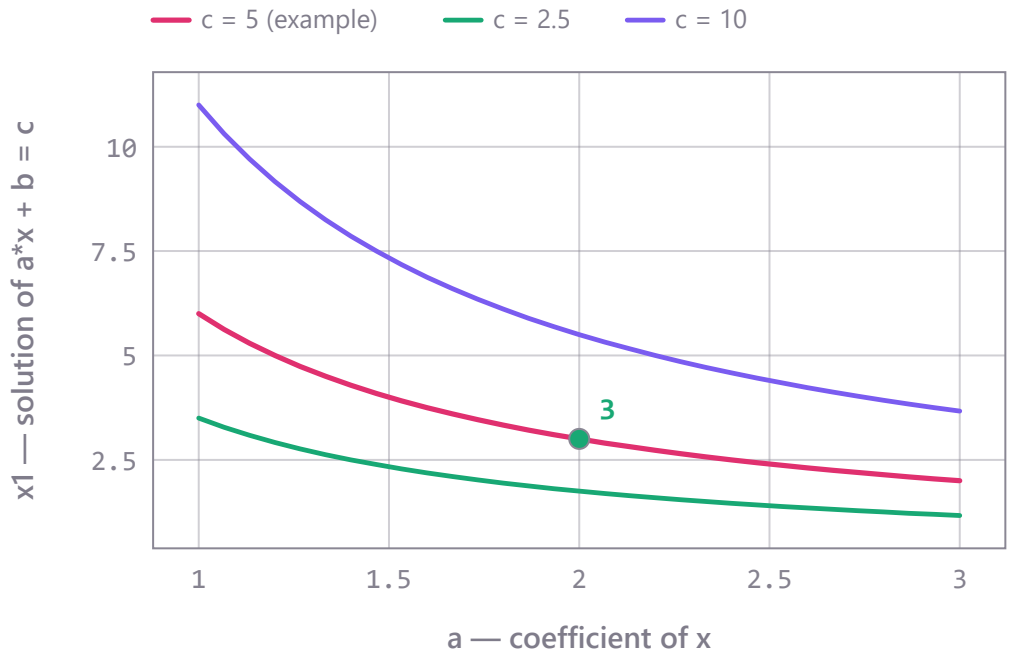
Solve $|ax + b|$ = c for x (two solutions when $c > 0$, one when c = 0).

$$ax_1 + b = c, \quad ax_2 + b = -c$$

| input | meaning | condition |
|---|---|---|
| a | coefficient of x | $a \neq 0$ |
| b | constant inside the absolute value | |
| c | right-hand side | $c \geq 0$ |

| output | meaning |
|---|---|
| x1 | solution of $ax + b = c$ |
| x2 | solution of $ax + b = -c$ |

- **Method:** the expression inside equals c or -c; solve each linear equation
- **Assumes:** $a \neq 0$; $c \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x1 (solution of a*x + b = c) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 5, 2.5, 10. Other inputs as in the worked example: b = -1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x1 = 3.

```
math_foundations_ai999x.solve('absolute_value_equation', a=2.0, b=-1.0, c=5.0)
# -> x1 = 3, x2 = -2   (SOLVED)
```

```
math_foundations_ai999x.solve('absolute_value_equation', a=1.0, b=3.0, c=0.0)
# -> x1 = -3, x2 = -3   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve \|2x - 1\| = 5. | SOLVED | x1 = 3, x2 = -2 |
| Solve \|x + 4\| = 6. | SOLVED | x1 = 2, x2 = -10 |

## Add rational expressions with same denominator — `add_rational_expressions_same_denom`

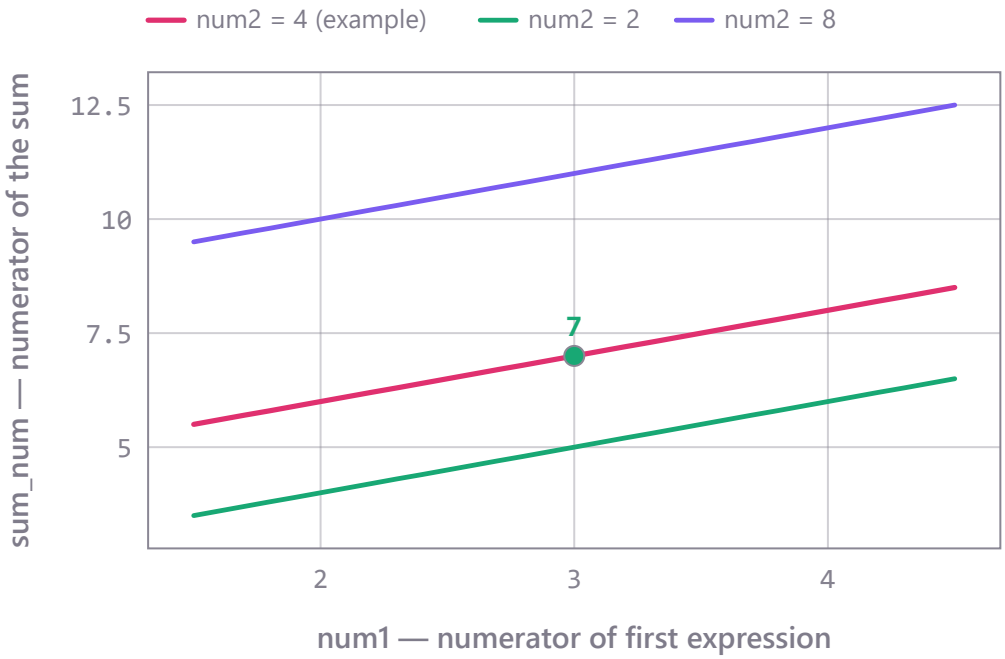
Add two rational expressions that share a common denominator.

$$\text{sum num} = \text{num}_1 + \text{num}_2, \quad \text{sum den} = \text{den}$$

| input | meaning | condition |
|---|---|---|
| `num1` | numerator of first expression | |
| `num2` | numerator of second expression | |
| `den` | common denominator | |

| output | meaning |
|---|---|
| `sum_num` | numerator of the sum |
| `sum_den` | denominator of the sum |

- **Method:** closed form verified against the defining equations
- **Assumes:** den != 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




sum_num (numerator of the sum) against num1 (numerator of first expression) from 1.5 to 4.5. One curve for each value of num2 (numerator of second expression): 4, 2, 8. Other inputs as in the worked example: den = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, num1 = 3, sum_num = 7.

```
math_foundations_ai999x.solve('add_rational_expressions_same_denom', den=5.0, num1=3.0, num2=4.0)
# -> sum_num = 7, sum_den = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('add_rational_expressions_same_denom', den=4.0, num1=10.0, num2=-2.0)
# -> sum_num = 8, sum_den = 4   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Add 1/6 and 2/6. | SOLVED | sum_num = 3, sum_den = 6 |

## Term and partial sum of an arithmetic sequence — `arithmetic_sequence`

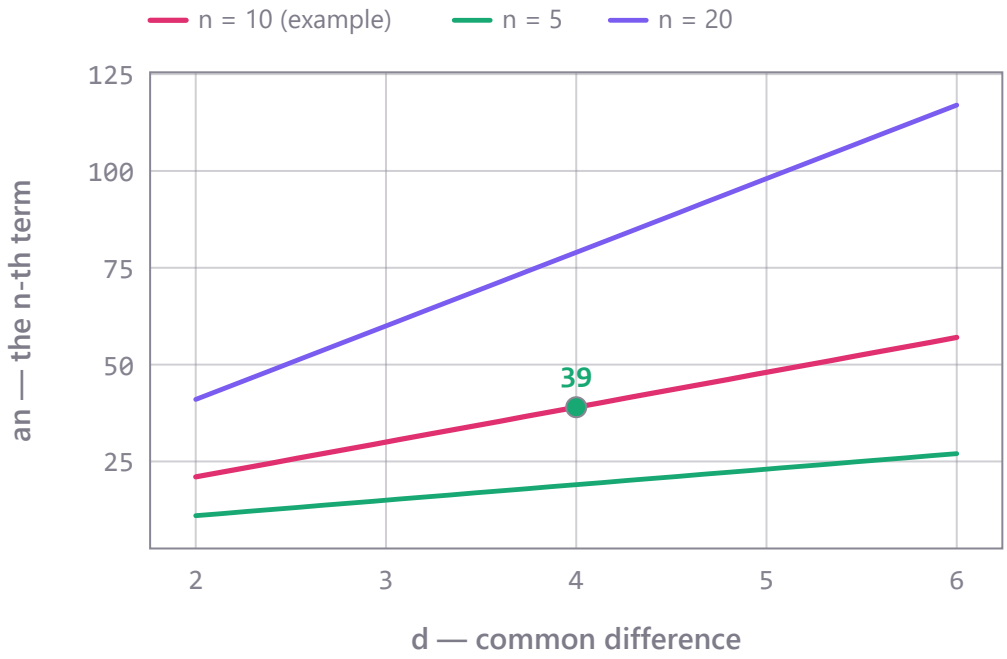
The n-th term and the sum of the first n terms of the arithmetic sequence with first term a1 and common difference d.

$$a_n = a_1 + (n-1)d, \quad 2s_n = n\,(a_1 + a_n)$$

| input | meaning | condition |
|---|---|---|
| `a1` | first term | |
| `d` | common difference | |
| `n` | position of the term (a positive whole number) | $n \geq 1$ |

| output | meaning |
|---|---|
| `an` | the n-th term |
| `sn` | the sum of the first n terms |

- **Method:** the n-th term adds d to the first term $n - 1$ times; the sum pairs first and last terms
- **Assumes:** $n \geq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




an (the n-th term) against d (common difference) from 2 to 6. One curve for each value of n (position of the term (a positive whole number)): 10, 5, 20. Other inputs as in the worked example: a1 = 3.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, d = 4, an = 39.

```
math_foundations_ai999x.solve('arithmetic_sequence', a1=3.0, d=4.0, n=10.0)
# -> an = 39, sn = 210   (SOLVED)
```

```
math_foundations_ai999x.solve('arithmetic_sequence', a1=100.0, d=-7.0, n=5.0)
# -> an = 72, sn = 430   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the 10th term of the arithmetic sequence with first term 3 and common difference 4. | SOLVED | an = 39, sn = 210 |
| Use the arithmetic sequence with a1 = 1, d = 2, n = 2.5. | INVALID_INPUT | Term and partial sum of an arithmetic sequence: n must be a whole number. |

### Average rate of change of a function between two inputs — `average_rate_of_change_definition`

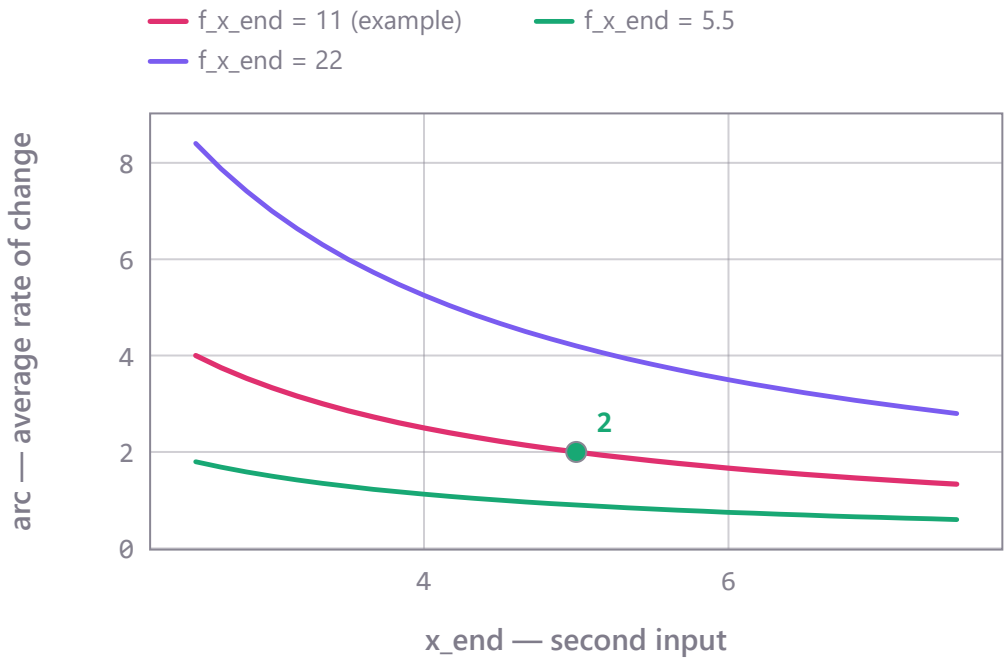
(f(x_end) - f(x_start)) / (x_end - x_start) from two values of a function.

$$\text{arc}\,(x_{\text{end}} - x_{\text{start}}) = f_{x,end} - f_{x,start}$$

| input | meaning | condition |
|---|---|---|
| `x_start` | first input | |
| `f_x_start` | value of the function at the first input | |
| `x_end` | second input | |
| `f_x_end` | value of the function at the second input | |

| output | meaning |
|---|---|
| `arc` | average rate of change |

- **Method:** change in output divided by change in input
- **Assumes:** x_end != x_start
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




arc (average rate of change) against x_end (second input) from 2.5 to 7.5. One curve for each value of f_x_end (value of the function at the second input): 11, 5.5, 22. Other inputs as in the worked example: x_start = 0.0, f_x_start = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, x_end = 5, arc = 2.

```
math_foundations_ai999x.solve('average_rate_of_change_definition', x_start=0.0, f_x_start=1.0,
x_end=5.0, f_x_end=11.0)
# -> arc = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('average_rate_of_change_definition', x_start=1.0, f_x_start=4.0,
x_end=3.0, f_x_end=0.0)
# -> arc = -2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| If f(0) = 1 and f(5) = 11, what is the average rate of change of f? | SOLVED | arc = 2 |
| If f(2) = 7 and f(2) = 9, what is the average rate of change of f? | INVALID_INPUT | Average rate of change of a function between two inputs: the condition x_end != x_start does not hold for these inputs. |

## Average rate of change between two points — `average_rate_of_change_linear`

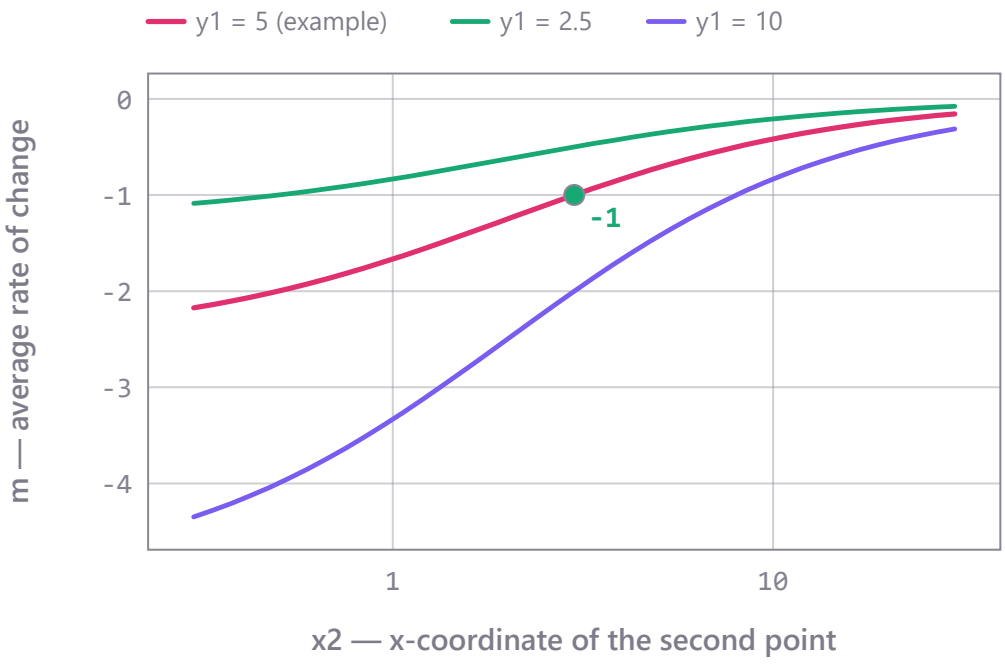
$(y_2 - y_1)/(x_2 - x_1)$ between the points (x1, y1) and (x2, y2).

$$m\,(x_2 - x_1) = y_2 - y_1$$

| input | meaning | condition |
|---|---|---|
| `x1` | x-coordinate of the first point | |
| `y1` | y-coordinate of the first point | |
| `x2` | x-coordinate of the second point | |
| `y2` | y-coordinate of the second point | |

| output | meaning |
|---|---|
| `m` | average rate of change |

- **Method:** change in y divided by change in x
- **Assumes:** $x_2 \neq x_1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




m (average rate of change) against x2 (x-coordinate of the second point) from 0.3 to 30 on a logarithmic axis. One curve for each value of y1 (y-coordinate of the first point): 5, 2.5, 10. Other inputs as in the worked example: x1 = -2.0, y2 = 0.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, x2 = 3, m = -1.

```
math_foundations_ai999x.solve('average_rate_of_change_linear', x1=-2.0, y1=5.0, x2=3.0, y2=0.0)
# -> m = -1   (SOLVED)
```

```
math_foundations_ai999x.solve('average_rate_of_change_linear', x1=0.0, y1=0.0, x2=4.0, y2=2.0)
# -> m = 0.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the average rate of change between the points (-2, 5) and (3, 0). | SOLVED | m = -1 |
| What is the average rate of change from (1, 2) to (1, 8)? | INVALID_INPUT | Average rate of change between two points: the condition x2 != x1 does not hold for these inputs. |

## Product of two binomials — `binomial_product`

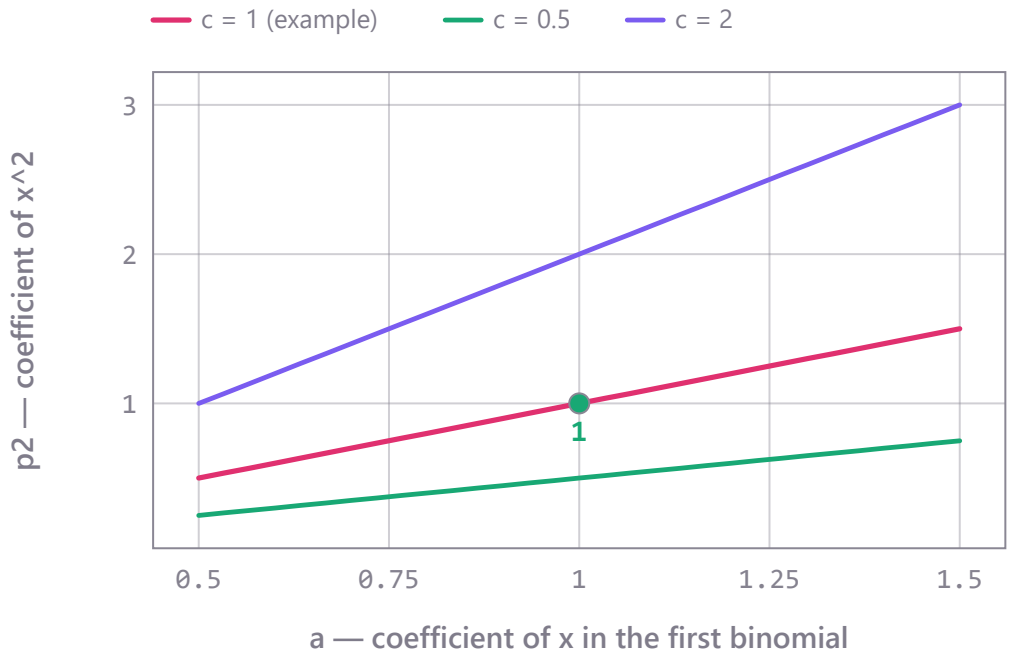
Expand $(ax + b)(cx + d)$ into $p_2x^2 + p_1x + p_0$.

$$p_2t^2 + p_1t + p_0 = (at + b)\,(ct + d)$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x in the first binomial | |
| `b` | constant of the first binomial | |
| `c` | coefficient of x in the second binomial | |
| `d` | constant of the second binomial | |

| output | meaning |
|---|---|
| `p2` | coefficient of $x^2$ |
| `p1` | coefficient of x |
| `p0` | constant term |

- **Method:** distribute each term of the first binomial over the second (FOIL) and collect like terms
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




p2 (coefficient of x^2) against a (coefficient of x in the first binomial) from 0.5 to 1.5. One curve for each value of c (coefficient of x in the second binomial): 1, 0.5, 2. Other inputs as in the worked example: b = 2.0, d = 3.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, p2 = 1.

```
math_foundations_ai999x.solve('binomial_product', a=1.0, b=2.0, c=1.0, d=3.0)
# -> p2 = 1, p1 = 5, p0 = 6   (SOLVED)
```

```
math_foundations_ai999x.solve('binomial_product', a=2.0, b=-1.0, c=3.0, d=4.0)
# -> p2 = 6, p1 = 5, p0 = -4   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Expand (x + 2)(x + 3). | SOLVED | p2 = 1, p1 = 5, p0 = 6 |
| Multiply (2x - 1) by (3x + 4). | SOLVED | p2 = 6, p1 = 5, p0 = -4 |

## Centre and radius of a circle from its general equation — `circle_from_general_form`

For $x^2 + y^2 + Dx + Ey + F = 0$, the centre (h, k) and radius rad.

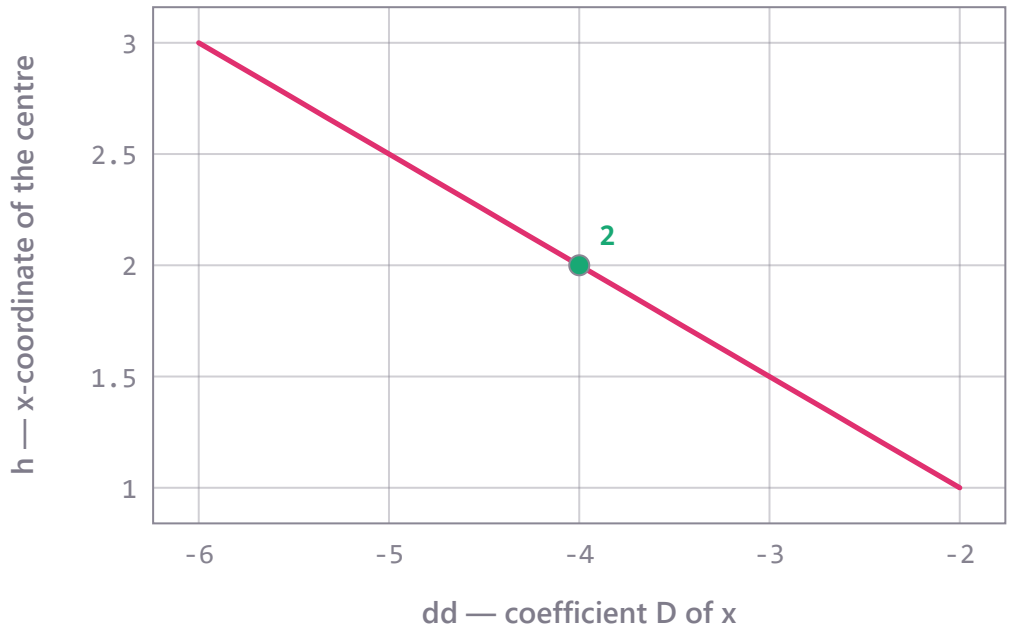
$$(t - h)^2 + (w - k)^2 - \mathrm{rad}^2 = t^2 + w^2 + d_d t + e_e w + f_f$$

| input | meaning | condition |
|---|---|---|
| `dd` | coefficient D of x | |
| `ee` | coefficient E of y | |
| `ff` | constant F | |

| output | meaning |
|---|---|
| `h` | x-coordinate of the centre |
| `k` | y-coordinate of the centre |

| output | meaning |
|---|---|
| `rad` | radius |

- **Method:** complete the square in x and in y
- **Assumes:** dd**$2/4 + ee$**$2/4 - ff > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




h (x-coordinate of the centre) against dd (coefficient D of x) from -6 to -2. Other inputs as in the worked example: ee = 6.0, ff = -3.0. 31 points computed by math-foundations-ai999x; marked: the worked example, dd = -4, h = 2.

```
math_foundations_ai999x.solve('circle_from_general_form', dd=-4.0, ee=6.0, ff=-3.0)
# -> h = 2, k = -3, rad = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('circle_from_general_form', dd=0.0, ee=0.0, ff=-25.0)
# -> h = 0, k = 0, rad = 5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the centre and radius of x^2 + y^2 - 4x + 6y - 3 = 0. | SOLVED | h = 2, k = -3, rad = 4 |
| Find the centre and radius of x^2 + y^2 + 2x + 2y + 5 = 0. | INVALID_INPUT | Centre and radius of a circle from its general equation: the condition dd**2/4 + ee**2/4 - ff > 0 does not hold for thes |

## Product and modulus of complex numbers — `complex_product_modulus`

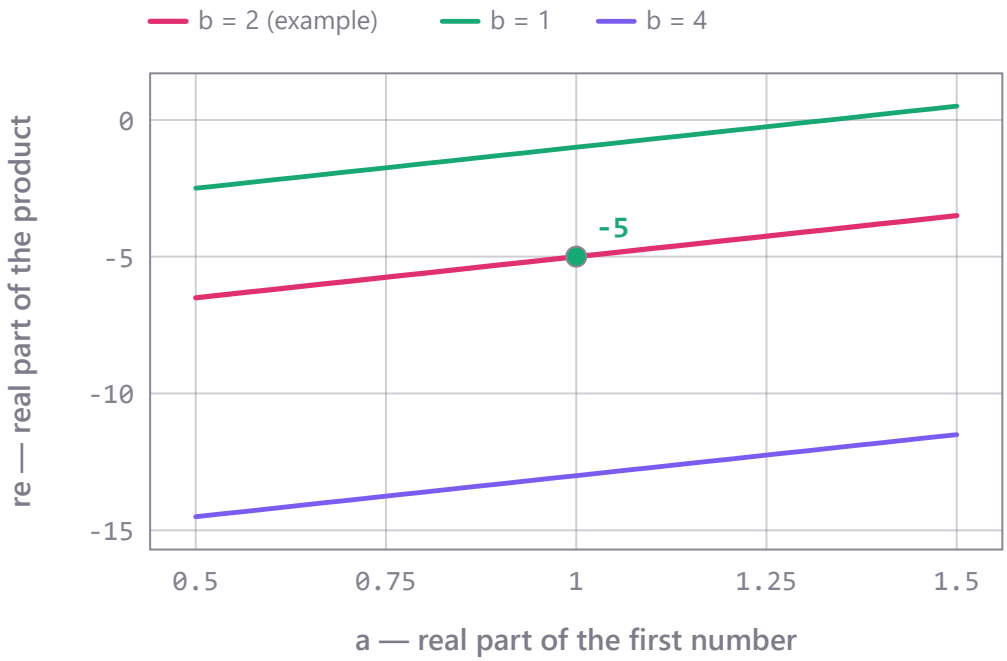
The product $(a + bi)(c + di) = re + imi$ and the modulus of the product.

$$\mathrm{Re} = ac - bd, \quad \mathrm{Im} = ad + bc, \quad \mathrm{modulus}^2 = \left(a^2 + b^2\right)\left(c^2 + d^2\right)$$

| input | meaning | condition |
|---|---|---|
| `a` | real part of the first number | |
| `b` | imaginary part of the first number | |
| `c` | real part of the second number | |
| `d` | imaginary part of the second number | |

| output | meaning |
|---|---|
| `re` | real part of the product |
| `im` | imaginary part of the product |
| `modulus` | modulus of the product |

- **Method:** distribute and use $i^2 = -1$; the modulus of a product is the product of the moduli
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




re (real part of the product) against a (real part of the first number) from 0.5 to 1.5. One curve for each value of b (imaginary part of the first number): 2, 1, 4. Other inputs as in the worked example: c = 3.0, d = 4.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, re = -5.

```
math_foundations_ai999x.solve('complex_product_modulus', a=1.0, b=2.0, c=3.0, d=4.0)
# -> re = -5, im = 10, modulus = 11.1803   (SOLVED)
```

```
math_foundations_ai999x.solve('complex_product_modulus', a=2.0, b=-1.0, c=2.0, d=1.0)
# -> re = 5, im = 0, modulus = 5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Multiply (1 + 2i)(3 + 4i). | SOLVED | re = -5, im = 10, modulus = 11.1803 |

### Composition of two linear functions — `compose_linear_functions`

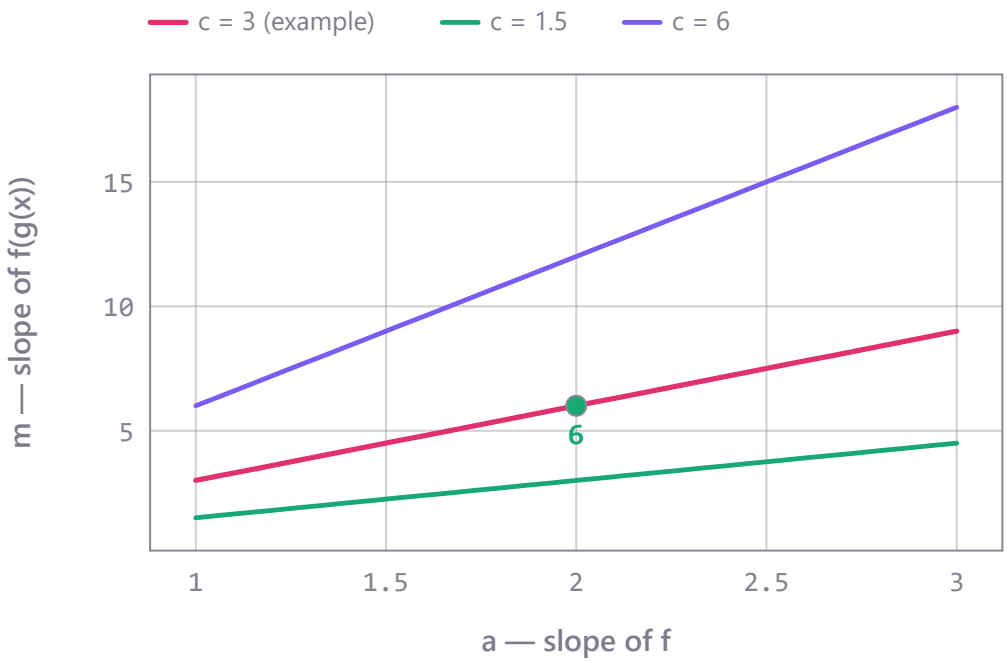
For f(x) = a*x + b and g(x) = c*x + d, the composition f(g(x)) = $mx + k$.

$$mt + k = a\,(ct + d) + b$$

| input | meaning | condition |
|---|---|---|
| `a` | slope of f | |
| `b` | intercept of f | |
| `c` | slope of g | |
| `d` | intercept of g | |

| output | meaning |
|---|---|
| `m` | slope of f(g(x)) |
| `k` | intercept of f(g(x)) |

- **Method:** substitute g(x) for x in f
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




m (slope of f(g(x))) against a (slope of f) from 1 to 3. One curve for each value of c (slope of g): 3, 1.5, 6. Other inputs as in the worked example: b = 1.0, d = -4.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, m = 6.

```
math_foundations_ai999x.solve('compose_linear_functions', a=2.0, b=1.0, c=3.0, d=-4.0)
# -> m = 6, k = -7   (SOLVED)
```

```
math_foundations_ai999x.solve('compose_linear_functions', a=-1.0, b=5.0, c=2.0, d=3.0)
# -> m = -2, k = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| If f(x) = 2x + 1 and g(x) = 3x - 4, find f(g(x)). | SOLVED | m = 6, k = -7 |

## Compound interest — `compound_interest`

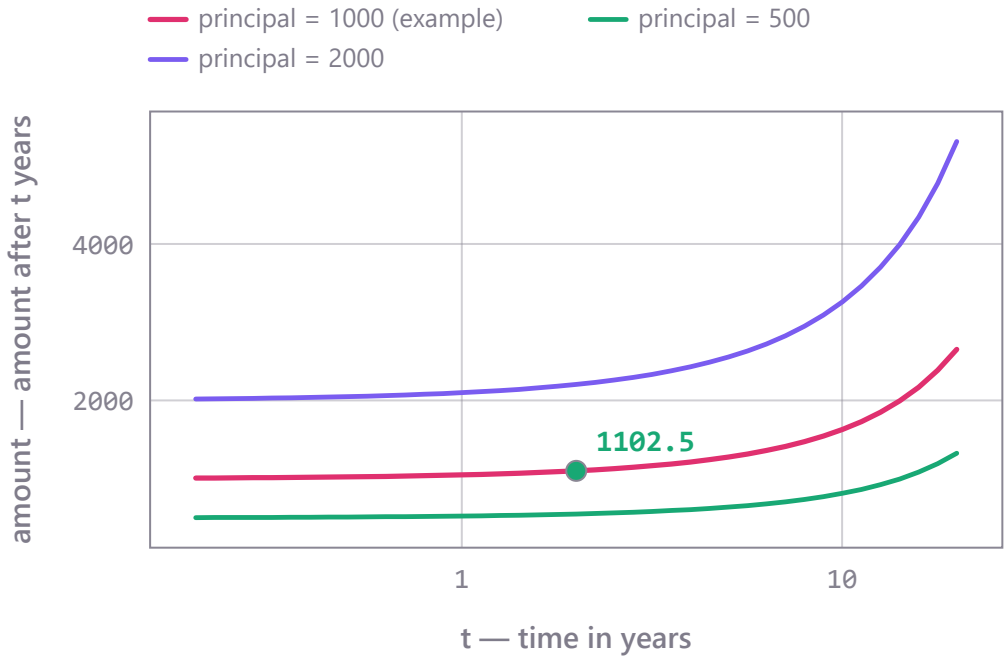
The amount after t years when a principal is compounded n times a year at an annual rate r (a decimal).

$$\ln(\text{amount}) = \ln(\text{principal}) + nt\ln\left(1 + \frac{r}{n}\right)$$

| input | meaning | condition |
|---|---|---|
| `principal` | starting amount | principal > 0 |
| `r` | annual interest rate as a decimal | $r > -1$ |
| `n` | compounding periods per year | $n \geq 1$ |
| `t` | time in years | $t \geq 0$ |

| output | meaning |
|---|---|
| `amount` | amount after t years |

- **Method:** each period multiplies the amount by $(1 + r/n)$, $nt$ times
- **Assumes:** principal > 0; $n \geq 1$; $t \geq 0$; $1 + r/n > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




amount (amount after t years) against t (time in years) from 0.2 to 20 on a logarithmic axis. One curve for each value of principal (starting amount): 1000, 500, 2000. Other inputs as in the worked example: r = 0.05, n = 1.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, t = 2, amount = 1102.5.

```
math_foundations_ai999x.solve('compound_interest', principal=1000.0, r=0.05, n=1.0, t=2.0)
# -> amount = 1102.5   (SOLVED)
```

```
math_foundations_ai999x.solve('compound_interest', principal=500.0, r=0.12, n=12.0, t=1.0)
# -> amount = 563.413   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the amount after 2 years if 1000 is invested at rate 0.05 compounded 1 times a year. | SOLVED | amount = 1102.5 |

## y-intercept of a line from its slope and a point — `convert_to_slope_intercept`

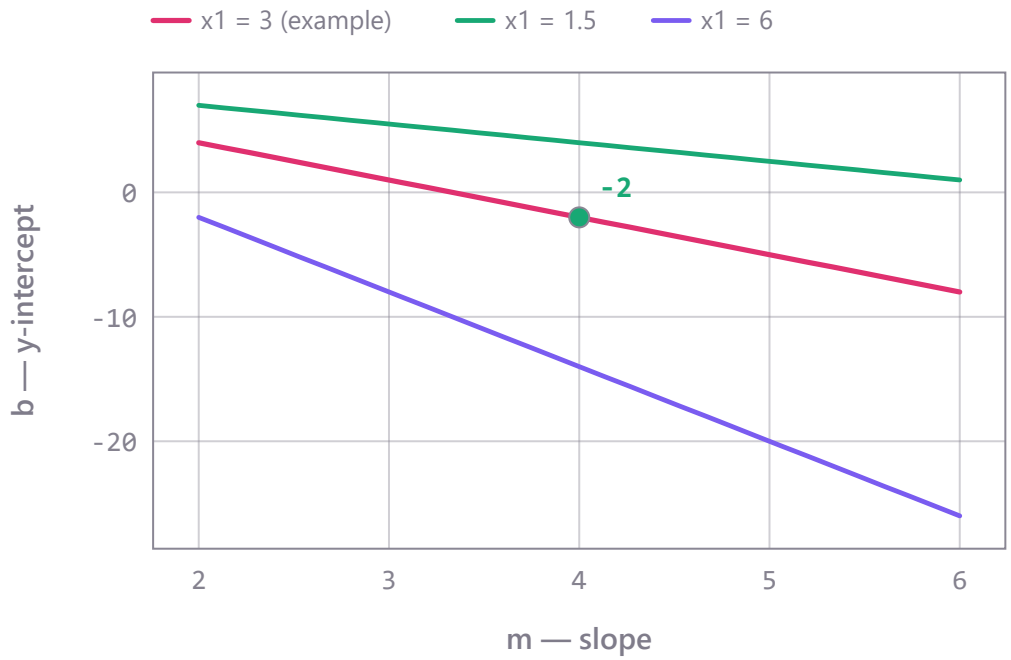
The constant b of $y = mx + b$ for the line with slope m through (x1, y1).

$$y_1 = mx_1 + b$$

| input | meaning | condition |
|---|---|---|
| `m` | slope | |
| `x1` | x-coordinate of the point | |
| `y1` | y-coordinate of the point | |

| output | meaning |
|---|---|
| `b` | y-intercept |

- **Method:** substitute the point into $y = mx + b$ and solve for b
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




b (y-intercept) against m (slope) from 2 to 6. One curve for each value of x1 (x-coordinate of the point): 3, 1.5, 6. Other inputs as in the worked example: y1 = 10.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, m = 4, b = -2.

```
math_foundations_ai999x.solve('convert_to_slope_intercept', m=4.0, x1=3.0, y1=10.0)
# -> b = -2   (SOLVED)
```

```
math_foundations_ai999x.solve('convert_to_slope_intercept', m=0.0, x1=5.0, y1=7.0)
# -> b = 7   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| A line with slope 4 passes through (3, 10). What is its y-intercept? | SOLVED | b = -2 |
| Find the y-intercept of the line with slope 0.5 through the point (2, 4). | SOLVED | b = 3 |

### Determinant of 2x2 matrix — `determinant_2x2_formula`

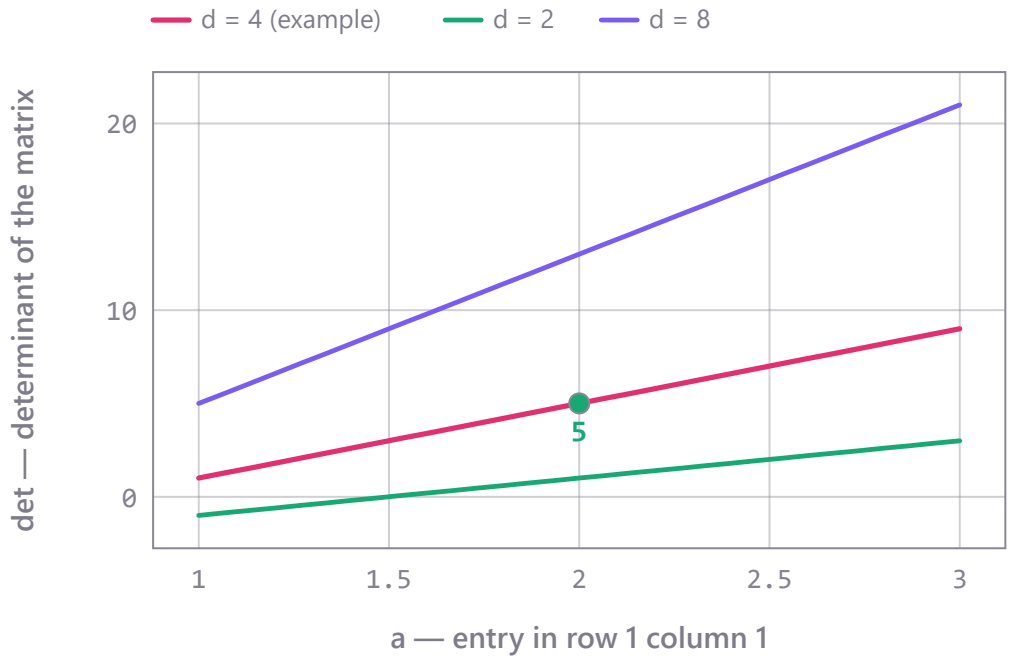
Computes the determinant of a 2x2 matrix using the standard formula.

$$\det = ad - bc$$

| input | meaning | condition |
|---|---|---|
| `a` | entry in row 1 column 1 | |
| `b` | entry in row 1 column 2 | |
| `c` | entry in row 2 column 1 | |
| `d` | entry in row 2 column 2 | |

| output | meaning |
|---|---|
| `det` | determinant of the matrix |

- **Method:** closed form verified against the defining equations
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




det (determinant of the matrix) against a (entry in row 1 column 1) from 1 to 3. One curve for each value of d (entry in row 2 column 2): 4, 2, 8. Other inputs as in the worked example: b = 3.0, c = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, det = 5.

```
math_foundations_ai999x.solve('determinant_2x2_formula', a=2.0, b=3.0, c=1.0, d=4.0)
# -> det = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('determinant_2x2_formula', a=5.0, b=2.0, c=4.0, d=8.0)
# -> det = 32   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the determinant of the matrix [[3, 1], [2, 4]]. | SOLVED | det = 10 |

## Distance between two points — `distance_between_points`

The distance between (x1, y1) and (x2, y2) in the plane.

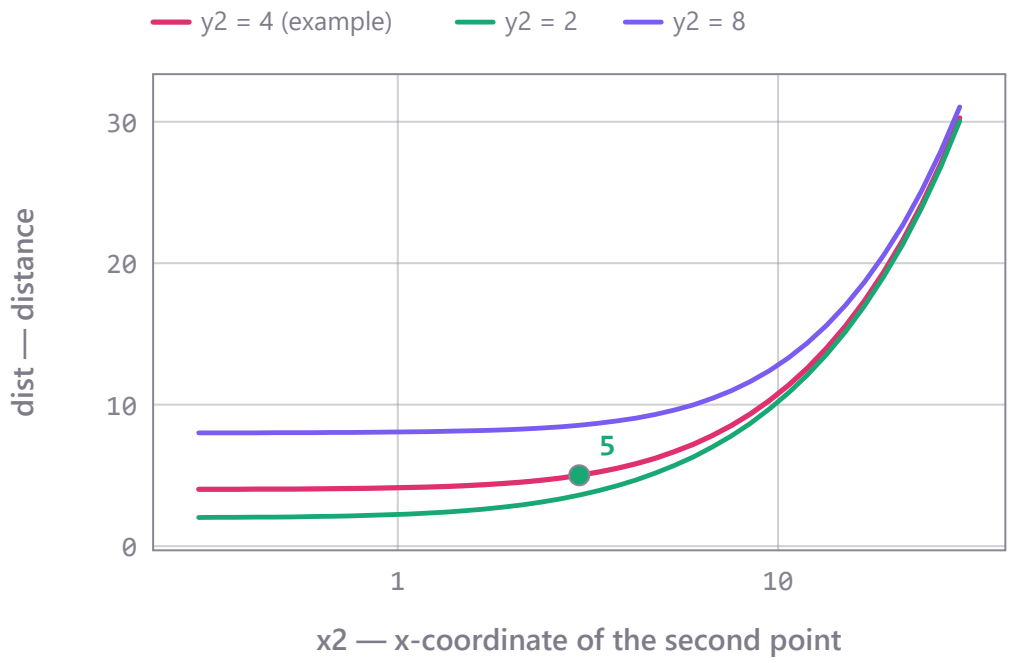
$$\text{dist}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

| input | meaning | condition |
|---|---|---|
| `x1` | x-coordinate of the first point | |
| `y1` | y-coordinate of the first point | |
| `x2` | x-coordinate of the second point | |
| `y2` | y-coordinate of the second point | |

| output | meaning |
|---|---|
| `dist` | distance |

- **Method:** the Pythagorean theorem on the horizontal and vertical differences

- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




dist (distance) against x2 (x-coordinate of the second point) from 0.3 to 30 on a logarithmic axis. One curve for each value of y2 (y-coordinate of the second point): 4, 2, 8. Other inputs as in the worked example: x1 = 0.0, y1 = 0.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, x2 = 3, dist = 5.

```
math_foundations_ai999x.solve('distance_between_points', x1=0.0, y1=0.0, x2=3.0, y2=4.0)
# -> dist = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('distance_between_points', x1=-1.0, y1=2.0, x2=5.0, y2=10.0)
# -> dist = 10   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the distance between (1, 1) and (4, 5). | SOLVED | dist = 5 |

## Doubling time of exponential growth — `doubling_time`

The time for a quantity growing as $N = N_0 e^{kt}$ to double, for a growth constant $k > 0$.

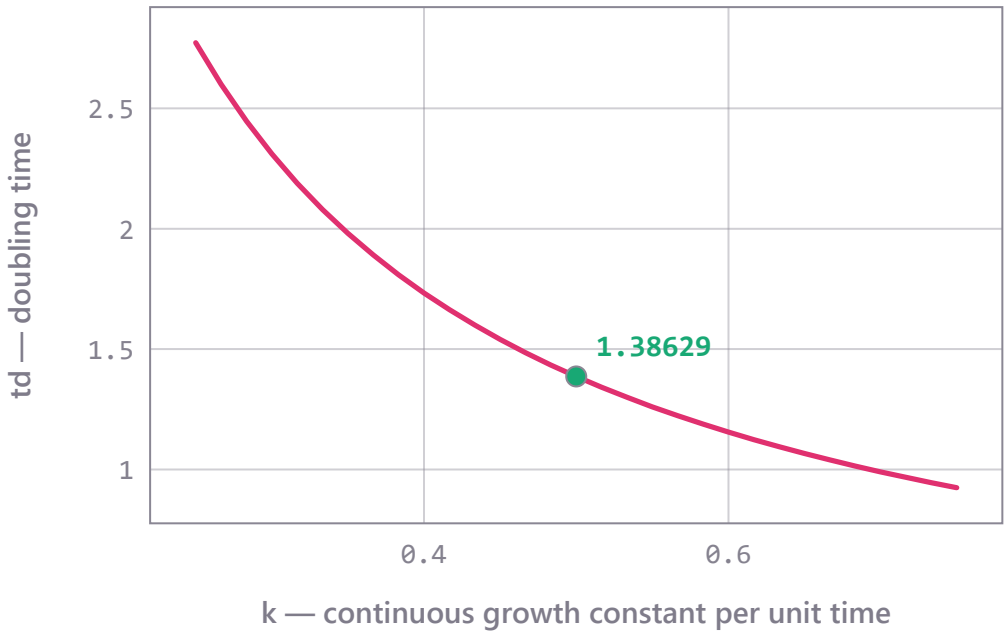
$$kt_d = \ln(2)$$

| input | meaning | condition |
|---|---|---|
| `k` | continuous growth constant per unit time | $k > 0$ |

| output | meaning |
|---|---|
| `td` | doubling time |

- **Method:** solve $e^{kt} = 2$ for t

- **Assumes:** $k > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




td (doubling time) against k (continuous growth constant per unit time) from 0.25 to 0.75. 31 points computed by math-foundations-ai999x; marked: the worked example, k = 0.5, td = 1.38629.

```
math_foundations_ai999x.solve('doubling_time', k=0.5)
# -> td = 1.38629   (SOLVED)
```

```
math_foundations_ai999x.solve('doubling_time', k=0.1)
# -> td = 6.93147   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| What is the doubling time for a growth constant of 0.1? | SOLVED | td = 6.93147 |

## Solution of an exponential equation — `exponential_equation`

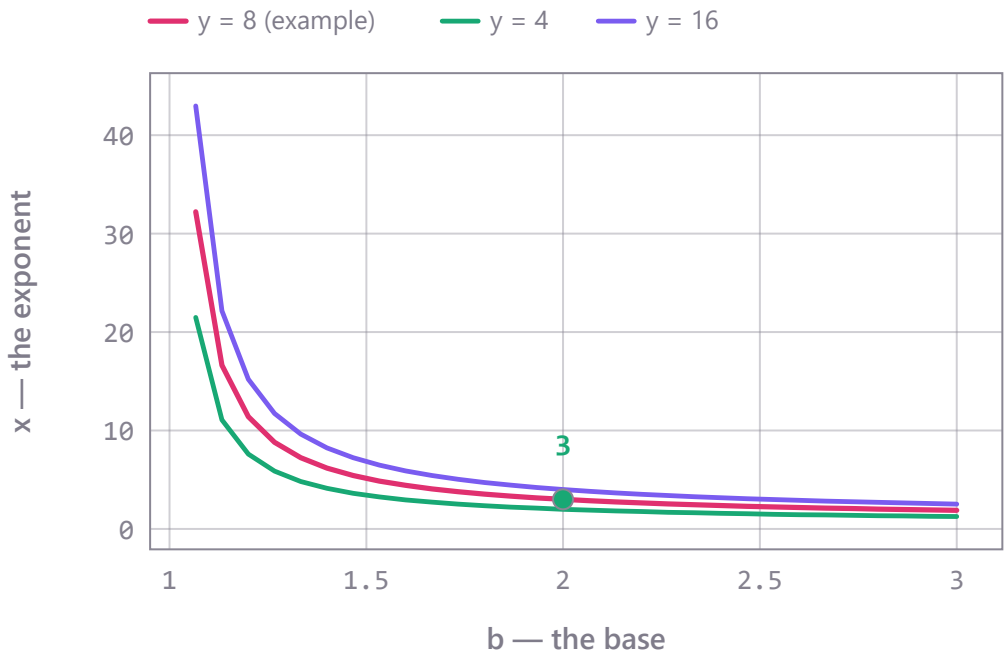
Solve $b^x = y$ for x, with $b > 0$, $b \neq 1$ and $y > 0$.

$$x \ln(b) = \ln(y)$$

| input | meaning | condition |
|---|---|---|
| b | the base | $b > 0$ |
| y | the value of the power | $y > 0$ |

| output | meaning |
|---|---|
| x | the exponent |

- **Method:** take logarithms of both sides: $x = \frac{\log(y)}{\log(b)}$
- **Assumes:** $b > 0$; $y > 0$; $b \neq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the exponent) against b (the base) from 1 to 3. One curve for each value of y (the value of the power): 8, 4, 16. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 3.

```
math_foundations_ai999x.solve('exponential_equation', b=2.0, y=8.0)
# -> x = 3   (SOLVED)
```

```
math_foundations_ai999x.solve('exponential_equation', b=10.0, y=0.001)
# -> x = -3   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve 2^x = 32. | SOLVED | x = 5 |
| Solve 1^x = 5. | INVALID_INPUT | Solution of an exponential equation: the condition b != 1 does not hold for these inputs. |

## Factoring a monic quadratic — `factor_monic_quadratic`

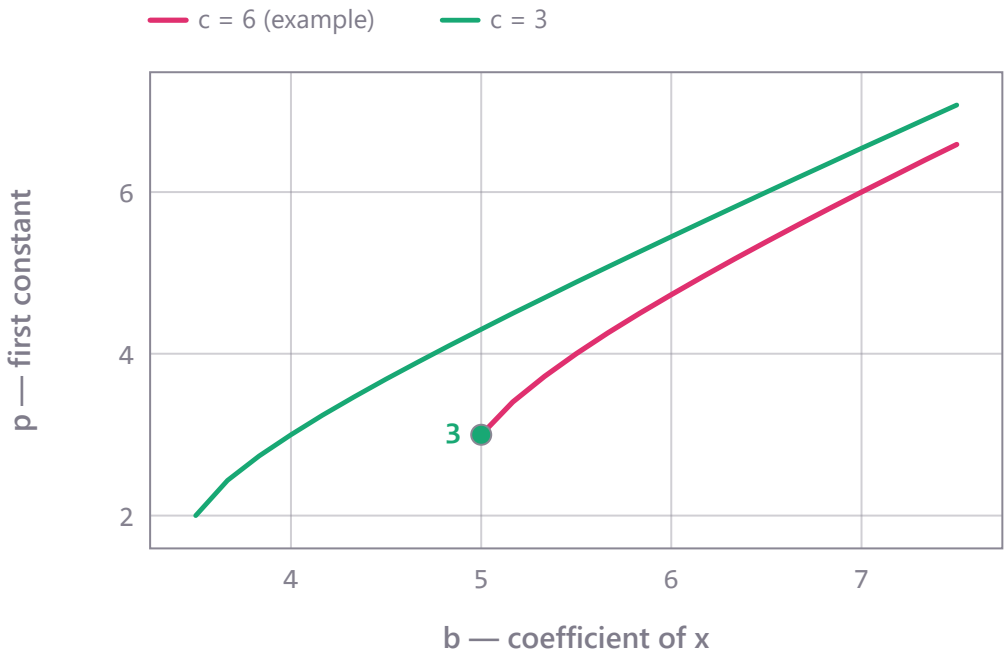
Write $x^2 + bx + c$ as $(x + p)(x + q)$ with real p and q.

$$(t + p)(t + q) = t^2 + bt + c$$

| input | meaning | condition |
|---|---|---|
| b | coefficient of x | |
| c | constant term | |

| output | meaning |
|---|---|
| p | first constant |
| q | second constant |

- **Method:** find p and q with $p + q = b$ and $pq = c$ (the roots of the quadratic are -p and -q)
- **Assumes:** b**$2 - 4c \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




p (first constant) against b (coefficient of x) from 2.5 to 7.5. One curve for each value of c (constant term): 6, 3. 16 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 5, p = 3.

```
math_foundations_ai999x.solve('factor_monic_quadratic', b=5.0, c=6.0)
# -> p = 3, q = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('factor_monic_quadratic', b=-1.0, c=-12.0)
# -> p = 3, q = -4   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Factor x^2 + 5x + 6. | SOLVED | p = 3, q = 2 |
| Factor x^2 - x - 12. | SOLVED | p = 3, q = -4 |
| Factor x^2 + 2x + 5. | INVALID_INPUT | Factoring a monic quadratic: the condition b**2 - 4*c >= 0 does not hold for these inputs. |

## Term and partial sum of a geometric sequence — `geometric_sequence`

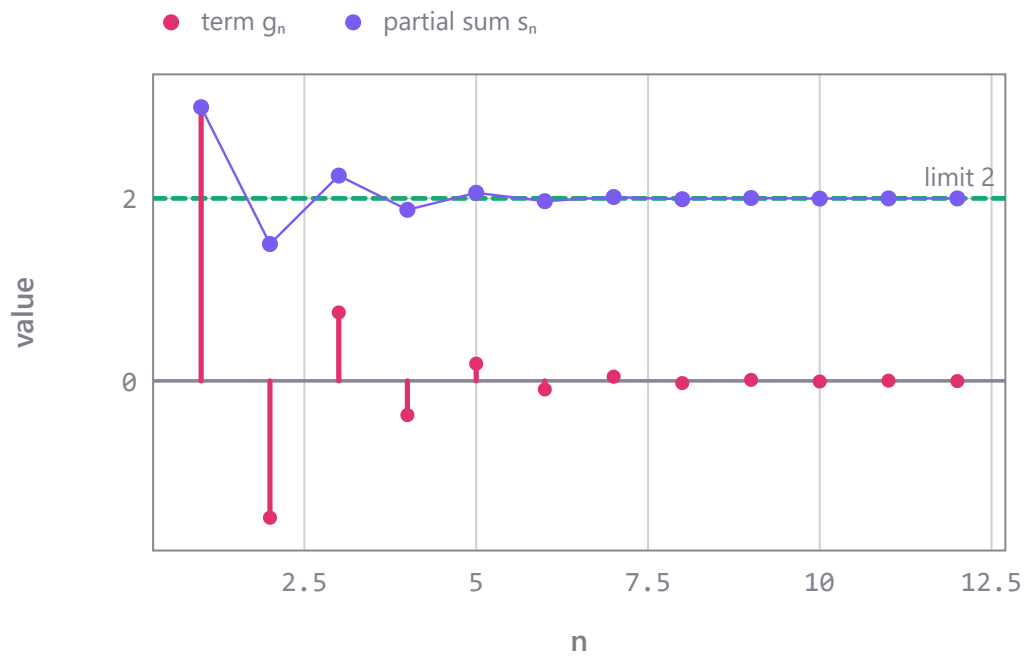
The n-th term and the sum of the first n terms of the geometric sequence with first term g1 and common ratio q $(q \neq 1)$.

$$g_n = g_1 q^{n-1}, \quad s_n\,(1-q) = g_1\,(1-q^n)$$

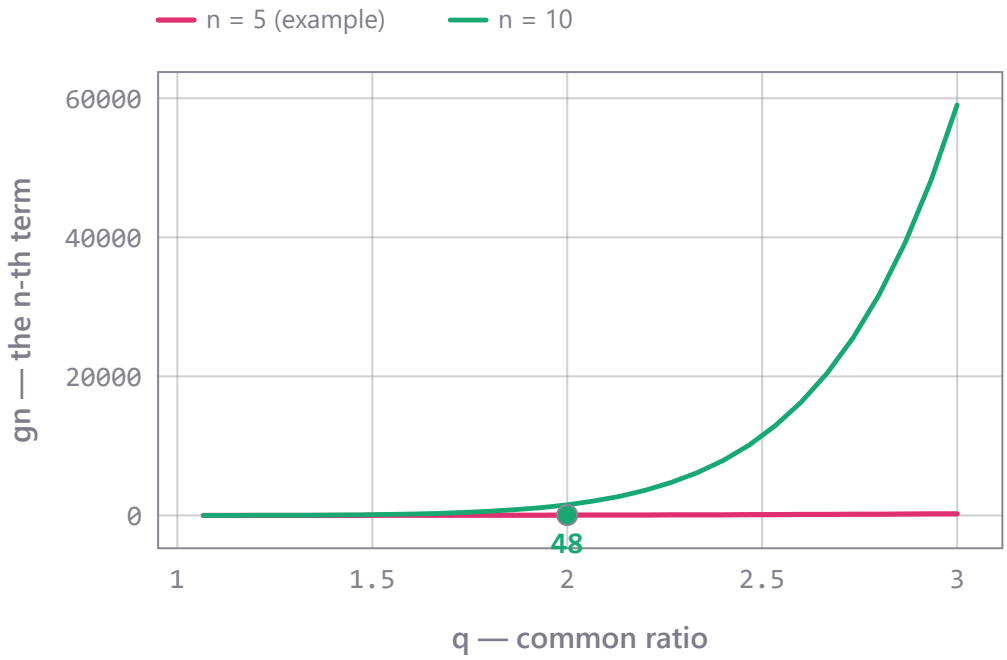
| input | meaning | condition |
|---|---|---|
| g1 | first term | |
| q | common ratio | $q \neq 1$ |
| n | position of the term | $n \geq 1$ |

| output | meaning |
|---|---|
| gn | the n-th term |
| sn | the sum of the first n terms |

- **Method:** each term multiplies the previous one by q; the sum telescopes when multiplied by $(1-q)$
- **Assumes:** $q \neq 1$; $n \geq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




The geometric sequence $g_1$ = 3, q = -0.5: the alternating terms (stems) and the partial sums (dots), each computed by the package, settle on the sum of the infinite series, 2; $g_{12}$ = -0.00146484, $s_{12}$ = 1.99951.




gn (the n-th term) against q (common ratio) from 1 to 3. One curve for each value of n (position of the term): 5, 10. Other inputs as in the worked example: g1 = 3.0. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, q = 2, gn = 48.

```
math_foundations_ai999x.solve('geometric_sequence', g1=3.0, q=2.0, n=5.0)
# -> gn = 48, sn = 93   (SOLVED)
```

```
math_foundations_ai999x.solve('geometric_sequence', g1=81.0, q=-0.3333333333333333, n=4.0)
# -> gn = -3, sn = 60   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find term 4 and the sum of the first 4 terms of the geometric sequence with first term 2 and ratio 3. | SOLVED | gn = 54, sn = 80 |

## Line through two points — `line_through_two_points`

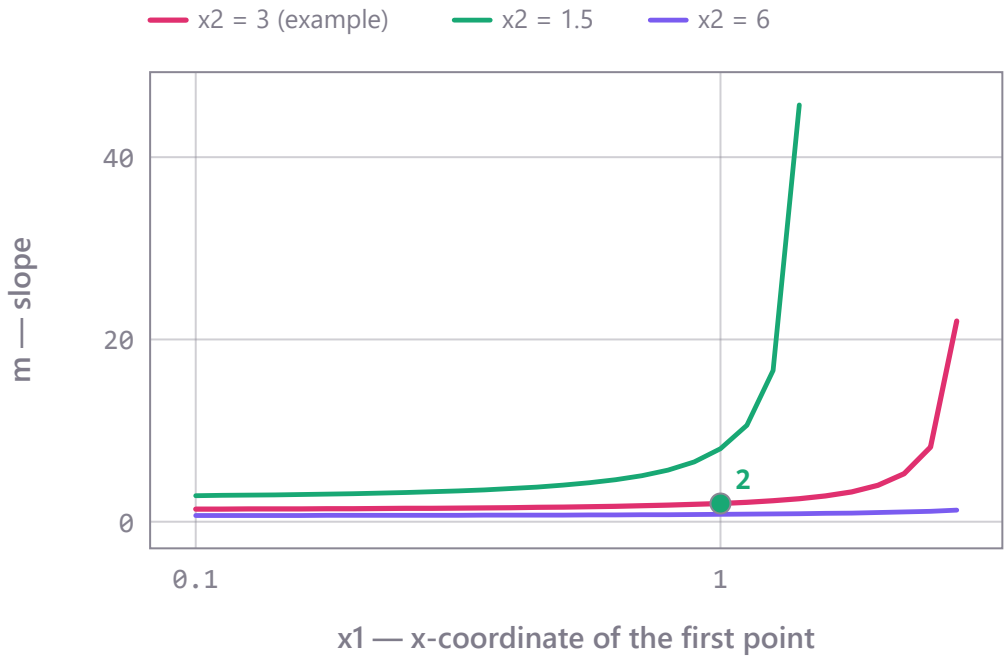
The slope m and y-intercept k of the line through (x1, y1) and (x2, y2), in the form $y = mx + k$.

$$mx_1 + k = y_1, \quad mx_2 + k = y_2$$

| input | meaning | condition |
|---|---|---|
| `x1` | x-coordinate of the first point | |
| `y1` | y-coordinate of the first point | |
| `x2` | x-coordinate of the second point | |
| `y2` | y-coordinate of the second point | |

| output | meaning |
|---|---|
| `m` | slope |
| `k` | y-intercept |

- **Method:** slope = rise over run; the intercept follows from one point
- **Assumes:** $x_2 - x_1 \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




m (slope) against x1 (x-coordinate of the first point) from 0.1 to 2.81838 on a logarithmic axis. One curve for each value of x2 (x-coordinate of the second point): 3, 1.5, 6. Other inputs as in the worked example: y1 = 3.0, y2 = 7.0. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, x1 = 1, m = 2.

```
math_foundations_ai999x.solve('line_through_two_points', x1=1.0, y1=3.0, x2=3.0, y2=7.0)
# -> m = 2, k = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('line_through_two_points', x1=-2.0, y1=5.0, x2=2.0, y2=-3.0)
# -> m = -2, k = 1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the equation of the line through (0, 1) and (2, 5). | SOLVED | m = 2, k = 1 |
| Find the equation of the line through (3, 1) and (3, 4). | INVALID_INPUT | Line through two points: the condition x2 - x1 != 0 does not hold for these inputs. |

## Root of a linear equation — `linear_equation_root`

Solve $ax + b = c$ for x.

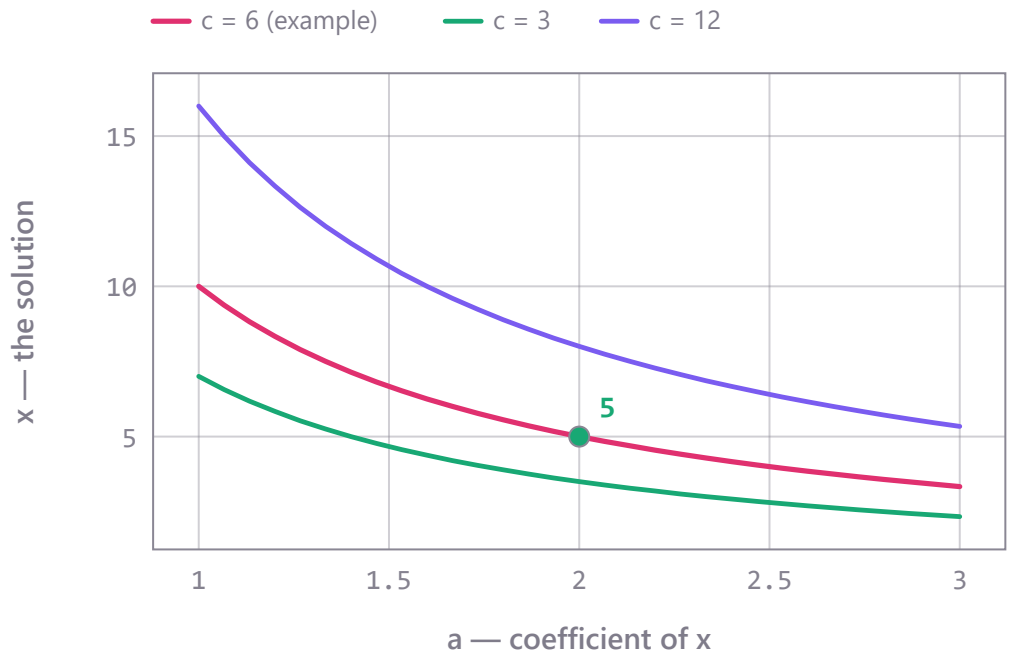
$$ax + b = c$$

| input | meaning | condition |
|---|---|---|
| a | coefficient of x | $a \neq 0$ |
| b | constant term on the left side | |

| input | meaning | condition |
|---|---|---|
| c | right-hand side | |

| output | meaning |
|---|---|
| x | the solution |

- **Method:** subtract b from both sides, then divide by a
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the solution) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 6, 3, 12. Other inputs as in the worked example: b = -4.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x = 5.

```
math_foundations_ai999x.solve('linear_equation_root', a=2.0, b=-4.0, c=6.0)
# -> x = 5   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_equation_root', a=-3.0, b=1.0, c=1.0)
# -> x = 0   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_equation_root', a=0.5, b=0.0, c=3.0)
# -> x = 6   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Solve 3x + 6 = 0. | SOLVED | x = -2 |
| Solve x - 4 = 10. | SOLVED | x = 14 |
| Find x if 5x + 1 = 2. | SOLVED | x = 0.2 |
| Solve the linear equation with a = 0, b = 1, c = 2. | INVALID_INPUT | Root of a linear equation: the condition a != 0 does not hold for these inputs. |

| question | outcome | result |
|---|---|---|
| Solve the linear equation with a = 4, b = 1. | NEEDS_INPUT | linear_equation_root needs c (right-hand side). Example: solve('linear_equation_root', a=2, b=-4, c=6). |

### Solution of a linear inequality — `linear_inequality`

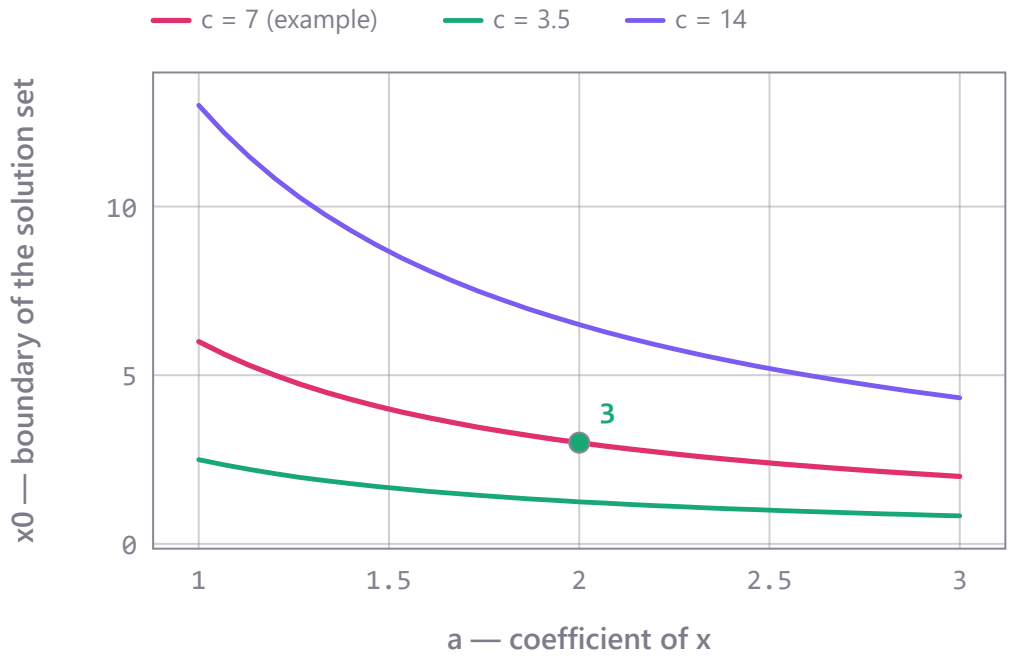
Solve $ax + b < c$ (or >): the boundary x0 and the direction s (s = 1 means the solution of the < form is $x < x_0$; $s = -1$ means it reverses to $x > x_0$ because a < 0).

$$ax_0 + b = c, \quad s\sqrt{a^2} = a$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x | $a \neq 0$ |
| `b` | constant on the left | |
| `c` | right-hand side | |

| output | meaning |
|---|---|
| `x0` | boundary of the solution set |
| `s` | 1 if the inequality keeps its direction, -1 if it reverses |

- **Method:** subtract b, divide by a; dividing by a negative number reverses the inequality
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x0 (boundary of the solution set) against a (coefficient of x) from 1 to 3. One curve for each value of c (right-hand side): 7, 3.5, 14. Other inputs as in the worked example: b = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, x0 = 3.

```
math_foundations_ai999x.solve('linear_inequality', a=2.0, b=1.0, c=7.0)
# -> x0 = 3, s = 1   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_inequality', a=-3.0, b=4.0, c=10.0)
# -> x0 = -2, s = -1   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve the inequality 2x + 1 < 7. | SOLVED | x0 = 3, s = 1 |
| Solve the inequality -3x + 4 < 10. | SOLVED | x0 = -2, s = -1 |

### Solution of two linear equations in two unknowns — `linear_system_2x2`

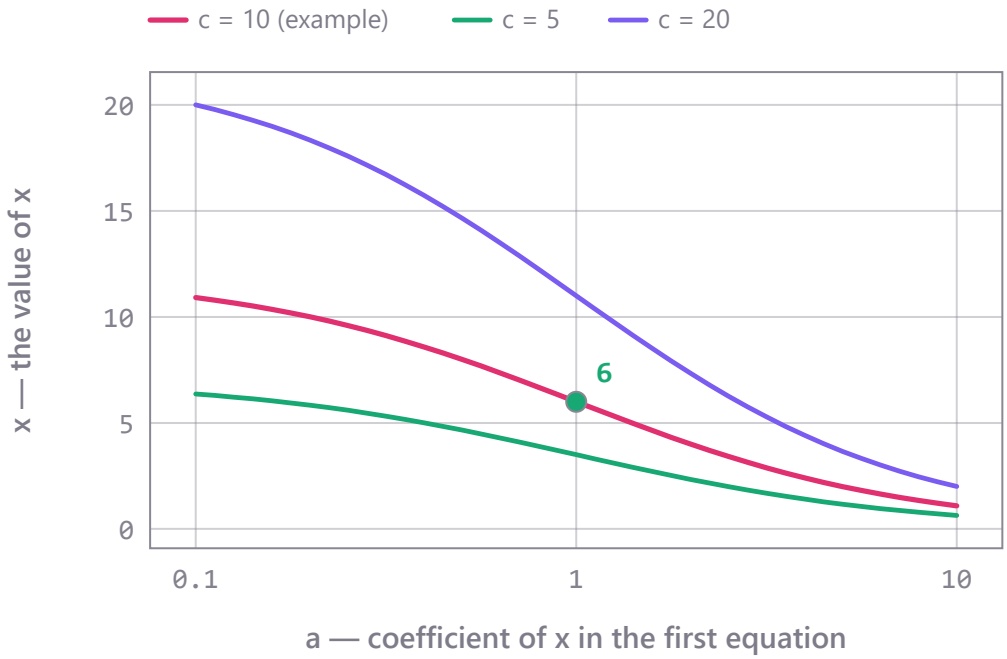
Solve $ax + by = c$ and $dx + ey = f$ for x and y.

$$ax + by = c, \quad dx + ey = f$$

| input | meaning | condition |
|---|---|---|
| `a` | coefficient of x in the first equation | |
| `b` | coefficient of y in the first equation | |
| `c` | right-hand side of the first equation | |
| `d` | coefficient of x in the second equation | |
| `e` | coefficient of y in the second equation | |
| `f` | right-hand side of the second equation | |

| output | meaning |
|---|---|
| `x` | the value of x |
| `y` | the value of y |

- **Method:** elimination (Cramer's rule); the determinant $ae - bd$ must not be zero
- **Assumes:** $ae - bd \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the value of x) against a (coefficient of x in the first equation) from 0.1 to 10 on a logarithmic axis. One curve for each value of c (right-hand side of the first equation): 10, 5, 20. Other inputs as in the worked example: b = 1.0, d = 1.0, e = -1.0, f = 2.0. 41 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, x = 6.

```
math_foundations_ai999x.solve('linear_system_2x2', a=1.0, b=1.0, c=10.0, d=1.0, e=-1.0, f=2.0)
# -> x = 6, y = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('linear_system_2x2', a=2.0, b=3.0, c=12.0, d=4.0, e=-1.0, f=10.0)
# -> x = 3, y = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve the system x + y = 10, x - y = 2. | SOLVED | x = 6, y = 4 |
| Solve the system 2x + 3y = 12, 4x - y = 10. | SOLVED | x = 3, y = 2 |
| Solve the system x + 2y = 3, 2x + 4y = 6. | INVALID_INPUT | Solution of two linear equations in two unknowns: the condition a*e - b*d != 0 does not hold for these inputs. |

## Solution of a basic logarithmic equation — `logarithmic_equation`

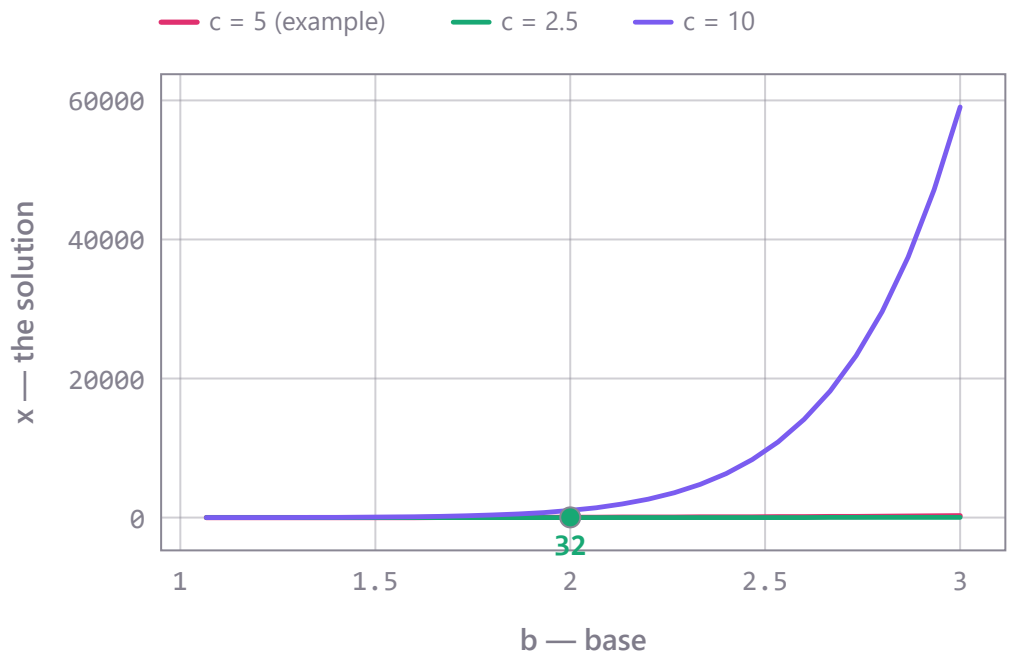
Solve log_b(x) = c for x.

$$\ln(x) = c\ln(b)$$

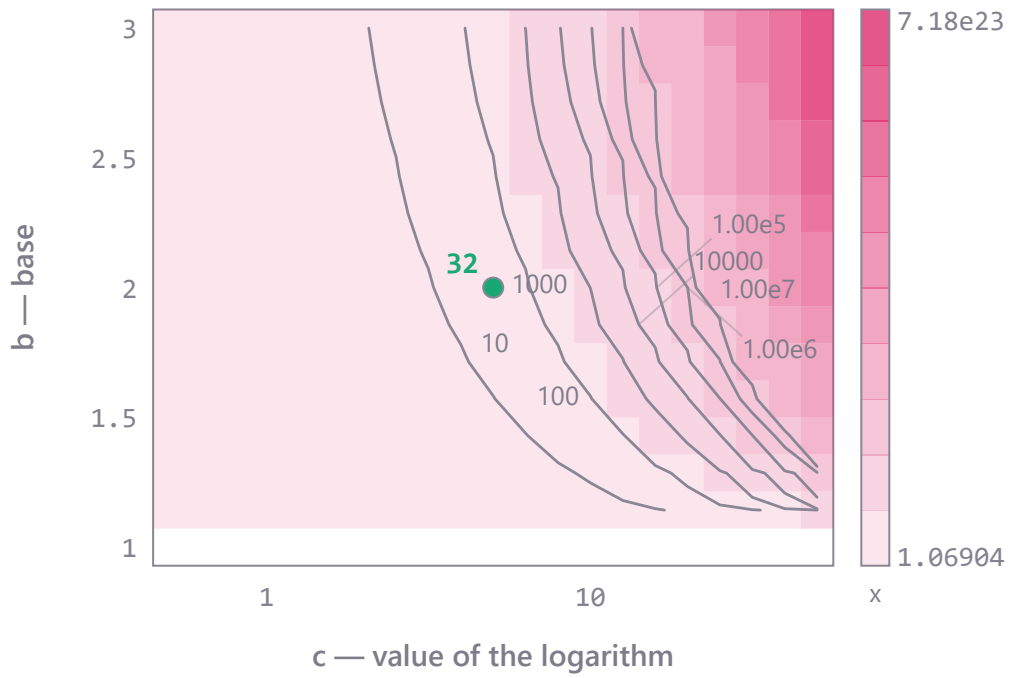
| input | meaning | condition |
|---|---|---|
| b | base | $b > 0$ |
| c | value of the logarithm | |

| output | meaning |
|---|---|
| x | the solution |

- **Method:** rewrite in exponential form: $x = b^c$
- **Assumes:** $b > 0$; $b \neq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




x (the solution) against b (base) from 1 to 3. One curve for each value of c (value of the logarithm): 5, 2.5, 10. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, b = 2, x = 32.




x (the solution) over c (value of the logarithm) from 0.5 to 50 and b (base) from 1 to 3 (logarithmic horizontal axis): each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 1.06904 to dark 7.1790e+23, logarithmic shading); iso-lines at x = 10, 100, 1000, 10000, 1.00e5, 1.00e6, 1.00e7; blank cells: inputs the capability refuses. Marked: the worked example, c = 5, b = 2, x = 32.

```
math_foundations_ai999x.solve('logarithmic_equation', b=2.0, c=5.0)
# -> x = 32   (SOLVED)
```

```
math_foundations_ai999x.solve('logarithmic_equation', b=10.0, c=-2.0)
# -> x = 0.01   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Solve log base 3 of x = 4. | SOLVED | x = 81 |
| Solve log base 1 of x = 2. | INVALID_INPUT | Solution of a basic logarithmic equation: the condition b != 1 does not hold for these inputs. |

## Determinant and inverse of a 2 by 2 matrix — `matrix_2x2_inverse`

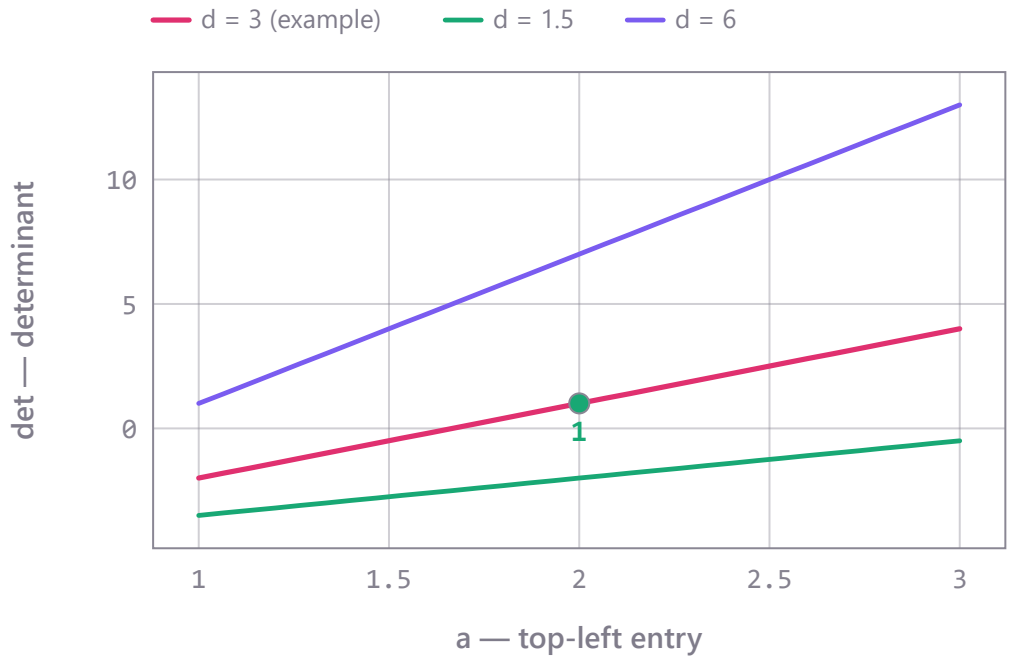
For the matrix [[a, b], [c, d]], its determinant and the entries of its inverse.

$$\det = ad - bc, \quad ai_{11} + bi_{21} = 1, \quad ai_{12} + bi_{22} = 0, \quad ci_{11} + di_{21} = 0, \quad ci_{12} + di_{22} = 1$$

| input | meaning | condition |
|---|---|---|
| `a` | top-left entry | |
| `b` | top-right entry | |
| `c` | bottom-left entry | |
| `d` | bottom-right entry | |

| output | meaning |
|---|---|
| `det` | determinant |
| `i11` | inverse, top-left |
| `i12` | inverse, top-right |
| `i21` | inverse, bottom-left |
| `i22` | inverse, bottom-right |

- **Method:** swap the diagonal, negate the off-diagonal, divide by the determinant
- **Assumes:** $ad - bc \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




det (determinant) against a (top-left entry) from 1 to 3. One curve for each value of d (bottom-right entry): 3, 1.5, 6. Other inputs as in the worked example: b = 1.0, c = 5.0. 30 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 2, det = 1.

```
math_foundations_ai999x.solve('matrix_2x2_inverse', a=2.0, b=1.0, c=5.0, d=3.0)
# -> det = 1, i11 = 3, i12 = -1, i21 = -5, i22 = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('matrix_2x2_inverse', a=4.0, b=7.0, c=2.0, d=6.0)
# -> det = 10, i11 = 0.6, i12 = -0.7, i21 = -0.2, i22 = 0.4   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the inverse of the matrix [[2, 1], [5, 3]]. | SOLVED | det = 1, i11 = 3, i12 = -1, i21 = -5, i22 = 2 |
| Find the inverse of the matrix [[1, 2], [2, 4]]. | INVALID_INPUT | Determinant and inverse of a 2 by 2 matrix: the condition a*d - b*c != 0 does not hold for these inputs. |

## Matrix scalar multiple definition — `matrix_scalar_multiple_definition`

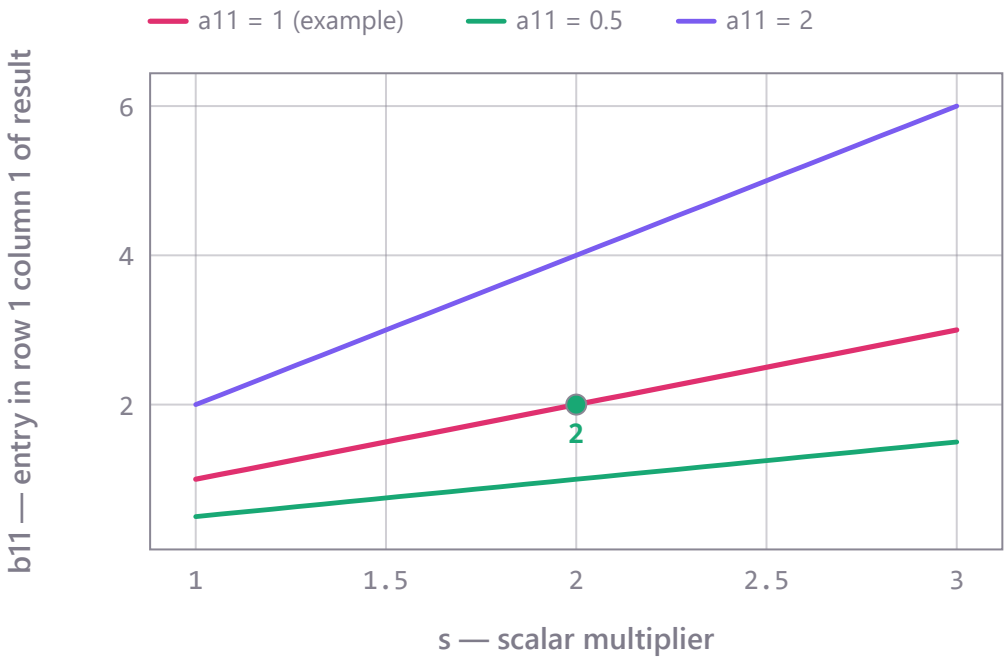
Computes the entries of a 2x2 matrix scaled by a scalar.

$$b_{11} = sa_{11}, \quad b_{12} = sa_{12}, \quad b_{21} = sa_{21}, \quad b_{22} = sa_{22}$$

| input | meaning | condition |
|---|---|---|
| `s` | scalar multiplier | |
| `a11` | entry in row 1 column 1 | |
| `a12` | entry in row 1 column 2 | |
| `a21` | entry in row 2 column 1 | |
| `a22` | entry in row 2 column 2 | |

| output | meaning |
|---|---|
| `b11` | entry in row 1 column 1 of result |
| `b12` | entry in row 1 column 2 of result |
| `b21` | entry in row 2 column 1 of result |
| `b22` | entry in row 2 column 2 of result |

- **Method:** closed form verified against the defining equations
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




b11 (entry in row 1 column 1 of result) against s (scalar multiplier) from 1 to 3. One curve for each value of a11 (entry in row 1 column 1): 1, 0.5, 2. Other inputs as in the worked example: a12 = 3.0, a21 = -4.0, a22 = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, s = 2, b11 = 2.

```
math_foundations_ai999x.solve('matrix_scalar_multiple_definition', a11=1.0, a12=3.0, a21=-4.0,
a22=5.0, s=2.0)
# -> b11 = 2, b12 = 6, b21 = -8, b22 = 10   (SOLVED)
```

```
math_foundations_ai999x.solve('matrix_scalar_multiple_definition', a11=0.0, a12=7.0, a21=2.0,
a22=9.0, s=-1.0)
# -> b11 = 0, b12 = -7, b21 = -2, b22 = -9   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Calculate 3 times the matrix [[1, 2], [3, 4]]. | SOLVED | b11 = 3, b12 = 6, b21 = 9, b22 = 12 |

## Midpoint of a line segment — `midpoint_of_segment`

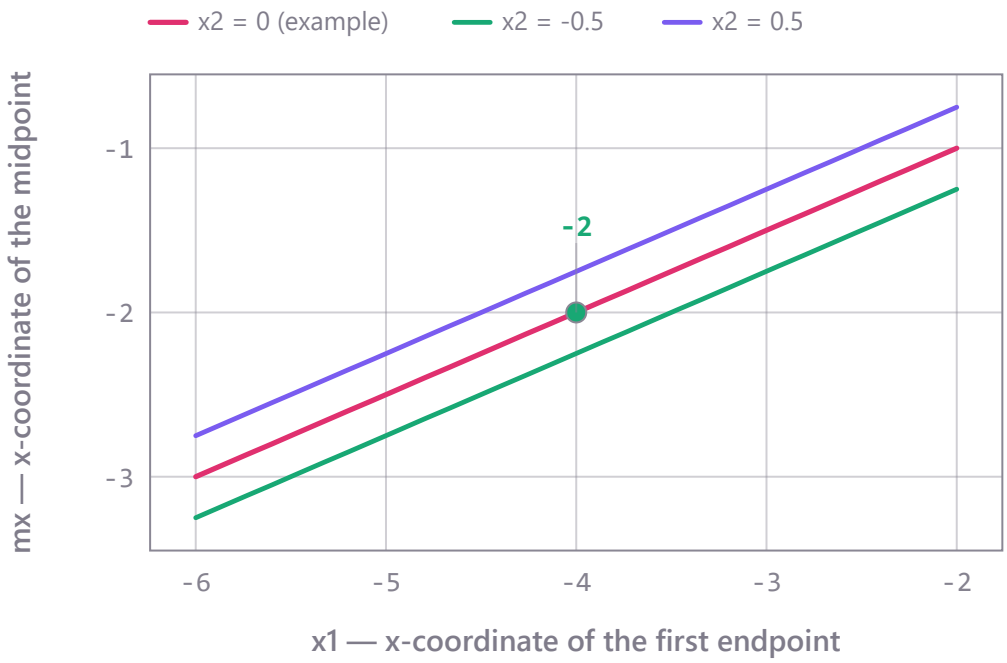
The midpoint (mx, my) of the segment from (x1, y1) to (x2, y2).

$$2m_x = x_1 + x_2, \quad 2m_y = y_1 + y_2$$

| input | meaning | condition |
|---|---|---|
| `x1` | x-coordinate of the first endpoint | |
| `y1` | y-coordinate of the first endpoint | |
| `x2` | x-coordinate of the second endpoint | |
| `y2` | y-coordinate of the second endpoint | |

| output | meaning |
|---|---|
| `mx` | x-coordinate of the midpoint |
| `my` | y-coordinate of the midpoint |

- **Method:** average the x-coordinates and the y-coordinates
- **Assumes:**
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




mx (x-coordinate of the midpoint) against x1 (x-coordinate of the first endpoint) from -6 to -2. One curve for each value of x2 (x-coordinate of the second endpoint): 0, -0.5, 0.5. Other inputs as in the worked example: y1 = 2.0, y2 = 6.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, x1 = -4, mx = -2.

```
math_foundations_ai999x.solve('midpoint_of_segment', x1=-4.0, y1=2.0, x2=0.0, y2=6.0)
# -> mx = -2, my = 4   (SOLVED)
```

```
math_foundations_ai999x.solve('midpoint_of_segment', x1=1.0, y1=1.0, x2=5.0, y2=3.0)
# -> mx = 3, my = 2   (SOLVED)
```

### Questions in words (documented forms), as executed:

| question | outcome | result |
|---|---|---|
| Find the midpoint of the segment from (-4, 2) to (0, 6). | SOLVED | mx = -2, my = 4 |

| question | outcome | result |
|---|---|---|
| What is the midpoint of the line segment joining (1, 1) and (5, 3)? | SOLVED | mx = 3, my = 2 |

### Multiply rational expressions — `multiply_rational_expressions`

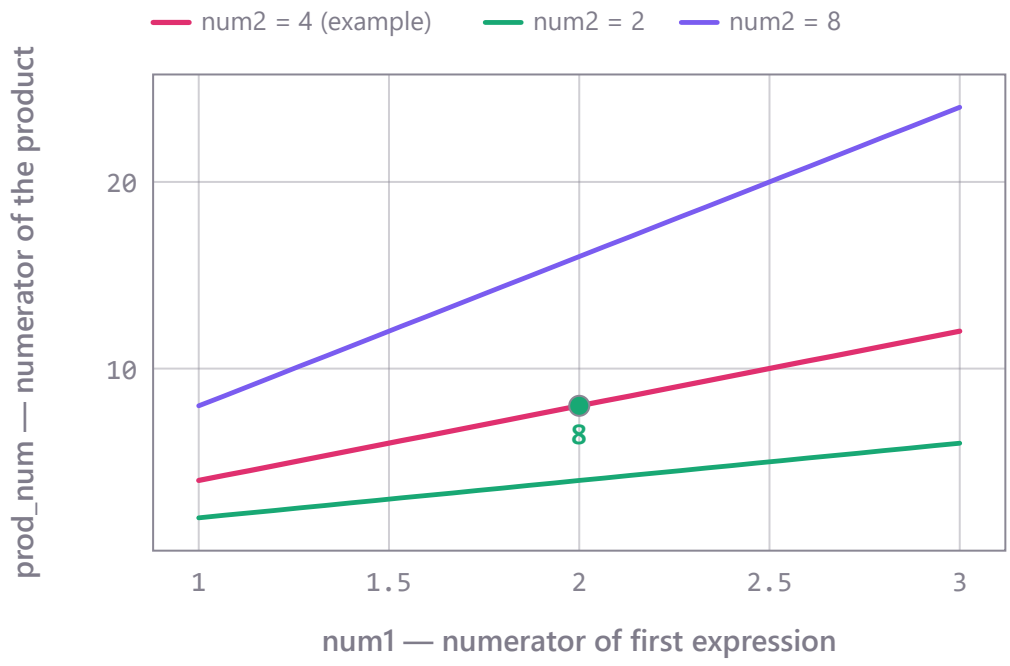
Compute the product of two rational expressions.

$$\mathrm{prod\ num} = \mathrm{num}_1\mathrm{num}_2, \quad \mathrm{prod\ den} = \mathrm{den}_1\mathrm{den}_2$$

| input | meaning | condition |
|---|---|---|
| `num1` | numerator of first expression | |
| `den1` | denominator of first expression | |
| `num2` | numerator of second expression | |
| `den2` | denominator of second expression | |

| output | meaning |
|---|---|
| `prod_num` | numerator of the product |
| `prod_den` | denominator of the product |

- **Method:** closed form verified against the defining equations
- **Assumes:** den1 != 0; den2 != 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




prod_num (numerator of the product) against num1 (numerator of first expression) from 1 to 3. One curve for each value of num2 (numerator of second expression): 4, 2, 8. Other inputs as in the worked example: den1 = 3.0, den2 = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, num1 = 2, prod_num = 8.

```
math_foundations_ai999x.solve('multiply_rational_expressions', den1=3.0, den2=5.0, num1=2.0,
num2=4.0)
# -> prod_num = 8, prod_den = 15   (SOLVED)
```

```
math_foundations_ai999x.solve('multiply_rational_expressions', den1=2.0, den2=4.0, num1=1.0,
num2=3.0)
# -> prod_num = 3, prod_den = 8   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Multiply 2/3 by 5/7. | SOLVED | prod_num = 10, prod_den = 21 |

## Perimeter of a rectangle — `perimeter_of_rectangle`

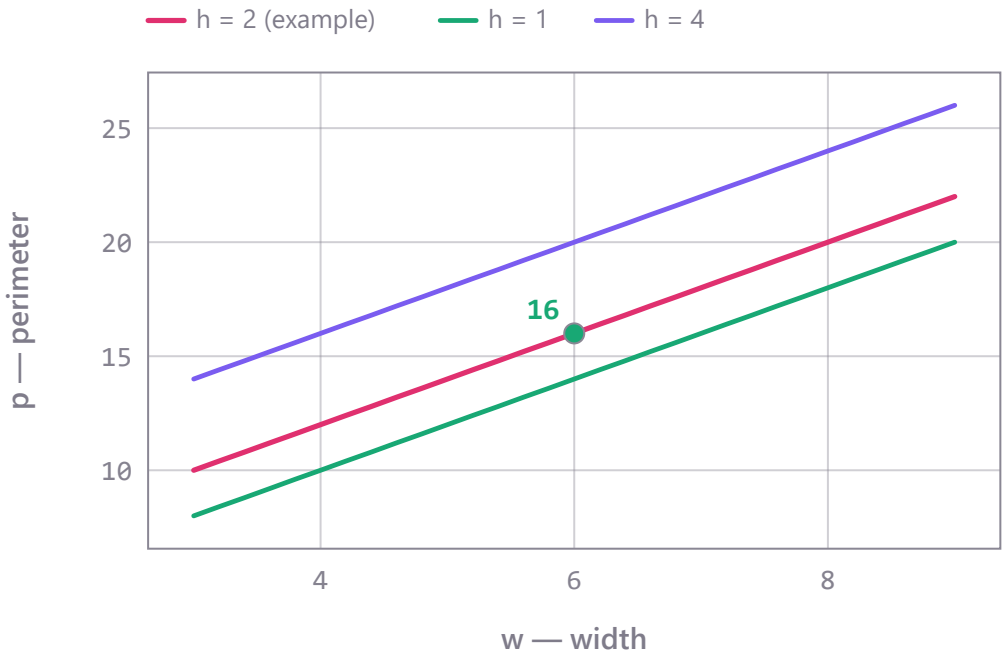
The perimeter p of a rectangle with width w and height h.

$$p - 2h = 2w$$

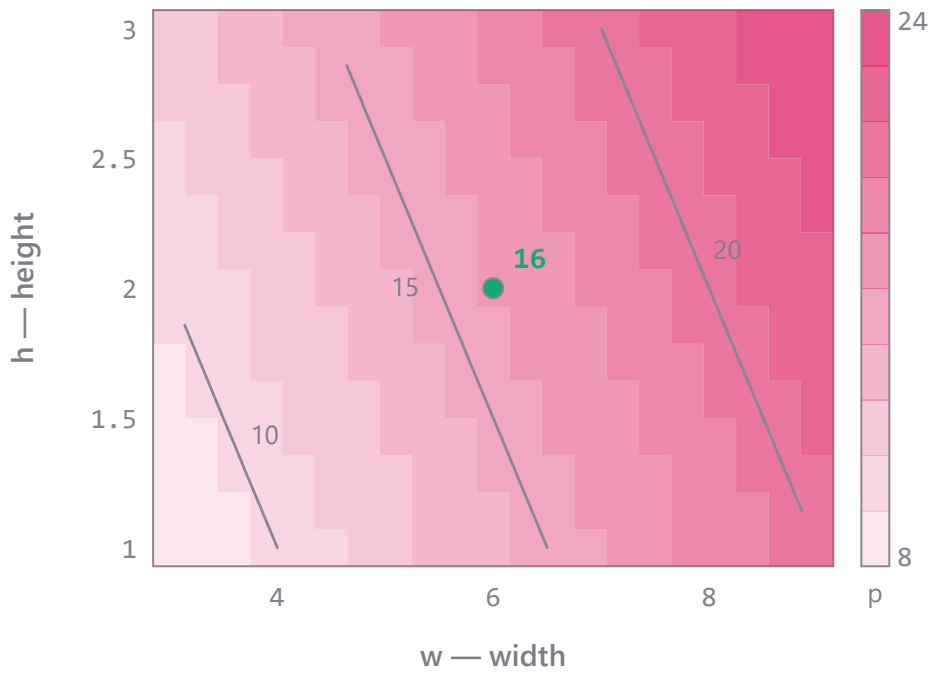
| input | meaning | condition |
|---|---|---|
| `w` | width | $w \geq 0$ |
| `h` | height | $h \geq 0$ |

| output | meaning |
|---|---|
| `p` | perimeter |

- **Method:** add the four sides
- **Assumes:** $w \geq 0$; $h \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




p (perimeter) against w (width) from 3 to 9. One curve for each value of h (height): 2, 1, 4. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, w = 6, p = 16.




p (perimeter) over w (width) from 3 to 9 and h (height) from 1 to 3: each of the 21 × 15 cells is one run of math-foundations-ai999x, shaded by its value (light 8 to dark 24); iso-lines at p = 10, 15, 20. Marked: the worked example, w = 6, h = 2, p = 16.

```
math_foundations_ai999x.solve('perimeter_of_rectangle', w=6.0, h=2.0)
# -> p = 16   (SOLVED)
```

```
math_foundations_ai999x.solve('perimeter_of_rectangle', w=2.5, h=4.0)
# -> p = 13   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| What is the perimeter of a rectangle with width 6 and height 2? | SOLVED | p = 16 |
| Find the perimeter of a rectangle of width 0 and height 3. | SOLVED | p = 6 |

## Slope of a perpendicular line — `perpendicular_lines_slope_product`

The slope m_perp of a line perpendicular to a line of nonzero slope m.

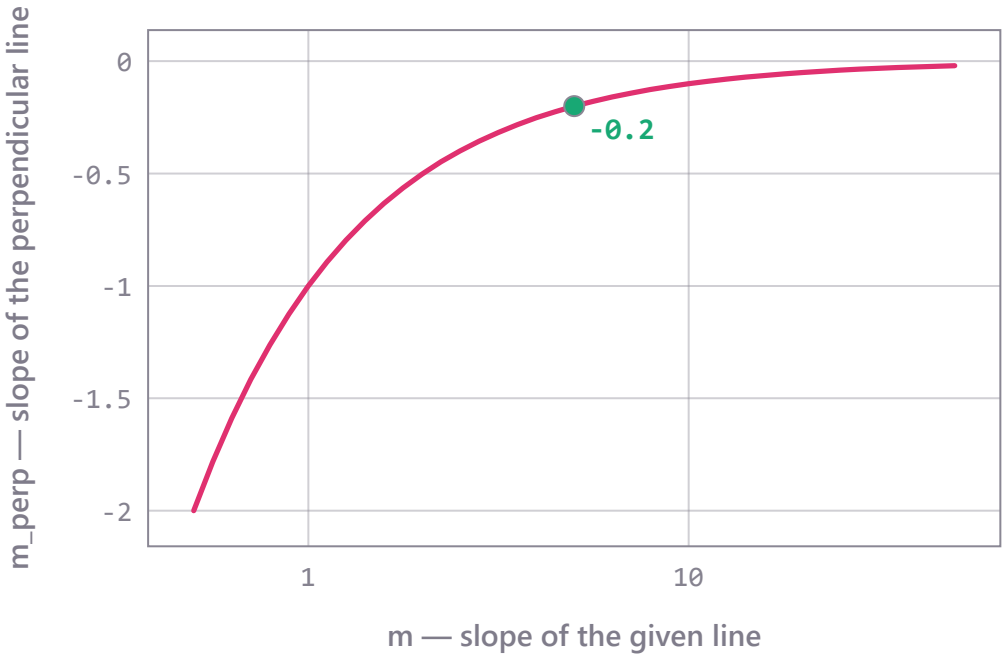
$$mm_{\text{perp}} = -1$$

| input | meaning | condition |
|---|---|---|
| `m` | slope of the given line | $m \neq 0$ |

| output | meaning |
|---|---|
| `m_perp` | slope of the perpendicular line |

- **Method:** the product of the slopes of perpendicular lines $is -1$
- **Assumes:** $m \neq 0$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




m_perp (slope of the perpendicular line) against m (slope of the given line) from 0.5 to 50 on a logarithmic axis. 41 points computed by math-foundations-ai999x; marked: the worked example, m = 5, m_perp = -0.2.

```
math_foundations_ai999x.solve('perpendicular_lines_slope_product', m=5.0)
# -> m_perp = -0.2   (SOLVED)
```

```
math_foundations_ai999x.solve('perpendicular_lines_slope_product', m=-0.5)
# -> m_perp = 2   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| A line has slope 5. What is the slope of a line perpendicular to it? | SOLVED | m_perp = -0.2 |
| A line has slope 0. What is the slope of a line perpendicular to it? | INVALID_INPUT | Slope of a perpendicular line: the condition m != 0 does not hold for these inputs. |

## Asymptotes of a linear rational function — `rational_function_asymptotes`

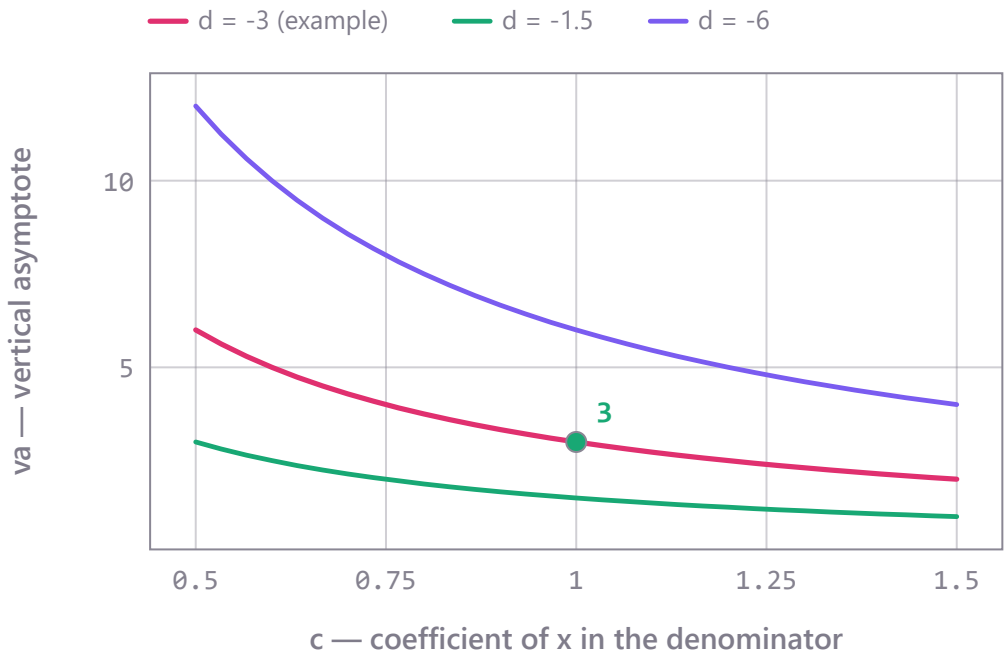
The vertical asymptote $x = va$ and horizontal asymptote $y = ha$ of f(x) = (a*x + b)/(c*x + d).

$$cv_a + d = 0, \quad ch_a = a$$

| input | meaning | condition |
|---|---|---|
| a | coefficient of x in the numerator | |
| b | constant of the numerator | |
| c | coefficient of x in the denominator | $c \neq 0$ |
| d | constant of the denominator | |

| output | meaning |
|---|---|
| `va` | vertical asymptote |
| `ha` | horizontal asymptote |

- **Method:** the denominator vanishes at $x = -d/c$; for large x the ratio tends to a/c (if $ad = bc$ the function is constant and has no vertical asymptote)
- **Assumes:** $c \neq 0$; $ad - bc \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




va (vertical asymptote) against c (coefficient of x in the denominator) from 0.5 to 1.5. One curve for each value of d (constant of the denominator): -3, -1.5, -6. Other inputs as in the worked example: a = 2.0, b = 1.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, c = 1, va = 3.

```
math_foundations_ai999x.solve('rational_function_asymptotes', a=2.0, b=1.0, c=1.0, d=-3.0)
# -> va = 3, ha = 2   (SOLVED)
```

```
math_foundations_ai999x.solve('rational_function_asymptotes', a=1.0, b=0.0, c=2.0, d=4.0)
# -> va = -2, ha = 0.5   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the asymptotes of f(x) = (2x + 1)/(x - 3). | SOLVED | va = 3, ha = 2 |
| Find the asymptotes of f(x) = (2x + 4)/(x + 2). | INVALID_INPUT | Asymptotes of a linear rational function: the condition a*d - b*c != 0 does not hold for these inputs. |

## Vertex form of a quadratic function — `vertex_form`

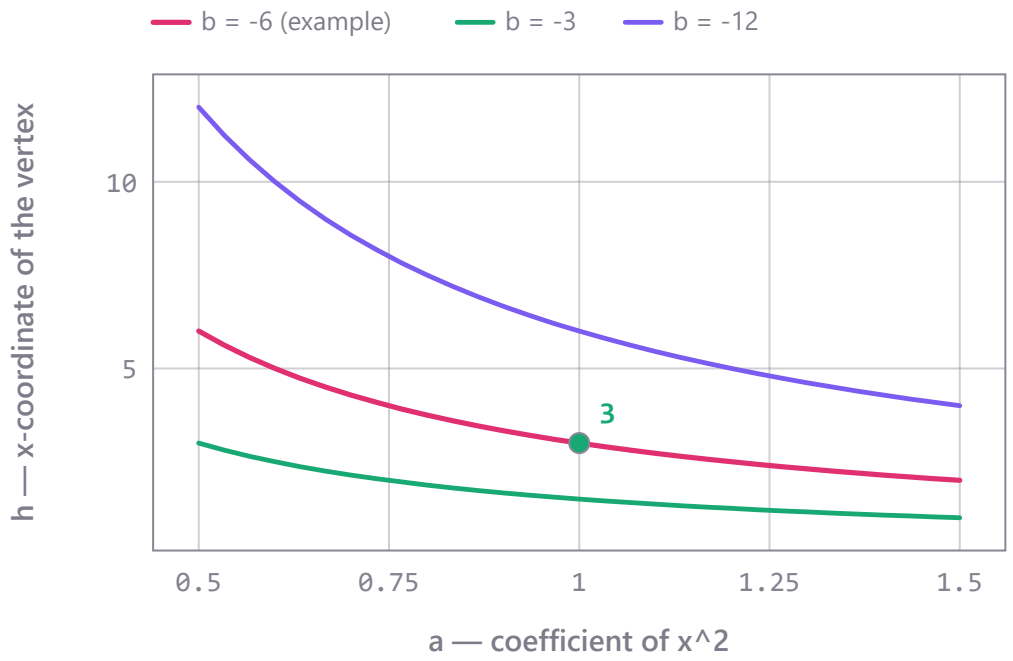
Write $ax^2 + bx + c$ as $a(x - h)^2 + k$, giving the vertex (h, k).

$$a(t-h)^2 + k = at^2 + bt + c$$

| input | meaning | condition |
|---|---|---|
| a | coefficient of $x^2$ | $a \neq 0$ |
| b | coefficient of x | |
| c | constant term | |

| output | meaning |
|---|---|
| h | x-coordinate of the vertex |
| k | y-coordinate of the vertex |

- **Method:** completing the square
- **Assumes:** $a \neq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)




h (x-coordinate of the vertex) against a (coefficient of x^2) from 0.5 to 1.5. One curve for each value of b (coefficient of x): -6, -3, -12. Other inputs as in the worked example: c = 5.0. 31 points per curve computed by math-foundations-ai999x; marked: the worked example, a = 1, h = 3.

```
math_foundations_ai999x.solve('vertex_form', a=1.0, b=-6.0, c=5.0)
# -> h = 3, k = -4   (SOLVED)
```

```
math_foundations_ai999x.solve('vertex_form', a=2.0, b=8.0, c=1.0)
# -> h = -2, k = -7   (SOLVED)
```

**Questions in words (documented forms), as executed:**

| question | outcome | result |
|---|---|---|
| Find the vertex of y = x^2 - 6x + 5. | SOLVED | h = 3, k = -4 |
| Write 2x^2 + 8x + 1 in vertex form. | SOLVED | h = -2, k = -7 |

## Proofs and derivations

Each derivation below was certified step by step: every step follows mechanically from the ones before and from what it rests on (a certified relation, an identity, a definition or a stated hypothesis), and a wrong variant of the result is refused. Ask in one of the documented forms, or call `math_foundations_ai999x.prove("<id>")`. A statement no certified derivation proves is OUT_OF_SCOPE; a conditional result asked without its hypotheses is NEEDS_INPUT and names them.

| documented question (executed) | id | steps |
|---|---|---|
| Prove that if A = l*w then A = l*w. | `area_of_rectangle` | 1 |
| Prove that (a + b) + c = a + (b + c). | `associative_property_of_addition` | 1 |
| Prove that if a*x + b*y = e and c*x + d*y = f then x = (e*d - b*f) / (a*d - b*c). | `cramer_rule_x_solution` | 8 |
| Prove that if a1*x + b1*y = c1 and a2*x + b2*y = c2 then x*(a1*b2 - a2*b1) = c1*b2 - c2*b1. | `determinant_cramer_two_variables` | 7 |
| Prove that a**3 - b**3 = (a - b)*(a**2 + a*b + b**2). | `difference_of_cubes_factorization` | 2 |
| Prove that (a + b) * (a - b) = a**2 - b**2. | `difference_of_squares` | 6 |
| Prove that a*(b + c) = a*b + a*c. | `distributive_property_verification` | 1 |
| Prove that if a*x + b*y = c and d*x + e*y = f then (a - k*d)*x + (b - k*e)*y = c - k*f. | `elimination_method_identity` | 4 |
| Prove that if a1*x + b1*y = c1 and a2*x + b2*y = c2 then x = (c1*b2 - c2*b1)/(a1*b2 - a2*b1). | `elimination_solution_two_variables` | 6 |
| Prove that if a*x + b = c then x = (c - b)/a. | `linear_equation_solution_formula` | 2 |
| Prove that if ((a*r1 + b)/(c*(r1 - r2)))/(x - r1) + ((a*r2 + b)/(c*(r2 - r1)))/(x - r2) = RHS_combined then (a*x + b)/(c*(x - r1)*(x - r2)) = ((a*r1 + b)/(c*(r1 - r2)))/(x - r1) + ((a*r2 + b)/(c*(r2 - r1)))/(x - r2). | `partial_fraction_decomposition_linear_factors` | 8 |
| Prove that (a + b)**2 = a**2 + 2*a*b + b**2. | `perfect_square_trinomial` | 6 |
| Prove that (a + b)*(c + d) = a*c + a*d + b*c + b*d. | `product_of_sums` | 2 |
| Prove that if h = -b/(2*a) and k = c - b**2/(4*a) then a*x**2 + b*x + c = a*(x - (-b/(2*a)))**2 + (c - b**2/(4*a)). | `quadratic_vertex_form_conversion` | 4 |
| Prove that if sqrt(x) = c then x = c**2. | `radical_equation_squaring` | 2 |
| Prove that (a - b)**2 = a**2 - 2*a*b + b**2. | `square_of_difference` | 4 |
| Prove that if a*x + b*y = c and x = d then a*d + b*y = c. | `substitution_method_identity` | 2 |
| Prove that if a1*x + b1*y = c1 and a2*x + b2*y = c2 then a1*((c1 - b1*((a1*c2 - a2*c1)/(a1*b2 - a2*b1)))/a1) + b1*((a1*c2 - a2*c1)/(a1*b2 - a2*b1)) = c1. | `substitution_solution_two_variables` | 7 |
| Prove that a**3 + b**3 = (a + b)*(a**2 - a*b + b**2). | `sum_of_cubes_factorization` | 5 |